Main Category	Category	Terms
Disease	OMIM_DISEASE	Glioblastoma, somatic
Disease	OMIM_DISEASE	Ovarian cancer, somatic
Disease	OMIM_DISEASE	Adenocarcinoma of lung, somatic
Disease	OMIM_DISEASE	Gastric cancer, somatic
Disease	OMIM_DISEASE	Visceral neuropathy, familial, 2, autosomal recessive
Disease	UP_KW_DISEASE	Disease variant
Functional Annotations	UP_KW_BIOLOGICAL_PROCESS	Transcription
Functional Annotations	UP_KW_BIOLOGICAL_PROCESS	Transcription regulation
Functional Annotations	UP_KW_CELLULAR_COMPONENT	Membrane
Functional Annotations	UP_KW_CELLULAR_COMPONENT	Nucleus
Functional Annotations	UP_KW_CELLULAR_COMPONENT	Cytoplasm
Functional Annotations	UP_KW_CELLULAR_COMPONENT	Cell projection
Functional Annotations	UP_KW_CELLULAR_COMPONENT	Endosome
Functional Annotations	UP_KW_CELLULAR_COMPONENT	Cell membrane
Functional Annotations	UP_KW_MOLECULAR_FUNCTION	Activator
Functional Annotations	UP_KW_MOLECULAR_FUNCTION	Kinase
Functional Annotations	UP_KW_MOLECULAR_FUNCTION	Receptor
Functional Annotations	UP_KW_MOLECULAR_FUNCTION	Transferase
Functional Annotations	UP_KW_MOLECULAR_FUNCTION	Tyrosine-protein kinase
Functional Annotations	UP_KW_PTM	Glycoprotein
Functional Annotations	UP_KW_PTM	Phosphoprotein
Functional Annotations	UP_KW_PTM	Disulfide bond
Functional Annotations	UP_SEQ_FEATURE	ACT_SITE:Proton acceptor

Main Category	Category	Terms
Functional Annotations	UP_SEQ_FEATURE	CARBOHYD:N-linked (GlcNAc...) asparagine
Functional Annotations	UP_SEQ_FEATURE	COMPBIAS:Basic residues
Functional Annotations	UP_SEQ_FEATURE	COMPBIAS:Pro residues
Functional Annotations	UP_SEQ_FEATURE	DOMAIN:Furin-like cysteine-rich
Functional Annotations	UP_SEQ_FEATURE	DOMAIN:Growth factor receptor
Functional Annotations	UP_SEQ_FEATURE	DOMAIN:Protein kinase
Functional Annotations	UP_SEQ_FEATURE	DOMAIN:Receptor L-domain
Functional Annotations	UP_SEQ_FEATURE	DOMAIN:Serine-threonine/tyrosine-protein kinase catalytic
Functional Annotations	UP_SEQ_FEATURE	MOTIF:Nuclear localization signal
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:LH->AA: Reduces dimerization with ERBB3.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:M->A: No effect on isoform production.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:M->A: Prevents synthesis of isoform 2.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:M->A: Prevents synthesis of isoform 3.
Functional Annotations	UP_SEQ_FEATURE	REGION:Disordered
Functional Annotations	UP_SEQ_FEATURE	REGION:Interaction with PIK3C2B
Functional Annotations	UP_SEQ_FEATURE	REGION:Required for interaction with KPNB1 and EEA1
Functional Annotations	UP_SEQ_FEATURE	TOPO_DOM:Cytoplasmic
Functional Annotations	UP_SEQ_FEATURE	TOPO_DOM:Extracellular
Functional Annotations	UP_SEQ_FEATURE	TRANSMEM:Helical
Gene Ontology	GOTERM_BP_DIRECT	protein phosphorylation
Gene Ontology	GOTERM_BP_DIRECT	signal transduction

Main Category	Category	Terms
Gene Ontology	GOTERM_BP_DIRECT	cell surface receptor signaling pathway
Gene Ontology	GOTERM_BP_DIRECT	enzyme-linked receptor protein signaling pathway
Gene Ontology	GOTERM_BP_DIRECT	cell surface receptor protein tyrosine kinase signaling pathway
Gene Ontology	GOTERM_BP_DIRECT	epidermal growth factor receptor signaling pathway
Gene Ontology	GOTERM_BP_DIRECT	nervous system development
Gene Ontology	GOTERM_BP_DIRECT	peripheral nervous system development
Gene Ontology	GOTERM_BP_DIRECT	heart development
Gene Ontology	GOTERM_BP_DIRECT	neuromuscular junction development
Gene Ontology	GOTERM_BP_DIRECT	motor neuron axon guidance
Gene Ontology	GOTERM_BP_DIRECT	cell population proliferation
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of gene expression
Gene Ontology	GOTERM_BP_DIRECT	Schwann cell development
Gene Ontology	GOTERM_BP_DIRECT	peptidyl-tyrosine phosphorylation
Gene Ontology	GOTERM_BP_DIRECT	neuron differentiation
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of cell growth
Gene Ontology	GOTERM_BP_DIRECT	regulation of microtubule-based process
Gene Ontology	GOTERM_BP_DIRECT	immature T cell proliferation in thymus
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of immature T cell proliferation in thymus
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of Rho protein signal transduction
Gene Ontology	GOTERM_BP_DIRECT	intracellular signal transduction
Gene Ontology	GOTERM_BP_DIRECT	ERBB2 signaling pathway

Main Category	Category	Terms
Gene Ontology	GOTERM_BP_DIRECT	ERBB2-ERBB3 signaling pathway
Gene Ontology	GOTERM_BP_DIRECT	ERBB2-EGFR signaling pathway
Gene Ontology	GOTERM_BP_DIRECT	ERBB2-ERBB4 signaling pathway
Gene Ontology	GOTERM_BP_DIRECT	wound healing
Gene Ontology	GOTERM_BP_DIRECT	regulation of cell population proliferation
Gene Ontology	GOTERM_BP_DIRECT	myelination
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of apoptotic process
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of MAP kinase activity
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of MAPK cascade
Gene Ontology	GOTERM_BP_DIRECT	phosphatidylinositol 3-kinase/protein kinase B signal transduction
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of translation
Gene Ontology	GOTERM_BP_DIRECT	regulation of angiogenesis
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of cell adhesion
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of transcription by RNA polymerase I
Gene Ontology	GOTERM_BP_DIRECT	cell development
Gene Ontology	GOTERM_BP_DIRECT	oligodendrocyte differentiation
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of epithelial cell proliferation
Gene Ontology	GOTERM_BP_DIRECT	regulation of developmental process
Gene Ontology	GOTERM_BP_DIRECT	regulation of ERK1 and ERK2 cascade
Gene Ontology	GOTERM_BP_DIRECT	cellular response to growth factor stimulus
Gene Ontology	GOTERM_BP_DIRECT	cellular response to epidermal growth factor stimulus

Main Category	Category	Terms
Gene Ontology	GOTERM_BP_DIRECT	semaphorin-plexin signaling pathway
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of protein targeting to membrane
Gene Ontology	GOTERM_BP_DIRECT	neurotransmitter receptor localization to postsynaptic specialization membrane
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of intracellular signal transduction
Gene Ontology	GOTERM_CC_DIRECT	semaphorin receptor complex
Gene Ontology	GOTERM_CC_DIRECT	nucleus
Gene Ontology	GOTERM_CC_DIRECT	nucleoplasm
Gene Ontology	GOTERM_CC_DIRECT	cytoplasm
Gene Ontology	GOTERM_CC_DIRECT	endosome
Gene Ontology	GOTERM_CC_DIRECT	early endosome
Gene Ontology	GOTERM_CC_DIRECT	cytosol
Gene Ontology	GOTERM_CC_DIRECT	plasma membrane
Gene Ontology	GOTERM_CC_DIRECT	basal plasma membrane
Gene Ontology	GOTERM_CC_DIRECT	endosome membrane
Gene Ontology	GOTERM_CC_DIRECT	endomembrane system
Gene Ontology	GOTERM_CC_DIRECT	membrane
Gene Ontology	GOTERM_CC_DIRECT	basolateral plasma membrane
Gene Ontology	GOTERM_CC_DIRECT	apical plasma membrane
Gene Ontology	GOTERM_CC_DIRECT	cytoplasmic vesicle
Gene Ontology	GOTERM_CC_DIRECT	neuromuscular junction
Gene Ontology	GOTERM_CC_DIRECT	ruffle membrane

Main Category	Category	Terms
Gene Ontology	GOTERM_CC_DIRECT	ERBB3:ERBB2 complex
Gene Ontology	GOTERM_CC_DIRECT	presynaptic membrane
Gene Ontology	GOTERM_CC_DIRECT	cell projection
Gene Ontology	GOTERM_CC_DIRECT	myelin sheath
Gene Ontology	GOTERM_CC_DIRECT	receptor complex
Gene Ontology	GOTERM_CC_DIRECT	perinuclear region of cytoplasm
Gene Ontology	GOTERM_MF_DIRECT	nucleotide binding
Gene Ontology	GOTERM_MF_DIRECT	RNA polymerase I core binding
Gene Ontology	GOTERM_MF_DIRECT	protein kinase activity
Gene Ontology	GOTERM_MF_DIRECT	protein tyrosine kinase activity
Gene Ontology	GOTERM_MF_DIRECT	transmembrane receptor protein tyrosine kinase activity
Gene Ontology	GOTERM_MF_DIRECT	transmembrane signaling receptor activity
Gene Ontology	GOTERM_MF_DIRECT	signaling receptor binding
Gene Ontology	GOTERM_MF_DIRECT	protein binding
Gene Ontology	GOTERM_MF_DIRECT	ATP binding
Gene Ontology	GOTERM_MF_DIRECT	coreceptor activity
Gene Ontology	GOTERM_MF_DIRECT	kinase activity
Gene Ontology	GOTERM_MF_DIRECT	transferase activity
Gene Ontology	GOTERM_MF_DIRECT	growth factor binding
Gene Ontology	GOTERM_MF_DIRECT	receptor tyrosine kinase binding
Gene Ontology	GOTERM_MF_DIRECT	identical protein binding

Main Category	Category	Terms
Gene Ontology	GOTERM_MF_DIRECT	ErbB-3 class receptor binding
Gene Ontology	GOTERM_MF_DIRECT	protein heterodimerization activity
General Annotations	CHROMOSOME	17
General Annotations	CYTOGENETIC_LOCATION	17q12
General Annotations	ENTREZ_GENE_SUMMARY	This gene encodes a member of the epidermal growth factor (EGF) receptor family of receptor tyrosine kinases. This protein has no ligand binding domain of its own and therefore cannot bind growth factors. However, it does bind tightly to other ligand-bound EGF receptor family members to form a heterodimer, stabilizing ligand binding and enhancing kinase-mediated activation of downstream signalling pathways, such as those involving mitogen-activated protein kinase and phosphatidylinositol-3 kinase. Allelic variations at amino acid positions 654 and 655 of isoform a (positions 624 and 625 of isoform b) have been reported, with the most common allele, Ile654/Ile655, shown here. Amplification and/or overexpression of this gene has been reported in numerous cancers, including breast and ovarian tumors. Alternative splicing results in several additional transcript variants, some encoding different isoforms and others that have not been fully characterized. [provided by RefSeq, Jul 2008]
General Annotations	UP_COMMENT_ACTIVITY_REGULATION	Activated by dimerization. Not activated by EGF, TGF-alpha and amphiregulin. Interaction with PTK6 increases its intrinsic kinase activity. {ECO:0000269|PubMed:18719096, ECO:0000269|PubMed:21454582}.
General Annotations	UP_COMMENT_ALTERNATIVE_PRODUCTS	Event=Alternative splicing, Alternative initiation; Named isoforms=6; Name=1; Synonyms=ERBB2, HER2; IsoId=P04626-1; Sequence=Displayed; Name=2; Synonyms=CTF-611; IsoId=P04626-2; Sequence=VSP_039249; Name=3; Synonyms=CTF-687; IsoId=P04626-3; Sequence=VSP_039250; Name=4; IsoId=P04626-4; Sequence=VSP_039248; Name=5; IsoId=P04626-5; Sequence=VSP_054787; Name=6; Synonyms=B; IsoId=P04626-6; Sequence=VSP_055902, VSP_055903, VSP_055904;
General Annotations	UP_COMMENT_CATALYTIC_ACTIVITY	Reaction=L-tyrosyl-[protein] + ATP = O-phospho-L-tyrosyl-[protein] + ADP + H(+); Xref=Rhea:RHEA:10596, Rhea:RHEA-COMP:10136, Rhea:RHEA- COMP:20101, ChEBI:CHEBI:15378, ChEBI:CHEBI:30616, ChEBI:CHEBI:46858, ChEBI:CHEBI:61978, ChEBI:CHEBI:456216; EC=2.7.10.1; Evidence={ECO:0000255|PROSITE-ProRule:PRU10028, ECO:0000269|PubMed:21454582};
General Annotations	UP_COMMENT_CATALYTIC_ACTIVITY	Reaction=L-tyrosyl-[protein] + ATP = O-phospho-L-tyrosyl-[protein] + ADP + H(+); Xref=Rhea:RHEA:10596, Rhea:RHEA-COMP:10136, Rhea:RHEA- COMP:20101,

Main Category	Category	Terms
		ChEBI:CHEBI:15378, ChEBI:CHEBI:30616, ChEBI:CHEBI:46858, ChEBI:CHEBI:61978, ChEBI:CHEBI:456216; EC=2.7.10.1; Evidence={ECO:0000256|ARBA:ARBA00051243};
General Annotations	UP_COMMENT_DISEASE	Gastric cancer (GASC) [MIM:613659]: A malignant disease which starts in the stomach, can spread to the esophagus or the small intestine, and can extend through the stomach wall to nearby lymph nodes and organs. It also can metastasize to other parts of the body. The term gastric cancer or gastric carcinoma refers to adenocarcinoma of the stomach that accounts for most of all gastric malignant tumors. Two main histologic types are recognized, diffuse type and intestinal type carcinomas. Diffuse tumors are poorly differentiated infiltrating lesions, resulting in thickening of the stomach. In contrast, intestinal tumors are usually exophytic, often ulcerating, and associated with intestinal metaplasia of the stomach, most often observed in sporadic disease. {ECO:0000269|PubMed:15457249, ECO:0000269|PubMed:17344846}. Note=The protein represented in this entry is involved in disease pathogenesis.
General Annotations	UP_COMMENT_DISEASE	Glioma (GLM) [MIM:137800]: Gliomas are benign or malignant central nervous system neoplasms derived from glial cells. They comprise astrocytomas and glioblastoma multiforme that are derived from astrocytes, oligodendrogliomas derived from oligodendrocytes and ependymomas derived from ependymocytes. {ECO:0000269|PubMed:15457249}. Note=The gene represented in this entry is involved in disease pathogenesis.
General Annotations	UP_COMMENT_DISEASE	Lung cancer (LNCR) [MIM:211980]: A common malignancy affecting tissues of the lung. The most common form of lung cancer is non-small cell lung cancer (NSCLC) that can be divided into 3 major histologic subtypes: squamous cell carcinoma, adenocarcinoma, and large cell lung cancer. NSCLC is often diagnosed at an advanced stage and has a poor prognosis. {ECO:0000269|PubMed:15457249}. Note=The gene represented in this entry is involved in disease pathogenesis.
General Annotations	UP_COMMENT_DISEASE	Note=Chromosomal aberrations involving ERBB2 may be a cause gastric cancer. Deletions within 17q12 region producing fusion transcripts with CDK12, leading to CDK12-ERBB2 fusion leading to truncated CDK12 protein not in-frame with ERBB2. {ECO:0000269|PubMed:21097718}.
General Annotations	UP_COMMENT_DISEASE	Ovarian cancer (OC) [MIM:167000]: The term ovarian cancer defines malignancies originating from ovarian tissue. Although many histologic types of ovarian tumors have been described, epithelial ovarian carcinoma is the most common form. Ovarian cancers are often asymptomatic and the recognized signs and symptoms, even of late-stage disease, are vague. Consequently, most patients are diagnosed with advanced disease. {ECO:0000269|PubMed:15457249,

Main Category	Category	Terms
		ECO:0000269|PubMed:17344846}. Note=The gene represented in this entry is involved in disease pathogenesis.
General Annotations	UP_COMMENT_DISEASE	Visceral neuropathy, familial, 2, autosomal recessive (VSCN2) [MIM:619465]: An autosomal recessive disorder characterized by intestinal dysmotility due to aganglionosis (Hirschsprung disease), hypoganglionosis, and/or chronic intestinal pseudoobstruction. Patients also show peripheral axonal neuropathy, hypotonia, mild developmental delay, unilateral ptosis, and sensorineural hearing loss. {ECO:0000269|PubMed:33497358}. Note=The disease is caused by variants affecting the gene represented in this entry.
General Annotations	UP_COMMENT_FUNCTION	In the nucleus is involved in transcriptional regulation. Associates with the 5'-TCAAATTC-3' sequence in the PTGS2/COX-2 promoter and activates its transcription. Implicated in transcriptional activation of CDKN1A; the function involves STAT3 and SRC. Involved in the transcription of rRNA genes by RNA Pol I and enhances protein synthesis and cell growth. {ECO:0000256|ARBA:ARBA00037619}.
General Annotations	UP_COMMENT_FUNCTION	In the nucleus is involved in transcriptional regulation. Associates with the 5'-TCAAATTC-3' sequence in the PTGS2/COX-2 promoter and activates its transcription. Implicated in transcriptional activation of CDKN1A; the function involves STAT3 and SRC. Involved in the transcription of rRNA genes by RNA Pol I and enhances protein synthesis and cell growth. {ECO:0000269|PubMed:10358079, ECO:0000269|PubMed:15380516, ECO:0000269|PubMed:21555369}.
General Annotations	UP_COMMENT_FUNCTION	Protein tyrosine kinase that is part of several cell surface receptor complexes, but that apparently needs a coreceptor for ligand binding. Essential component of a neuregulin-receptor complex, although neuregulins do not interact with it alone. GP30 is a potential ligand for this receptor. Regulates outgrowth and stabilization of peripheral microtubules (MTs). Upon ERBB2 activation, the MEMO1-RHOA-DIAPH1 signaling pathway elicits the phosphorylation and thus the inhibition of GSK3B at cell membrane. This prevents the phosphorylation of APC and CLASP2, allowing its association with the cell membrane. In turn, membrane-bound APC allows the localization of MACF1 to the cell membrane, which is required for microtubule capture and stabilization. {ECO:0000305}.
General Annotations	UP_COMMENT_INTERACTION	P04626; P00519: ABL1; NbExp=2; IntAct=EBI-641062, EBI-375543; P04626; P42684: ABL2; NbExp=6; IntAct=EBI-641062, EBI-1102694; P04626; P15309: ACP3; NbExp=2; IntAct=EBI-641062, EBI-1222012; P04626; P60709: ACTB; NbExp=10; IntAct=EBI-641062, EBI-353944; P04626; Q92625: ANKS1A; NbExp=2; IntAct=EBI-641062, EBI-1048612; P04626; O00213: APBB1; NbExp=2; IntAct=EBI-641062, EBI-81694; P04626; O75815: BCAR3; NbExp=2; IntAct=EBI-641062, EBI-702336; P04626; Q9HB71: CACYBP; NbExp=2;

Main Category	Category	Terms
		IntAct=EBI-641062, EBI-1047302; P04626; Q16543: CDC37; NbExp=5; IntAct=EBI-641062, EBI-295634; P04626; Q9NSE2: CISH; NbExp=4; IntAct=EBI-641062, EBI-617866; P04626; Q7Z7G1: CLNK; NbExp=2; IntAct=EBI-641062, EBI-7878194; P04626; P46109: CRKL; NbExp=2; IntAct=EBI-641062, EBI-910; P04626; Q93034: CUL5; NbExp=2; IntAct=EBI-641062, EBI-1057139; P04626; Q99704: DOK1; NbExp=2; IntAct=EBI-641062, EBI-1384360; P04626; Q8TEW6: DOK4; NbExp=2; IntAct=EBI-641062, EBI-6918542; P04626; Q15075: EEA1; NbExp=5; IntAct=EBI-641062, EBI-298113; P04626; P98172: EFNB1; NbExp=12; IntAct=EBI-641062, EBI-538287; P04626; P00533: EGFR; NbExp=25; IntAct=EBI-641062, EBI-297353; P04626; P04626: ERBB2; NbExp=12; IntAct=EBI-641062, EBI-641062; P04626; P21860: ERBB3; NbExp=30; IntAct=EBI-641062, EBI-720706; P04626; Q15303: ERBB4; NbExp=5; IntAct=EBI-641062, EBI-80371; P04626; Q9UJM3: ERRFI1; NbExp=4; IntAct=EBI-641062, EBI-2941912; P04626; P09769: FGR; NbExp=3; IntAct=EBI-641062, EBI-1383732; P04626; P06241: FYN; NbExp=2; IntAct=EBI-641062, EBI-515315; P04626; O75791: GRAP2; NbExp=2; IntAct=EBI-641062, EBI-740418; P04626; P62993: GRB2; NbExp=7; IntAct=EBI-641062, EBI-401755; P04626; Q14451: GRB7; NbExp=5; IntAct=EBI-641062, EBI-970191; P04626; P07900: HSP90AA1; NbExp=6; IntAct=EBI-641062, EBI-296047; P04626; P08238: HSP90AB1; NbExp=5; IntAct=EBI-641062, EBI-352572; P04626; P14625: HSP90B1; NbExp=2; IntAct=EBI-641062, EBI-359129; P04626; P11021: HSPA5; NbExp=3; IntAct=EBI-641062, EBI-354921; P04626; P46940: IQGAP1; NbExp=5; IntAct=EBI-641062, EBI-297509; P04626; P35568: IRS1; NbExp=2; IntAct=EBI-641062, EBI-517592; P04626; Q08881: ITK; NbExp=2; IntAct=EBI-641062, EBI-968552; P04626; P23458: JAK1; NbExp=2; IntAct=EBI-641062, EBI-1383438; P04626; Q14974: KPNB1; NbExp=14; IntAct=EBI-641062, EBI-286758; P04626; Q96JA1: LRIG1; NbExp=7; IntAct=EBI-641062, EBI-2865191; P04626; O75367: MACROH2A1; NbExp=6; IntAct=EBI-641062, EBI-2868511; P04626; O75367-3: MACROH2A1; NbExp=3; IntAct=EBI-641062, EBI-6250866; P04626; Q9UQF2: MAPK8IP1; NbExp=4; IntAct=EBI-641062, EBI-78404; P04626; Q13387: MAPK8IP2; NbExp=3; IntAct=EBI-641062, EBI-722813; P04626; P42679: MATK; NbExp=2; IntAct=EBI-641062, EBI-751664; P04626; Q9Y316: MEMO1; NbExp=6; IntAct=EBI-641062, EBI-1104564; P04626; O43639: NCK2; NbExp=2; IntAct=EBI-641062, EBI-713635; P04626; Q02297-7: NRG1; NbExp=2; IntAct=EBI-641062, EBI-2460927; P04626; O00750: PIK3C2B; NbExp=2; IntAct=EBI-641062, EBI-641107; P04626; P27986: PIK3R1; NbExp=12; IntAct=EBI-641062, EBI-79464; P04626; O00459: PIK3R2; NbExp=7; IntAct=EBI-641062, EBI-346930; P04626; Q92569: PIK3R3; NbExp=10; IntAct=EBI-641062, EBI-79893; P04626; P19174: PLCG1; NbExp=5; IntAct=EBI-641062, EBI-79387; P04626; P16885: PLCG2; NbExp=3; IntAct=EBI-641062, EBI-617403; P04626; O95602: POLR1A; NbExp=16; IntAct=EBI-641062, EBI-359472; P04626; Q13882:

Main Category	Category	Terms
		PTK6; NbExp=4; IntAct=EBI-641062, EBI-1383632; P04626; Q06124: PTPN11; NbExp=3; IntAct=EBI-641062, EBI-297779; P04626; Q05209: PTPN12; NbExp=4; IntAct=EBI-641062, EBI-2266035; P04626; Q99952: PTPN18; NbExp=6; IntAct=EBI-641062, EBI-1384210; P04626; Q99952-1: PTPN18; NbExp=5; IntAct=EBI-641062, EBI-12739708; P04626; P23467: PTPRB; NbExp=2; IntAct=EBI-641062, EBI-1265766; P04626; P08575: PTPRC; NbExp=2; IntAct=EBI-641062, EBI-1341; P04626; Q12913: PTPRJ; NbExp=2; IntAct=EBI-641062, EBI-2264500; P04626; Q15262: PTPRK; NbExp=2; IntAct=EBI-641062, EBI-474052; P04626; Q16827: PTPRO; NbExp=2; IntAct=EBI-641062, EBI-723739; P04626; Q15256: PTPRR; NbExp=2; IntAct=EBI-641062, EBI-2265659; P04626; P49792: RANBP2; NbExp=3; IntAct=EBI-641062, EBI-973138; P04626; P20936: RASA1; NbExp=8; IntAct=EBI-641062, EBI-1026476; P04626; O95980: RECK; NbExp=4; IntAct=EBI-641062, EBI-2823742; P04626; Q9NP31: SH2D2A; NbExp=2; IntAct=EBI-641062, EBI-490630; P04626; P29353: SHC1; NbExp=11; IntAct=EBI-641062, EBI-78835; P04626; P98077: SHC2; NbExp=3; IntAct=EBI-641062, EBI-7256023; P04626; Q92529: SHC3; NbExp=2; IntAct=EBI-641062, EBI-79084; P04626; Q9H6Q3: SLA2; NbExp=2; IntAct=EBI-641062, EBI-1222854; P04626; O15524: SOCS1; NbExp=2; IntAct=EBI-641062, EBI-968198; P04626; P12931: SRC; NbExp=11; IntAct=EBI-641062, EBI-621482; P04626; P42224: STAT1; NbExp=3; IntAct=EBI-641062, EBI-1057697; P04626; P40763: STAT3; NbExp=9; IntAct=EBI-641062, EBI-518675; P04626; P31948: STIP1; NbExp=2; IntAct=EBI-641062, EBI-1054052; P04626; Q7KZ85: SUPT6H; NbExp=2; IntAct=EBI-641062, EBI-2515547; P04626; P43405: SYK; NbExp=7; IntAct=EBI-641062, EBI-78302; P04626; Q9Y490: TLN1; NbExp=3; IntAct=EBI-641062, EBI-2462036; P04626; Q63HR2: TNS2; NbExp=4; IntAct=EBI-641062, EBI-949753; P04626; Q68CZ2: TNS3; NbExp=2; IntAct=EBI-641062, EBI-1220488; P04626; Q96D37: VAV1; NbExp=2; IntAct=EBI-641062, EBI-7875353; P04626; P52735: VAV2; NbExp=3; IntAct=EBI-641062, EBI-297549; P04626; O14980: XPO1; NbExp=3; IntAct=EBI-641062, EBI-355867; P04626; P62258: YWHAE; NbExp=2; IntAct=EBI-641062, EBI-356498;
General Annotations	UP_COMMENT_MISCELLANEOUS	[Isoform 2]: Produced by alternative initiation at Met-611 of isoform 1. {ECO:0000305}.
General Annotations	UP_COMMENT_MISCELLANEOUS	[Isoform 3]: Produced by alternative initiation at Met-687 of isoform 1. {ECO:0000305}.
General Annotations	UP_COMMENT_MISCELLANEOUS	[Isoform 4]: Produced by alternative splicing of isoform 1. {ECO:0000305}.
General Annotations	UP_COMMENT_POLYMORPHISM	There are four alleles due to the variations in positions 654 and 655. Allele B1 (Ile-654/Ile-655) has a frequency of 0.782; allele B2 (Ile-654/Val-655) has a frequency of 0.206; allele B3 (Val- 654/Val-655) has a frequency of 0.012.

Main Category	Category	Terms
General Annotations	UP_COMMENT_PTM	Autophosphorylated. Autophosphorylation occurs in trans, i.e. one subunit of the dimeric receptor phosphorylates tyrosine residues on the other subunit (Probable). Ligand-binding increases phosphorylation on tyrosine residues (PubMed:27134172, PubMed:33497358). Signaling via SEMA4C promotes phosphorylation at Tyr-1248 (PubMed:17554007). Dephosphorylated by PTPN12 (PubMed:27134172). {ECO:0000269|PubMed:17554007, ECO:0000269|PubMed:27134172, ECO:0000269|PubMed:33497358, ECO:0000305}.
General Annotations	UP_COMMENT_SIMILARITY	Belongs to the protein kinase superfamily. Tyr protein kinase family. EGF receptor subfamily. {ECO:0000255|PROSITE- ProRule:PRU00159}.
General Annotations	UP_COMMENT_SIMILARITY	Belongs to the protein kinase superfamily. Tyr protein kinase family. EGF receptor subfamily. {ECO:0000256|PIRNR:PIRNR000619}.
General Annotations	UP_COMMENT_SUBCELLULAR_LOCATION	Cell envelope {ECO:0000256|ARBA:ARBA00004196}. Membrane {ECO:0000256|ARBA:ARBA00004479}; Single-pass type I membrane protein {ECO:0000256|ARBA:ARBA00004479}.
General Annotations	UP_COMMENT_SUBCELLULAR_LOCATION	Cell membrane {ECO:0000269|PubMed:32381043}; Single-pass type I membrane protein {ECO:0000255}. Cell projection, ruffle membrane {ECO:0000269|PubMed:34380438}; Single-pass type I membrane protein {ECO:0000255}. Note=Internalized from the cell membrane in response to EGF stimulation. {ECO:0000269|PubMed:32381043}.
General Annotations	UP_COMMENT_SUBCELLULAR_LOCATION	Cell projection, ruffle membrane {ECO:0000256|ARBA:ARBA00004199}; Single-pass type I membrane protein {ECO:0000256|ARBA:ARBA00004199}. Cytoplasm, perinuclear region {ECO:0000256|ARBA:ARBA00004556}. Early endosome {ECO:0000256|ARBA:ARBA00004412}. Nucleus {ECO:0000256|ARBA:ARBA00004123}.
General Annotations	UP_COMMENT_SUBCELLULAR_LOCATION	Endomembrane system {ECO:0000256|ARBA:ARBA00004308}. Membrane {ECO:0000256|ARBA:ARBA00004479}; Single-pass type I membrane protein {ECO:0000256|ARBA:ARBA00004479}.
General Annotations	UP_COMMENT_SUBCELLULAR_LOCATION	Membrane {ECO:0000256|ARBA:ARBA00004479}; Single- pass type I membrane protein {ECO:0000256|ARBA:ARBA00004479}.
General Annotations	UP_COMMENT_SUBCELLULAR_LOCATION	[Isoform 1]: Cell membrane {ECO:0000269|PubMed:31138794, ECO:0000269|PubMed:33497358}; Single-pass type I membrane protein {ECO:0000255}. Early endosome {ECO:0000269|PubMed:31138794}. Cytoplasm, perinuclear region. Nucleus. Note=Translocation to the nucleus requires endocytosis, probably endosomal sorting and is mediated by importin beta-1/KPNB1. Also detected in VPS35-positive endosome-to-TGN retrograde vesicles (PubMed:31138794). {ECO:0000269|PubMed:31138794}.

Main Category	Category	Terms
General Annotations	UP_COMMENT_SUBCELLULAR_LOCATION	[Isoform 2]: Cytoplasm. Nucleus.
General Annotations	UP_COMMENT_SUBCELLULAR_LOCATION	[Isoform 3]: Cytoplasm. Nucleus.
General Annotations	UP_COMMENT_SUBUNIT	Homodimer (PubMed:21454582). Heterodimer with EGFR, ERBB3 and ERBB4 (PubMed:10358079, PubMed:15093539, PubMed:16978839, PubMed:21190959). Part of a complex with EGFR and either PIK3C2A or PIK3C2B. May interact with PIK3C2B when phosphorylated on Tyr-1196 (PubMed:10805725). Interacts with PLXNB1 (PubMed:15210733). Interacts (when phosphorylated on Tyr-1248) with MEMO1 (PubMed:15156151). Interacts with MUC1; the interaction is enhanced by heregulin (HRG) (PubMed:12939402). Interacts (when phosphorylated on Tyr-1139) with GRB7 (via SH2 domain) (PubMed:12975581). Interacts (when phosphorylated on Tyr-1248) with ERBIN (PubMed:12444095). Interacts with KPNB1, RANBP2, EEA1, CRM1 and CLTC (PubMed:16314522). Interacts with PTK6 (PubMed:18719096). Interacts with RPA194 and ACTB (PubMed:21555369). Interacts with PRKCABP, SRC and MYOC (By similarity). Interacts (preferentially with the tyrosine phosphorylated form) with CPNE3; this interaction occurs at the cell membrane and is increased in a growth factor heregulin-dependent manner (PubMed:20010870). Interacts with HSP90AA1 and HSP90AB1 in an ATP-dependent manner; the interaction suppresses ERBB2 kinase activity (PubMed:26517842). Interacts with SORL1; this interaction regulates ERBB2 subcellular distribution by promoting its recycling after internalization from endosomes back to the plasma membrane, hence stimulates ERBB2-mediated signaling (PubMed:31138794). Interacts with SH3BGRL (PubMed:32381043). Interacts with ROR1 (PubMed:36949068). {ECO:0000250|UniProtKB:P70424, ECO:0000269|PubMed:10358079, ECO:0000269|PubMed:10805725, ECO:0000269|PubMed:12939402, ECO:0000269|PubMed:12975581, ECO:0000269|PubMed:15093539, ECO:0000269|PubMed:15156151, ECO:0000269|PubMed:15210733, ECO:0000269|PubMed:16314522, ECO:0000269|PubMed:16978839, ECO:0000269|PubMed:18719096, ECO:0000269|PubMed:20010870, ECO:0000269|PubMed:21190959, ECO:0000269|PubMed:21454582, ECO:0000269|PubMed:21555369, ECO:0000269|PubMed:26517842, ECO:0000269|PubMed:31138794, ECO:0000269|PubMed:32381043, ECO:0000269|PubMed:36949068}.
General Annotations	UP_COMMENT_TISSUE_SPECIFICITY	Expressed in a variety of tumor tissues including primary breast tumors and tumors from small bowel, esophagus, kidney and mouth. {ECO:0000269|PubMed:15380516}.
Identifiers	ENTREZ_GENE_ID	2064
Identifiers	ENSEMBL_GENE_ID	ENSG00000141736

Main Category	Category	Terms
Identifiers	UNIPROT_ID	X5DNK3_HUMAN
Identifiers	UNIPROT_ID	X5DQX2_HUMAN
Identifiers	UNIPROT_ID	Q9NP09_HUMAN
Identifiers	UNIPROT_ID	Q4H1F1_HUMAN
Identifiers	UNIPROT_ID	Q4H1F2_HUMAN
Identifiers	UNIPROT_ID	L8E8G2_HUMAN
Identifiers	UNIPROT_ID	J3QRX1_HUMAN
Identifiers	UNIPROT_ID	J3QRJ7_HUMAN
Identifiers	UNIPROT_ID	J3QL83_HUMAN
Identifiers	UNIPROT_ID	J3QLU9_HUMAN
Identifiers	UNIPROT_ID	J3QLV2_HUMAN
Identifiers	UNIPROT_ID	J3KRI9_HUMAN
Identifiers	UNIPROT_ID	J3KTI5_HUMAN
Identifiers	UNIPROT_ID	F5H1T4_HUMAN
Identifiers	UNIPROT_ID	ERBB2_HUMAN
Identifiers	UNIPROT_ID	E2I6F8_HUMAN
Identifiers	UNIPROT_ID	B4DLA2_HUMAN
Identifiers	UNIPROT_ID	B4DTR1_HUMAN
Identifiers	UNIPROT_ID	A0A7T8JKQ4_HUMAN
Identifiers	UNIPROT_ID	A0A0R9RWK2_HUMAN
Interactions	DRUGBANK	Trastuzumab
Interactions	DRUGBANK	Lapatinib
Interactions	DRUGBANK	IGN311
Interactions	DRUGBANK	Trastuzumab emtansine
Interactions	DRUGBANK	Varlitinib
Interactions	DRUGBANK	AV-412
Interactions	DRUGBANK	Pertuzumab
Interactions	DRUGBANK	Afatinib
Interactions	DRUGBANK	Tucatinib
Interactions	DRUGBANK	Tesevatinib
Interactions	DRUGBANK	Fostamatinib
Interactions	DRUGBANK	Brigatinib

Main Category	Category	Terms
Interactions	DRUGBANK	Margetuximab
Interactions	DRUGBANK	Zanubrutinib
Interactions	UP_KW_LIGAND	ATP-binding
Interactions	UP_KW_LIGAND	Nucleotide-binding
Literature	GENERIF_SUMMARY	peptide library generation for HER2/neu ligand identification
Literature	GENERIF_SUMMARY	HER-2 gene amplification and protein overexpression has been associated with increased risk of advanced-stage breast cancer and poor prognosis.
Literature	GENERIF_SUMMARY	Epidermal growth factor receptor (HER1) tyrosine kinase inhibitor ZD1839 (Iressa) inhibits HER2/neu (erbB2)-overexpressing breast cancer cells in vitro and in vivo.
Literature	GENERIF_SUMMARY	overexpression not associated with in vitro drug resistance to CMF or FEC chemotherapy combinations in primary breast cancer
Literature	GENERIF_SUMMARY	Expression and gene copy number analysis of ERBB2 oncogene in prostate cancer.
Literature	GENERIF_SUMMARY	Identification of a minimal c-erbB-2 promoter region that mediates preferential expression of a linked foreign gene in human breast cancer cells
Literature	GENERIF_SUMMARY	NH(2)-terminal truncated HER-2 protein but not full-length receptor is associated with nodal metastasis in human breast cancer
Literature	GENERIF_SUMMARY	HER-2-positive breast carcinomas as a particular subset with peculiar clinical behaviors.
Literature	GENERIF_SUMMARY	Characterization of the HER-2/neu oncogene by immunohistochemical and fluorescence in situ hybridization analysis in oral and oropharyngeal squamous cell carcinoma
Literature	GENERIF_SUMMARY	Automated electrorotation to reveal dielectric variations related to HER-2/neu overexpression in MCF-7 sublines
Literature	GENERIF_SUMMARY	ErbB-2 demonstrates a strong tendency toward stable self-association of transmembrane domains identifiable as coexisting populations of peptides whose associations are thought to modulate signal transduction.
Literature	GENERIF_SUMMARY	lack of amplification in nasopharyngeal neoplasms
Literature	GENERIF_SUMMARY	feasibility of fluorescence in situ hybridization analysis of HER-2/neu amplification in oral mucosa brushings and to compare the HER-2/neu status with the history and smoking and drinking habits of healthy subjects
Literature	GENERIF_SUMMARY	Data suggest that amplification of Her-2/neu appears to be mainly involved in initiation of breast oncogenesis and that its role in progression of breast cancers is

Main Category	Category	Terms
		uncertain.
Literature	GENERIF_SUMMARY	Except in a certain subset of cases, aneusomy 17 probably is not a significant factor for HER-2/neu protein expression or for clinical assessment of HER-2/neu status.
Literature	GENERIF_SUMMARY	Bombesin antagonists inhibit growth of MDA-MB-435 estrogen-independent breast cancers and decrease the expression of the ErbB-2/HER-2 oncoprotein and c-jun and c-fos oncogenes
Literature	GENERIF_SUMMARY	Increased c-erbB-2 expression contributes to the development of cholangiocarcinogenesis into an advanced stage associated with tumour metastasis.
Literature	GENERIF_SUMMARY	expression of cyclooxygenase 2 in HE-2 positive breast cancer
Literature	GENERIF_SUMMARY	Epidermal growth factor contains both positive and negative determinants for interaction with ErbB-2/ErbB-3 heterodimers
Literature	GENERIF_SUMMARY	expression correlated with increased event-free and overall survival in high-grade osteosarcoma
Literature	GENERIF_SUMMARY	ErbB-beta-catenin complexes are associated with human infiltrating ductal breast and murine mammary tumor virus (MMTV)-Wnt-1 and MMTV-c-Neu transgenic carcinomas.
Literature	GENERIF_SUMMARY	p53 mutational pathway may favor selection for ErbB2 gene amplification during tumor progression in breast cancer
Literature	GENERIF_SUMMARY	Mechanism of 17-beta-estradiol-induced Erk1/2 activation in breast cancer cells. A role for HER2 AND PKC-delta
Literature	GENERIF_SUMMARY	Cyclooxygenase-2 (COX-2), epidermal growth factor receptor (EGFR), and Her-2/neu are expressed in ovarian cancer.
Literature	GENERIF_SUMMARY	gene is essential in preventing dilated cardiomyopathy
Literature	GENERIF_SUMMARY	overexpression seen in 16% of NSCLC tumors, most frequently in adenocarcinomas and large cell carcinomas
Literature	GENERIF_SUMMARY	identification of a dimerization motif for ErbB homomeric association
Literature	GENERIF_SUMMARY	ErbB2 activation of ESX gene expression
Literature	GENERIF_SUMMARY	ErbB2 membrane RTK can confer resistance to taxol-induced apoptosis by directly phosphorylating Cdc2.
Literature	GENERIF_SUMMARY	HER-2/neu peptides can activate T cells in draining lymph nodes from women with invasive breast cancer.
Literature	GENERIF_SUMMARY	S-erbB-2 serum levels above 40 U/ml independently predicted unfavorable response to 2d-line hormone or

Main Category	Category	Terms
		chemotherapy in advanced metastatic breast cancer. 1st-line drugs may select for overexpression of erbB-2 genes and lesser response to 2d-line drugs.
Literature	GENERIF_SUMMARY	Expression of c-erbB-2 in node negative breast cancer does not correlate with estrogen receptor status, predictors of hormone responsiveness, or PCNA expression
Literature	GENERIF_SUMMARY	Role of the N-terminus of epidermal growth factor in binding studied by phage display.
Literature	GENERIF_SUMMARY	Her-2/neu expression and gene amplification in gastrinomas is correlated with tumor biology, growth, and aggressiveness.
Literature	GENERIF_SUMMARY	Overexpression of erbb2 increases expression of VEGF A, C and D in breast carcinoma
Literature	GENERIF_SUMMARY	the expression of mitochondria-encoded COXII is HRG-responsive. The levels of ErbB2 expression are decisive for the diverse biological activities of HRG.
Literature	GENERIF_SUMMARY	ERBB2 binds to the SH2 domain of CHK and inhibits cell growth in human breast tumor cell lines
Literature	GENERIF_SUMMARY	The expression of this molecule and its correlation with prognostic markers in patients with head and neck tumors
Literature	GENERIF_SUMMARY	ErbB1 and ErbB2 employ different mechanisms of plasma membrane targeting during keratinocyte differentiation; cytoskeletal association may facilitate the coupling of activated ErbB1 and ERK.
Literature	GENERIF_SUMMARY	ERBB-2 overexpression and cyclooxygenase-2 up-regulation in human cholangiocarcinoma and risk conditions.
Literature	GENERIF_SUMMARY	HER-2 tyrosine phosphorylation is a component of cytotoxic t-lymphocyte stimulation by tumor cells
Literature	GENERIF_SUMMARY	ErbB2 overexpression in an ovarian cancer cell line confers sensitivity to the inhibitor geldanamycin (HSP90)
Literature	GENERIF_SUMMARY	uPA and its receptor are not upregulated by the circulating fraction of this proto-oncogene in advanced NSCLC
Literature	GENERIF_SUMMARY	Data report the identification of signaling pathways required for suppression of integrin alpha2beta1 function by c-erbB2.
Literature	GENERIF_SUMMARY	Overexpression of c-erbB-2 was significantly associated with poor survival & can serve as a prognostic marker. C-erbB-2 is related with tumor progression in colorectal cancer which can be seen on protein level & reflects chromosomal gain at the 17q locus.
Literature	GENERIF_SUMMARY	Wound closure could be delayed up to 50% by antisense c-erbB2.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	ILC did not display ERBB2 overexpression . The observed discrepancy in the pattern of the human oncogenes CCND1 and ERBB2, which are involved in the process of carcinogenesis of ductal tumors.
Literature	GENERIF_SUMMARY	significant HER2 expression seen in high-grade, muscle-invasive urothelial carcinoma; but HER2 expression in context of paclitaxel-based chemotherapy associated with significantly reduced risk of death
Literature	GENERIF_SUMMARY	Prognostic value of the quantitative measurement of the oncoprotein p185(Her-2/neu) in a group of patients with breast cancer and positive node involvement.
Literature	GENERIF_SUMMARY	HER2 expression is regulated by the interaction of DRIP130 and ESX
Literature	GENERIF_SUMMARY	c-myc and c-erbB2 amplification in breast cancer
Literature	GENERIF_SUMMARY	differential gene expression patterns in positive and negative breast cancer cell lines and tissues
Literature	GENERIF_SUMMARY	High-level coexpression of ERBB2 and ERBB4 was significantly related to tumor proliferative activity in ependymoma
Literature	GENERIF_SUMMARY	marked intratumoral heterogeneity of c-myc and cyclinD1 but not this gene in breast cancer
Literature	GENERIF_SUMMARY	C-erb-2 overexpression is not statistically related to either proliferation or cancer specific death in upper urinary tract urothelial tumors.
Literature	GENERIF_SUMMARY	contribution of HER2 transcription factor binding site to ERBB2 overexpression
Literature	GENERIF_SUMMARY	Cell proliferation, nuclear ploidy, and EGFr and HER2/neu tyrosine kinase oncoproteins in infiltrating ductal breast carcinoma.
Literature	GENERIF_SUMMARY	Kaplan-Meier curves showed a significantly worse disease-related survival (p = 0.034) in patients with HER2-overexpressing tumors compared to those without HER2 overexpression.
Literature	GENERIF_SUMMARY	Signaling by HER-2/neu has a critical role in the early stages of breast tumorigenesis.
Literature	GENERIF_SUMMARY	relationship between the mast cell density and the context of clinicopathological parameters and expression of p185, estrogen receptor, and proliferating cell nuclear antigen in gastric carcinoma
Literature	GENERIF_SUMMARY	selective binding and sequestration of this residue in its unphosphorylated state by the Erbin PDZ provides a novel mechanism for regulation of the ErbB2-mediated signaling and oncogenicity
Literature	GENERIF_SUMMARY	may be activated in the early stage of pathogenesis of cervical carcinoma in geriatric patients
Literature	GENERIF_SUMMARY	Computational exploration of conformation space of

Main Category	Category	Terms
		erbB2 homodimer transmembrane segments revealed 2 stable conformations
Literature	GENERIF_SUMMARY	Overexpressed erbB-2 can cause EGF independent transformation of nonmalignant MCF-10A breast cells.
Literature	GENERIF_SUMMARY	Differential expression follows BCG immunotherapy in superficial papillary transitional cell carcinoma of the bladder
Literature	GENERIF_SUMMARY	role of overexpression in lung cancer development in conjunction with erb-B-3 overexpression
Literature	GENERIF_SUMMARY	identification of domain motifs required for geldanamycin-induced degradation
Literature	GENERIF_SUMMARY	Transcriptional analysis reveals a molecular connection to fatty acid synthesis.
Literature	GENERIF_SUMMARY	up-regulates S100A4 and several other prometastatic genes in medulloblastoma
Literature	GENERIF_SUMMARY	examination of amplification in breast cancer
Literature	GENERIF_SUMMARY	reduced ER/PR expression may be one mechanism to explain the relative resistance of HER-2/neu-positive
Literature	GENERIF_SUMMARY	Her-2/neu overexpression in patients with osteosarcoma is associated with an increased risk of lung metastasis
Literature	GENERIF_SUMMARY	UROC28 mRNA expression was greater in breast cancers than in noncancerous tissues (p
Literature	GENERIF_SUMMARY	HER-2/neu overexpression does not adversely and may favorably influence response to adjuvant oophorectomy and tamoxifen treatment in patients with estrogen receptor-positive breast tumors
Literature	GENERIF_SUMMARY	ErbB2 is a target of CHIP ubiquitin ligase activity; CHIP E3 controls both the association of Hsp70/Hsp90 chaperones with ErbB2 and the down-regulation of ErbB2 induced by inhibitors of Hsp90
Literature	GENERIF_SUMMARY	Tumors expressing this protein produced more in the affected breast than in the unaffected breast in unilateral breast cancer.
Literature	GENERIF_SUMMARY	results may suggest the contribution of c-erbB-1 molecule in progression of gastric carcinomas in southern Iranian patients
Literature	GENERIF_SUMMARY	results demonstrate that tamoxifen resistant MCF-7 cell growth is mediated by the autocrine release and action of an epidermal growth factor receptor-specific ligand inducing preferential epidermal growth factor receptor/ c-erbB2 dimerization
Literature	GENERIF_SUMMARY	Inhibition of erbB receptor family members protects HaCaT keratinocytes from ultraviolet-B-induced apoptosis. This inhibition was specific for the erbB receptor family and specific for ultraviolet-B-induced

Main Category	Category	Terms
		apoptosis.
Literature	GENERIF_SUMMARY	Expression of vascular endothelial growth factor in invasive ductal carcinoma of the breast and the relation to angiogenesis and p53 and HER-2/neu protein expression.
Literature	GENERIF_SUMMARY	Crystal structures of the entire extracellular regions of rat HER2 at 2.4 A and human HER2 complexed with the Herceptin antigen-binding fragment (Fab) at 2.5 A
Literature	GENERIF_SUMMARY	model in which ErbB2 is already in the activated conformation and ready to interact with other ligand-activated ErbB receptors
Literature	GENERIF_SUMMARY	Segregation of receptor and ligand regulates activation of epithelial growth factor receptor; following a mechanical injury, heregulin-alpha activates erbB2 in cells at the edge of the wound, and this process hastens restoration of epithelial integrity
Literature	GENERIF_SUMMARY	erbB-2-mediated invasiveness is dependent on p38MAPK induces cell surface alpha4 integrin downregulation
Literature	GENERIF_SUMMARY	Expression of c-erbB-2 occurs more often in breast cancer cases than in cervical cancer and shows a significant relationship with histologic grading of the tumor, in a study group of Indian women.
Literature	GENERIF_SUMMARY	EGFR/HER2 heterodimers traffic as single entities; levels of HER2 in normal cells are barely at the threshold necessary to drive efficient heterodimerization
Literature	GENERIF_SUMMARY	T cells transduced with the same chimeric gene might be utilised in the treatment of patients with HER-2/neu(+) tumours.
Literature	GENERIF_SUMMARY	HER2 expression together with PgR negativity may serve as the counterpart of the proliferation marker in predicting the in vivo response to DXR.
Literature	GENERIF_SUMMARY	This gene is amplified in breast cancer.
Literature	GENERIF_SUMMARY	ERBB-2 signaling is regulated by a MUC4/sialomucin complex
Literature	GENERIF_SUMMARY	Phosphatidylinositol 3'-kinase signals cell neoplastic transformation by erbb2.
Literature	GENERIF_SUMMARY	results demonstrated significant positive staining of c-erbB-2 in the salivary tumorigenic tissue but not in the surrounding non-tumorigenic tissue, pointing to a biological role in the tumorigenic process
Literature	GENERIF_SUMMARY	Expression of epidermal growth factor receptor, erbB2, and erbB3, but not erbB4, was detected throughout the epidermis. Labeling for erbB2 and erbB3 accentuated in upper spinous layers
Literature	GENERIF_SUMMARY	in MCF-7 breast cancer cell line, elevated MAPK activity

Main Category	Category	Terms
		results from enhanced ERBB2 expression in the estrogen deprived cells
Literature	GENERIF_SUMMARY	MUC1 expression was inhibited by NGF treatment in breast cancer tumor cell lines, suggesting that its expression can be regulated by signals resulting from the homodimerization of Erb-B2.
Literature	GENERIF_SUMMARY	no evidence that the phosphorylation of TuM2-PK is initiated by the oncoprotein HER-2/neu in breast cancer patients
Literature	GENERIF_SUMMARY	Levels of this protein in blood are positive predictors of the treatment of breast cancers with herceptin.
Literature	GENERIF_SUMMARY	Levels of this receptor in blood serve as a predictive marker for the clinical course of breast cancer.
Literature	GENERIF_SUMMARY	Data support that EGFR and HER-2/neu play an important role in cell cycle control in ductal carcinoma in situ.
Literature	GENERIF_SUMMARY	Aberrations of these genes is regulated in tumor microenvironments.
Literature	GENERIF_SUMMARY	ErbB2/ErbB3 dimer functions as an oncogenic unit to drive breast tumor cell proliferation.
Literature	GENERIF_SUMMARY	HER-2 expression and cell proliferation seem to provide prognostic information for node-negative breast cancer patients.
Literature	GENERIF_SUMMARY	overexpression of HER-2/neu gene could induce NF-kappaB activity in human breast cancer cells
Literature	GENERIF_SUMMARY	The presence of increased levels of HER2/neu in synovial sarcoma is associated with a more favorable clinical course.
Literature	GENERIF_SUMMARY	her2/neu contributes to Wilms tumor angiogenesis in vivo by regulating VEGF
Literature	GENERIF_SUMMARY	ERBB2 and ERBB3 expression is inhibited by quercetin, resulting in decreased autophosphorylation and cell growth
Literature	GENERIF_SUMMARY	The review evaluates the status of serum-based testing for circulating HER-2/neu receptor protein and its ability to predict disease outcome and therapy response.
Literature	GENERIF_SUMMARY	expression of nm 23 and c-erbB-2 in primary tumor and metastases of colorectal adenocarcinoma showed that the incidence and expression of both protein markers in primary tumor tissue tended to increase after the appearance of liver metastases
Literature	GENERIF_SUMMARY	ER81 is acetylated by two coactivators/ acetyltransferases, p300 and p300- and CBP-associated factor (P/CAF) . Whereas p300 acetylates two lysine residues (K33 and K116) within the ER81 N-terminal transactivation domain, P/CAF targets only K116.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	different transcriptional and post-transcriptional mechanisms are responsible for the increased levels of erbB-2 transcript and protein in breast and non-breast cancer cells
Literature	GENERIF_SUMMARY	our results suggest that estradiol treatment results in binding to membrane ER-alpha and interaction with a heterodimer containing ErbB2, leading to tyrosine phosphorylation.
Literature	GENERIF_SUMMARY	overexpression of HER2/neu is associated with the development of muscle-invasive transitional cell carcinoma of the bladder
Literature	GENERIF_SUMMARY	A single nucleotide polymorphism in the transmembrane domain coding region of HER-2 is associated with development and malignant phenotype of gastric cancer
Literature	GENERIF_SUMMARY	High levels of serum HER-2/neu reflect aggressiveness of metastatic breast cancer
Literature	GENERIF_SUMMARY	HER2 has a role in reducing apoptosis in breast cancer cells treated wtih 4-HPR
Literature	GENERIF_SUMMARY	In cervical carcinogenesis, the overexpression of bcl-2 protein may be activated in the early stage of pathogenesis of cervical carcinoma. Expression of c-erbB2 was the signal of cell malignant change.
Literature	GENERIF_SUMMARY	role of urokinase plasminogen activator receptor (uPAR) and c-erbB-2 in breast and ovarian cancer
Literature	GENERIF_SUMMARY	data suggest that HER-2 amplifications are frequently linked to alterations of the TOP2A gene in bladder cancer
Literature	GENERIF_SUMMARY	value of this tumor marker regarding relapse, metastasis and death in resectable non-small cell lung cancer
Literature	GENERIF_SUMMARY	a detailed characterization of the molecular events occurring at the ERBB2 amplicon in primary breast tumors
Literature	GENERIF_SUMMARY	This study highlights a new pathway by which HER-2 may modify cancer behaviour.
Literature	GENERIF_SUMMARY	Immunization Her-2/neu peptides successfully induced humoral immune response with anti-tumor activity in an animal model of breast cancer.
Literature	GENERIF_SUMMARY	findings suggest that the coexpression of c-erbB-2 oncogene protein, epidermal growth factor receptor, and TGF-beta1 in pancreatic ductal adenocarcinoma is related to the histopathological grades and clinical stages of tumors
Literature	GENERIF_SUMMARY	HER2/neu predominantly uses Akt to suppress RARE binding activity, which may be one mechanism by which HER2/neu induces ATRA resistance in breast cancer cells

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	HER2/neu disrupts cell-cycle regulation. (review)
Literature	GENERIF_SUMMARY	Overexpression of ErbB2 is associated with transitional cell carcinoma of the bladder
Literature	GENERIF_SUMMARY	Her2/neu contributes to the growth of some Wilms' tumors, and an important mechanism of its action is promotion of angiogenesis.
Literature	GENERIF_SUMMARY	c-erbb-3 is overexpressed and amplified as a single and combined prognostic parameter in breast cancer.
Literature	GENERIF_SUMMARY	HER-2 protein expression can be a useful tool in differentiating a primary cutaneous appendageal neoplasm from HER-2 expressing metastatic breast carcinoma.
Literature	GENERIF_SUMMARY	HER-2/neu has no role in melanoma
Literature	GENERIF_SUMMARY	HER-2/neu overexpression correlated with AKT activation (P=0.015).
Literature	GENERIF_SUMMARY	HER2 signaling is negatively regulated by the PEST-type protein-tyrosine phosphatase BDP1
Literature	GENERIF_SUMMARY	expression of c-erbB-2 protein may reflect biological behaviour of the tumour and may appear to be an important factor in development of colorectal cancer metastases
Literature	GENERIF_SUMMARY	There is a clear-cut difference in Her-2/neu amplification between screen-detected and interval breast carcinomas.
Literature	GENERIF_SUMMARY	Study indicate that the coexistence of HER2 over-expression and p53 protein accumulation is a strong prognostic molecular marker in breast cancer.
Literature	GENERIF_SUMMARY	immunohistochemistry in different thyroid cancers
Literature	GENERIF_SUMMARY	In Paget's disease of breast, ERBB-2 protein overexpression is caused by amplification of the ERBB2 gene. ERBB-2 protein overexpression is caused by amplification of the ERBB2 gene.
Literature	GENERIF_SUMMARY	Strong membrane immunoreactivity is associated with high levels of HER-2 mRNA and gene amplification whereas cytoplasmic HER-2 is detected frequently and seems to be a marker of gastric tumor differentiation
Literature	GENERIF_SUMMARY	ErbB2 receptor has a role in the inhibition of the IGF-I-induced Shc-MAPK signaling pathway in breast cancer cells
Literature	GENERIF_SUMMARY	EGFR and ErbB-2 have roles in ligand-dependent apoptosis that could be a natural mechanism to protect tissues from unrestricted proliferation
Literature	GENERIF_SUMMARY	expression and prognostic significance of HER-2 in colorectal cancer and its relationship with clinicopathological parameters

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	research suggests that E6/E7 cooperates with ErbB-2 in head and neck carcinogenesis, at least in part, via the conversion of beta-catenin from a cell adhesion to a nuclear function, that is, to act as a potential transcriptional regulator
Literature	GENERIF_SUMMARY	overexpression of ErbB2 enhanced NF-kappaB activation induced by ionizing radiation in human breast carcinoma MCF-7 cells transfected with ErbB2 genes (MCF-7/ErbB2)
Literature	GENERIF_SUMMARY	Overexpression of HER2 is associated with recurrent non-small cell lung cancer
Literature	GENERIF_SUMMARY	regulation by hereulin through heterodimer formation with ErbB3
Literature	GENERIF_SUMMARY	ErbB2 is a remarkably internalization-resistant receptor due to the firm association of ErbB2 with protrusions
Literature	GENERIF_SUMMARY	A method validating ERBB2 gene expression in breast neoplasms has been developed.
Literature	GENERIF_SUMMARY	COX-2 up-regulates VEGF-C and promotes lymphangiogenesis in human lung adenocarcinoma via the EP(1)/Src/HER-2/Neu signaling pathway.
Literature	GENERIF_SUMMARY	HER2/Neu and ACTR may synergize to orchestrate mammary tumorigenesis through the dysregulation of the transcription factor ER81 and its target genes
Literature	GENERIF_SUMMARY	C-erbB-2 gene may be an important regulating gene in the coal miners with pneumoconiosis complicated by pulmonary cancer, and as a reference index to determine lymph node metastasis and prognosis.
Literature	GENERIF_SUMMARY	Her-2/NEU oncogene is essential for the regulation of VEGF-C in ovarian carcinoma; p38 MAPK and NF-kappa B are critically involved in the transcriptional activation of the VEGF-C gene by Her-2/NEU.
Literature	GENERIF_SUMMARY	Shc is a critical angiogenic switch for VEGF production downstream from the HGF and ErbB2 RTKs.
Literature	GENERIF_SUMMARY	overexpression of ERBB2 is associated with gastric cancer
Literature	GENERIF_SUMMARY	Chimeric immunoGrB molecule has therapeutic potential against HER2 tumors, especially in cases in which caspase-dependent apoptosis is inhibited.
Literature	GENERIF_SUMMARY	Monitoring ERBB2 blood levels represents a valuable tool for early prediction of the probability of response and progression-free survival to trastuzumab-based treatment of breast cancer.
Literature	GENERIF_SUMMARY	Association between HER-2/neu and VEGF expression supports the use of combination therapies directed against both HER-2/neu and VEGF for treatment of breast cancers.
Literature	GENERIF_SUMMARY	Cell surface expression of Her-2 is associated with

Main Category	Category	Terms
		osteosarcoma pathogenesis
Literature	GENERIF_SUMMARY	overexpression of HER2 in nontumorigenic mammary epithelial is permissive for the ability of TGF-beta to induce cell motility and Rac1 activity and HER2 and TGF-beta signaling cooperate in the induction of cellular events associated with tumor progression
Literature	GENERIF_SUMMARY	the extracellular domain of Serum HER-2 has a role in progression of breast neoplasms
Literature	GENERIF_SUMMARY	Predictive value of a polymorphism in HER-2 (Val655Ile) to determine the risk of developing prostate cancer in patients with benign prostatic hyperplasia
Literature	GENERIF_SUMMARY	evaluation of gene amplification in a subset of colorectal tumors and normal colonic mucosa
Literature	GENERIF_SUMMARY	transactivation occurs through an indirect interaction between erbB2 and prolactin or leptin receptors
Literature	GENERIF_SUMMARY	HER2 signaling enhances 5'UTR-mediated translation of c-Myc mRNA.
Literature	GENERIF_SUMMARY	Data provide the distribution frequency of HER2 protein expression and gene amplification in invasive ductal and lobular breast cancer.
Literature	GENERIF_SUMMARY	Overexpression of HER-2 was observed in only a small fraction of colorectal carcinomas, but were frequently accompanied by gene amplification.
Literature	GENERIF_SUMMARY	Activation of a set of signalling molecules, including MAPK, phosphatidylinositol-3-OH kinase (PI(3)K) and Src, is required for Neu/ErbB2-dependent lamellipodia formation and for motility of breast carcinoma cells.
Literature	GENERIF_SUMMARY	estrogen receptor redistribution to the cytoplasm and its interaction with HER2 are important downstream effects of HER2 overexpression
Literature	GENERIF_SUMMARY	p21Cip1/WAF1 and HER2/neu togehter have a role in progression of breast neoplasms
Literature	GENERIF_SUMMARY	only a few cases with aberrant Her2/neu expression in melanoma, many of them being primary cutaneous lesions rather than recurrent or metastatic lesions
Literature	GENERIF_SUMMARY	overexpression of HER-2/neu in pancreatic adenocarcinoma seems to be a result of increased transcription rather than gene amplification
Literature	GENERIF_SUMMARY	Circulating tumor cells in breast camcer were used to measure HER-2 gene amplification as an indicator of cancer progession.
Literature	GENERIF_SUMMARY	Expression of HER2/neu is not a poor prognostic factor in low-grade osteosarcoma.
Literature	GENERIF_SUMMARY	Relation between expression, DNA ploidy and human papillomavirus infection in cervical carcinoma.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	ErbB-2 is an important component of the plexin-B receptor system.
Literature	GENERIF_SUMMARY	In pancreatic ductal adenocarcinoma, membrane HER-2 overexpression was more frequent in intraductal than in invasive components but the incidence of cytoplasmic HER-2 overexpression did not differ between intraductal & invasive components.
Literature	GENERIF_SUMMARY	HER-2-amplified breast cancers have increased amounts of angiogenesis, decreased amounts of hypoxia, and increased markers of fibrin degradation
Literature	GENERIF_SUMMARY	c-erbB-2 and EGF-R are overexpressed in breast neoplasms and have an inverse association with Estrogen Receptor expression
Literature	GENERIF_SUMMARY	Data suggest that HER2 may act as the key molecular sensor of energy imbalance after the perturbation of tumor-associated fatty acid synthase hyperactivity in cancer cells.
Literature	GENERIF_SUMMARY	The extracellular domain of HER-2/neu, the p105 fraction, which is found in the circulation, does not appear to have any regulatory influence on uPA in patients with cervical cancer.
Literature	GENERIF_SUMMARY	EGFR, c-erbB-2, VEGF and MMP-2 and MMP-9 play an important role in tumor growth, invasion and metastasis in squamous cell carcinoma of the head and neck
Literature	GENERIF_SUMMARY	Significant correlation was observed between HER-2/neu overexpression and clinical outcome in germ-cell testicular tumors.
Literature	GENERIF_SUMMARY	In patients with breast cancer, most CNS metastatic tumor deposits showed expression for either EGFR or HER-2/neu, and less often for both.
Literature	GENERIF_SUMMARY	This study demonstrates the presence of ErbB-2 in the nucleus and identifies the function of ErbB-2 as a transcriptional regulator.
Literature	GENERIF_SUMMARY	Genetic polymorphisms may affect the development of prostate cancer. The frequency of Val655 in HER-2 was significantly lower in Japanese prostate cancer patients.
Literature	GENERIF_SUMMARY	serum epidermal growth factor receptor and HER2 have roles in response of advanced non-small cell lung cancer to chemotherapy
Literature	GENERIF_SUMMARY	kinase domain mutations identified in human tumors
Literature	GENERIF_SUMMARY	PEA3 represses the transcriptional activity of two fragments of the ErbB-2 promoter in a dose-dependent manner and decreases the endogenous ErbB-2 mRNA in pancreatic cancer cells.
Literature	GENERIF_SUMMARY	we show that HER2 enhances the expression of CXCR4, which is required for HER2-mediated invasion in vitro

Main Category	Category	Terms
		and lung metastasis in vivo.
Literature	GENERIF_SUMMARY	ErbB2 serves as a critical component that couples erbB receptor tyrosine kinase to the migration machinery of corneal epithelial cells.
Literature	GENERIF_SUMMARY	HER-2/neu overexpression may be linked with overall increased tumor viability and a significant increase in the population of viable hypoxic cells, which is not due to differences in tumor vascularization.
Literature	GENERIF_SUMMARY	Expression of MMP-2 and MMP-9 in breast cancer seems to be partly related to expression of AP-2 and HER2
Literature	GENERIF_SUMMARY	HER2-mediated effects on EGFR dimerization and trafficking were sufficient to explain the observed HER2-mediated amplification of epidermal growth factor-induced ERK signaling
Literature	GENERIF_SUMMARY	the alpha6beta4-erbB2 heterodimer is an important signaling complex for the formation of cohesive keratinocyte layers
Literature	GENERIF_SUMMARY	HER2/neu has roles in neoplasm progression and therapy [review]
Literature	GENERIF_SUMMARY	c-erbB-2 related aggressiveness in breast cancer is hypoxia inducible factor-1alpha dependent
Literature	GENERIF_SUMMARY	Amplification of EGFR was found in 10.37% of cases. The analysis revealed a lack of correlation between amplification of the oncogenes and the tumor phenotype.
Literature	GENERIF_SUMMARY	ERBB2 interacts with genes involved in angiogenesis
Literature	GENERIF_SUMMARY	The HER-2/neu, a proto-oncogene, was found in about 30% of human breast cancers, promoting cancer growth and making cancer cells resistant to chemo- and radio-therapy.
Literature	GENERIF_SUMMARY	Hyaluronan regulates ErbB2 activity and its interactions with other signaling factors in carcinoma cells.
Literature	GENERIF_SUMMARY	c-erb-B2 and Bcl-xl expression can be useful for the histopatologic diagnosis of Barrett exophagux and correct interpretation of dysplasia.
Literature	GENERIF_SUMMARY	c-erbB-2 oncoprotein expressed in membrane may have a role in progression of gastric cancer
Literature	GENERIF_SUMMARY	HER2/neu +3 may have a role in progression of breast cancer
Literature	GENERIF_SUMMARY	data suggest that alterations of the Her2 gene can occur in breast cancer, although not usually after primary or neoadjuvant chemotherapy
Literature	GENERIF_SUMMARY	HER2/neu engages Akt to increase WT1 expression, regulating cell cycle progression and apoptosis in HER2/neu-overexpressing breast cancer cells.

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Literature	GENERIF_SUMMARY	erbB receptor inactivation by unknown mechanisms results in altered splicing of bcl-x towards enhanced formation of proapoptotic Bcl-xS, thereby contributing to enhanced apoptotic susceptibility of failing human myocardium
Literature	GENERIF_SUMMARY	Applicability of employing FOXO4 regulation as a therapeutic intervention in HER2-overexpressing cancers.
Literature	GENERIF_SUMMARY	resveratrol downregulates HER-2/neu expression and induces apoptosis in tumor cells
Literature	GENERIF_SUMMARY	a splice variant of decay-accelerating factor is expressed in c-erbB-2-positive mammary carcinoma cells showing increased transendothelial invasiveness
Literature	GENERIF_SUMMARY	HER-2 overexpression is highly correlated with the expression of the apoptosis-suppressing gene bcl-xL.
Literature	GENERIF_SUMMARY	Inhibition of HER2/HER3 signaling protects against pulmonary fibrosis and improves survival.
Literature	GENERIF_SUMMARY	HER-2/neu over-expression correlates with more advanced disease in Iranian colorectal neoplasm patients.
Literature	GENERIF_SUMMARY	X-ray irradiation could lead to overexpression of HER-2 receptors on breast tumor cells
Literature	GENERIF_SUMMARY	PARP-1 is involved in expression of ERBB2 in concert with NF-kappaB, which might be associated with proliferation of rheumatoid arthritis synovial cells.
Literature	GENERIF_SUMMARY	c-MYC amplification is an early event in breast cancer progression, while ZNF217 and Her2/neu amplification may play a role in the later stage of tumor development
Literature	GENERIF_SUMMARY	integrin alpha5/beta1 exerts its tumor suppressor-like activity in colon cancer cells by inhibiting HER-2 signaling
Literature	GENERIF_SUMMARY	hyaluronan, phosphoinositide 3-kinase, and ErbB2 receptor kinase form a positive feedback loop that strongly amplifies multidrug resistance 1(MDR1) expression and regulates drug resistance in human breast carcinoma cells
Literature	GENERIF_SUMMARY	low-level amplification of TIIalpha gene locus may be sporadic mechanism of increased TIIalpha protein expression in pediatric non-brainstem glioma, which can coincide with low-level amplification of Her-2/neu
Literature	GENERIF_SUMMARY	Direct-double-differential PCR determines HER2 gene dosage and amplification in breast cancer.
Literature	GENERIF_SUMMARY	loss of RALT expression cooperates with ERBB2 gene amplification to drive full oncogenic signalling by the ErbB-2 receptor
Literature	GENERIF_SUMMARY	Her-2/neu is overexpressed in 25-35% of all mammary carcinomas in humans

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Single nucleotide polymorphism is associated wiwth cancer risk.
Literature	GENERIF_SUMMARY	Transcriptional activity of eztrogen receptor beta was altered in a manner similar to ERalpha by activation of ErbB2/ErbB3.
Literature	GENERIF_SUMMARY	Diets enriched with corn or fish reduced mammary tumorigenesis in Her-2 transgenic mice.
Literature	GENERIF_SUMMARY	YY1 cooperates with AP-2 to stimulate ERBB2 promoter activity through the AP-2 binding sites
Literature	GENERIF_SUMMARY	antibody trapping of ErbB2 in the endoplasmic reticulum showed inactivation of ErbB2 tyrosine phosphorylation and reduced heterodimerization of ErbB2 and ErbB3 and an increase in EGFR expression and activation
Literature	GENERIF_SUMMARY	topo2a and HER-2 status might have therapeutic and prognostic implications.
Literature	GENERIF_SUMMARY	In operable non-small lung cell cancers, there may be different relationship of this protein with paatient outcome.
Literature	GENERIF_SUMMARY	Glioma BK channels are a downstream target of erbB2/neuregulin signaling.
Literature	GENERIF_SUMMARY	Results suggest that the currently identified genetic polymorphisms of HER-2 are not associated with an increased risk of breast cancer in Korean women, whereas one haplotype does affect protein expression of the tumor and disease outcome.
Literature	GENERIF_SUMMARY	HER-2-expressing esophageal squamous cell carcinoma cells could be killed by trastuzumab-mediated antibody-dependent cellular cytotoxicity.
Literature	GENERIF_SUMMARY	The ErbB2 is imported into the nucleus through a nuclear localization signal (NLS)-mediated mechanism.
Literature	GENERIF_SUMMARY	findings suggest that HER-2/neu signaling may result, directly or indirectly, in enhanced activation of various metabolic, stress-responsive, antioxidative, and detoxification processes within the breast tumor microenvironment
Literature	GENERIF_SUMMARY	Elevated cytosolic HER-2/neu levels (> or =1,400 NHU/mg protein) were associated with a high probability of both shortened relapse-free, and overall survival.
Literature	GENERIF_SUMMARY	unusual stability is main mechanism that raises levels of AP-2 proteins; defective ubiquitin-dependent proteasomal-degradation pathway is possibly prime cause that affects the HER-2/neu gene and culminates in breast cancer
Literature	GENERIF_SUMMARY	We hypothesize that HER2 genotypes can be predictive biomarkers in ovarian cancer, contributing to a genetic individual profile of great interest in clinical oncology.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Distal ERBB2 promoter fragment displays specific transcriptional and nuclear binding activities in ERBB2 overexpressing breast cancer cells.
Literature	GENERIF_SUMMARY	These results demonstrate that C-erb-B2 (HER2/neu) may play a role in the tumorigenesis of synovial sarcoma.
Literature	GENERIF_SUMMARY	Her-2 overexpression was seen in 24% of breast cancer affecting Jordanian females.
Literature	GENERIF_SUMMARY	Systemic therapy with scFv(FRP5)-ETA can be safely administered up to a maximum tolerated dose of 12.5 microg/kg in patients with ErbB2-expressing tumors, justifying further clinical development.
Literature	GENERIF_SUMMARY	The co-expression of wt rat neu/ErbB2 transgene and mouse ErbB3, with physical and functional interactions between these two species of RTK receptors, was demonstrated in mouse mammary tumors.
Literature	GENERIF_SUMMARY	HER-2/neu and p53 are likely to be involved in the regulation of COX-2 expression in invasive ductal carcinomas of the breast.
Literature	GENERIF_SUMMARY	The movement of Arg784 in HER-2 appears to result from the absence of an anchoring residue like Asp746 in EGFR, which has been changed to Gly778 in HER-2.
Literature	GENERIF_SUMMARY	Subset of synovial sardcomsa with Her-2 amplification has a better overall prognosis
Literature	GENERIF_SUMMARY	Cell proliferation was increased when ErbB2 and ErbB3 were both overexpressed.
Literature	GENERIF_SUMMARY	Both p53 and c-erbB-2 proteins appear to be involved at an early stage of malignization of pleomorphic adenoma.
Literature	GENERIF_SUMMARY	The overexpression of HER-2/neu protein was found in recurrent superficial urothelial carcinoma.
Literature	GENERIF_SUMMARY	The positivity of c-erbB2 oncoprotein was correlated with the size of the breast tumor; the cancers which are not infiltrative ductal, knows as having a better prognostic, were c-erbB2 negative.
Literature	GENERIF_SUMMARY	the most notable changes consisted in the overexpression of ErbB2 by Schwann cells of nerves from Charcot-Marie-tooth disease type 1A patients
Literature	GENERIF_SUMMARY	A novel mechanism of ErbB-2 nuclear localization that involves interaction with the transport receptor importin beta1, nuclear pore protein Nup358, and a host of players in endocytic internalization was reported.
Literature	GENERIF_SUMMARY	These results suggest that HRG might be a new member of the growth factor family involved in the COX-2-dependent ulcer repair process.
Literature	GENERIF_SUMMARY	IHC and FISH together showed HER-2 overexpression/ gene amplification in 21% of breast invasive

Main Category	Category	Terms
		carcinomas.
Literature	GENERIF_SUMMARY	HER-2/neu induces the binding of Sp proteins and HDAC1 to the RECK promoter to inhibit RECK expression and to promote cell invasion
Literature	GENERIF_SUMMARY	c-erbB-2 oncoprotein overexpression is related to the clinical course of papillary thyroid carcinoma.
Literature	GENERIF_SUMMARY	ERBB3 is the pivotal element of the Erbb pathway promoting tumorigenesis by heterodimerization with NEU or EGFR, but the NEU/EGFR dimer does not appear to play a significant role in prostate cancer
Literature	GENERIF_SUMMARY	HER-2/neu overexpression or gene amplification does not correlate with histological tumour type, stage, grade or prognosis.
Literature	GENERIF_SUMMARY	Trophic maintenance of the mammary gland in pregnant bitransgenic mice provides the additional events required for tumor formation and maintains the population of cells that are targeted by ErbB2/Neu for transformation.
Literature	GENERIF_SUMMARY	Our observations suggest potential prognostic significance of p185(HER2) overexpression with PTEN loss in gastric adenocarcinoma patients. This opens up the possibility of considering p185(HER2)and PTEN as a therapeutic target in gastric cancer.
Literature	GENERIF_SUMMARY	HER-2 overexpression promotes the growth and malignancy of mammary epithelial cells, in part, by conferring resistance to the growth inhibitory effects of TGF-beta
Literature	GENERIF_SUMMARY	These data show that erbB2 stimulation is required for maintaining epithelial differentiation.
Literature	GENERIF_SUMMARY	Data show that c-erbB-1 and c-erbB-2 are differentially regulated in cancers of the papilla of Vater and pancreas, suggesting that the two types of cancer are biologically different.
Literature	GENERIF_SUMMARY	ERBB2 gene showed an agreement in amplification and polysomy and the level of protein expression in gastric adenocarcinoma.
Literature	GENERIF_SUMMARY	p185 probably plays an important role during the carcinogenesis and progression process of laryngeal squamous cell carcinoma.
Literature	GENERIF_SUMMARY	HER-2/neu has a role in progression of a unique tumor type of breast carcinoma
Literature	GENERIF_SUMMARY	Study provides some evidence of a prognostic effect of HER2-positive circulating tumor cel in stage I to III breast cancer.
Literature	GENERIF_SUMMARY	Infection of human dendritic cells with recombinant vaccinia virus produces human Her2/neu.
Literature	GENERIF_SUMMARY	Overexpression of HER-2 was associated with sentinel

Main Category	Category	Terms
		lymph node micro-metastasis in breast cancer patients
Literature	GENERIF_SUMMARY	The Increased expression of erbB-2 was associated with increased levels of Ki-67 and MCM2 expression, and combined overexpression of these receptors was associated with the highest levels of proliferation, suggesting a synergistic effect.
Literature	GENERIF_SUMMARY	HER2 overexpression was the best single predictive marker, but combinations of HER3, HER2 markers provided additional predictive information.
Literature	GENERIF_SUMMARY	The ability of Muc4 to segregate ErbB2 and other ErbB receptors and to alter downstream signaling cascades in polarized epithelial cells suggests that it has a role in regulating ErbB2 in differentiated epithelia.
Literature	GENERIF_SUMMARY	Blocking HER-2 cleavage with selective ADAM inhibitors may represent a novel therapeutic approach for treating HER-2 overexpressing breast cancer patients.
Literature	GENERIF_SUMMARY	ADAM10 is a major determinant of HER2 shedding, the inhibition of which, may provide a novel therapeutic approach for treating a variety of breast cancers with active HER2 signaling.
Literature	GENERIF_SUMMARY	HER2 appears to lack a critical role in the progression from ductal carcinoma in situ and invasive ductal carcinoma and HER2 status is maintained in metastatic lesions.
Literature	GENERIF_SUMMARY	the incidence of HER-2/neu positivity in colorectal cancers; the relationship of the HER-2/neu expression and patients' survival
Literature	GENERIF_SUMMARY	The Val allele was overrepresented in the subset of HER2-positive breast cancers and the Ile allele in serous ovarian cancer.
Literature	GENERIF_SUMMARY	Data suggest that acquired resistance to lapatinib is mediated by a switch in cell survival dependence from ErbB2 alone to codependence upon ER and ErbB2, rather than loss of ErbB2 expression or insensitivity of ErbB2 signaling to lapatinib.
Literature	GENERIF_SUMMARY	Subset of pancreatic ductal adenocarcinomas is characterized by HER2/neu gene amplification. (Her 2 protooncogene protein)
Literature	GENERIF_SUMMARY	HER3 and HER4 have been related to a poor prognosis in bladder cancer.
Literature	GENERIF_SUMMARY	The activation of HER-2/CXCR4/ Akt signaling pathway in primary breast tumors may contribute to the formation of bone metastases in breast cancer.
Literature	GENERIF_SUMMARY	Pathways up- or down-regulated in the presence of ERBB2 over-expression, including genes not previously implicated in breast cancer that could be considered as candidate markers for prognosis and therapy.
Literature	GENERIF_SUMMARY	p53 and c-erbB-2 may have independent role in

Main Category	Category	Terms
		carcinogenesis of gall bladder cancer
Literature	GENERIF_SUMMARY	Muc4 potentiates neuregulin signaling by increasing the cell-surface populations of ErbB2 and ErbB3
Literature	GENERIF_SUMMARY	HER-2 codon 655 SNP and/or c-erbB-2 overexpression may also be used as a poor prognostic indicator for gastric carcinomas.
Literature	GENERIF_SUMMARY	ErbB2 has few interaction partners; of them Shc is the most common.
Literature	GENERIF_SUMMARY	Several known receptor tyrosine kinases (RTKs) are upregulated in gastric cancer being prime targets of a tailored therapy.
Literature	GENERIF_SUMMARY	There was a significant relationship between the expression of HER-2/neu and clinical stages and lymph node metastasis in laryngeal squamous cell carcinoma.
Literature	GENERIF_SUMMARY	PEA3 expression is not correlated with HER-2/neu expression in breast cancer tumor tissues
Literature	GENERIF_SUMMARY	cyclin E expression in breast cancer cells was associated with negative steroid receptor status, HER2 presence, higher tumor grade and higher proliferation index
Literature	GENERIF_SUMMARY	stable silencing of Her2 gene expression with plasmid expressing shRNA may hold great promise as a novel therapy for Her2 expressing breast cancers alone or in combination with anthracycline chemotherapy
Literature	GENERIF_SUMMARY	Observed consistent increases in persistence associated with HER2 overexpression indicate a prospective mechanism for invasiveness previously documented in HER2-overexpressing human breast tumors.
Literature	GENERIF_SUMMARY	PEA3 and c-Jun stimulate synergistically the HER2/neu gene transcription with p300
Literature	GENERIF_SUMMARY	The kinase domain is necessary for the activity of HER2 CTFs and the presence of these HER2 fragments could account for the resistance to treatment with antibodies.
Literature	GENERIF_SUMMARY	HER2(neu/ErbB2) is an effector of an apical EGF receptor complex mispolarization and its inhibition should be considered a candidate for clinical therapy of polycystic kidney disease.
Literature	GENERIF_SUMMARY	The prevalence of ER, PR and HER2/neu positivity in breast cancer of Jordanian women is lower than that of the western populations and close to other populations such as the Chinese and the minor ethnic groups of Northern America (African Americans).
Literature	GENERIF_SUMMARY	Data suggest that mutant HER2 containing a G776(YVMA) insertion activates cellular substrates more potently than wild-type HER2.
Literature	GENERIF_SUMMARY	HER-2/neu and p53 may have a role in chemoresistance

Main Category	Category	Terms
		in patients with non-small cell lung cancer
Literature	GENERIF_SUMMARY	PC cell-derived growth factor stimulates proliferation and confers Trastuzumab resistance to Her-2-overexpressing breast cancer cells
Literature	GENERIF_SUMMARY	The purpose of this study was to evaluate amplification of HER2 genes as predictors of response to chemotherapy in advanced breast cancer. HER2 gene amplification in 38 (45%) of the primary tumours.
Literature	GENERIF_SUMMARY	HER-2/neu amplification may constitute an independent prognostic factor in gastric cancer patients
Literature	GENERIF_SUMMARY	The lack of gene amplification and robust HER-2/neu protein expression in small cell lung carcinoma (SCLC) tumour cells does not suggest a prominent role for the HER-2/neu gene in SCLC tumour progression.
Literature	GENERIF_SUMMARY	Among the glassy cell carcinoma (GCC) cases, estrogen receptor (ER), progesterone receptor (PR) and Her2/neu were positive in 2 (18.1%), 1 (9.1%) and 5 (45.4%) cases, respectively.
Literature	GENERIF_SUMMARY	Intraductal trastuzumab treatment may be effective in the management of ductal carcinoma in situ of the breast with HER-2 overexpression.
Literature	GENERIF_SUMMARY	Combining endocrine therapy with EGF receptor/HER-2/neu inhibitors should be tested in clinical breast cancer.
Literature	GENERIF_SUMMARY	HER-2/neu overexpression was detected in 47.4% of colorectal cancer patients; tumors with HER-2/neu overexpression showed higher postoperative recurrence; HER-2/neu overexpression may constitute an independent prognostic factor in colorectal cancer
Literature	GENERIF_SUMMARY	These results suggest a relevant role for STAT5 and Bcl-xL as apoptosis-regulatory proteins in the pathogenesis of lung cancer, and overexpression of both Neu and activated STAT3, could be related with the proliferation rate in lung carcinoma cells.
Literature	GENERIF_SUMMARY	The development and characterization of 4 PEGylated antibodies to p185HER-2 (HER-2) are reported.
Literature	GENERIF_SUMMARY	The in vitro selection of DARPins against human epidermal growth factor receptor 2 (Her2), an important target for cancer therapy and diagnosis is reported.
Literature	GENERIF_SUMMARY	Significantly higher serum HER2 level was associated with bone metastasis of prostate cancer patients
Literature	GENERIF_SUMMARY	Akt activ correlates with HER2 overexpr or LOH at the PTEN gene locus while inversely correlating with PR expression. When LOH at the PTEN gene locus and HER2 overexpr occurred simultaneously, incidence of Akt activ and reduced PR expr was significant.
Literature	GENERIF_SUMMARY	The phosphorylation sites most strongly correlated with migration & proliferation were identified. HER2 Y1248 is correlated with cell migration.

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Literature	GENERIF_SUMMARY	In HER2-overexpressing cells, TGF-beta, via PI3K, recruits actin & actinin to HER2, which then joins Vav2, Rac1, & Pak1 at lamellipodia, prolonging Rac1 activation, enhancing motility, survival & invasiveness, phosphorylating Bad, & uncoupling Bad/Bcl-2.
Literature	GENERIF_SUMMARY	HSP90 binding to ErbB2 participates in regulation of src kinase activity as well as kinase stability
Literature	GENERIF_SUMMARY	HER2 overexpression and amplification were present in tumors of high grade.
Literature	GENERIF_SUMMARY	Expression of HER-2 and plasminogen activator, urokinase receptor in primary tumors predicted gene status in 100 and 92% of patients, respectively
Literature	GENERIF_SUMMARY	The importance of this biomarker was validated in a longitudinal study comparing pre- and post-operative levels and was shown to revert to normal levels after breast csncer surgery.
Literature	GENERIF_SUMMARY	N-WASP plays a pivotal role in regulating hyaluronan-mediated CD44-ErbB2 interaction, beta-catenin signaling, and actin cytoskeleton functions that are required for tumor-specific behaviors and ovarian cancer progression
Literature	GENERIF_SUMMARY	study shows that newly diagnosed HER-2-overexpressing breast cancer patients are at increased risk for brain metastases
Literature	GENERIF_SUMMARY	Newly described mutations in ERBB2 may play a role in predicting response to epidermal growth factor receptor (EGFR)-targeted therapy in hepatoma.
Literature	GENERIF_SUMMARY	To offer prognostic information and to direct appropriate treatment it is important to provide accurate laboratory assessment of the status of HER2 in breast cancer.[REVIEW]
Literature	GENERIF_SUMMARY	HER2 has a role in development of breast carcinoma in patients treated for Hodgkin's lymphoma or other pediatric solid tumors with radiation during breast maturation
Literature	GENERIF_SUMMARY	enhanced signaling from the HER2/neu-HER3 pathway has a role in growth of tumors treated with fulvestrant in the presence of physiologic estradiol
Literature	GENERIF_SUMMARY	Transgenic mice develop mammary tumors overexpressing rat Erbb2. Immunization with human Erbb2 peptide vaccine prevented growth of these tumors. Recombinant mouse IL-12 enhanced vaccine efficacy.
Literature	GENERIF_SUMMARY	In women over age 50 or with known postmenopausal status with an operable breast cancer, there is an inverse association between BMI and HER-2 over-expression.
Literature	GENERIF_SUMMARY	In summary, results indicate that Heregulin beta-induced store-operated calcium channels were

Main Category	Category	Terms
		regulated by c-erbB2 level and dependent on activation of phospholipase C in human breast cancer cells.
Literature	GENERIF_SUMMARY	type of TP53 mutation, especially missense mutation, is a strong prognostic indicator for disease-free survival and disease-specific survival in node-negative breast cancer, particularly in combination with ERBB2 amplification
Literature	GENERIF_SUMMARY	Amplicon size is associated with response to trastuzumab in metastatic breast cancer.
Literature	GENERIF_SUMMARY	The efficiency of anti-HER2 adjuvant therapy could be evaluated, in a clinical trial, by sequential detection of HER2-positive micrometastatic cells within the bone marrow, before and after treatment.
Literature	GENERIF_SUMMARY	ErbB2 and Muc4 bind in signet ring carcinoma cells, which was not seen in highly differentiated adenocarcinoma cell lines.
Literature	GENERIF_SUMMARY	Detection of HER2 mRNA-positive circulating tumor cells after chemotherapy was associated with reduced disease-free survival but not with overall survival.
Literature	GENERIF_SUMMARY	The results of this study suggest that some cases of primary breast carcinoma are heterogeneous in regard to HER2/neu gene amplification or protein overexpression
Literature	GENERIF_SUMMARY	Findings suggest that the ERBB2 mutation is rare in Korean lung cancer patients.
Literature	GENERIF_SUMMARY	Her-2/neu protein overexpression by HER-2 gene amplification may occur in extrahepatic cholangiocarcinoma and constitute an independent prognostic factor in patients with lymph node metastases.
Literature	GENERIF_SUMMARY	The use of RNA expression profiles and RT-PCR to analyze HER2 in frozen or formalin-fixed breast cancer samples may be an alternate approach to immunohistochemistry in combination with fluorescence- and chromogenic in situ hybridization.
Literature	GENERIF_SUMMARY	Human cancers overexpressing HER-2 might benefit from trastuzumab therapy.
Literature	GENERIF_SUMMARY	Review of the molecular interconnection data argue for cooperation between HER2 and Fhit in breast carcinogenesis.
Literature	GENERIF_SUMMARY	Our results show that c-erbB-2 over-expression correlates with poor histological grade and negative ER/PR status, and predicts poor overall survival in Asian women with breast cancer.
Literature	GENERIF_SUMMARY	propose that the non-incidental coamplification of Myc and either ERBB2 or EGFR occurred through translocation and subsequent rearrangement
Literature	GENERIF_SUMMARY	Breast tumors that express truncated HER2 are

Main Category	Category	Terms
		resistant to trastuzumab and may require alternative or additional anti-HER2-targeting strategies
Literature	GENERIF_SUMMARY	The erbB2+ cluster of the intrinsic gene set predicts tumor response of breast cancer patients receiving neoadjuvant chemotherapy with docetaxel, doxorubicin and cyclophosphamide
Literature	GENERIF_SUMMARY	HER-2/neu overexpression and amplification is associated with uterine serous papillary carcinoma
Literature	GENERIF_SUMMARY	These findings point to the existence of an association of ERBB2 allelic variants at both loci with specific breast tumor phenotypes.
Literature	GENERIF_SUMMARY	40 breast carcinomas from BRCA1 mutation carriers and 40 sporadic breast carcinomas were examined for HER-2 and TOP2A amplification status
Literature	GENERIF_SUMMARY	Results describe a designed ankyrin repeat protein evolved to picomolar affinity to Her2.
Literature	GENERIF_SUMMARY	HER2 overexpression is associated with higher rate of pathologic complete response to preoperative paclitaxel/FAC chemotherapy.
Literature	GENERIF_SUMMARY	Treatment with gefitinib, trastuzumab and pertuzumab to block signals from all HER homo- and heterodimers inhibited growth of HER2-overexpressing xenografts statistically significantly better than single agents and dual combinations.
Literature	GENERIF_SUMMARY	Expression of c-erbB-2 and p53 has no prognostic value in patients with early-stage breast cancer in which axillary lymph node metastasis is absent.
Literature	GENERIF_SUMMARY	progesterone receptor and HER2 levels in breast cancer vary by BMI, suggesting a heterogeneous etiology for tumors related to these markers
Literature	GENERIF_SUMMARY	HER2 gene status remains highly conserved as breast cancers metastasise but discrepant results occur because of interpretational difficulties of HER2 amplification.
Literature	GENERIF_SUMMARY	No GBM demonstrates HER-2/neu protein by IHC or amplification of the HER-2/neu gene by FISH. The HER-2/neu oncogene does not seem to play a role in the pathogenesis of GBM
Literature	GENERIF_SUMMARY	Brain-metastasizing breast cancer belongs predominantly to the basal type or Her-2/neu type.
Literature	GENERIF_SUMMARY	HER2 overexpression in HER2 2+ carcinomas is associated with neither an increase in gene transcription nor a deregulation in the ubiquitin-dependent pathways, but instead seems to be regulated by protein kinase Calpha (PKCalpha) activity.
Literature	GENERIF_SUMMARY	Overexpression of Her-2 is associated with Lymph node metastasis in bladder cancer

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	High C-erbB-2 expression is associated with recurrence in breast cancer
Literature	GENERIF_SUMMARY	Our data suggest that erbB-1/erbB-2 overexpression is a direct effect of higher than normal transcriptional activity of the encoding genes in a subset of human endometrial carcinomas
Literature	GENERIF_SUMMARY	BAG-1 negativity in association with p53 and c-erbB2 positivity identified a subgroup of tongue cancer patients with an aggressive phenotype
Literature	GENERIF_SUMMARY	HER-2 protein overexpression as evaluated by immunohistochemistry and HER-2 gene amplification assessed by FISH were found to be relatively common in intestinal-type gastric carcinomas
Literature	GENERIF_SUMMARY	Borderline for HER-2 protein expression at immunohistochemical assay (2+) were assessed for HER-2 gene amplification by real-time PCR and by FISH. PCR might be proposed as a highly predictive positive test in HER-2 assessment.
Literature	GENERIF_SUMMARY	HER-2/neu status of the primary breast cancer is determined by immunohistochemistry and fluorescent in situ hybridization.
Literature	GENERIF_SUMMARY	The overexpression rate of c-erbB-2 was higher in recurrence group than in non-recurrence group of Dukes'A and B colorectal carcinoma.
Literature	GENERIF_SUMMARY	ErbB-2, via PYK2-MAP kinase, upregulates the adhesive ability of androgen receptor human prostate cancer cells.
Literature	GENERIF_SUMMARY	Study shows a significant association between HER2 expression and the type of glioblastoma (GBM), with influence on survival rate. GBM with low-HER2 expression are more likely to be secondary GBM, carrying a better prognosis than de novo GBM.
Literature	GENERIF_SUMMARY	study of EGFR, HER2, TP53& KRAS mutations of p14arf expression of non-small cell lung cancers in relation to smoking
Literature	GENERIF_SUMMARY	Imbalance in c-Met expression between tumour and surrounding normal tissue is associated with aggressive ductal carcinoma in situ phenotype. c-Met and HGF/SF may contribute to tumour development by different means than those controlled by Her2/neu.
Literature	GENERIF_SUMMARY	Activated HER2 is dependent on the spatial relation to a malignant tumor.
Literature	GENERIF_SUMMARY	Levels of HER-2/neu extracellular domain were above the cut-off value in breast cancer patients.
Literature	GENERIF_SUMMARY	In the absence of EGF, p38MAPK-activated AKT is necessary for HER-2 overexpressing human breast cancer cells to survive and to form colonies in soft agar.
Literature	GENERIF_SUMMARY	Results determine the level of MMP-2, its natural

Main Category	Category	Terms
		inhibitor TIMP-2, their ratio and HER-2/neu as diagnostic and prognostic factors in breast cancer.
Literature	GENERIF_SUMMARY	The combination of cetuximab and trastuzumab could induce synergistic antiproliferative effects in several oesophageal Squamous cell carcinoma cell lines with EGFR and HER-2 expression.
Literature	GENERIF_SUMMARY	C-terminal cleavage of ErbB2 releases the receptor from a retention mechanism and causes endocytosis and lysosomal degradation of ErbB2.
Literature	GENERIF_SUMMARY	Neither mutation of EGFR nor increased copy number of EGFR or HER2 was diagnostic of response to gefitinib.
Literature	GENERIF_SUMMARY	HER-2 and Bax are related to poor prognosis of breast cancer, while Bcl-2 is related to good prognosis.
Literature	GENERIF_SUMMARY	the major mechanism of HER2-mediated induction of FASN and ACCalpha in the breast cancer cells used in this study is translational regulation primarily through the mTOR signaling pathway.
Literature	GENERIF_SUMMARY	We showed that the mean HER-2 amplification index in carcinomas determined as 2+ was significantly lower as compared to carcinomas defined as 3+ (p
Literature	GENERIF_SUMMARY	patients with HER2/neu-positive cancers benefit more from -based and taxane-based adjuvant chemotherapy than those with HER2/neu-negative cancers.
Literature	GENERIF_SUMMARY	no treatment interaction was found between HER2 status and tamoxifen in ER+ tumours.
Literature	GENERIF_SUMMARY	These data suggest that the distinct topography of receptors and their docking partners modulates signaling activities.
Literature	GENERIF_SUMMARY	Abnormal gene copy number of HER2 is associated with breast tumor progression
Literature	GENERIF_SUMMARY	analysis of EGFR, HER2 and HER3 expression in esophageal primary tumours and corresponding metastases
Literature	GENERIF_SUMMARY	Expression of C-erbB2 was significantly higher in nasal squamous cell carcinoma than that in nasal polyps.
Literature	GENERIF_SUMMARY	Ile655Val polymorphism of the ERBB2 gene do not alter its activity and may not be associated with increased breast cancer risk among Chinese women.
Literature	GENERIF_SUMMARY	Findings suggest a strong age-related selective growth advantage for breast tumour cells belonging to the ER+PR+HER-2+ subgroup.
Literature	GENERIF_SUMMARY	c-erbB-2 is highly correlated with differentiation grade and metastasis of the extrahepatic cholangiocarcinoma tumor
Literature	GENERIF_SUMMARY	a difference between the two HER2 isoforms regarding their tumorogenic potential with an advantage for the

Main Category	Category	Terms
		Val/HER2 isoform. In breast cancer patients treated with trastuzumab, the Val allele may constitute a risk factor for cardiac toxicity.
Literature	GENERIF_SUMMARY	Wwox reduces ErbB2 protein expression in vitro and expression of Wwox protein inversely correlates with expression of ErbB2 protein in prostate cancer tissues.
Literature	GENERIF_SUMMARY	A statistically significant correlation between serum level concentration and tissue HER-2 status, was observed.
Literature	GENERIF_SUMMARY	polysomy of chromosome 17 does not affect the ERBB2 expression and topoisomerase IIalpha mRNA expression is not related to gene status
Literature	GENERIF_SUMMARY	Data suggest that MnSOD may be useful in treating HER2/neu-mediated human breast tumor malignancy.
Literature	GENERIF_SUMMARY	disruption of ERBB2 signaling with the selective inhibitor AG825 severely inhibited hESC proliferation and promoted apoptosis.
Literature	GENERIF_SUMMARY	HER2 protein overexpression and/or gene amplification are rare in colorectal carcinomas.
Literature	GENERIF_SUMMARY	Epitope mapping and structural analysis of an anti-ErbB2 antibody A21
Literature	GENERIF_SUMMARY	p53 and Ki-67, but not bcl-2, cyclin D1 or HER-2 may have roles in the process of tumor genesis in non-small cell lung carcinoma
Literature	GENERIF_SUMMARY	data demonstrate that the full-length ErbB2 was localized at the cytoplasmic membrane and ErbB2 intracellular domain localized in the nucleus predominantly
Literature	GENERIF_SUMMARY	Measurement of CA 15-3 serum values in conjunction with sHER2 and CA 15-3 can increase sensitivity in metastasis detection.
Literature	GENERIF_SUMMARY	This results suggest that FISH testing should be considered for tumours with 50% HER2 positive stained cells and that polysomy 17 without amplification is not associated with poor prognostic features.
Literature	GENERIF_SUMMARY	Significant heterogeneity exists between Her-2/neu expression in the in situ component and invasive components of adenocarcinoma of the breast.
Literature	GENERIF_SUMMARY	It was observed that HER-2/neu expression was significantly associated with high grade ovarian tumours, however intensity of positivity did not correlate with the grade of tumour.
Literature	GENERIF_SUMMARY	HER2, is a member of the ErbB family of receptor tyrosine kinases, that plays a key role in the pathogenesis of breast cancer, and when overexpressed, can correlate with a particularly aggressive clinical phenotype.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	HER2 gene amplification is a complex event
Literature	GENERIF_SUMMARY	HER-2/neu transcriptionally activates Jab1 expression to promote proliferation of breast cancer cells.
Literature	GENERIF_SUMMARY	Activation of HER-2/Neu phosphorylation is associated with increase in tyrosine phosphorylation of phosphoinositide-specific lipase C-gamma-1 and recruitment of PLC-gamma-1 to HER-2/Neu protein molecule
Literature	GENERIF_SUMMARY	Association of the HER-2 genotype with clinicopathologic characteristics and HER-2 expression may indicate its possible implication on the more aggressive phenotype in breast cancer.
Literature	GENERIF_SUMMARY	TOP2A status cannot be predicted from the HER-2 status and evaluation of the TOP2A status only in patients with HER-2 overexpression may lead to missing cases with TOP2A deletion with possible resistance to therapy in breast neoplasms.
Literature	GENERIF_SUMMARY	HER-2/neu showed overexpression, caused by gain/amplification in 50%, and may be involved in progression of Fallopian tube carcinoma
Literature	GENERIF_SUMMARY	HER2/neu protein is frequently overexpressed in SDC, and in contrast to previous reports, overexpression of the protein is associated in most cases with HER2/neu gene amplification.
Literature	GENERIF_SUMMARY	HER2 immunoreactivity might have a limited prognostic value for advanced urothelial carcinoma patients with adjuvant M-VEC chemotherapy.
Literature	GENERIF_SUMMARY	The expression of C-erb B-2 contributes to angiogenesis in nasopharygeal carcinoma.
Literature	GENERIF_SUMMARY	we have shown an increased expression of ErbB-2 vestibular schwannomas
Literature	GENERIF_SUMMARY	evaluation of HER-2 status should be performed in neoplastic tissue from metastatic site, whenever possible.
Literature	GENERIF_SUMMARY	HER-2 amplification has a role in response to trastuzumab-based neoadjuvant therapy
Literature	GENERIF_SUMMARY	cellular PTPN13 inhibits Her2 activity by dephosphorylating the signal domain of Her2 and plays a role in attenuating invasiveness and metastasis of Her2 overactive tumors.
Literature	GENERIF_SUMMARY	Expression of HER-2/neu oncogene has a direct relationship iwth the grade of bladder transitional cell carcinoma.
Literature	GENERIF_SUMMARY	Genistein at > or =1 microM inhibited HER2 protein expression, phosphorylation, and promoter activity through an estrogen receptor-independent mechanism. In the presence of ERalpha, genistein mimicked E2 and inhibited HER2 protein phosphorylation.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	early-stage HER2+ cancers associated with lymphocytic infiltration are a biologically distinct subtype with an improved natural history
Literature	GENERIF_SUMMARY	downregulation of SKP2 was critical for FOXP3-mediated growth inhibition in breast cancer cells that do not overexpress ERBB2/HER2.
Literature	GENERIF_SUMMARY	a novel mechanism by which plexin-mediated signaling can be regulated and explains how Sema4D can exert different biological activities through the differential association of its receptor with ErbB-2 and Met.
Literature	GENERIF_SUMMARY	Cyclin G2 expression is modulated by HER2 signaling through multiple pathways including phosphoinositide 3-kinase, c-jun NH(2)-terminal kinase, and mTOR signaling.
Literature	GENERIF_SUMMARY	A tight interaction between HER-2 and EPIL gene expression in invasive breast cancer cells is probable.
Literature	GENERIF_SUMMARY	Overexpresion of Her-2 is associated with micrometastatic disease in men with prostate cancer
Literature	GENERIF_SUMMARY	These findings reveal a new kinase regulatory mechanism in which the alphaC-beta4 loop functions as an intramolecular switch that controls ErbB2 activity.
Literature	GENERIF_SUMMARY	A small group of hormone receptor-negative breast tumors with poorer 5-year survival express HER2 and basal markers simultaneously.
Literature	GENERIF_SUMMARY	Explore QSAR of ErbB2 inhibitors.
Literature	GENERIF_SUMMARY	disruption of surface HSP90/HER-2 interaction leads to inhibition of heregulin-induced HER-2-HER-3 heterodimer formation, reduced HER-2 phosphorylation, and impaired downstream kinase signaling.
Literature	GENERIF_SUMMARY	Our results indicate that the Val carrier was associated with increased risks in patients with breast cancer in Taiwan. The association was more apparent in patients who were younger than or equal to 45 years of age.
Literature	GENERIF_SUMMARY	investigate the status of Her2, EGFR and cyclin D1 in 95 primary breast carcinomas
Literature	GENERIF_SUMMARY	Solamargine enhances HER2 expression and increases the susceptibility of human lung cancer cells to trastuzumab and epirubicin.
Literature	GENERIF_SUMMARY	EphA2 cooperates with ErbB2 to promote tumor progression in mice and may provide a novel therapeutic target for ErbB2-dependent tumors in humans.
Literature	GENERIF_SUMMARY	estrogen receptor-positive tumors with Akt-1 expression...with the presence of HER2-positive tumors with strong Notch-2 expression, support the role of Notch and Akt in breast cancer progression
Literature	GENERIF_SUMMARY	PF00299804 is a highly effective inhibitor of both the

Main Category	Category	Terms
		wild-type ERBB2 and the gefitinib-resistant oncogenic ERBB2 mutation identified in lung cancer cells.
Literature	GENERIF_SUMMARY	HER-2 polymorphism, especially in a homozygous form might play some role in the etiology of breast fibroadenoma formation
Literature	GENERIF_SUMMARY	Salivary duct carcinomas in this study show HER-2/neu overexpression on both the protein and gene levels in approximately 30% of cases.
Literature	GENERIF_SUMMARY	efficacy of molecular targeted therapy can be expected even for patients with metastatic lymph nodes as long as the primary tumors are positive for HER-2 expression.
Literature	GENERIF_SUMMARY	c-erbB-2 expression was significantly associated with lymph node metastasis.
Literature	GENERIF_SUMMARY	NRG1alpha-induced adhesion variation is blocked by erbB2, PI3K, and Akt inhibitors, but not by PLC, ROCK, MLCK, or MEK inhibitors, implicating the erbB2/PI3K/Akt1 signaling pathway in NRG1-stimulated, integrin-mediated cell adhesion
Literature	GENERIF_SUMMARY	The expression rate of c-erbB-2 was higher in adenomas than adenocarcinomas.
Literature	GENERIF_SUMMARY	HER2 overexpression was found in only four cases (10.5%) of the studied primary tumors and in all cases the HER2 expression was retained in the paired metastases.
Literature	GENERIF_SUMMARY	RIG1 plays a role in IFN-gamma-mediated therapy by downregulating p185 and its downstream PI3K/Akt/mTOR/VEGF-signaling pathway
Literature	GENERIF_SUMMARY	UVA enhances the expression of ErbB2 via cAMP- and protein kinase-dependent Transcription Factor AP2 alpha activation in keratinocytes
Literature	GENERIF_SUMMARY	analysis of the structure of the dimeric transmembrane domain of the growth factor receptor ErbB2 that may correspond to the receptor active state
Literature	GENERIF_SUMMARY	Genomic alteration of the HER2-neu and EGFR genes is frequent (25%) in ovarian cancer.
Literature	GENERIF_SUMMARY	those with an increase in both HER2 and pAkt expression had the worst disease free survival.
Literature	GENERIF_SUMMARY	nine strong possible ligands can be identified for binding by mycobacterium leprae
Literature	GENERIF_SUMMARY	Results show that disease stage, tumor grade, SES, race/ethnicity, negative ER and PR status, rather than negative HER2 status, were risk factors for survival.
Literature	GENERIF_SUMMARY	There can be differences in radiosensitivity, which, if they also exist between patient breast cancer cells, are important to consider for both conventional radiotherapy and for HER2-targeted radionuclide

Main Category	Category	Terms
		therapy.
Literature	GENERIF_SUMMARY	increased HER2 copy numbers may have a role in breast cancer
Literature	GENERIF_SUMMARY	Cell line genetic heterogeneity and/or multiple determinants modulate the role played by EGFR/HER2 in regulating cell proliferation.
Literature	GENERIF_SUMMARY	HER-2/neu may have roles in the agressive phenotype and high proliferation in endometrial carcinomas
Literature	GENERIF_SUMMARY	Her-2, together with cyclooxygenase-1 (Cox-1), may be involved in the early stage of endometrial cancer development.
Literature	GENERIF_SUMMARY	All cases were wild-type for HER2. Mutations of EGFR and K-ras genes represent an early event in lung adenocarcinomagenesis, and AAH convincingly seems to be a precursor lesion in a subset of cases of adenocarcinoma.
Literature	GENERIF_SUMMARY	examination of role of lipids in HER1/HER2-driven progression of human breast epithelial cells towards malignancy
Literature	GENERIF_SUMMARY	Data show that serum level of soluble HER2/neu in patients with tumors characterized by high expression of this protein was significantly higher than in patients with low expression of HER2/neu and in women with benign diseases of the mammary glands.
Literature	GENERIF_SUMMARY	This study highlights the role of both AP-2alpha and YY1 transcription factors in ERBB2 oncogene overexpression in breast tumors.High ERBB2 expression may result either from gene amplification or from increased transcription factor levels.
Literature	GENERIF_SUMMARY	In human uterine leiomyomas, ERBB2 were also overexpressed.
Literature	GENERIF_SUMMARY	Findings strongly demonstrate that retinoblastoma (RB) and cyclin-dependent kinase 2 (CDK2) on one side and cytokeratin 8 (CK8) and epidermal growth factor receptor 2 (HER2) on the other may affect the clinical course of the disease in 56% of patients.
Literature	GENERIF_SUMMARY	polymorphic alleles of HER2 and CCND1 may not play an important role as genetic markers for breast cancer risk, but presence of Val allele may be useful for tumor prognosis.
Literature	GENERIF_SUMMARY	HER-2 Ile655Val SNP does affect serum HER-2 levels and it can be regarded as a predictive biomarker for breast cancer. patients with poor prognosis.
Literature	GENERIF_SUMMARY	Her2 positive rate was 40%
Literature	GENERIF_SUMMARY	Neoadjuvant therapy resulted in a significant decrease in mitotic count and an increase in the proportion of patients with HER2/neu overexpression.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	not a suitable marker that could play a primary role in the clinical-therapeutic management of squamous cell carcinoma of the tongue
Literature	GENERIF_SUMMARY	We found a positive correlation between FOXM1 expression and HER2 status, pointing to a potential role of FOXM1 as a new drug target in HER2 resistant breast tumour, as FOXM1 inhibitors for cancer treatment were described recently.
Literature	GENERIF_SUMMARY	Suggest that endocytosis defect restricts the degradation of ErbB2, and that c-Cbl-catalysed mono-ubiquitination is not involved in the impairment in ligand-induced degradation of ErbB2.
Literature	GENERIF_SUMMARY	Macrophage inhibitory cytokine-1may participate in the malignant progression of certain human cancer cells that overexpress ErbB2 through the transactivation of ErbB2 tyrosine kinase.
Literature	GENERIF_SUMMARY	Here we show that epidermal growth factor (EGF) and heregulin induce CD44 shedding in JIMT-1, an ErbB2-overexpressing cell line resistant to trastuzumab, accompanied by internalization and intramembrane proteolysis of CD44 and enhanced cellular motility
Literature	GENERIF_SUMMARY	The researchers found that estrogen receptors, progesterone receptors, and Her/2neu correlated with tumor grade and age of breast carcinoma diagnosis as in previous research.
Literature	GENERIF_SUMMARY	HER-2-mediated endocytosis of magnetic nanospheres and the implications in cell targeting and particle magnetization are reported.
Literature	GENERIF_SUMMARY	Findings substantiate the notion that breast cancer progression is often associated with alterations of HER-2/neu expression.
Literature	GENERIF_SUMMARY	HER2/neu was overexpressed in human bronchial epithelial cells treated with anti-BPDE.
Literature	GENERIF_SUMMARY	Overexpression of ERBB-2 is associated with malignant than benign pheochromocytomas
Literature	GENERIF_SUMMARY	restricting CDC25A can limit tumorigenesis induced by the HER2/neu-RAS oncogenic pathway without compromising normal cell division or viability [review]
Literature	GENERIF_SUMMARY	Met receptor contributes to trastuzumab resistance of Her2-overexpressing breast cancer cells
Literature	GENERIF_SUMMARY	HER2 and estrogen receptor alpha expression depends upon nuclear localization of Y-box binding protein-1 in human breast cancers
Literature	GENERIF_SUMMARY	analysis of insulin-like growth factor I receptor and HER2 in breast cancer
Literature	GENERIF_SUMMARY	Data presented here indicate that endogenous HER-2/neu-specific humoral and T-cell immunity is greater in patients with +3 protein overexpression in their tumors

Main Category	Category	Terms
		than in patients with lower levels of HER-2/neu expression
Literature	GENERIF_SUMMARY	The androgen receptor signaling pathways may contribute to development of metastatic disease in prostate cancer.
Literature	GENERIF_SUMMARY	HER2 status was the only primary tumor characteristic that correlated with the presence of circulating tumor cells.
Literature	GENERIF_SUMMARY	First detailed genome-wide characterisation of HER2/TOP2A-amplified breast cancers. The 17q12 amplicon is complex and harbours multiple genes that may be associated with breast cancer development and progression, and potentially therapeutic targets.
Literature	GENERIF_SUMMARY	Lipid raft-disrupting agents inhibited raft-associated CXCL12/CXCR4 transactivation of the HER2 and cellular invasion in prostate cancer cells.
Literature	GENERIF_SUMMARY	Knowledge of molecular mechanisms involved in geldanamycin-induced down-regulation of ErbB2 is important for future design of cancer treatment.
Literature	GENERIF_SUMMARY	Down-regulation of NOTCH3 significantly suppressed proliferation and promoted apoptosis of the ErbB2-negative tumor cell lines. Down-regulation of NOTCH3 did not have a significant effect on the ErbB2-positive tumor cell lines.
Literature	GENERIF_SUMMARY	The protein product of the ERBB2 oncogene is overexpressed in 33.3% of small cell lung carcinomas and is associated with the presence of disseminated disease
Literature	GENERIF_SUMMARY	These findings suggested a dynamic change in Her2/neu expression during the development of oral squamous cell carcinoma(OSCC).
Literature	GENERIF_SUMMARY	PAR1-stimulated EGFR and ErbB2 transactivation leads to prolonged extracellular signal-regulated kinase-1 and -2 signaling and promotes breast carcinoma cell invasion.
Literature	GENERIF_SUMMARY	The interaction of human MUC4 with the receptor tyrosine kinase HER2 in pancreatic adenocarcinoma cells is showm by reciprocal coimmunoprecipitation and cocapping studies.
Literature	GENERIF_SUMMARY	More than 25% of HER2 overexpression identified by immunohistochemistry assays in this Hong Kong cohort could not be verified by confirmatory in-situ hybridisation assays.
Literature	GENERIF_SUMMARY	Hpa may up-regulate the expression of C-erbB-2 in invasive ductal breast carcinoma.
Literature	GENERIF_SUMMARY	findings suggest that MUC4 is up-regulated and interacts with erbB2 in human gallbladder carcinoma, and thereby support the potential implication of MUC4 in erbB2 activation.

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Literature	GENERIF_SUMMARY	BIBW2992, an anilino-quinazoline designed to irreversibly bind EGFR and HER2, potently suppresses the kinase activity of wild-type and activated EGFR and HER2 mutants
Literature	GENERIF_SUMMARY	Differential regulation of ErbB2 expression by cAMP-dependent protein kinase in tamoxifen-resistant breast cancer cells.
Literature	GENERIF_SUMMARY	HER2-dependent transcriptional upregulation and protein secretion of MMP-7 are mediated by activated STAT3.
Literature	GENERIF_SUMMARY	There was no strong association between HER-2/neu overexpression and gene amplification in invasive urothelial carcinomas, and polysomy 17 was higher in tumors showing HER-2/neu overexpression.
Literature	GENERIF_SUMMARY	ERBB2 coding sequence reveals an absence of activating mutations in ERBB2 amplified breast cancer
Literature	GENERIF_SUMMARY	caspases activate a previously unrecognized proapoptotic function of HER-2 by releasing a Bad-like cell death effector
Literature	GENERIF_SUMMARY	Her-2/neu has a negative prognostic role in breast cancer survival among a Chinese population, irrespective of whether FISH or CISH is used to detect amplification of the Her-2/neu gene
Literature	GENERIF_SUMMARY	Dock7 functions as an intracellular substrate for ErbB2 to promote Schwann cell migration.
Literature	GENERIF_SUMMARY	the infection rate of HPV16 E6 and the positive expression rate of c-erbB-2 were significantly higher in the BCs than in the normal tissues
Literature	GENERIF_SUMMARY	ADAM15 catalyzes the cleavage of E-cadherin to generate a soluble fragment that in turn binds to and stimulates ErbB receptor signaling
Literature	GENERIF_SUMMARY	the extracellular domain of HER2 (ECD/HER2) may have a role in progression of breast cancer
Literature	GENERIF_SUMMARY	HER2 codon 655 polymorphism is associated with breast cancer.
Literature	GENERIF_SUMMARY	study investigated the internalization of erbB2 in SKBr3 and SKOv3 cells, both overexpressing erbB2; two different mechanisms, which are cell type dependent for the internalization of erbB2, are proposed
Literature	GENERIF_SUMMARY	Patients with HER2 FISH-positive LA-NSCLC had a poorer outcome even when treated with cisplatin-based chemoradiotherapy.
Literature	GENERIF_SUMMARY	We conclude that although HER-2/Neu amplification is not the primary mechanism in the development of liver tumors, it might play a role in one of the steps of multistage carcinogenesis.
Literature	GENERIF_SUMMARY	Although a significant percentage (25%) of pancreatic

Main Category	Category	Terms
		cancers demonstrate amplification of the HER2 gene, there is no association of HER2 gene amplification or chromosome 17 hyperploidy with poorer survival.
Literature	GENERIF_SUMMARY	Compared to PAI-1 protein levels, Chalkley counts and MIB-1, HER2+ and mutations of TP53 were the strongest independent markers of poor prognosis irrespective of nodal statusin breast cancer.
Literature	GENERIF_SUMMARY	A possible mechanism of role for nuclear HER2 in breast cancer patients.
Literature	GENERIF_SUMMARY	ErbB-2 overexpression suppresses Notch-1 activity, which can be reversed by trastuzumab or erbb2 tyrosine kinase inhibitors.
Literature	GENERIF_SUMMARY	HER2 expression was found in 16 (18%) of patients with gastric cancer.
Literature	GENERIF_SUMMARY	Estrogen receptor and HER2/neu status affect epigenetic differences of tumor-related genes in primary breast tumors
Literature	GENERIF_SUMMARY	HER-2-targeting recombinant protein with truncated pseudomonas exotoxin A translocation domain efficiently kills breast cancer cells.
Literature	GENERIF_SUMMARY	Overexpression of HER2 is encountered in approximately 20% of invasive breast cancers.
Literature	GENERIF_SUMMARY	A possible correlation between overexpression of p53, proliferating cell nuclear antigen (PCNA), and c-erbB-2, and the clinicopathologic features of laryngeal squamous cell carcinoma, was investigated.
Literature	GENERIF_SUMMARY	we propose that SNPid rs4252633 could be the most deleterious nsSNP for HER2 receptor
Literature	GENERIF_SUMMARY	ErbB2 inhibition leads to a persistent phosphorylation of ERK1/2, which negatively regulates the downstream signaling and cell growth
Literature	GENERIF_SUMMARY	ErbB2 is found at the cell surface of differentiating keratinocytes and in the cytoplasm of poorly differentiated tumor cells.
Literature	GENERIF_SUMMARY	HER-2(10(85))-specific human CTL recognized the HER-2/neu+ HLA-A2.1+ tumor cell line SKBR3.A2
Literature	GENERIF_SUMMARY	HER2/neu was exclusively overexpressed in serous carcinomas of endometrial origin.
Literature	GENERIF_SUMMARY	This study shows superior survival in HER2+ group following whole brain radiotherapy for cerebral metastases of breast cancer as compared to HER2- group.
Literature	GENERIF_SUMMARY	CK19 signal was occasionally observed in the peripheral blood of normal controls, and the HER2 signal was frequently present in the peripheral blood of both normal controls and breast cancer patients.

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Literature	GENERIF_SUMMARY	HER2-positive groups showed worse prognosis in stage III of breast cancer, but not in stage I or II
Literature	GENERIF_SUMMARY	Review of HER2-positive breast cancer treatment strategies.
Literature	GENERIF_SUMMARY	effects of HER2 amplification on carcinogenesis, tumorigenesis and invasion may be due to its effects on normal and malignant mammary stem/progenitor cells.
Literature	GENERIF_SUMMARY	HST may act as a paracrine factor in the adult breast.
Literature	GENERIF_SUMMARY	ABCC3 is a mediator of taxane resistance in HER2-amplified breast cancer.
Literature	GENERIF_SUMMARY	HRG-beta-induced MMP-7 expression was regulated by HER2-mediated AP-1 activation in MCF-7 cells.
Literature	GENERIF_SUMMARY	HER-2 may not have any role in gastrointestinal stromal tumor pathogenesis
Literature	GENERIF_SUMMARY	Datas show that transforming growth factor beta engages TACE and ErbB3 to activate phosphatidylinositol-3 kinase/Akt in ErbB2-overexpressing breast cancer and desensitizes cells to trastuzumab.
Literature	GENERIF_SUMMARY	Coamplification of HER2 and MYC occurs in a subset of patients with metastatic UCC.
Literature	GENERIF_SUMMARY	we emphasize the important short-term prognostic value of currently used immunohistochemical markers for ER, PR, and HER-2
Literature	GENERIF_SUMMARY	inhibition of HER3 may be more clinically relevant than inhibition of EGFR in HER2-amplified breast cancer and adding pertuzumab to trastuzumab may augment therapeutic benefit by blocking HER2/HER3 signaling
Literature	GENERIF_SUMMARY	The mRNA stability factor HuR was shown to support ERBB2 transcript integrity, bind and endogenously associate with a conserved U-rich element within the ERBB2 transcript 3' UTR, and colocalize with HDAC6.
Literature	GENERIF_SUMMARY	Overexpression of HER2 protein in cancer tissue might suggest an increased risk of dying from prostate cancer
Literature	GENERIF_SUMMARY	EGFR and erbB2 are potential targets in treatment of Malignant peripheral nerve sheath tumors
Literature	GENERIF_SUMMARY	Inhibition of tyrosine kinase activity of Her-2/neu by quercetin could indicate an lateration in the Her-2/neu structure which promotes CHIP recruitments and down-regulation of Her-2/neu
Literature	GENERIF_SUMMARY	Results describe the engineering of a retargeted herpes simplex virus 1 recombinant capable of entering cells through human HER2.
Literature	GENERIF_SUMMARY	Systematic review with statistical analysis of immunohistochemistry and FISH studies of HER2 testing and selection of patients suitable for

Main Category	Category	Terms
		trastuzumab therapy
Literature	GENERIF_SUMMARY	analysis of EGFR and HER2 expression in primary cervical cancers and corresponding lymph node metastases
Literature	GENERIF_SUMMARY	Akt1 activation accelerates MMTV-c-ErbB2 mammary tumourigenesis in mice without activation of ErbB3
Literature	GENERIF_SUMMARY	TOP2A deletion is associated with poor prognosis in ERBB2-amplified breast carcinomas.
Literature	GENERIF_SUMMARY	PTK6 is coamplified and coexpressed with ERBB2 in human breast cancers. ERBB2 interacts with PTK6 and increases its intrinsic kinase activity
Literature	GENERIF_SUMMARY	I655V polymorphism of HER2 transmembrane domain-coding region at codon 655 is associated with breast cancer.
Literature	GENERIF_SUMMARY	WWP1 may promote cell proliferation and survival partially through suppressing RNF11-mediated ErbB2 and EGFR downregulation in human cancer cells.
Literature	GENERIF_SUMMARY	A selective allosteric inhibitor of Akt kinase was used to interrogate a panel of breast cancer cell lines characterized for genetic lesions that activate HER2 amplification.
Literature	GENERIF_SUMMARY	Findings suggest that aberrant c-erbB2 expression in cell clusters overlying focally disrupted breast myoepithelial cell layers may trigger or signify the emergence of biologically more aggressive cell clones.
Literature	GENERIF_SUMMARY	HER2/Neu synergized with ETV1 to upregulate the endogenous Rcl.
Literature	GENERIF_SUMMARY	DNA methylation of tumor suppressor genes is a frequent event in ovarian cancer, and in some cases is associated with Her-2/neu overexpression.
Literature	GENERIF_SUMMARY	Her-2/neu oncogene and nuclear roundness variance were shown to be significant in the prediction of progression-free survival in prostate cancer.
Literature	GENERIF_SUMMARY	Her-2/neu expression and %DNA index were significant prognosticators for progression
Literature	GENERIF_SUMMARY	Patients with HER2 nonamplified and HER1 through HER3-negative tumors showed significantly increased relapse-free and overall survival rates when treated with epirubucin-CMF.
Literature	GENERIF_SUMMARY	combined 17-AAG and Trastuzumab treatment of ErbB2-overexpressing breast cancer cell lines leads to enhanced ubiquitinylation, downregulation from the cell surface and lysosomal degradation of ErbB2
Literature	GENERIF_SUMMARY	PCR-free L-FISH method can be used to evaluate ERBB2 amplification in both breast cancer cell-containing (paraffin-embedded tissue) and cell-free (serum) samples

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	a trivalent bispecific antibody targets ErbB2 and CD16 in tumor cells
Literature	GENERIF_SUMMARY	EGFRvIII with ErbB-2 enhances tumorigenesis
Literature	GENERIF_SUMMARY	Protein kinase A-mediated gating of neuregulin-dependent ErbB2-ErbB3 activation has a role in the synergistic action of cAMP on Schwann cell proliferation
Literature	GENERIF_SUMMARY	Grade 1 invasive ductal carcinoma have a higher incidence of lymph node metastasis and may have Her-2-neu overexpression compared to tubular carcinoma.
Literature	GENERIF_SUMMARY	EBNA1-NT may act as a repressor of the HER2/neu oncogene
Literature	GENERIF_SUMMARY	This study suggests a pivotal role of PI3 K, MAPK and mTOR pathways, via tuberin, in post-transcriptional control of CXCR4, initiated through estrogen-stimulated crosstalk between ER and HER2.
Literature	GENERIF_SUMMARY	Erb-B2 mRNA levels were not grade- or histotype-related in patients with meningioma
Literature	GENERIF_SUMMARY	in some infiltrating squamous cervical carcinoma there is a rupture of the ERBB2 receptor
Literature	GENERIF_SUMMARY	These data support the cooperative function of Pyk2 and FAK in breast cancer progression and suggest that dual inhibition of FAK and Pyk2 is an efficient therapeutic approach for targeting invasive breast cancer.
Literature	GENERIF_SUMMARY	Results suggest that HER-2 codon 655 polymorphism is associated with progession of oral squamous cell carcinoma in Taiwanse patients.
Literature	GENERIF_SUMMARY	This study confirms that Ets-1 is a downstream effector of oncogenic HER2, associated with MMP-1.
Literature	GENERIF_SUMMARY	The frequency of allelic imbalance was significantly higher (P
Literature	GENERIF_SUMMARY	findings indicate that serum HER2/neu levels are clinically valuable in monitoring metastatic breast cancer and non-small cell lung cancer patients.
Literature	GENERIF_SUMMARY	Suppression of the negative regulator LRIG1 contributes to ErbB2 overexpression in breast cancer.
Literature	GENERIF_SUMMARY	CK7, bax, CCND1, and HER2 represent marker proteins and frequently amplified genes in carcinomas of the ampulla of Vater.
Literature	GENERIF_SUMMARY	ERBB3 receptors are apparently segregated from ERBB2 receptors in their resting state, and both ligand-activated ERBB3 and ERBB2 do not share the same microenvironment as inactive ERBB3
Literature	GENERIF_SUMMARY	we have demonstrated an increase in ERBB2 receptor

Main Category	Category	Terms
		activation in incompletely resected preinvasive breast cancer
Literature	GENERIF_SUMMARY	Increased serum HER-2 is associated with advanced breast cancer.
Literature	GENERIF_SUMMARY	Its expression is prognostic parameter in breast carcinomas.
Literature	GENERIF_SUMMARY	EGFR as well as HER2 are prognostic factors of worse clinical outcomes, in high-risk early breast cancer
Literature	GENERIF_SUMMARY	These results suggest an oncogenic role for HER2/neu in anti-BPDE-induced carcinogenesis.
Literature	GENERIF_SUMMARY	Examine human epidermal growth factor receptor 2 overexpression as a prognostic factor in a large tissue microarray series of node-negative breast cancers.
Literature	GENERIF_SUMMARY	Results showed that median age, menarche age, childbearing age, number of children, menopause age, and body-mass indexes were similar in both HER-2 pos and neg groups. # of involved lymph nodes and HER-2 status found to be prog. factors for survival.
Literature	GENERIF_SUMMARY	study found ERBB2 mutations (6%) are prevalent among ovarian serous low malignant potential tumors
Literature	GENERIF_SUMMARY	study showed in ovarian cancer cells LHR expression & activation upregulate ErbB-2 expression; data establish role for LH & LHR in regulation of ErbB-2 expression & suggest ErbB-2 upregulation alone is insufficient in producing aggressive phenotype
Literature	GENERIF_SUMMARY	HER2 mRNA could be a valuable prognostic marker for the detection of circulating tumor cells in early breast cancer patients
Literature	GENERIF_SUMMARY	erbB2 plays a role in cardiomyocyte survival
Literature	GENERIF_SUMMARY	that p38 MAPK and Akt signaling pathways are crucial for the ErbB2-induced MMP-9 up-regulation, invasion and migration of MCF10A cells
Literature	GENERIF_SUMMARY	HER2 affects glial-cell migration by modulating EGFR-HER2 signal transduction, and that this effect is mediated by N-cadherin.
Literature	GENERIF_SUMMARY	Ectopically overexpressed HER2 in medulloblastoma cell lines significantly increased sensitivity to gefitinib's antitumor effects in vivo
Literature	GENERIF_SUMMARY	Overexpression of HER-2 oncoprotein is associated with gastric carcinoma.
Literature	GENERIF_SUMMARY	Results demonstrated that in ER+ breast cancer, the HER2 receptor is commonly activated, but its low prevalence in ER+ tumors does not render it a useful prognostic parameter in tamoxifen-treated patients.
Literature	GENERIF_SUMMARY	analysis of the effect of HER2 pre-mRNA alternative splicing on breast cancer cells

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	polymorphisms of the HER-2 gene are associated with an increased susceptibility to lung cancer in females, non-smokers and non-drinkers subgroups in the Korean population
Literature	GENERIF_SUMMARY	These data highlight the role of PAR4 as a new important player in the control of colon tumors and underline the critical role of ErbB-2 transactivation.
Literature	GENERIF_SUMMARY	Her2 immunostaining >and SBR grade 3 were independent factors predicting HER2 amplification status irrespective to cutoff guidelines
Literature	GENERIF_SUMMARY	stabilization and accumulation of HER2 is induced by lapatinib, which potentiates trastuzumab-dependent cell cytotoxicity
Literature	GENERIF_SUMMARY	HER-2/neu, AR, and p53 are expressed in a subset of histologically and clinically benign pleomorphic adenomas. These markers cannot be used to reliably predict early carcinomatous transformation in pleomorphic adenoma.
Literature	GENERIF_SUMMARY	gene amplification correlates with protein overexpression in Chinese patients with gastric carcinoma; chromosome 17 polysomy is likely not to be the main cause
Literature	GENERIF_SUMMARY	Pin1 is a novel regulator of erbB2 that modulates the ubiquitin-mediated degradation of erbB2.
Literature	GENERIF_SUMMARY	HER2 and Topo IIalpha overexpression could be predictors of the response to neoadjuvant chemotherapy in both the CEF and CMF arms.
Literature	GENERIF_SUMMARY	Results revealed significantly increased levels of HER-2 neu in both serum and bronco-alveolar lavage.
Literature	GENERIF_SUMMARY	Combination therapy with trastuzumab was not effective in the HER2 or luminal B group
Literature	GENERIF_SUMMARY	HER-2 overexpression represent important alterations that are related to the molecular pathways underpinning colorectal carcinogenesis
Literature	GENERIF_SUMMARY	A significant MHC class I-restricted cytotoxic T-lymphocyte killing was demonstrated against Her-2/neu-positive ovarian cancer cells.
Literature	GENERIF_SUMMARY	Gene amplification and protein expression of HER-2/neu is correlated in esophageal squamous cell carcinoma.
Literature	GENERIF_SUMMARY	The aim of the study was to evaluate the correlation between clinical characteristics, histopatologic features and c-erbB-2 as well as p53 expression in cancer tissues.
Literature	GENERIF_SUMMARY	Our results cannot support the hypothesis of a transactivation of HER-2 and thus a possible therapeutic benefit of trastuzumab in HER-2 negative breast cancers.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	statistically significant association was found between ERBB2 amplification and worse survival in patients with expansive gastric carcinomas (P=0.011). We conclude that ERBB2 status may have clinical significance in subsets of gastric cancer patients
Literature	GENERIF_SUMMARY	frequent methylation had a strong association with HER2 amplification in breast cancers and suggested that frequent methylation can be a determinant of various characteristics in a fraction of human breast cancers.
Literature	GENERIF_SUMMARY	ErbB2 signalling regulated focal adhesion turnover in invasive breast cancer cells.
Literature	GENERIF_SUMMARY	Patients with a HER2 positive tumor receiving trastuzumab more frequently develop brain metastases than patients with a HER2 negative tumor but have a more favorable prognosis.
Literature	GENERIF_SUMMARY	Data indicate that the level of HCCR-1 in breast cancer tissues is correlated with the HER2 overexpression, p53 mutation, and ER/PR status, and determination of HCCR-1 levels as options for HER2 testing is promising although it needs further evaluation.
Literature	GENERIF_SUMMARY	ER and PR showed inverse association with HER-2/neu while positive association was seen with lymph node metastases in breast carcinoma patients in Pakistan
Literature	GENERIF_SUMMARY	namely the ability of mAb combinations to instigate ErbB-2 endocytosis, is demonstrated
Literature	GENERIF_SUMMARY	Memo was detected in complexes with cofilin, ErbB2 and PLCgamma1.
Literature	GENERIF_SUMMARY	The use of the small scaffold targeting Affibody provides better contrast in HER2 imaging than does the monoclonal antibody.
Literature	GENERIF_SUMMARY	Human epidermal growth factor receptor 2 (HER2) status significantly affects the survival in breast cancer.
Literature	GENERIF_SUMMARY	Paients with HER2-overexpressing breast cancer treated with trastuzumab had a high incidence of brain metastasis.
Literature	GENERIF_SUMMARY	mutations in EGFR and copy number gains in EGFR and HER2 were independent factors related to gefitinib sensitivity, in descending order of importance
Literature	GENERIF_SUMMARY	Even low-level HER2 expression may be associated with worse outcome in node-positive breast cancer.
Literature	GENERIF_SUMMARY	we conclude that there is currently insufficient evidence to support the use of serum HER-2 extracellular domain in the routine management of individual patients with breast cancer
Literature	GENERIF_SUMMARY	we cannot recommend using serum HER2 extracellular domain levels to make trastuzumab or other treatment decisions for individual patients with advanced/

Main Category	Category	Terms
		metastatic breast cancer.
Literature	GENERIF_SUMMARY	Data show that an eukaryotic plasmid encoding HER2 (pcDNA6/v5-his-HER2) has been constructed and a cell line expressing HER2 stably has been prepared successfully.
Literature	GENERIF_SUMMARY	SHP2 promotes HER2-induced signaling and transformation at least in part by dephosphorylating a negative regulatory autophosphorylation site.
Literature	GENERIF_SUMMARY	Poor downregulation of ErbB2 receptors is not due to sequence variations in the Proto-Oncogene Proteins Cbl binding site of these receptors.
Literature	GENERIF_SUMMARY	The high level of HER-2 amplification in combination with the homogeneity of its expression in primary and metastatic tumors argues for a possible therapeutic utility of trastuzumab in HER-2-amplified gastric adenocarcinomas.
Literature	GENERIF_SUMMARY	Invasive micropapillary carcinoma more commonly showed immunoreactivity for MUC1, CA125, and Her2Neu compared to invasive urothelial carcinoma with retraction artifact
Literature	GENERIF_SUMMARY	Although c-erbB-2 expression is important from a biological standpoint, multivariate analysis showed that it is not an independent prognostic indicator in breast carcinoma in the local population.
Literature	GENERIF_SUMMARY	It is proposed that the negative feedback loop linking ERBB2 and GATA4 plays a role in the transcriptional dysregulation of ERBB2 gene expression in breast cancer.
Literature	GENERIF_SUMMARY	no cases of sinonasal undifferentiated carcinoma were positive for HER2/neu
Literature	GENERIF_SUMMARY	ErbB-1/-2 receptors are inhibited by lapatinib in bladder neoplasms
Literature	GENERIF_SUMMARY	a new model for the airway epithelial damage response, in which the MUC4-ErbB2 complex is a key element in the sensor mechanism and phosphorylation of the receptors.
Literature	GENERIF_SUMMARY	HER-2 overexpression/amplification in Barrett's oesophagus has a role in transition from dysplasia to adenocarcinoma
Literature	GENERIF_SUMMARY	Our results indicated that HER2 status is independent of a response to a taxanes/cisplatin combination and is also not a prognostic factor for survival.
Literature	GENERIF_SUMMARY	HER1 and 2 do not seem to have a role in endometriosis physiopathology.
Literature	GENERIF_SUMMARY	Osteoblast-induced EGFR/ERBB2 signaling in androgen-sensitive prostate carcinoma cells is characterized by multiplex kinase activity profiling

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Findings indicated that dqRT-PCR was more precise and reproducible than IHC to detect HER-2, while CISH was an alternative and useful procedure for investigating amplifications involving the HER-2 gene.
Literature	GENERIF_SUMMARY	Inhibition of both AR isoforms did not affect EGFR and ERBB2 transcript levels but decreased EGFR and increased ERBB2 protein levels. Proliferation of 22Rv1 cells in SDM was inhibited in the absence of AR and ARDeltaCTD.
Literature	GENERIF_SUMMARY	Study shows that genomic instability due to Myc amplification may cause specific amplification of EGFR and/or ERBB2 in gallbladder adenocarcinoma.
Literature	GENERIF_SUMMARY	Heritable genetic variation of ERBB2 is associated with breast cancer.
Literature	GENERIF_SUMMARY	Overexpression of HER2 is associated with gastric cancer.
Literature	GENERIF_SUMMARY	These data provided direct evidence that HER2/neu signaling is essential for tumorigenicity of non-small cell lung cancer cells
Literature	GENERIF_SUMMARY	Her-2/neu (47.1%), p-53 (31.74%), and their combined expression (15.87%) were not related to grade or stage of tumor in osteosarcoma
Literature	GENERIF_SUMMARY	Most breast cancer patients with various levels of HER2/neu expression responded immunologically and seemed to benefit from vaccination.
Literature	GENERIF_SUMMARY	Expression of tyrosine-phosphorylated PP2A (pY307-PP2A) was highly increased in the HER-2/neu positive breast tumours, and significantly correlated to tumour progression, thus enhancing its potential prognostic value.
Literature	GENERIF_SUMMARY	A transcriptomic analysis revealed that HER2 carboxy-terminal fragment 611-CTF specifically controlled the expression of genes that we found to be correlated with poor prognosis in breast cancer.
Literature	GENERIF_SUMMARY	The routine assessment of HER2/neu amplification using the ASCO/CAP criteria on whole tissue sections is not significantly confounded by intratumoral heterogeneity in breast cancer with high-grade amplification.
Literature	GENERIF_SUMMARY	Amplification of the Her2/neu gene is a rare event in cutaneous hidradenocarcinoma.
Literature	GENERIF_SUMMARY	directed towards this region of domain III of the HER2, distinct from the well-known monoclonal antibodies trastuzumab and pertuzumab, bind to the HER2 on living cells and exhibit growth inhibition.
Literature	GENERIF_SUMMARY	Her-2 positive breast carcinomas were larger and involved more lymph nodes than Her-2 negative carcinomas

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	miR-21 is up-regulated via the MAPK (ERK1/2) pathway upon stimulation of HER2/neu signaling in breast cancer cells
Literature	GENERIF_SUMMARY	expression of phospho-4E-BP1 and HER2 overexpression were the only characteristics with prognostic significance in endometrial carcinoma
Literature	GENERIF_SUMMARY	HER-2/neu amplification could be regarded as an additional genetic factor playing role in meningioma pathogenesis together with known chromosomal abnormalities
Literature	GENERIF_SUMMARY	HER2 overexpression correlates with survival after curative resection of pancreatic cancer
Literature	GENERIF_SUMMARY	Activation of ErbB2, but not ErbB1, increases PTP1B expression, and increased expression of PTP1B activates Src and induces a Src-dependent transformed phenotype.
Literature	GENERIF_SUMMARY	Expression of ER, progesterone receptor, and HER2 proteins and the Ki67 index appear to distinguish luminal A from luminal B breast cancer subtypes
Literature	GENERIF_SUMMARY	findings seem to suggest that obese carriers with minor variant allele of the HER-2 polymorphism may have an increased risk of endometrial cancer and may be consistent with that the HER-2 I655V minor allele variant may be functionally important.
Literature	GENERIF_SUMMARY	Pten functions downstream of ErbB-2 to restrict prostate epithelial transformation by blocking full activation of the PDK1 signaling cascade
Literature	GENERIF_SUMMARY	although HER2/neu amplification, found in high grade and invasive tumors, is a rare event in transitional cell carcinoma of urinary bladder
Literature	GENERIF_SUMMARY	The use of anti-HER-2 ribozymes may be a beneficial strategy for gene therapy of breast cancer.
Literature	GENERIF_SUMMARY	Immunodetection of HER2/neu receptor correlated positively with both ERalpha and PR immunolabelling. The Ki-67 proliferation index correlated only with ERalpha immunopositivity.
Literature	GENERIF_SUMMARY	retrospectively examined a series of breast cancer patients with brain metastases for the presence of HER-2
Literature	GENERIF_SUMMARY	Results showed that 45.8% of typical carcinoid and 28.6% of atypical carcinoid tumors express EGFR, 100% of the tumors lack expression of ErbB2, and 100% have moderate to intense staining for ErbB3 and ErbB4.
Literature	GENERIF_SUMMARY	HER-2 was negative in all NHLs studied using the chromogenic in situ hybridisation test
Literature	GENERIF_SUMMARY	The 1170 A-P single-nucleotide polymorphism (SNP) in the Her-2/neu protein (HER2) acts as a minor histocompatibility antigen in Acute Myeloid Leukemia.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	HER-2/neu, and p53 do not seem to play a significant role in the process of high grade transformation of AciCC
Literature	GENERIF_SUMMARY	Suppression of ErbB-2 in androgen-independent human prostate cancer cells enhances cytotoxic effect by gemcitabine in an androgen-reduced environment.
Literature	GENERIF_SUMMARY	Both ER-beta positive and HER2 positive status was associated with poorer overall survival (OS) by univariate analysis.
Literature	GENERIF_SUMMARY	these results validate the role of miR-548d-3p and miR-559 in regulating the ERBB2 expression.
Literature	GENERIF_SUMMARY	ERBB2 and LRIG1 copy number is increased in breast cancer
Literature	GENERIF_SUMMARY	findings show that EGFR and ErbB2 can negatively regulate androgen receptor (AR) mRNA and may provide an approach to suppress AR expression in CRPC
Literature	GENERIF_SUMMARY	high HER2 and EGF protein levels were significantly correlated with TOB1 expression in breast cancer, whereas EGFR and EGF levels correlated with TOB1 phosphorylation
Literature	GENERIF_SUMMARY	Data demonstrate a more stable binding of heregulin to cells co-expressing HER3 and HER2 than those expressing HER3 alone.
Literature	GENERIF_SUMMARY	Findings highlight an important role for Rac1 in trastuzumab resistance of human breast cancer cells and identify the impaired trastuzumab-mediated endocytic down-regulation of ErbB2 as a novel mechanism of trastuzumab resistance.
Literature	GENERIF_SUMMARY	FoxM1 is a downstream target and marker of HER2 overexpression in breast cancer.
Literature	GENERIF_SUMMARY	Radiation could induce ERBB2 nuclear transport in breast cancer cells.
Literature	GENERIF_SUMMARY	Data show that overexpression of shrew-1 leads to preformation of an E-cadherin/EGF receptor/HER2/src-kinase/shrew-1 signaling complex and accelerated E-cadherin internalization.
Literature	GENERIF_SUMMARY	HER-2/neu expression in extramammary Paget disease
Literature	GENERIF_SUMMARY	Our results suggest that the described (18)F-FBEM-Z(HER2
Literature	GENERIF_SUMMARY	Studies define an interaction between Neu and SLK signaling in the regulation of cancer cell motility.
Literature	GENERIF_SUMMARY	lapatinib inhibits the nuclear translocation of EGFR and HER2 and downregulates TS, thus sensitizing cancer cells to fluoropyrimidine
Literature	GENERIF_SUMMARY	HER-2 is expressed in breast cancer in the Finnish

Main Category	Category	Terms
		population at unchanged levels from 1982-2005
Literature	GENERIF_SUMMARY	In this study there was no association between HER2 plasma levels and cardiac function or left ventricular wall thickness.
Literature	GENERIF_SUMMARY	c-erbB-2 has a complex role in early breast carcinomas, being associated with both favorable (triple-positive) & unfavorable (c-erb-B2 positive after excluding triple-positive and triple-negative tumors)prognosis due to activating different pathways.
Literature	GENERIF_SUMMARY	ER, PR and Her2 have roles in progression of breast cancer
Literature	GENERIF_SUMMARY	X-ray diffraction data set was collected to 2.45 A resolution for antibody-antigen complex consisting of the single-chain variable fragment (scFv) of chA21 and an N-terminal fragment (residues 1-192, named EP I) of the ErbB2 extracellular domain
Literature	GENERIF_SUMMARY	results indicate that HER2 status may be a predictive and prognostic factor for cisplatin- based therapy response and disease survival.
Literature	GENERIF_SUMMARY	not over-expressed or amplified in pancreatic adenocarcinoma
Literature	GENERIF_SUMMARY	miR-331-3p regulates ERBB-2 expression and androgen receptor signaling in prostate cancer
Literature	GENERIF_SUMMARY	HER2 carboxyl-terminal fragments regulate cell migration and cortactin phosphorylation
Literature	GENERIF_SUMMARY	a subset of patients with HER2-negative primary tumors develops HER2-positive circulating tumor cells during disease progression.
Literature	GENERIF_SUMMARY	HER2/neu expression in Wilms tumor could be a marker for epithelial and homologous differentiation and its expression could be a good predictor for overall survival and longer recurrence free survival.
Literature	GENERIF_SUMMARY	Data show that aberrant expression of ErbB2, ErbB3 and ErbB4 in the two resistant cell lines, suggesting the existence of an autocrine loop leading to constitutive activation of MAPK and PI3K/Akt survival pathways.
Literature	GENERIF_SUMMARY	Increased expression of ErbB-2 is associated with liver cancer.
Literature	GENERIF_SUMMARY	(111)In-DTPA-pertuzumab sensitively imaged HER2 downregulation after 3 d of treatment with trastuzumab and detected a reduction in viable HER2-positive tumor cells after 3 wk of therapy in MDA-MB-361 human breast cancer xenografts.
Literature	GENERIF_SUMMARY	Non-small cell lung carcinoma brain metastases have some significant differences in HER family receptor-related abnormalities from primary lung tumors.
Literature	GENERIF_SUMMARY	Growth of most HER2-amplified cells was inhibited by

Main Category	Category	Terms
		LY294002, regardless of PIK3CA genotype.
Literature	GENERIF_SUMMARY	Nuclear Y-box binding protein-1 is correlated with expression of HER2/ErbB2 and HER3/ErbB3 in non-small cell lung cancer.
Literature	GENERIF_SUMMARY	LMO4 is a novel cell cycle regulator with a key role in mediating ErbB2-induced proliferation, a hallmark of ErbB2-positive disease.
Literature	GENERIF_SUMMARY	HER-2/neu increased in high-grade ductal breast tumors compared to DCIS.
Literature	GENERIF_SUMMARY	HER-2 IHC3+/FISH- breast cancer could be a subgroup showing lower histological/nuclear grade and higher expression of ERs/PRs.
Literature	GENERIF_SUMMARY	In univariate analyses, Tau- and ER-positive status were both associated with better DFS and OS (P
Literature	GENERIF_SUMMARY	ErbB2 promotes glycolysis at least partially through the HSF1-mediated upregulation of LDH-A.
Literature	GENERIF_SUMMARY	Synovial sacromas did not show HER2 protein overexpression or gene amplification.
Literature	GENERIF_SUMMARY	Data show that in multivariate analyses incorporating tumor size, tumor grade, lymph node status, ER, PR and HER2 status, low STARD10 expression was an independent predictor of death from breast cancer.
Literature	GENERIF_SUMMARY	detachment of mammary epithelial cells from ECM causes an ATP deficiency owing to the loss of glucose transport; overexpression of ERBB2 rescues the ATP deficiency by restoring glucose uptake
Literature	GENERIF_SUMMARY	In the absence of inhibitors, 3D cultures had substantial impact on HER2 downstream signaling and induced a switch between PI3K-AKT- and RAS-MAPK-pathway activation in all cell lines studied, including cells lacking HER2 amplification and overexpression.
Literature	GENERIF_SUMMARY	HER-2-positive status tended to be associated with a worse 2-year survival, but the overall results were not formally statistically significant.
Literature	GENERIF_SUMMARY	Studies indicate that neither HER-2 status nor topoisomerase IIalpha status can be considered clinically valuable in guiding prescription of anthracyclines.
Literature	GENERIF_SUMMARY	normal and pathogenic ErbB2 signalling may be regulated by ligands in the same way as dEGFR
Literature	GENERIF_SUMMARY	Her2 overexpression may increase the hTERT transcriptional activation.
Literature	GENERIF_SUMMARY	Serum HER-2 is a useful marker for the detection of recurrence of breast cancer
Literature	GENERIF_SUMMARY	no difference in metastatic breast cancer survival between HER2-nonamplified and HER2-amplified

Main Category	Category	Terms
		patients for subgroup as whole
Literature	GENERIF_SUMMARY	14-3-3zeta cooperates with ErbB2 to promote ductal carcinoma in situ progression to invasive breast cancer by inducing epithelial-mesenchymal transition.
Literature	GENERIF_SUMMARY	A subset of human cancers maintain telomere length by a recombination mechanism. This study describes evidentce for this phenotype in a subset of breast cancers with HER-2 overexpression.
Literature	GENERIF_SUMMARY	Depending on the strength of ErbB2 signaling there is a differential regulation of CHOP and eIF2alpha phosphorylation.
Literature	GENERIF_SUMMARY	Survivin-c-erbB2-cox2 co-expression score has independent prognostic value for endometrial cancer.
Literature	GENERIF_SUMMARY	The aim of the current study was to assess the expression levels of c-Src and phosphorylated Src kinase in human breast cancers and to establish if these are linked to oestrogen receptor, progesterone receptor, and HER2 status or patient survival.
Literature	GENERIF_SUMMARY	The incidence of CNS metastases in trastuzumab-treated patients is similar to that in all patients with HER2-positive disease
Literature	GENERIF_SUMMARY	HER2/neu plays no major role in the development and progression of cervical neoplasia.
Literature	GENERIF_SUMMARY	Her2 is expressed in bilary tract cancer
Literature	GENERIF_SUMMARY	The ERBB2 gene was neither overexpressed nor amplified in any Wilms tumor case.
Literature	GENERIF_SUMMARY	HER2 phenotype upregulates hTERT through c-myc activation and confers radio-resistance to breast cancer cells.
Literature	GENERIF_SUMMARY	Plasma TNF-alpha and IL-2 levels were positively correlated in ERBB2 positive breast cancer patients.
Literature	GENERIF_SUMMARY	breast tumors with apocrine differentiation are most often of ERBB2 type.
Literature	GENERIF_SUMMARY	HER2 overexpression represents a prognostic factor for adverse urothelial bladder cancer outcome.
Literature	GENERIF_SUMMARY	Report immunogenicity of allo-vesicle carrying ERBB2 tumor antigen for dendritic cell-based anti-tumor immunotherapy.
Literature	GENERIF_SUMMARY	HER2 overexpression was not associated with higher local or regional recurrence in breast cancer.
Literature	GENERIF_SUMMARY	Data identify a novel function for Shc as a regulator of apoptosis and drug resistance in mammary epithelial cells.
Literature	GENERIF_SUMMARY	Aberrant expression of HER2/neu apparently does not contribute to carcinogenesis in the oral epithelium.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	triple negative (estrogen receptor, progesterone receptor, and human epidermal growth factor receptor 2) breast cancer in young women has a significantly poorer prognosis for both disease-free survival rate and overall survival rate.
Literature	GENERIF_SUMMARY	our results support the role of Her-2/neu in acquired hormone resistance.
Literature	GENERIF_SUMMARY	Data show that both full-length and p95-HER2 interact with the HSP90 chaperone protein and are degraded in tumor cells exposed to HSP90 inhibitors.
Literature	GENERIF_SUMMARY	Studies indicate that capsaicin is a novel modulator of the EGFR/HER-2 pathway in both ER-positive and -negative breast cancer cells with a potential role in the treatment and prevention of human breast cancer.
Literature	GENERIF_SUMMARY	SNP HER2 I655V, but not the EGFR R497K, was associated with thyroid cancer risk.
Literature	GENERIF_SUMMARY	Mannosylated ECD/Her2 elicited a humoral response with HER2/neu specific antibodies in vaccinated mice.
Literature	GENERIF_SUMMARY	HER2 receptors are activated at high levels after EGF stimulation in all HER2 expressing human mammary epithelial cell lines.
Literature	GENERIF_SUMMARY	Management of HER2-positive breast cancer in Asia
Literature	GENERIF_SUMMARY	No relationship of Akt phosphorylation with HER2, ER-alpha or ER-beta expression in breast cancer was found
Literature	GENERIF_SUMMARY	estrogen receptor (ER), progesterone receptor (PR), and HER2 have roles in breast cancer
Literature	GENERIF_SUMMARY	Study examined the potential significance of HER2 nuclear expression in DNA repair in breast cancer.
Literature	GENERIF_SUMMARY	5.1% of invasive bladder carcinomas had a HER2 gene amplification.
Literature	GENERIF_SUMMARY	Ku70 and Ku80 proteins in interaction with AP-2 (alpha and gamma) contribute to increased ERBB2 mRNA and protein levels in breast cancer cells.
Literature	GENERIF_SUMMARY	qRT-PCR showed HER-2/neu amplification in 13 cases of breast cancer, while the standard IHC-FISH combined approach identified 17 HER-2/neu-positive cases.
Literature	GENERIF_SUMMARY	The high expressions of VEGF-C and C-erbB-2 are closely related to lymph node metastasis in breast cancer patients.
Literature	GENERIF_SUMMARY	The expression of 8 proteins, including epidermal growth factor receptor, glutathione S-transferase pi (GST-pi), cyclin D1, Neu/Her-2, C-myc, p53, p27 and p21, were evaluated in 110 cases with cardiac cancer and 101 cases with distal gastric cancer
Literature	GENERIF_SUMMARY	the HER2 signaling pathway may have a role in radiosensitization in breast neoplasms

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	a significant association was found between the short tandem repeat (STR) [H(AC)I4] and breast cancer risk at both genotypic and allelic levels
Literature	GENERIF_SUMMARY	Inhibition of PI3K and mTOR are required for the growth-inhibitory effect of HER2 antagonists
Literature	GENERIF_SUMMARY	High HER2 is associated with high vascular endothelial growth factor-C and lymphangiogenesis in lymph node-positive breast cancer.
Literature	GENERIF_SUMMARY	simple co-amplification is not the main mechanism leading to patients with both ERBB2 and TOP2A aberrations and the term simultaneous amplification is more precise. ERBB2 amplification is of higher level than TOP2A amplification.
Literature	GENERIF_SUMMARY	The synergism deriving from EGFR and ErbB-2 blockade is mediated by alterations in signal transduction and in the expression of cell cycle regulatory proteins, including transcriptional and posttranscriptional regulation of p27(kip1) expression.
Literature	GENERIF_SUMMARY	inhibition of Notch signaling could antagonize survival signal of Her2-negative breast cancer-initiating cells carrying genomic damage
Literature	GENERIF_SUMMARY	Our study shows that HER2-overexpressing breast cancer displays characteristic patterns of relapse and metastatic spread depending on estrogen receptor status
Literature	GENERIF_SUMMARY	Toker cells and mammary Paget cells differ substantially in their expression of ER, c-erbB-2 and Ki-67.
Literature	GENERIF_SUMMARY	Trastuzumab therapy is a treatment option for patients with mucinous carcinoma when the tumor has HER2 amplification and overexpression.
Literature	GENERIF_SUMMARY	Data suggest that Copine-III is a novel player in the regulation of ErbB2-dependent cancer cell motility.
Literature	GENERIF_SUMMARY	Mutations were absent in all 85 cases. Out of 57 cases available for IHC, Her2 protein expression was negative (0) in 40 tumors
Literature	GENERIF_SUMMARY	ERBB2 were genes activated in late endometrial endometrioid carcinoma (stages III-IV).
Literature	GENERIF_SUMMARY	Data demonstrate that sequence differences on the C-lobe surface influence the heterodimerization and activation of ErbB2/4 kinase domains.
Literature	GENERIF_SUMMARY	Metastatic cancer-related hyperprolactinemia and lack of surgery-induced hyperprolactinemia are statistically more frequent in HER2-positive patients.
Literature	GENERIF_SUMMARY	In breast cancer discordant HER2 protein/gene expression is a rare event that is not always explained by the actions of AP-2 and YY1.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	The HER2 655Val variant may be associated with the incidence of advanced cervical cancer.
Literature	GENERIF_SUMMARY	A relationship between high levels of EGFR and HER-2 in the tumors of patients with esophageal squamous-cell carcinoma and intravascular tumor invasion and poor outcome of the disease was detected.
Literature	GENERIF_SUMMARY	Increased HER-2 is associated with prostatic carcinoma.
Literature	GENERIF_SUMMARY	Studies indicate that lapatinib - a tyrosine kinase inhibitor targets not only EGFR (erb B1) but also erb B2 (HER2) receptor tyrosine kinase signalling.
Literature	GENERIF_SUMMARY	In non-anaplastic thyroid tumours, HER2 was significantly associated with SRC-1, and these proteins were associated with poorly differentiated tumours, capsular invasion and disease recurrence.
Literature	GENERIF_SUMMARY	HER2/neu amplification/overexpression was associated with worse survival in patients with advanced gastric carcinoma.
Literature	GENERIF_SUMMARY	EGFR mutation combined with HER2/3 expression is a significant predictor for gefitinib efficacy on Chinese patients with advanced non-small cell lung cancer.
Literature	GENERIF_SUMMARY	HER2/neu expression is uncommon in these tumors.
Literature	GENERIF_SUMMARY	The expression of HER2 increased after each surgical procedure. However it was lower after CO(2) pneumoperitoneum than after laparotomy.
Literature	GENERIF_SUMMARY	Although the frequency of T cell receptor (TCR) alpha chains with dominant HER2/neu peptide recognition is not known, they may represent interesting tools for TCR optimization resulting in enhanced functionality when paired to novel partner chains.
Literature	GENERIF_SUMMARY	overexpression of ErbB2 promotes invasive and migratory abilities of H-Ras-activated MCF10A human breast epithelial cells through upregulation of matrix metalloproteinase (MMP)-13 and urokinase-type plasminogen activator (uPA).
Literature	GENERIF_SUMMARY	Interpretive guidelines for HER2 status assay in circulating cancer cells (CTCs) and raise great opportunities for using CTCs as non-invasive and 'real-time' biopsy to examine and monitor the status of tumor markers.
Literature	GENERIF_SUMMARY	The vulnerability of the heart as a pluricellular paracrine organ
Literature	GENERIF_SUMMARY	Simultaneous downregulation of HER2 and uPAR may offer an effective tool for breast cancer therapy.
Literature	GENERIF_SUMMARY	Use anti-HER2 immunoliposomes to deliver high concentrations of nitroxides selectively to HER2-overexpressing tumor cells for high-contrast electron paramagnetic resonance imaging.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	ERBB2 influences the expression of Bcl-2 family members to induce an antiapoptotic phenotype.
Literature	GENERIF_SUMMARY	Significant correlation was found between ErbB-2 status and lymphnode status, tumor size and ductal carcinoma
Literature	GENERIF_SUMMARY	To determine the prevalence of Her-2/Neu-cerbb-2 in the gastric mucosa of patients with gastric adenocarcinoma in a brazilian patient group. Low prevalence of Her-2/Neu was observed.
Literature	GENERIF_SUMMARY	The expression of p53, c-erbB-2 and Ki67 proteins in 20 simple intestinal metaplasia (SIM), 27 atypical intestinal metaplasia (AIM), and 37 gastric carcinomas showed significant differences between SIM and gastric carcinomas.
Literature	GENERIF_SUMMARY	Quantified/identified differences in protein levels between tumor and adjacent normal breast tissue from the same breast in 18 women with stage I/II ER positive/Her2/neu negative invasive breast cancer.
Literature	GENERIF_SUMMARY	gene status in endometrial carcinomas
Literature	GENERIF_SUMMARY	Results suggest that trastuzumab resistance in breast cancer might be overcome by therapeutic strategies that jointly target erbB3, erbB2, and IGF-IR.
Literature	GENERIF_SUMMARY	HER2 overexpression was inversely related to ER expression and PR expression and was associated with a higher tumour grade, lymphovascular positivity and infiltrating ductal carcinoma subtype.
Literature	GENERIF_SUMMARY	ErbB2 tyrosine kinase inhibitor resistance in breast cancer is mediated by calcium-dependent activation of RelA
Literature	GENERIF_SUMMARY	Data are consistent with a preferential localization of KLF6 in the nuclear compartment of early stage and small HER2-ERBB2 overexpressing ductal breast tumor cells, also presenting lymph node metastatic events
Literature	GENERIF_SUMMARY	TOP2A gene amplification may define a subset of HER2-amplified breast cancers that are responsible for the markedly improved chemosensitivity seen in HER2-positive breast cancer.
Literature	GENERIF_SUMMARY	both high and low levels of HER2 are associated with poor outcome in breast cancer
Literature	GENERIF_SUMMARY	findings sustain the ganglioside GM(3) as an essential molecule for EGFR/ErbB2 heterodimer stability and important regulator of EGFR tyrosine phosphorylation, but it is not crucial for tyrosine phosphorylation of the heterodimerization partner ErbB2
Literature	GENERIF_SUMMARY	The expression of HER-2/neu may play a role in the invasion, metastasis and progression of hepatocellular carcinoma.
Literature	GENERIF_SUMMARY	150 genes are co-overexpressed with ERBB2; these

Main Category	Category	Terms
		genes represent factors influencing ERBB2-positive tumors. RNAi based analysis of these genes identifies transcriptional regulators of fat synthesis and storage as critical for the cell survival.
Literature	GENERIF_SUMMARY	The expression of c-erbB-2 in lymph nodes was statistically significantly related to the expressions of EGF and EGFR both in the main mass of tumour and in lymph nodes.
Literature	GENERIF_SUMMARY	Altogether, these findings support the notion that HPSE is a critical downstream target of HER2 mechanisms driving BMBC and is potentially relevant for BMBC therapeutic interventions.
Literature	GENERIF_SUMMARY	Although our study provides additional evidence of the potential prognostic role of c-erbB-2 in endometrioid carcinoma, further prospective and controlled studies are required to validate their clinical usefulness.
Literature	GENERIF_SUMMARY	Breast cancer tissues predefined for HER2 status were subjected to MALDI-IMS, and protein profiles were obtained.We found that specific protein/peptide expression changes strongly correlated with the HER2 overexpression.
Literature	GENERIF_SUMMARY	HER-2 alterations specifically correlate with clinicopathological characteristics of Bulgarian patients with breast cancer.
Literature	GENERIF_SUMMARY	HER2 may have a role in response to trastuzumab in patients with metastatic breast cancer
Literature	GENERIF_SUMMARY	Pin1 plays an important role in the overexpression of HER-2 through Pin1-MEK1-activator protein-2alpha signaling in breast cancer.
Literature	GENERIF_SUMMARY	The effect of lapatinib on ERalpha signaling varied markedly depending on the nature of the HER2/ERalpha cross-talk
Literature	GENERIF_SUMMARY	Interaction of syndecan and alpha6beta4 integrin cytoplasmic domains
Literature	GENERIF_SUMMARY	Real-time RT-PCR analysis for evaluating the HER2/neu status in breast cancer is reported.
Literature	GENERIF_SUMMARY	Cyclin A2 and HER2 are associated with proliferation and high recurrence, particularly when combined.
Literature	GENERIF_SUMMARY	Suggest that HER-2 shRNA-based combined therapy with paclitaxel could be a novel strategy to inhibit the progression of HER-2-positive nasopharyngeal neoplasms.
Literature	GENERIF_SUMMARY	Ethanol enhances the adhesion of breast cancer cells to fibronectin in an ErbB2-dependent manner.
Literature	GENERIF_SUMMARY	is expressed in breast cancer and metastatic lymph nodes
Literature	GENERIF_SUMMARY	EGFR, HER-2 and HER-4 expression is not associated

Main Category	Category	Terms
		with poorer survival in neuroendocrine cancer
Literature	GENERIF_SUMMARY	Statistically significant correlation between mRNA and protein expression for HER2 in breast cancer patients.
Literature	GENERIF_SUMMARY	No relationship was found between HER2 expression and patient survival.
Literature	GENERIF_SUMMARY	We observed overexpression of HER-2/neu receptor in teratocarcinoma subtype of germ cell tumor
Literature	GENERIF_SUMMARY	The study reports the first evidence for a correlation between cPLA(2)alpha enzymatic activity and overexpression of the HER2 receptor.
Literature	GENERIF_SUMMARY	Studies indicate a possible role of Her-2 as a prognostic factor in bladder carcinoma.
Literature	GENERIF_SUMMARY	The cells with high Her2/neu protein gene amplification and/or protein overexpression can be detected and analysed even if their amount in the sample is small. Chromosome 17 polyploidy is important in some cases.
Literature	GENERIF_SUMMARY	Presence of Foxp3+ tumor cells represents a new independent prognostic factor of improved outcome in HER2-overexpressing breast carcinoma, which could help identify high-risk patients for additional therapies after neoadjuvant chemotherapy.
Literature	GENERIF_SUMMARY	HER-2 over-expressing breast cancer cells resistant to trastuzumab can be targeted for selective gene expression and destruction by viruses with envelope-proteins engineered to bind to this antibody.
Literature	GENERIF_SUMMARY	beta2-adrenergic receptor is up-regulated in MCF-7 cells overexpressing Her2 and Her2 overexpression plus phosphorylation of ERK cause epinephrine autocrine release resulting in up-regulation of beta2-adrenergic receptor in a positive feedback loop.
Literature	GENERIF_SUMMARY	Overexpression of c-erbB-2 is associated with lung cancer.
Literature	GENERIF_SUMMARY	ILK has a critical role in the initiation phase of ErbB2 tumor induction.
Literature	GENERIF_SUMMARY	Data show that HER-2 gene amplification is a potential prognostic marker for advanced stage breast cancer.
Literature	GENERIF_SUMMARY	Elevated expression of C-erb-B2 may reflect pre-neoplastic liver cell proliferation, cellular necrosis associated with HCV related chronic liver disease and alternatively from HCV carcinogens which enhance malignant transformation.
Literature	GENERIF_SUMMARY	TRMT2A protein expression as a biomarker of increased risk of recurrence in HER2+ breast cancer patients.
Literature	GENERIF_SUMMARY	ER+/PR- tumors more frequently showed HER2 (human epidermal growth factor receptor) overexpression and represent a distinct subset in familial breast cancer

Main Category	Category	Terms
		patients.
Literature	GENERIF_SUMMARY	c-erbB2 overexpression seems to predict aggressive behaviour in supratentorial anaplastic astrocytomas.
Literature	GENERIF_SUMMARY	Integrin alpha5 serves as a key mediator of Src and ErbB2-survival signaling in low adhesion states, which are necessary to block the pro-anoikis mediator Bim.
Literature	GENERIF_SUMMARY	The positive correlations in positivity rate between the PRL-R and ER/PR expressions are found only in CerbB-2 positive patients with breast cancer.
Literature	GENERIF_SUMMARY	data suggest PTPN9 as a negative regulator of breast cancer cells by targeting ErbB2 and EGFR and inhibiting STAT activation
Literature	GENERIF_SUMMARY	Combining CEA with Cyfra 21-1 analysis of malignant pleural effusion increases diagnostic sensitivity of lung adenocarcinomato nearly 100%
Literature	GENERIF_SUMMARY	tumor marker for breast cancer
Literature	GENERIF_SUMMARY	In the adjuvant breast cancer setting, primary fluorescence in situ hybridization testing with subsequent trastuzumab treatment of HER-2-positive patients is a cost-effective and preferable approach.
Literature	GENERIF_SUMMARY	HER-2 gene amplification and protein high expression is associated with salivary duct carcinoma.
Literature	GENERIF_SUMMARY	Patients with HER2 positive breast cancer had increased pathological complete remission rates, but survival rates were worse, especially in those with residual disease after preoperative therapy.
Literature	GENERIF_SUMMARY	The P53 gene codon 72 and Her-2 gene polymorphisms were not associated with the risk of breast neoplasms in Turkish women.
Literature	GENERIF_SUMMARY	In HER2-positive breast cancer, basal co-expression of PTEN and p-Akt might identify those patients who are more likely to benefit from trastuzumab-based therapies.
Literature	GENERIF_SUMMARY	Meta-analysis suggests that HER2 Ile 655Val polymorphism may contribute to breast cancer risk.
Literature	GENERIF_SUMMARY	Data show that information on the HER2 status of CTC might be helpful for stratification and monitoring of HER2-directed therapies.
Literature	GENERIF_SUMMARY	HER2 gene amplification activates lamellipodia formation in breast cancer cells, which initiates their metalloproteinase-independent migration
Literature	GENERIF_SUMMARY	The combination of S-1 and HER2-targeting agents is a promising treatment option for gastric cancer with HER2 amplification.
Literature	GENERIF_SUMMARY	EGFR and Her-2 may not have a role in oral squamous cell carcinoma, but there are low levels of EGF in saliva

Main Category	Category	Terms
		during disease progression
Literature	GENERIF_SUMMARY	Results suggest that the expression of c-erbB-2 and EGF protein, but not necessarily ERG receptors, can help predict the postoperative survival time in stomach cancer.
Literature	GENERIF_SUMMARY	Results indicate that Muc4 modulates ErbB2 signaling potential significantly by stabilizing and directly interacting with the ErbB2-ErbB3 heterodimer.
Literature	GENERIF_SUMMARY	This paper briefly reviews the assessment of ERalpha, PgR, and HER2 in breast cancer, emphasizing recent progress and persistent controversies.
Literature	GENERIF_SUMMARY	Testing of Her2 status in gastric cancer predicts outcome, compared with outcome in breast cancer.
Literature	GENERIF_SUMMARY	Cyclin D1 functions downstream as a genetic target of Notch1, amplifies Notch1 activity by repressing Numb, and identifies a novel pathway by which ErbB2 induces Notch1 activity via the induction of cyclin D1.
Literature	GENERIF_SUMMARY	The dimerization partner of EGFR, ERBB2 is also a possible low penetrance gene candidate associated with risk of glioblastoma development.
Literature	GENERIF_SUMMARY	High levels of HER2 and Ki-67 or TK1 expression correlate with the increase of tumor grades and tumor recurrence in meningiomas.
Literature	GENERIF_SUMMARY	HER2-mediated signaling effects low molecular weight cyclin E expression, which in turn deregulates the cell cycle.
Literature	GENERIF_SUMMARY	data demonstrate that aberrant Notch regulation is an early event in mammary carcinogenesis and is associated with the HER-2 molecular subtype of breast cancer
Literature	GENERIF_SUMMARY	Lapatinib, a dual tyrosine kinase inhibitor of EGFR and HER-2 receptors demonstrated antiangiogenic/ antitumor effects in a lung cancer model.
Literature	GENERIF_SUMMARY	in gastric cancer cells, HER2 knockdown resulted in the downregulation of the expression of MMP-1, while HER2 overexpression improved the transcription of MMP-1 through the activation of an MMP-1 promoter
Literature	GENERIF_SUMMARY	Results describe the structure of the transmembrane domain heterodimer of ErbB1 and ErbB2 and suggest an important contribution of the transmembrane helix-helix interaction to signal transduction.
Literature	GENERIF_SUMMARY	While for endocrine responsive and Her2-positive disease different molecular targeted therapies are available, up to now no specific targeted approach for triple-negative breast cancer has been developed.
Literature	GENERIF_SUMMARY	Our study indicates for the first time, that oncogenic ras and loss of Smad signaling cooperate to upregulate EGFR and erbB2, which plays a role in promoting

Main Category	Category	Terms
		invasion.
Literature	GENERIF_SUMMARY	In HER2-positive breast tumor patients, trastuzumab added to epirubicin/cyclophosphamide/docetaxel leads to increased pCR rates.
Literature	GENERIF_SUMMARY	The use of estrogen receptor/progesterone receptor/ HER2 subtypes and correlation with molecular classification of breast cancer is recommended.
Literature	GENERIF_SUMMARY	There is no correlation between HER2/neu overexpression and the tumor stage.
Literature	GENERIF_SUMMARY	prostatic acid phosphatase, an authentic tyrosine phosphatase, dephosphorylates ErbB-2 and regulates prostate cancer cell growth
Literature	GENERIF_SUMMARY	Results support the biological relevance of elevated HER2/ECD levels in the outcome of HER2-disease and in the susceptibility to Trastuzumab-based therapy.
Literature	GENERIF_SUMMARY	Meta-analysis suggests that HER2 codon 655 polymorphism is not associated with the increased breast cancer risk.
Literature	GENERIF_SUMMARY	Analysis revealed an association between HER2 protein expression and gene amplification in breast carcinoma. 14% of colon carcinomas, 2% of prostate carcinoma, 4% of lung carcinomas and 16% ovarian carcinomas revealed gene amplification.
Literature	GENERIF_SUMMARY	HER2 and TUBB3 status might be a good biomarker for determining the most appropriate therapeutic modality in extramammary Paget disease.
Literature	GENERIF_SUMMARY	High expression of ErbB-2 found in brains of meningioma patients implicates a potential role for ErbB-2 in benign tumorigenesis of meningioma consistent with interactions of ErbB-2 homodimers that may be associated with low signal transduction.
Literature	GENERIF_SUMMARY	Case Report
Literature	GENERIF_SUMMARY	Data show that the concordance rates of histological type, estrogen receptor and progesterone receptor status, and HER2 were 84.4%, 81.3%, 92.9%, and 89.3%, respectively between CNB and surgical specimens.
Literature	GENERIF_SUMMARY	HER-2/neu and TOPOIIalpha amplification/ overexpression might discriminate between Barrett's oesophagus, dysplasia and adenocarcinoma.
Literature	GENERIF_SUMMARY	An overexpression of Her2 was found in one anorectal melanoma.
Literature	GENERIF_SUMMARY	HER2, but not EGFR gene amplification, is frequently observed in KRAS and BRAF wild type colorectal cancer patients
Literature	GENERIF_SUMMARY	htid is a tissue independent and evolutionarily conserved suppressor of ErbB-2.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Data show that a highly significant positive correlation of icb-1 transcript levels with c-erbB2 (HER2) expression (p
Literature	GENERIF_SUMMARY	c-erbB2 was altered in 45% of the oral squamous cell carcinoma samples from a Serbian population. Statistical association with tumour natural history was poor, indicating high clonal heterogeneity and multiple pathways of carcinogenesis.
Literature	GENERIF_SUMMARY	observations suggest that PI3K mutants enhance HER2-mediated transformation by amplifying the ligand-induced signaling output of the ErbB network
Literature	GENERIF_SUMMARY	Phase II study of nab-paclitaxel, bevacizumab, and gemcitabine as first-line therapy for patients with HER2-negative metastatic breast cancer.
Literature	GENERIF_SUMMARY	The ability of apical ErbB2 to initiate an altered downstream cascade suggests that subcellular localization of the receptor plays an important role in regulating ErbB2 signaling in polarized epithelia.
Literature	GENERIF_SUMMARY	analysis of CEA-, Her2/neu-, BCRP- and Hsp27-positive microparticles in breast cancer patients
Literature	GENERIF_SUMMARY	In some oral squamous cell carcinomas Her-2 immunohistochemical expression may be associated with chromosome 17 polysomy and not Her-2 amplification.
Literature	GENERIF_SUMMARY	controls expression and cellular localization of ZO-1 tight junction protein
Literature	GENERIF_SUMMARY	Data showed a positive correlation between HER-2 and COX-2 or VEGF-C mRNA levels in breast cancer.
Literature	GENERIF_SUMMARY	Monitoring serum HER2 levels in the presence of anti-HER2 treatment might be an adjunct to the clinical evaluation during neoadjuvant chemotherapy of breast cancer.
Literature	GENERIF_SUMMARY	HER2-overexpressing esophageal squamous cell carcinomas tumour cells showed a reduced sensitivity for CTLs through the downregulation of MHC class I.
Literature	GENERIF_SUMMARY	study investigates the correlation of intra-individual HER2 status between primary breast cancers and corresponding recurrences in a population derived cohort
Literature	GENERIF_SUMMARY	Her-2/neu protein over-expression is significantly associated with advanced surgical stage uterine papillary serous carcinoma and poor survival outcome.
Literature	GENERIF_SUMMARY	this study did not find HER-2 expression to be an independent factor of either tumour progression or cancer-specific survival.
Literature	GENERIF_SUMMARY	We describe a tool (a customized Excel spreadsheet) for easy and reproducible diagnosis of HER2 genetic heterogeneity according to ASCO/CAP criteria.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Study does not confirm the hypothesis that sunitinib consolidation therapy can lead to a predefined clinically relevant proportion of HER2-negative breast cancer patients with PFS of > or =5 months after an objective response to taxanes.
Literature	GENERIF_SUMMARY	The aim of our study was to develop and optimize methods for relative and absolute protein quantifications in formalin-fixed and paraffin-embedded (FFPE) tissues with special emphasis on HER mediated pathways in breast cancer.
Literature	GENERIF_SUMMARY	the ratio of ErbB1 and ErbB2 in their heterocluster
Literature	GENERIF_SUMMARY	High-level HER-2 amplification occurs in a small fraction of colorectal cancers.
Literature	GENERIF_SUMMARY	The combination of nab-paclitaxel and cyclophosphamide, with or without trastuzumab, is feasible and well tolerated in patients with early stage HER-2 positive breast cancer.
Literature	GENERIF_SUMMARY	The effect of lapatinib on HER2 protein extracellular domain shedding and its effect on trastuzumab and geftinib pharmacology in breast cancer cells are reported.
Literature	GENERIF_SUMMARY	Report determination of the Her-2/neu gene amplification status in cytologic breast cancer specimens using automated silver-enhanced in-situ hybridization.
Literature	GENERIF_SUMMARY	There is a consistent association of HER-2/neu over expression and Gleason less than 7 with a higher RR of death and recurrence in patients with prostate cancer
Literature	GENERIF_SUMMARY	Pre-clinical models have indicated a role for HER2 signalling in initiating Chromosomal instability and defective cell-cycle control.
Literature	GENERIF_SUMMARY	The HER2 positivity rate of breast tumors (DIN1C-3, invasive carcinoma) may be as low as 9%. Immunohistochemistry as firstline testing for HER2 may result in false positive and false negative results.
Literature	GENERIF_SUMMARY	Data suggest a role for VRK2A in ErbB2-MAPK signaling.
Literature	GENERIF_SUMMARY	Studies indicate that tumor grade and race were associated with HER-2 for women aged >/= 50 years after controlling for other variables.
Literature	GENERIF_SUMMARY	Results, besides corroborating existing reports that discrepancies exist between HER2 ISH and IHC assays, also suggest the need to rigorously evaluate CISH as an independent reference standard for assessment of HER2 amplification in gastric cancers.
Literature	GENERIF_SUMMARY	Neu/V664E TM domain does not affect the phosphorylation levels of full-length FGFR3/A3
Literature	GENERIF_SUMMARY	The addition of progesterone receptor and HER2 to

Main Category	Category	Terms
		Nottingham Prognostic Index increases its 5-year prognostic accuracy in breast cancer.
Literature	GENERIF_SUMMARY	HER2 over-expressing tumors may be more prone to escape from HLA-A2 restricted cytotoxic T lymphocytes (CTLs).
Literature	GENERIF_SUMMARY	Data revealed the significance of HER2 in the radio-resistance of HER2(+) breast cancer cells through induction of transcription factors NF-kappaB and c-myc, leading to activation of survivin, a downstream target oncogene preventing apoptosis.
Literature	GENERIF_SUMMARY	HER2 was up regulated in prostate cancer epithelium while CMTM5 was down regulated. Overexpression of CMTM5 in PC3 cells could lower the HER2 and Cyclin D1 protein level.
Literature	GENERIF_SUMMARY	Data indicate that high ErbB2 expression enhances cholangiocarcinoma invasion, motility and proliferation via the AKT/p70S6K pathway.
Literature	GENERIF_SUMMARY	Insights into biological processes that come under the control of HER-2 after it acquires full oncogenic potential in a breast cancer cell line.
Literature	GENERIF_SUMMARY	FISH is recommended as the primary HER2 testing modality for women with breast cancer who are candidates for HER2-targeted therapies.
Literature	GENERIF_SUMMARY	High HER2 gene copy number in tumor epithelial cells is associated with non-small cell lung cancer.
Literature	GENERIF_SUMMARY	Findings suggest that breast cancer developing on a background of an inherited TP53 mutation is highly likely to present with amplification of HER2.
Literature	GENERIF_SUMMARY	Data show that S1P4 uses HER2 to regulate extracellular signal regulated kinase-1/2 MDA-MB-453 breast cancer cells.
Literature	GENERIF_SUMMARY	ILK regulates the expression of Her2/neu through TWIST and YB-1, lending support to the use of ILK inhibitors in the treatment of aggressive Her2/neu-positive tumors.
Literature	GENERIF_SUMMARY	Her-2 and basal-like subtypes of breast cancer are correlated with factors associated with a poor prognosis.
Literature	GENERIF_SUMMARY	While HER2 amplifications appear to be prevalent in select non-breast tumors, lapatinib monotherapy is associated with modest activity.
Literature	GENERIF_SUMMARY	Compare HER2 status of circulating tumor cells in patients with metastatic breast cancer using two different methods.
Literature	GENERIF_SUMMARY	Her-2 and VEGF may play synergistic roles in the occurrence and development of breast cancer. Over-expressions of Her-2 and VEGF predict poor prognosis.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Patients with Ile-Val & Val-Val genotypes had significant TNF-alpha elevation compared with homozygous Ile-Ile patients. TNFA's negative impact on survival seems to be enhanced through the interaction with HER-2 Ile655Val SNP.
Literature	GENERIF_SUMMARY	Studies indicate that breast carcinomas Oncotype DX Recurrence Score (RS) was correlated with six pathobiologic features
Literature	GENERIF_SUMMARY	The mechanism of HER2Delta16 therapeutic evasion, involving tamoxifen-induced upregulation of BCL-2 and suppression of miR-15a/16.
Literature	GENERIF_SUMMARY	PTEN expression at both mRNA and protein levels were not significantly affected by erbB2 silencing (P>0.05).
Literature	GENERIF_SUMMARY	Mutations in ERBB2 is associated with adenocarcinoma of the lung.
Literature	GENERIF_SUMMARY	Results propose a model for HER2-EGFR dimerization and possible blocking by HERP5 peptidomimetic using a protein-protein docking method.
Literature	GENERIF_SUMMARY	estrogen receptor, progesterone receptor, and HER-2 may have a role in progression in breast cancer despite response to doxorubicin/cyclophosphamide therapy
Literature	GENERIF_SUMMARY	Information on presence or absence of HER2 overexpression can be obtained in circulating tumor cells.
Literature	GENERIF_SUMMARY	HER2/neu is a marker of prognostic significance that is routinely assessed in invasive cancer.
Literature	GENERIF_SUMMARY	HER2 gene genetic heterogeneity was associated with HER2 protein expression and ER expression.
Literature	GENERIF_SUMMARY	ER+ and ER- ERBB2-amplified BCs are different, and ERBB2 amplicons in inflammatory breast cancer and non-inflammatory breast cancer can be distinguished
Literature	GENERIF_SUMMARY	Data suggest that TGF-alpha plays a role in resistance to lapatinib and capecitabine therapy among HER2-positive breast cancer.
Literature	GENERIF_SUMMARY	ACF7 targeting to the plasma membrane is both required and sufficient for microtubule capture downstream of ErbB2.
Literature	GENERIF_SUMMARY	JNK1 promotes cell survival in Her2/neu-positive breast cancer
Literature	GENERIF_SUMMARY	Focusing on the mechanisms that confer resistance to individual HER2 targeting agents, this review envisage therapeutic approaches to delay or overcome the evolvement of resistance in patients. [review]
Literature	GENERIF_SUMMARY	HER2 molecular subtype is the dominant, but not the sole subtype seen in Paget's cells of the nipple
Literature	GENERIF_SUMMARY	Analyses indicate that p130Cas expression in ErbB2

Main Category	Category	Terms
		positive human breast cancers significantly correlates with higher risk to develop distant metastasis, thus underlying the value of the p130Cas/ErbB2 synergism in regulating breast cancer invasion.
Literature	GENERIF_SUMMARY	ERRalpha and PGC-1beta are key players in the etiology of breast cancer by coordinating the transcriptional regulation of genes located in the 17q12 region, a process that also involves interference with the repressive function of ERalpha on ERBB2 expression.
Literature	GENERIF_SUMMARY	EGFR and HER-2 expression is variable in embryonal brain tumors
Literature	GENERIF_SUMMARY	ErbB-2 signaling via the phosphatidylinositol 3-kinase pathway results in up-regulated matriptase zymogen activity, which contributes to prostate cancer cell invasion.
Literature	GENERIF_SUMMARY	This review discusses the molecular mechanisms leading to the development of trastuzumab resistance, including circulating HER2 extracellular domain, loss of PTEN, activation of alternative pathways (e.g. IGFR), and receptor-antibody interaction block.
Literature	GENERIF_SUMMARY	Fatty acid synthase expression and activity has emerged as a common phenotype in most human carcinomas, including breast cancer.
Literature	GENERIF_SUMMARY	Her-2 expression is an important prognostic factor in patients with node-positive breast cancer, but not for patients with node-negative disease in Chinese women.
Literature	GENERIF_SUMMARY	ER, PR and Her-2 are independent prognostic factors for locoregional recurrence and survival in node-positive breast cancer patients treated by mastectomy.
Literature	GENERIF_SUMMARY	The study also showed that HER2/neu amplification is higher in Saudi patients than other western populations.
Literature	GENERIF_SUMMARY	HER2 may induce breast cancer by increasing the frequency of tumor stem cells and upregulating the expression of COX-2 and HDAC6 that play pivotal roles in tumor progression.
Literature	GENERIF_SUMMARY	in ER negative basal and HER2+ breast cancer, gene expression modules reflecting T-cell helper-1 (Th1) and T-cell helper-2 (Th2) mediated immune responses play antagonistic roles as major risk factors for distant metastasis
Literature	GENERIF_SUMMARY	Data suggest that HER2 overexpression may constitute a unique molecular event independent of the tumor site.
Literature	GENERIF_SUMMARY	Concomitant expression of HER2 (protein tyrosine kinase-type cell surface receptor HER2) and HIF-1alpha (hypoxia inducible factor 1 alpha) is a predictor of poor prognosis in uterine cervical carcinoma treated with concurrent chemoradiotherapy
Literature	GENERIF_SUMMARY	Study shows the pivotal roles of Dmp1 in HER2/neu-p53 signaling and breast carcinogenesis.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	COX-2 activation is associated with Akt phosphorylation and poor survival in ER-negative, HER2-positive breast cancer
Literature	GENERIF_SUMMARY	Carbamazepine promotes Her-2 protein degradation in breast cancer cells by modulating HDAC6 activity and acetylation of Hsp90.
Literature	GENERIF_SUMMARY	Targeting EGFR/HER2 pathways enhances the antiproliferative effect of gemcitabine in biliary tract and gallbladder carcinomas.
Literature	GENERIF_SUMMARY	HER2 protein overexpression and gene amplification are infrequent events, if they occur at all, in small intestinal adenocarcinoma.
Literature	GENERIF_SUMMARY	HER2 overexpression or gene amplification is involved in the transition from DCIS to invasive disease.
Literature	GENERIF_SUMMARY	Findings reveal a novel mechanism through which HER2 silences tumor suppression in a concerted manner by switching expression of C/EBPss isoforms.
Literature	GENERIF_SUMMARY	The breast cancer patients with co-overexpression of ErbB-1 and ErbB-2 had a shorter disease free survival.
Literature	GENERIF_SUMMARY	Prolactinoma HER2 potently induces prolactin and regulates experimental prolactinoma cell proliferation.
Literature	GENERIF_SUMMARY	HER2 seems not to be of such interest for radionuclide therapy due to rare expression in colorectal carcinoma
Literature	GENERIF_SUMMARY	Adding carboplatin did not enhance TH antitumor activity.TH (docetaxel, 100 mg/m(2)) and TCH (docetaxel, 75 mg/m(2)) demonstrated efficacy with acceptable toxicity in women with HER2-amplified MBC.
Literature	GENERIF_SUMMARY	the HER pathway is blocked by lapatinib and trastuzumab in HER2/neu-overexpressing breast tumor xenografts
Literature	GENERIF_SUMMARY	Studies indicate that aUPD is a common and non-random molecular feature of breast cancer that is most prominent in HER2/ER/PR triple negative cases.
Literature	GENERIF_SUMMARY	meta-analysis suggests that HER2 overexpression is a poor prognostic factor in lung cancer
Literature	GENERIF_SUMMARY	Trastuzumab-DM1 has robust single-agent activity in patients with heavily pretreated, HER2-positive metastatic breast cancer and is well tolerated at the recommended phase II dose.
Literature	GENERIF_SUMMARY	The Ventana HER2 dual-color ISH assay is robust and reproducible and shows good concordance with a standard FISH assay.
Literature	GENERIF_SUMMARY	Inhibition of CaM function or deletion of the CaM binding sites from HER2 significantly decreased both HER2 phosphorylation and HER2-stimulated cell growth.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	These data indicate that the activation and/or overexpression of SST receptors along with the inhibition of EGFR will serve as an important therapeutic approach in the treatment of ErbB-positive tumors.
Literature	GENERIF_SUMMARY	HER-2 expression significantly correlates with Survivin at the protein level in primary invasive ductal breast cancer tissues
Literature	GENERIF_SUMMARY	HER receptor family plays an important role in embryonic development and in pathogenesis and progression of neuroblastic tumors.
Literature	GENERIF_SUMMARY	No statistically significant correlation was found between HER-2/neu codon 655 polymorphism and breast cancer.
Literature	GENERIF_SUMMARY	hMena and HER2 signalling has a role in breast cancer
Literature	GENERIF_SUMMARY	epidermal growth factor receptor-2
Literature	GENERIF_SUMMARY	HER-2 overexpression is associated with M1b prostate cancer.
Literature	GENERIF_SUMMARY	Expression of ADAM 10 in gastric cancer is significantly associated with lymph node and distant metastasis, high C-erbB-2 expression, and poor prognosis.
Literature	GENERIF_SUMMARY	No correlation was noted between HER2/neu, ER or PR status, and blood group type. Among this cohort, HER2/neu positivity was less than 20%
Literature	GENERIF_SUMMARY	More positivity HER2 was found in advanced stages (III/IV) vs. early stages (I/II); intestinal histology subtype vs diffuse/ mixed and gastro-oesophageal junction cancer vs GC
Literature	GENERIF_SUMMARY	HER2-positive Circulating Tumor Cells were detected in DCIS/LCIS or M0 breast cancer irrespective of the primary tumor HER2 status. Nevertheless, their presence was more common in women with HER2-positive disease
Literature	GENERIF_SUMMARY	Her-2/neu gene amplification is associated with esophageal adenocarcinoma.
Literature	GENERIF_SUMMARY	quantifying EGFR/HER2 heterodimers using our antibody-based TR-FRET assay may represent a useful method to predict the efficacy and explain the mechanisms of action of therapeutic mAbs
Literature	GENERIF_SUMMARY	HER2 amplification seems to drive genomic instability along chromosome 17q, leading to different patterns of gene amplification.
Literature	GENERIF_SUMMARY	Our study does not support a role for human epidermal growth factor receptor 2 as a prognostic marker in osteosarcoma.
Literature	GENERIF_SUMMARY	Polymalic acid-based nanobiopolymer provides efficient systemic breast cancer treatment by inhibiting both

Main Category	Category	Terms
		HER2/neu receptor synthesis and activity.
Literature	GENERIF_SUMMARY	Patients who changed HER2 status from negative to positive, presented significant longer time to progression when treated with trastuzumab compared to those who were untreated
Literature	GENERIF_SUMMARY	Our results showed that dual-color SISH is a viable alternative to FISH for assessing HER2 gene amplification in breast cancer.
Literature	GENERIF_SUMMARY	HER2 gene amplification and protein over-expression is associated with esophageal carcinoma.
Literature	GENERIF_SUMMARY	HER overexpression orchestrates broad changes in the tumor microenvironment by altering the secretion of a diverse variety of biologically active proteins.
Literature	GENERIF_SUMMARY	Results support the hypothesis that Her-2/neu overexpression contributes to tamoxifen resistance.
Literature	GENERIF_SUMMARY	HER2-positivity is present in a significant proportion of oesophago-gastric adenocarcinomas (24%), but at a lower rate in the distal stomach
Literature	GENERIF_SUMMARY	HER2 gene amplification was found in 41% of Vietnamese breast cancers.
Literature	GENERIF_SUMMARY	Trastuzumab plus chemotherapy in gastric cancer overexpressing HER-2 and EGFR
Literature	GENERIF_SUMMARY	Coexisting HER2 gene amplification and novel mutations in the HER2 protein-overexpression are associated with mucinous epithelial ovarian cancer
Literature	GENERIF_SUMMARY	gene mutationa are not associated with the prognosis of patients with lung cancers
Literature	GENERIF_SUMMARY	In hormone-independent breast cancer cases negative for CerbB-2, Ki-67 and P53 expressions were significantly higher than that in hormone-dependent CerbB-2-negative cases.
Literature	GENERIF_SUMMARY	COX-2 overexpression correlates with HER2 overexpression in invasive breast carcinomas and may provide a clinically useful biomarker for estimating tumor aggressiveness.
Literature	GENERIF_SUMMARY	Data suggest that since PTK6 plays an important role in HER receptor signal transduction, its down-regulation might be suitable for future therapy approaches in breast cancer.
Literature	GENERIF_SUMMARY	Bevacizumab combination therapy improves clinical benefit in terms of increased progression-free survival in first-line treatment of Her-2 negative metastatic breast cancer.
Literature	GENERIF_SUMMARY	therapeutic inhibitors of HER3 should be used in combination with HER2 inhibitors and PI3K pathway inhibitors in patients with HER2- and PI3K-dependent cancers

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	We found a significant correlation of HER2 with human epidermal growth factor receptor 3 (HER3/erbB3), epidermal growth factor receptor 1 (EGFR/HER1/erbB1) and urokinase plasminogen receptor (uPAR) in breast cancer tissues.
Literature	GENERIF_SUMMARY	Hunk is required for HER2/neu-induced mammary tumorigenesis
Literature	GENERIF_SUMMARY	Results indicate a significant association between HER2 expression and age.
Literature	GENERIF_SUMMARY	HER-2 expression, however, might play an important role in predicting the prognosis of lipid-rich carcinoma of the breast.
Literature	GENERIF_SUMMARY	Data show that moderate- to high-affinity mAbs targeting the same HER2 epitope with monovalent affinity above 23 nmol/L had equal tumor accumulation of residualizing radiolabel over 120 hours.
Literature	GENERIF_SUMMARY	148 Penile squamous cell carcinoma were microarrayed and immunostained for phosphorylated EGFR (pEGFR), HER2, HER3, HER4, phosphorylated Akt (pAkt), Akt1 and PTEN proteins.
Literature	GENERIF_SUMMARY	HER2 was overexpressed in 30/44 pT1GIII lesions & predicted recurrence in this subgroup of bladder transitional cell carcinomas. Expression was negative in 26/40 pT1GII tumors
Literature	GENERIF_SUMMARY	As malignancy further progressed, we identified a subset of cells harboring HER2 amplification
Literature	GENERIF_SUMMARY	Conserved intramolecular non-covalent bond networks across ErbB2 family kinase are identified that emerge from sequence homology and lead to conserved dynamic characteristics as well as function.
Literature	GENERIF_SUMMARY	The existence of amplified HER2 in cfDNA in the follow-up of breast cancer patients who are otherwise disease free.
Literature	GENERIF_SUMMARY	A comprehensive immunohistochemical study was performed to investigate the protein expression of eight CT genes in 454 invasive ductal carcinomas, including 225 ER/PR/HER2-negative (triple-negative) carcinomas.
Literature	GENERIF_SUMMARY	in ileal neuroendocrine cancer, HER-2/neu overexpression plays a role in the carcinogenetic process and by triggering the altered expression of c-Met and MTA-1, may activate the molecular pathway(s) promoting tumor progression and metastasis development.
Literature	GENERIF_SUMMARY	Majority of patients with HER-2/neu negative breast cancer had HER-2/neu-specific IFN-gamma producing T cell responses which was stronger than those in patients with HER-2/neu positive tumors.
Literature	GENERIF_SUMMARY	Structural analysis of the mechanism of inhibition and allosteric activation of the kinase domain of HER2

Main Category	Category	Terms
		protein.
Literature	GENERIF_SUMMARY	HER2 was identified as a novel mediator of eosinophil peroxidase signaling. Eosinophil peroxidase, at noncytotoxic levels, can drive cell-cycle progression and proliferation.
Literature	GENERIF_SUMMARY	we have found that HER2 amplification does not associate with poor prognosis in total 232 esophageal adenocarcinoma patients
Literature	GENERIF_SUMMARY	These findings point to the role of HER2-positive exosomes in modulating sensitivity to trastuzumab, and, consequently, to HER2-driven tumor aggressiveness.
Literature	GENERIF_SUMMARY	High HER2 protein expression is associated with gastric carcinomas.
Literature	GENERIF_SUMMARY	Detection of HER-2 gene amplification can assess the malignant biological behaviors and prognosis of gastric cancer.
Literature	GENERIF_SUMMARY	C-erbB-2 and EGFR might have a synergetic effect in the development and progress of nasopharyngeal carcinoma. Co-expression closely correlates with cell proliferation status.
Literature	GENERIF_SUMMARY	It is unlikely that HER2 rs1136201 is a risk factor for breast cancer.
Literature	GENERIF_SUMMARY	Data show that (99m)Tc-labeled 2Rs15d has a high tumor uptake in HER2(+) tumor models.
Literature	GENERIF_SUMMARY	plasma copy number variation can be used as prognostic marker for esophageal cancer
Literature	GENERIF_SUMMARY	Suggest PTPN12 and ERK5 mediate HER2-driven cell motility in ovarian cancer cells, through FAK targeting.
Literature	GENERIF_SUMMARY	Studies characterized the anti-human epidermal growth factor receptor 2 (HER2) mAb produced in a glycoengineered Pichia.
Literature	GENERIF_SUMMARY	ERBB2 juxtamembrane domain (trastuzumab binding site) gene mutation is a rare event in breast cancer
Literature	GENERIF_SUMMARY	In invasive breast cancer ER is negatively associated with HER-2 amplification, whereas tumor emboli and lymph node status are positively associated with HER-2.
Literature	GENERIF_SUMMARY	We evaluated HER2 protein expression and gene amplification in micropapillary carcinoma
Literature	GENERIF_SUMMARY	we demonstrate that a significant number of oesophageal adenocarcinomas are positive for ErbB2
Literature	GENERIF_SUMMARY	IHC can be used firstly to screen the HER-2/neu status, and FISH can be used as a supplementary role to IHC and 2+ and some negative cases.
Literature	GENERIF_SUMMARY	HER-2/neu gene amplification by CISH was found in 10

Main Category	Category	Terms
		(20%) out of 50 patients with borderline (2+) IHC score.
Literature	GENERIF_SUMMARY	Copy number increase of ERBB2 is observed in OSCCs and correlates with gene overexpression in these tumors. In addition, overexpression of ERBB2 is also observed in some OSCCs that lack copy number changes, indicating involvement of another mechanism.
Literature	GENERIF_SUMMARY	These studies indicate that TSP1 disrupts the endothelial barrier through EGFR/ErbB2 activation although additional signals are necessary in cells with low receptor expression.
Literature	GENERIF_SUMMARY	prognostic significance and impact of p53 protein and human epidermal growth factor receptor 2 (HER2) overexpression on 5-year survival in young patients with stage II breast cancer
Literature	GENERIF_SUMMARY	The pre-therapeutic beta-catenin and Her2/neu protein expression seem to be valuable predictive and prognostic markers in the multimodality treatment of advanced rectal cancer.
Literature	GENERIF_SUMMARY	we report on the production and purification of a soluble anti-HER2 scFv antibody secreted by Pichia pastoris.
Literature	GENERIF_SUMMARY	These findings illustrate appropriate and effective implementation of a HER2 testing strategy in a managed care setting.
Literature	GENERIF_SUMMARY	The association for ER-/PR- tumors did not differ by expression of HER2 (p(heterogeneity) = 0.3).
Literature	GENERIF_SUMMARY	Testicular HER2 was tyrosine phosphorylated and strongly expressed in spermatogonia, early spermatocytes, elongating/elongated spermatids, Sertoli cells, Leydig cells, and peritubular cells.
Literature	GENERIF_SUMMARY	findings indicate that nuclear ErbB2 functions as a regulator of rRNA synthesis and cellular translation, which may contribute to tumor development and progression
Literature	GENERIF_SUMMARY	the Delta16HER2 splice variant represents the transforming form of the HER2 oncoprotein
Literature	GENERIF_SUMMARY	p185 as a naturally occurring autoantigen mimotope
Literature	GENERIF_SUMMARY	elevated levels of HER2 are a negative prognostic factor and patients overexpressing HER2 show a significant increase of overall survival when treated with trastuzumab
Literature	GENERIF_SUMMARY	HER-2 and p53 have important prognostic value in breast cancer and should be considered collectively.
Literature	GENERIF_SUMMARY	Devised a novel flow cytometric method to quantify the expression of ERalpha, HER2, and the activation of ERK1/2 simultaneously in single cells.
Literature	GENERIF_SUMMARY	HER2 expression level is directly related to breast

Main Category	Category	Terms
		cancer invasion.
Literature	GENERIF_SUMMARY	Expression of HER2 is associated with clinicopathological features in gastroesophageal cancer.
Literature	GENERIF_SUMMARY	HER2 expression in gastric and gastroesophageal junction adenocarcinoma in a US population
Literature	GENERIF_SUMMARY	estrogen receptor(ER) and progesterone receptor(PR) expressions were inversely correlated with HER2 protein expression and ER but not PR expression is inversely correlated with Her2 gene amplification in ovarian cancer
Literature	GENERIF_SUMMARY	following irradiation, HER2 receptor activates hTERT/ telomerase, increasing the breast cancer cells' survival potential, through sequential induction of transcription factors NF-kappaBeta and c-myc
Literature	GENERIF_SUMMARY	Expression of survivin, CD117, and C-erbB-2 in neuroendocrine carcinoma of the uterine cervix.
Literature	GENERIF_SUMMARY	In situ detection of HER2 and HER2
Literature	GENERIF_SUMMARY	Her2 amplification in metastasising bladder cancer is relatively frequent, is homogeneous in each tumour component, and predicts early death.
Literature	GENERIF_SUMMARY	For women
Literature	GENERIF_SUMMARY	A change between primary breast tumour and corresponding relapse is frequent for progesterone receptor, infrequent for estrogen receptor and rare for HER-2.
Literature	GENERIF_SUMMARY	The study included a total of 28 cases of urothelial carcinomas, which were analyzed histopathologically and immunohistochemically using anti-human p16, c-erbB2 and Ki67 antibodies.
Literature	GENERIF_SUMMARY	Expression of ER, HER2, and TOP2A displayed discordance with a sufficient frequency to emphasize the role of confirmatory biopsies from metastatic lesions in future management of recurrent breast cancer.
Literature	GENERIF_SUMMARY	HER2 overexpression caused by gene amplification was observed in about 20% of patients with salivary duct cancers but was rare in salivary adenomas and other carcinomas.
Literature	GENERIF_SUMMARY	truncated ErbB2 rendered cells resistant to lapatinib-induced apoptosis.
Literature	GENERIF_SUMMARY	study concluded that in breast cancer, HER2 status was correlated with HER2 protein expression levels; frequency of polysomy 17 was correlated with the HER2 copy number and partially contributed to HER2/neu amplification but not HER2 protein expression
Literature	GENERIF_SUMMARY	Menopausal status played a very important role in

Main Category	Category	Terms
		determining HER2 and PgR status in ER(+) breast cancer patients. HER2 was independently inversely associated with PgR only in the postmenopausal women with ER(+) breast cancers.
Literature	GENERIF_SUMMARY	Results indicate that amplification of TOP2A in breast cancer is not confined to those who are concomitantly HER-2/neu(+), and suggest that a significant proportion of HER-2/neu(-) patients exhibit high levels of TOP2A.
Literature	GENERIF_SUMMARY	There was no association between HER2/neu and VEGF expression among 317 colon cancer patients with all tumor stages. HER-2/neu expression was significantly correlated with tumor size & distant metastases.
Literature	GENERIF_SUMMARY	GRP94, FN14, and inhibin have roles in brain and non-brain metastases in ErbB-2+ and ErbB-2- breast neoplasms
Literature	GENERIF_SUMMARY	HER2 amplification was significantly related to worse survival in gastric cancer.
Literature	GENERIF_SUMMARY	our data suggest that Tid1 and CHIP play pivotal roles in affecting the levels of ErbB2 protein, and that both are significant prognostic indicators of breast cancer patient survival.
Literature	GENERIF_SUMMARY	Iionizing Radiation-induced, but not EGF-induced phosphorylation of cytoplasmic and/or nuclear Akt-S473 is erbB2 dependent.
Literature	GENERIF_SUMMARY	IQGAP1 protein binds human epidermal growth factor receptor 2 (HER2) and modulates trastuzumab resistance.
Literature	GENERIF_SUMMARY	The first quantitative and qualitative analysis of the poly (A) transcriptome of two human mammary cell lines, differentially expressing human epidermal growth factor receptor, is reported.
Literature	GENERIF_SUMMARY	Multifocal/multicentric invasive ductal carcinomas usually have a single phenotype in terms of hormone receptors, human epidermal growth factor receptor 2, and molecular subtypes.
Literature	GENERIF_SUMMARY	Using immunohistochemistry, our data showed that the association between HER2/neu overexpression and patient survival provides a good indirect validation for quality of this investigation and it may act as a prognostic parameter in gastric cancer.
Literature	GENERIF_SUMMARY	Data show that HER2 overexpression enhances Notch3 promoter activity in mammary epithelial cells.
Literature	GENERIF_SUMMARY	Data show that the triple-negative and HER2 subtypes were associated with poorer outcomes compared with the luminal A subtype.
Literature	GENERIF_SUMMARY	Sarcosine may induce prostate cancer progression by increased HER2/neu expression.
Literature	GENERIF_SUMMARY	There was a significantly positive association between

Main Category	Category	Terms
		HER2 expression status and tumor differentiation and histological type in Chinese patients with gastric carcinoma.
Literature	GENERIF_SUMMARY	Dual-colour chromogenic in situ hybridization is an excellent alternative for the evaluation of HER2 genomic status in gastric cancers.
Literature	GENERIF_SUMMARY	percentage of tumor cells with HER2/CEP17 ratios more than 2.20 does not identify cases with heterogeneous amplification or poor outcome
Literature	GENERIF_SUMMARY	Early onset HER2-positive breast cancer is associated with germline TP53 mutations.
Literature	GENERIF_SUMMARY	Data show that triple-negative breast cancer (TNBC) are larger in size compared with other subtypes and almost all grade 3.
Literature	GENERIF_SUMMARY	Her-2 over-expression was found to be more common in esophageal adenocarcinomas. (Review)
Literature	GENERIF_SUMMARY	Inhibition of EGFR and Her2 phosphorylation by Lapatinib is shown to be associated with the inhibition of Akt phosphorylation.
Literature	GENERIF_SUMMARY	These results suggest that down-regulation of miR-205 in erbB2-overexpressing breast epithelial cells is essential for erbB2-induced tumorigenesis.
Literature	GENERIF_SUMMARY	Her2 gene amplification was detected in 0 of IHC score 1+, 24/86 (27.91%) 2+, and 8/13 (61.54%) 3+.
Literature	GENERIF_SUMMARY	These results support GDF15-mediated activation of TGF beta receptor-Src-HER2 signaling crosstalk as a novel mechanism of trastuzumab resistance.
Literature	GENERIF_SUMMARY	Data show that C-erbB-2 protooncogene alteration is an early phenomenon in glial tumor development and progression.
Literature	GENERIF_SUMMARY	mathematical model predicting the fraction of proliferating breast cancer cells in G1, S, and G2/M phases of the cell cycle as a function of EGFR and HER2
Literature	GENERIF_SUMMARY	Alterations of tumor suppressor genes such as CDKN2A and oncogenes such as ERBB2 play a role in the progression of intestinal metaplasia to adenocarcinoma in patients with Barrett's esophagus.
Literature	GENERIF_SUMMARY	A significant proportion (21.3%) of patients with equivocal IHC(2+) test results show amplification of the HER2 gene.
Literature	GENERIF_SUMMARY	We confirmed evidence that discordance of ER/Her2 expression between primary and metastatic breast carcinoma does occur.
Literature	GENERIF_SUMMARY	Data show that overexpression of ErbB2 maintains PDH flux by suppressing PDK4 expression in an Erk-dependent manner.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	multivariate analysis, which used luminal A subtype in old patients as reference, indicated that HER2 subtype in young patients was associated with increased risk of ipsilateral breast tumor recurrence
Literature	GENERIF_SUMMARY	Because IHC is recommended before gene amplification studies in HER2 testing, gastric cancer scoring system should be used for evaluating HER2 expression in gastric cancers.
Literature	GENERIF_SUMMARY	Metformin decreased erbB2/insulin-like growth factor-1 receptor complexes without altering erbB2 expression.
Literature	GENERIF_SUMMARY	Breast cancers HER2-indeterminate (2+) by HercepTest IHC can be strongly separated into those probably HER2 amplified, a core indeterminate group and those probably not HER2 amplified.
Literature	GENERIF_SUMMARY	Data show that hierarchical clustering based on mRNA expression of estrogen receptors, progesterone receptor and neu receptor by quantitative RT-PCR identified patient groups that were comparable to the subtypes identified by immunohistochemistry.
Literature	GENERIF_SUMMARY	demonstrated that ER-alpha36 positively regulates HER2 expression and the population of ALDH1 positive breast cancer cells, and suggested that non-genomic estrogen signaling mediated by ER-alpha36 is involved in maintenance of breast cancer stem cells
Literature	GENERIF_SUMMARY	HER2 oncogenicity is dependent, not only on its constitutively active nature, but also on its ability to muster collaborative signaling from family members through modulation of ligand-induced receptor regulation
Literature	GENERIF_SUMMARY	results indicated that expression patterns of ER, NM23-H1, and HER2 in primary breast cancer lesions warn that cells might have metastatic potential
Literature	GENERIF_SUMMARY	Meta-analysis
Literature	GENERIF_SUMMARY	The expression of leptin, OB-Rb and HER-2 was significantly higher in non-small-cell lung cancer (NSCLC) tissues than in normal lung tissues. The expression of leptin is an independent prognostic factor for NSCLC.
Literature	GENERIF_SUMMARY	Results suggest that Rab7 is likely involved in protecting EGFR and Her2 from being degraded by the proteosome and in maintaining optimal tumor cell Akt survival signal.
Literature	GENERIF_SUMMARY	data suggest that stratification of CaP patients for HER2/3 and PTEN status could identify patients with aggressive CaP who may respond favorably to MEK inhibition
Literature	GENERIF_SUMMARY	This study revealed a novel mechanism of Akt2 regulation by ErbB2 in prostate cancer cells.
Literature	GENERIF_SUMMARY	The results indicated a significant correlation between

Main Category	Category	Terms
		HER-2/neu and grade, but an inverse relationship between HER-2/neu and hormone receptors in breast carcinoma.
Literature	GENERIF_SUMMARY	Data show that Rac1 was highly activated in PTEN deficient and IGF-IR overexpressing Trastuzumab-resistant cells in a HER2-independent manner.
Literature	GENERIF_SUMMARY	Report population pharmacokinetics of trastuzumab emtansine, a HER2-targeted antibody-drug conjugate, in patients with HER2-positive metastatic breast cancer.
Literature	GENERIF_SUMMARY	Her2-Neu was expressed in 36.1% of 122 Mexican breast cancer patients. There was a close with relation the tumour size & the presence of lymph node metastases but not with age, differentiation or estrogen receptors.
Literature	GENERIF_SUMMARY	Results suggest for the development of a Her2-targeted treatment option for this new patient subgroup of oesophageal adenocarcinoma.
Literature	GENERIF_SUMMARY	Cinical study indicates that co-overexpression of Her2 with GEP100 in primary lung adenocarcinomas of patients is correlated with the presence of their node-metastasis with a statistical significance.
Literature	GENERIF_SUMMARY	HuR and miR-331-3p participate in the overexpression of ERBB-2 observed in some PCas.
Literature	GENERIF_SUMMARY	The tumor was finally discovered to overexpress the ERBB2 protein and the patient responded well to trastuzumab plus 2 cytotoxic agents.
Literature	GENERIF_SUMMARY	For patients with HER2-overexpressing gastric cancer, there is an improved prognosis with additional trastuzumab to chemotherapy with a platinum compound and a fluoropyrimidin in first-line therapy.
Literature	GENERIF_SUMMARY	A positive feedback loop between HER2 and ADAM12 in human head and neck cancer cells increases migration and invasion.
Literature	GENERIF_SUMMARY	This study shows that the HER2 codon 655 G-allele was associated with a larger reduction in plasma HDL-C levels in breast cancer patients under tamoxifen therapy.
Literature	GENERIF_SUMMARY	the feasible dose for lapatinib, a dual HER2/EGFR tyrosine kinase inhibitor, with paclitaxel and gemcitabine as a neoadjuvant treatment in HER2 positive patients.
Literature	GENERIF_SUMMARY	HER2 genetic heterogeneity is frequent in breast carcinoma, and is most often present in carcinomas with an equivocal (2+) HER2 score
Literature	GENERIF_SUMMARY	Antisense oligodeoxynucleotide targeting HER2 mRNA sensitized breast cancer cells to docetaxcel treatment.
Literature	GENERIF_SUMMARY	We have shown that TGF-beta1 expression in gastric tumor tissue with HER2/neu-negative status is of

Main Category	Category	Terms
		prognostic relevance in gastric cancer.
Literature	GENERIF_SUMMARY	Gene expression signatures comprising 24, 51 and 14 genes predictive of ER, PR and ERBB2 status respectively, were investigated.
Literature	GENERIF_SUMMARY	There was no significant correlation between patient age, the various histological types, tumor grade, and Her-2/neu positivity in breast carcinoma
Literature	GENERIF_SUMMARY	Data show that cyclin D3 is significantly upregulated in breast cancers and frequently exceeds the expression of Cyclin D1 in ErbB2-positive cases.
Literature	GENERIF_SUMMARY	Results suggest that co-expression of androgen and estrogen receptors can provide a protective effect based on phenotypical presentation of HER2-positive carcinomas.
Literature	GENERIF_SUMMARY	The gene expressions of PCNA, c-erbB-2, and p53 gradually decreased with the increased distance from primary tumor (P
Literature	GENERIF_SUMMARY	HER-2/neu overexpression may be used as an independent prognostic factor for intestinal-type and early-stage gastric cancer patients
Literature	GENERIF_SUMMARY	The soluble production of the novel scFv800E6-EGFP protein in the cytoplasm of Escherichia coli was investigated at different induction temperatures (25, 30 and 37 degrees C)
Literature	GENERIF_SUMMARY	Multivariate analysis identified HER2 overexpression as an independent unfavourable prognostic variable for adenocarcinomas as a whole and also for the hepatoid variant.
Literature	GENERIF_SUMMARY	HER2 mutations in lung adenocarcinoma predominantly occurred in women and never smokers.
Literature	GENERIF_SUMMARY	SLP-2 and HER2/neu can play a role in lymph node/ distant metastases of breast cancers
Literature	GENERIF_SUMMARY	Report role of systems biology in identifying drug targets and biomarkers of ErbB2/3-mediated drug resistance in ovarian neoplasms.
Literature	GENERIF_SUMMARY	the discordant HER2 amplification observed in metastatic gastric carcinoma lesions is explained substantially by heterogeneity within primary tumours.
Literature	GENERIF_SUMMARY	Report remarkable response rate in trial of preoperative paclitaxel, gemcitabine, and trastuzumab combination therapy in HER2 positive stage II/III breast cancer.
Literature	GENERIF_SUMMARY	aberrant HER2 protein expression in carcinoma in situ seems to be more commonly associated with polysomy than with gene amplification.
Literature	GENERIF_SUMMARY	novel 5-alkynyl-4-anilinopyrimidines are potent, orally active dual inhibitors of EGFR and Her-2 tyrosine kinases

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	ErbB2 overexpression is associated with chemoradiation resistance in muscle-invasive bladder cancer.
Literature	GENERIF_SUMMARY	This is the first report of a positive response to trastuzumab in a patient with HER2-overexpressing metastatic gastric cancer that was accompanied by bone marrow involvement and severe thrombocytopenia.
Literature	GENERIF_SUMMARY	In response to ErbB2 activation, the biosynthesis and secretion of soluble interleukin-1 receptor-like 1 (sST2) were increased. sST2-blocking antibodies reduced ErbB2-induced motility.
Literature	GENERIF_SUMMARY	patients with HER2-overexpressing locally advanced breast cancer show a more limited tumor-related immune suppression.
Literature	GENERIF_SUMMARY	Serum HER2/ECD concentration of >/=7.7 ng/ml has possible diagnostic value in stage I and II breast cancer. It should not be used as a determinant of HER2 positivity.
Literature	GENERIF_SUMMARY	There was a significant correlation between serum HER2 ECD levels and HER2 tissue status (P
Literature	GENERIF_SUMMARY	The decreased Th/Tr cells ratio indicates impaired immune function, suggesting that the stage IV breast cancer and the Her-2/VEGF-positive breast cancer patients have lower immune function.
Literature	GENERIF_SUMMARY	Invasive ductal mammary carcinoma type was analyzed for comparative investigation of hormonal receptors (estrogen receptor ER and progesterone receptor PR) and Her2/neu oncoprotein expression.
Literature	GENERIF_SUMMARY	The HER2/neu supraexpression was encountered in 58/156.
Literature	GENERIF_SUMMARY	Data show that DeltaNErbB2 expression elicited Na(+), HCO(3)(-) cotransporter NBCn1 upregulation, Ser(703)-phosphorylation of Na(+)/H(+) exchanger NHE1.
Literature	GENERIF_SUMMARY	in breast cancer patents, HER-2/neu amplification was found to be significantly associated with increased risk of death; findings support a negative prognostic role for HER-2/neu in breast-cancer survival
Literature	GENERIF_SUMMARY	a gene expression signature based on the HER2/EGF-activated transcriptional program can predict poorer prognosis of a subgroup of HER2-overexpressing patients.
Literature	GENERIF_SUMMARY	Amplification of HER2 and deletion of TOP2A were frequently observed in tumors with a high ploidy level.
Literature	GENERIF_SUMMARY	HER2-positive breast cancer patients given pertuzumab and trastuzumab plus docetaxel had significantly improved pathological complete response rate compared with those given trastuzumab plus docetaxel, without substantial differences in tolerability.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	and increased gene copy number of EGFR and HER2 characterize high-grade malignancy with unfavorable prognosis in salivary gland cancer
Literature	GENERIF_SUMMARY	Suggest that responses to preoperative chemotherapy, including trastuzumab, are biologically driven and that a specific nomogram or predictor for HER2-positive patients has to be developed.
Literature	GENERIF_SUMMARY	No significant correlations were observed between the clinicopathological variables and PIK3CA, PTEN, and pHER2 status in breast cnacer.
Literature	GENERIF_SUMMARY	Case Report
Literature	GENERIF_SUMMARY	HER2 seems to be a valid therapeutic target of EGFR inhibitors in HER2-addicted lung carcinomas
Literature	GENERIF_SUMMARY	analysis of HER2 overexpression in non-small cell lung cancer
Literature	GENERIF_SUMMARY	Protein interactions with HER2 can have different characteristics depending on the hosting cell line.
Literature	GENERIF_SUMMARY	Negative/low HER2 expression alone or combined with E-cadherin positivity is predictive of better prognosis in Tunisian patients with breast carcinoma.
Literature	GENERIF_SUMMARY	Data indicate that MET-mediated resistance to lapatinib is a novel mechanism of resistance to HER2-targeted agents in gastric cancer cells.
Literature	GENERIF_SUMMARY	This study confirms the favorable prognostic value of HER2/TOP2A co-amplification.
Literature	GENERIF_SUMMARY	Findings show that the bispecific antibody MM-111 forms a trimeric complex with ErbB2 and ErbB3, effectively inhibiting ErbB3 signaling and showing antitumor activity in preclinical models that is dependent on ErbB2 overexpression.
Literature	GENERIF_SUMMARY	esophageal adenocarcinomas with Barrett esophagus had nearly twice the odds of being HER2 positive and, within this subgroup, HER2 positivity was independently associated with improved survival.
Literature	GENERIF_SUMMARY	Data suggest that iIn gastric cancer, HER2 gene amplification is the main mechanism for HER2 protein overexpression and is preserved in lymph node metastases.
Literature	GENERIF_SUMMARY	Consider it appropriate to evaluate HER2 expression using gastric cancer biopsy specimens.
Literature	GENERIF_SUMMARY	Report the interplay of HER2/HER3/PI3K and EGFR/HER2/PLC-gamma1 signalling in breast cancer cell migration and dissemination.
Literature	GENERIF_SUMMARY	The findings uncovered an unexpected link between PGC-1a-mediated nutrient availability, unfolded protein response, and ErbB2/Neu levels.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	HER2 expression of circulating tumor cells in peripheral blood of patients prior to radical cystectomy predicts early disease recurrence and cancer-specific and overall mortality.
Literature	GENERIF_SUMMARY	Data show that EphrinB1, a PTPN13 substrate, interacts with ErbB2, and Src kinase mediates EphrinB1 phosphorylation and subsequent MAP Kinase signaling.
Literature	GENERIF_SUMMARY	HER-2/neu and TOPIIa are over expressed in gastric tumors and may promote disease progression.
Literature	GENERIF_SUMMARY	ER/PR and HER2 statuses between primary and liver metastatic lesions of breast carcinoma can be modified after treatment but are stable in most cases during the natural metastatic process.
Literature	GENERIF_SUMMARY	Signaling via the HER2/HER3 pathway in breast carcinoma cell lines may lead to enhanced natural killer cell (NKG2D)-histocompatibility class I-related chains A/B (MICA/B) recognition by NK cells and T cells.
Literature	GENERIF_SUMMARY	Data indicate that 266 (76.2%) tumors harbored EGFR mutations, 16 (4.6%) HER2 mutations, 15 (4.3%) EML4-ALK fusions, 7 (2.0%) KRAS mutations, and 2 (0.6%) BRAF mutations.
Literature	GENERIF_SUMMARY	Mutations in the kinase domain of HER2 is associated with treatment response in lung adenocarcinoma.
Literature	GENERIF_SUMMARY	29 out of 83 (35%) samples tested by dPCR and 28 out of 86 (32.5%) samples tested by real time PCR have HER-2/neu gene amplification.
Literature	GENERIF_SUMMARY	Immunoreactivity for erbB1 was strong and that for erbB2 and 3 was moderate, while that for erbB4 was faint in nasal mucosa chronic rhinitis.
Literature	GENERIF_SUMMARY	HER-2 is stably expressed in primary gastric cancer and metastatic lymph nodes. HER-2 and leptin play an important role in the progression and angiogenesis of gastric cancer
Literature	GENERIF_SUMMARY	The expression of Survivin in combination with BCRP and HER-2 is associated with clinical response to TE neoadjuvant chemotherapy in breast cancer.
Literature	GENERIF_SUMMARY	AEG-1 might facilitate the proliferation and invasion of breast cancer cells by upregulating HER2/neu expression.
Literature	GENERIF_SUMMARY	HER2 and CK20 were seen exclusively in the mucinogenic proliferations in congenital pulmonary airway malformations.
Literature	GENERIF_SUMMARY	This study shows a very high frequency of HER2 overexpression in breast cancers in patients with anti-Yo-associated paraneoplastic cerebellar degeneration.
Literature	GENERIF_SUMMARY	Women with a HER2/neu/neu+/Ki67+ ductal carcinoma in situ (DCIS) have a higher risk of developing DCIS local recurrence after breast-conserving surgery.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	c-erbB2 oncogene expression is a valuable marker in the differential diagnosis and prognostic evaluation of patients with intraductal proliferative lesions.
Literature	GENERIF_SUMMARY	HER2 overexpression is associated with recurrence in breast cancer.
Literature	GENERIF_SUMMARY	ErbB-2 overexpression in human breast cancer cell lines leads to phosphorylation & activation of Plexin-B1. This was required for ErbB-2-dependent activation of RhoA & RhoC & promoted invasive behavior.
Literature	GENERIF_SUMMARY	linkage between NRF2 and ErbB2 was confirmed in the ErbB2-positive breast cancer cell line BT-474
Literature	GENERIF_SUMMARY	intratumoral heterogeneity of HER2 gene amplification is present in a subset of HER2-amplified breast cancers
Literature	GENERIF_SUMMARY	ErbB2 and MUC4, which interact physically, activate different intracellular signalling pathways to regulate biological properties of CAPAN-2 pancreatic cancer cells
Literature	GENERIF_SUMMARY	EGFR overexpression relates to triple negative profile (ER-/PR-/HER2-) and poor prognosis in breast cancer patients in Tunisia
Literature	GENERIF_SUMMARY	ERBB2 overexpression triggers transient high mechanoactivity of breast tumor.
Literature	GENERIF_SUMMARY	HER2 expression was less frequent in resectable gastric cancer than in metastatic gastric cancer; the impact of HER2 expression on survival was limited
Literature	GENERIF_SUMMARY	HER2-negative (1+) breast cancer with unfavorable prognostic features.
Literature	GENERIF_SUMMARY	Addition of sorafenib to capecitabine improved progression-free survival in patients with HER2-negative advanced breast cancer.
Literature	GENERIF_SUMMARY	68.3% of patients were ER-positive, and 71.5% were PR-positive. Among the 28.3% of HER2-positive tumors, the majority (19.2%) stained +++. Almost 50% of the tumors were ER+ PR+ HER2-, which makes this the predominant tumor type.
Literature	GENERIF_SUMMARY	These findings indicate that in the absence of HER2 amplification, chromosome 17 centromere gains do not have a significant effect on HER2 protein expression.
Literature	GENERIF_SUMMARY	No significant difference was seen between grade I and grade II/III meningiomas, primary and recurrent tumors, and males and females from the point of view of HER2 expression.
Literature	GENERIF_SUMMARY	identified two dominant expression clusters corresponding to samples belonging to the ERBB2+ and ERBB2- groups of pheochromocytomas and functional paragangliomas
Literature	GENERIF_SUMMARY	Focal HER2 gene amplification is more common in

Main Category	Category	Terms
		gastric carcinomas with no or low expression of HER2.
Literature	GENERIF_SUMMARY	Membrane-bound trafficking regulates nuclear transport of integral epidermal growth factor receptor (EGFR) and ErbB-2.
Literature	GENERIF_SUMMARY	Aberrantly expressed miR-125b contributes to HEC1B cells invasion partly through directly down-regulating ERBB2 protein expression in endometrioid endometrial cancer.
Literature	GENERIF_SUMMARY	Her2-tumor-initiating cells-enriched signature was specific to HER2(+)
Literature	GENERIF_SUMMARY	signaling network activates the transcription of cathepsin B gene via myeloid zinc finger-1 transcription factor that binds to an ErbB2-responsive enhancer element in the first intron of cathepsin B
Literature	GENERIF_SUMMARY	The up-regulation of HER2 and B(1)R in precursor lesions of gallbladder carcinoma suggests cross-talk between these two receptors that may be of importance in the modulation of cell proliferation in gallbladder carcinogenesis.
Literature	GENERIF_SUMMARY	This review focuses on mechanisms of intrinsic and/or acquired resistance to HER2-targeted therapies that have been identified in preclinical and clinical studies.
Literature	GENERIF_SUMMARY	HER2 GH and HER2-positive tumors have distinctly different impacts on the clinicopathologic features of tumors. HER2 GH tumors may be closer to HER2 non-amplified tumors in their biologic behavior and is different from that of HER2 gene-amplified tumors.
Literature	GENERIF_SUMMARY	Pretreatment of tumor cells with STAT3 specific inhibitors or knocking down of STAT3 by SiRNA makes the tumor cell more susceptible to apoptosis and dendritic cell mediated inhibition of both CD24 and HER-2/neu
Literature	GENERIF_SUMMARY	Data indicate that C-erbB2 gene plays important roles in the carcinogenesis and development of colon cancer.
Literature	GENERIF_SUMMARY	Expression of pSTAT3 correlates with Her-2 status and CAIX expression and is associated with tumor progression and worse outcome in esophageal cancer
Literature	GENERIF_SUMMARY	HER2 may play an important role in the progression of carcinoma ex pleomorphic adenoma, and that HER2 amplification may be an additional prognostic indicator of CXPA.
Literature	GENERIF_SUMMARY	3
Literature	GENERIF_SUMMARY	overexpression correlates with the urothelial carcinoma of the upper urinary tract stage but not with survival or recurrence
Literature	GENERIF_SUMMARY	This study demonstrates the role of C-erbB2 in the pathogenesis of gallbladder cancer as a late event in north Indian population which is a igh prevalence region

Main Category	Category	Terms
		for gallbladder cancer.
Literature	GENERIF_SUMMARY	HER-2 gene amplification is associated with breast cancer.
Literature	GENERIF_SUMMARY	Letter/Case Report
Literature	GENERIF_SUMMARY	Results show strong correlations between mRNA levels of PRA and PRB, protein levels of hormone receptors, HER/ErbB receptors and ligands network, and suggest that crosstalks between PR and the HER family are a hallmark of breast cancer growth.
Literature	GENERIF_SUMMARY	The nanoparticle effectively and significantly inhibited colony formation of the target cells and silenced the Her2 gene expression at 5 nM
Literature	GENERIF_SUMMARY	Combinations of trastuzumab plus docetaxel-based regimens were well tolerated and effective in previously treated metastatic gastric cancer of Chinese patients with HER2 over-expression or gene amplification.
Literature	GENERIF_SUMMARY	Mutation of this amino acid led to increased ERBB receptor activation and upregulation of the ERBB3/PI3K/AKT signaling pathway, which was no longer responsive to MEK inhibition
Literature	GENERIF_SUMMARY	data presented here demonstrate a reciprocal expression pattern between EGFR family members during keratinocyte maturation in the skin, and altered expression of EGFR and ErbB2 was identified in atopic dermatitis lesional skin as compared to healthy skin.
Literature	GENERIF_SUMMARY	the relationship between HER2/neu, breast cancer subtypes, and a number of novel targets located at the HER2/neu amplicon
Literature	GENERIF_SUMMARY	This study revealed that the previously described up-regulation of EGFR and down-regulation of ERBB4 occurred in all analyzed renal cell carcinoma types, whereas down-regulation of ERBB2 and LRIG1 was only present in clear cell renal cell carcinoma.
Literature	GENERIF_SUMMARY	Amplification of Her-2/neu and/or topoisomerase IIalpha is associated with poor overall survival in breast cancer.
Literature	GENERIF_SUMMARY	Report use of somatostatin receptor/Her2 immunohistochemical scoring schemes in gastroenteropancreatic neuroendocrine tumors.
Literature	GENERIF_SUMMARY	HER-2 copy number but not the HER-2/CEP-17 ratio determines taxane-containing chemotherapy sensitivity in locally advanced breast cancer.
Literature	GENERIF_SUMMARY	Overexpression of HER2 and pHER2 to be associated with a significant reduction in overall survival in early breast cancer.
Literature	GENERIF_SUMMARY	HER2 gene amplification is significantly associated with patient survival in gastric cancers.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	This study demonstrates that there is a cross-regulation network between FOXA1 and ErbB2 signaling that connects FOXA1 to some of the key signaling pathways in estrogen receptor-negativebreast cancer.
Literature	GENERIF_SUMMARY	Bgn inhibits the major properties of HER-2/neu-transformed cells, which is inversely modulated by the PKC signaling cascade.
Literature	GENERIF_SUMMARY	GRB7 is a context-dependent oncogene, which modulates the ERBB2 signaling pathway through enhanced phosphorylation of ERBB2 and Akt.
Literature	GENERIF_SUMMARY	A population-based cancer registry study demonstrated that HER2-positive tumors account for a substantial proportion of mammographic screening failure.
Literature	GENERIF_SUMMARY	rs4919510
Literature	GENERIF_SUMMARY	The combined inhibition of HER2 and EGFR induced overt, long-lasting tumor regression.
Literature	GENERIF_SUMMARY	Within the macro domain of mH2A1.2, a trinucleotide insertion (-EIS-) sequence not found in mH2A1.1 was essential for the interaction between HER-2 and mH2A1.2 as well as mH2A1.2-induced HER-2 expression and cell proliferation.
Literature	GENERIF_SUMMARY	The ratio of c-erbB-2 staining increased as the grade and stage increased in colorectal adenocarcinoma.
Literature	GENERIF_SUMMARY	A dominant-negative N-terminal fragment of HER2 frequently expressed in breast cancers.
Literature	GENERIF_SUMMARY	Lack of c-erbB2 expression in oral leukoplakia suggested that it does not play a pathogenetic role in the early stages of carcinogenesis in this Indian population.
Literature	GENERIF_SUMMARY	HER-2 expression in immunohistochemistry has no prognostic significance in gastric cancer patients
Literature	GENERIF_SUMMARY	human epidermal growth factor receptor 2 (HER2/neu) protein, a 185-kDa transmembrane tyrosine kinase receptor, is associated with tumor proliferation, migration, and differentiation
Literature	GENERIF_SUMMARY	The HER2 proto-oncogene is located on chromosome 17q21 and encodes a 185-kDa transmembrane tyrosine kinase receptor, HER2 (also known as HER2/neu, ERBB2, p185)
Literature	GENERIF_SUMMARY	hormone receptor-negative breast carcinomas are undifferentiated and diagnosed at advanced stages and HER2 expression is not associated with overall survival
Literature	GENERIF_SUMMARY	Report immunohistochemistry/in situ hybridization assay for determination of HER2 protein/gene in breast cancer sections.
Literature	GENERIF_SUMMARY	the up-regulation of the Pcdhbeta gene cluster as a mechanism for reduced her-2-mediated tumorigenesis

Main Category	Category	Terms
		resulting from GnT-V deletion.
Literature	GENERIF_SUMMARY	These data provide new insight into HER2 trafficking at the plasma membrane and the changes induced by polyclonal HER2-specific antibodies.
Literature	GENERIF_SUMMARY	STAT1 expression is upregulated by nuclear EGFR and HER2, and that STAT3 synergizes with the three receptors to further enhance STAT1 expression.
Literature	GENERIF_SUMMARY	Tethered conformation of unliganded ErbBs evolved to prevent crosstalk among ErbBs, and establish a molecular basis for both negative cooperativity of ligand binding to vertebrate ErbBs and the absence of active ErbB2/HER2 homodimers in normal conditions.
Literature	GENERIF_SUMMARY	Immunohistochemical study of adhesive molecules Her-2/neu in nonendometrial carcinoma of corpus uteri
Literature	GENERIF_SUMMARY	Compared to the United States, low age standardized and age specific breast cancer incidence rates were found in Kurdish women; nevertheless, the proportional expression of HR and HER2 for both Kurds and Arabs was comparable to that of US white women.
Literature	GENERIF_SUMMARY	A-62176 is a dual inhibitor that impairs the expression of HER2.
Literature	GENERIF_SUMMARY	Higher order complexes of nonoverexpressed ERBB receptors are an integral and qualitatively distinct part of normal ERBB2/ERBB3 signaling.
Literature	GENERIF_SUMMARY	Lymph node ratio is a new feature for defining risk category of HER-2 node-positive breast cancer patients.
Literature	GENERIF_SUMMARY	HER2 may be related with the pathogenesis of NSCLC and act as an unfavorable prognostic factor in T1N0M0 and T2N0M0 NSCLC, especially in stages IB and IIA diseases.
Literature	GENERIF_SUMMARY	analysis of multiorgan metastasis of human HER-2+ breast cancer in Rag2-/-;Il2rg-/- mice and treatment with PI3K inhibitor
Literature	GENERIF_SUMMARY	The frequency of HER2 overexpression in early onset breast cancer patients was 28.1%, with a higher proportion of whites with overexpression of HER2 at 34.1% compared with 22.9% and 25% for African-Americans and Asians.
Literature	GENERIF_SUMMARY	study concludes that JAM-A is co-expressed with HER2 and associates with aggressive breast cancer phenotypes; speculate that JAM-A may regulate HER2 proteasomal degradation and activity
Literature	GENERIF_SUMMARY	Among the 139 HER2-positive cases, the relapse rate was associated with the intensity of stromal hyaluronan staining as most of the relapses occurred in the cases with intense stromal hyaluronan staining.
Literature	GENERIF_SUMMARY	Data indicate that regulatory T cells (T-regs) cell number of Her-2/neu-positive patients were higher than in

Main Category	Category	Terms
		samples from patients with hormone receptor positive, Her-2/neu-negative cancers.
Literature	GENERIF_SUMMARY	HER2 mutations identify a distinct subset of lung ADCs. Given the high prevalence of lung cancer worldwide and the availability of standard and investigational therapies targeting HER2, routine clinical genotyping of lung ADC should include HER2.
Literature	GENERIF_SUMMARY	The probe developed for RNA-ISH represents a viable, effective possible alternative to FISH and IHC for analysing HER2 status in primary breast tumours.
Literature	GENERIF_SUMMARY	ErbB2 amplification results in aberrant activation of cap-dependent mRNA translation in esophageal cancer cell lines.
Literature	GENERIF_SUMMARY	HER-2 receptors that can act as a starting point to trigger intracellular signaling pathways are strongly involved in the progression of cancer.
Literature	GENERIF_SUMMARY	The endosomal recycling of the ErbB2 and EGFR receptors is associated with efficient K63-polyubiquitination, but significantly decreased Hrs tyrosine phosphorylation as well as impaired deubiquitination by AMSH.
Literature	GENERIF_SUMMARY	Binding of L1CAM to erbB enhances its response to neuregulins.
Literature	GENERIF_SUMMARY	Quantifying HER2 expression and dimerization by TR-FRET assays might help identifying novel clinical markers for optimizing patients' treatment in oncology.
Literature	GENERIF_SUMMARY	Correlation of HER2, p95HER2 and HER3 expression and treatment outcome of lapatinib plus capecitabine in her2-positive metastatic breast cancer.
Literature	GENERIF_SUMMARY	ErbB2 is a direct substrate of PTPRO and decreased expression of PTPRO predicts poor prognosis for ErbB2-positive breast cancer patients.
Literature	GENERIF_SUMMARY	In ER-positive breast cancer, correlations were found in the expression of activated HER-receptors (pHER1, pHER2, pHER3), indicating a potential role of HER1 and HER3 in the activation of HER2. HER2 expression was associated with poor prognosis.
Literature	GENERIF_SUMMARY	the low number of patients in this study, genetic profile of the patient population, possible weight changes after treatment and the inverse relationship between body mass index and c-erbB2 expression.
Literature	GENERIF_SUMMARY	we describe how flotillin-1 and -2 contribute to the stabilization of ErbB2 at the cell surface in breast cancer tissue
Literature	GENERIF_SUMMARY	MMP-9 significantly increased HER2/neu expression on the surface of human mammary epithelial cells.
Literature	GENERIF_SUMMARY	Her-2/neu-testing in breast cancer

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Different Her-2 status in infiltrative and intraductal components of the breast cancer
Literature	GENERIF_SUMMARY	There is early involvement of HER2 dysregulation (amplification and protein overexpression) in both gastric (intestinal-type) and Barrett's oncogenesis.
Literature	GENERIF_SUMMARY	Data suggest that HER-2 and leptin (but not leptin receptor) are up-regulated in gastric cancer and are associated with tumor invasiveness, lymphatic metastasis, and angiogenesis.
Literature	GENERIF_SUMMARY	Results show that elevated baseline sHER2 levels (>15 ng ml(1)) and a decrease of sHER2 levels (>20%) early after therapy initiation are both relevant criteria to predict response to lapatinib-based treatment.
Literature	GENERIF_SUMMARY	High-level ERBB2 amplification occurs in a small fraction of lung cancers with a strong propensity to high-grade adenocarcinomas.
Literature	GENERIF_SUMMARY	Propose that HER2 represents a reasonable target for molecular therapy in HER2- positive metastatic or recurrent mucinous ovarian carcinoma.
Literature	GENERIF_SUMMARY	c-erbB-2 overexpression may have a prognostic significance in patients with metastatic gastric cancer.
Literature	GENERIF_SUMMARY	High c-ErbB-2 expression is associated with ovarian cancer.
Literature	GENERIF_SUMMARY	These patients had significantly longer TTBM than those without anti-HER2 treatment.
Literature	GENERIF_SUMMARY	TNF-alpha treatment led to intracellular Her-2 cleavage via a caspase-8-dependent signaling pathway in MCF-7 cells defective for NFkappa B activation.
Literature	GENERIF_SUMMARY	study found no overexpression of HER2/neu in head and neck squamous cell carcinomas; findings suggested that the use of Her-2 as a salivary marker of head and neck squamous cell carcinoma cannot be recommended
Literature	GENERIF_SUMMARY	The simultaneous gain of NUPR1 and ERBB2 can be a significant predictor of poor prognosis in early-stage breast cancers.
Literature	GENERIF_SUMMARY	study found that HER-2 2 /HER-2- tumors carry a prognosis that is worse than that of tumors with no or weak HER-2 expression
Literature	GENERIF_SUMMARY	clinical data analyses revealed a strong correlation between CD151 and ErbB2 expression and metastasis-free survival of breast cancer patients.
Literature	GENERIF_SUMMARY	These results implicate HER2 as a novel protein involved in the sensitivity or resistance of EGFR-mutant lung cancer
Literature	GENERIF_SUMMARY	MED1 is recruited to the HER2 gene and required for its expression.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Patients with HER2(+) metastatic breast cancers have epithelial-to-mesenchymal transition-circulating tumor cells.
Literature	GENERIF_SUMMARY	positivity, but not HER2 positivity, was associated with poor patient outcomes after curative resection of stage II/III gastric cancer.
Literature	GENERIF_SUMMARY	intracellular domains play a crucial role in HER3's function as an allosteric activator and its role in downstream signaling
Literature	GENERIF_SUMMARY	Downregulation of HER2 via siRNA resulted in a significant decrease in ERK phosphorylation and a 50% reduction in IL-8 secretion.
Literature	GENERIF_SUMMARY	inhibition of EGFR expression reduced HER2-induced anchorage-independent growth and tumorigenesis
Literature	GENERIF_SUMMARY	Incorporation of this engineered Fc into trastuzumab, resulted in a 75% increase in tumor cell phagocytosis by macrophages compared to that of the parental glycosylated trastuzumab with both medium and low Her2-expressing cancer cells.
Literature	GENERIF_SUMMARY	Data indicate that the monoclonal antibodies were directed to extracellular domain of HER2.
Literature	GENERIF_SUMMARY	Levels of C-erbB-2, CXCR4 and ER-alpha in patients decreased after they received neo-adjuvant chemotherapy.
Literature	GENERIF_SUMMARY	Our findings demonstrate that HER2 and HER3 co-expression induces IL8 autocrine signaling, leading to the invasion of mammary cells.
Literature	GENERIF_SUMMARY	This could be an important mechanism leading to trastuzumab resistance in HER2-overexpressing breast cancer cells.
Literature	GENERIF_SUMMARY	The positive correlation between HER2- and CXCR4-expression was validated in esophageal cancer patients.
Literature	GENERIF_SUMMARY	HER2 transcription levels were found to be significantly related to pT class, pN class, and tumor recurrence. mRNA expression was well correlated with protein overexpression and gene amplification
Literature	GENERIF_SUMMARY	Polymorphisms in the HER2 gene (Ile655Val) and an ATCTT insertion at intron 4 of PTEN, are associated with breast cancer susceptibility.
Literature	GENERIF_SUMMARY	A subset of high-grade non-muscle-invasive bladder cancers contain Her2 amplification and are associated with markedly aggressive behaviour.
Literature	GENERIF_SUMMARY	a profound activation of the EGFR system in breast cancer
Literature	GENERIF_SUMMARY	HER2-mediated prosurvival signaling network is responsible for the aggressive phenotype of breast

Main Category	Category	Terms
		cancer stem cells.
Literature	GENERIF_SUMMARY	High ErbB2 expression is associated with breast cancer.
Literature	GENERIF_SUMMARY	Reflex HER2 immunohistochemistry testing after initially ambiguous HER2 FISH testing provides definitive HER2 status in a majority of cases (70%).
Literature	GENERIF_SUMMARY	Plasma levels of HER2 extracellular domain predicts response to trastuzumab regimens in breast cancer.
Literature	GENERIF_SUMMARY	Chromogenic in situ hybridization could provide an accurate and practical alternative to fluorescence in situ hybridization for the clinical diagnosis of HER2 gene amplification.
Literature	GENERIF_SUMMARY	Breast cancer mRNA levels of ESR1 and HER2 predict response to neoadjuvant chemotherapy and are significantly associated with long-term outcome.
Literature	GENERIF_SUMMARY	Results suggest a possible participation of erbB1-4 and claudin-1 in tissue remodelling in chronic hypertrophic rhinitis.
Literature	GENERIF_SUMMARY	Data suggest that the traditional Chinese medicine (TCM) candidate potential three-in-one inhibitors for three drug target proteins epidermal growth factor receptor (EGFR), Her2, and uroporphyrinogen decarboxylase (UROD) against head and neck cancer.
Literature	GENERIF_SUMMARY	These findings reveal that ERBB2 and ERBB3 expression is regulated by ROS via miR-199a and miR-125b downregulation and DNA hypermethylation.
Literature	GENERIF_SUMMARY	Data show that HER2 gene copy by quantitative PCR on tissue correlated well with those by FISH technique, but plasma HER2 level was not.
Literature	GENERIF_SUMMARY	ErbB2 levels inversely impact cell clathrin-mediated endocytosis capacity in breast cancer cells treated with geldanamycin.
Literature	GENERIF_SUMMARY	Data indicate that the concordance between HER2 status of the paired samples by local immunohistochemistry testing and central in-situ hybridization were 78% and 99%, respectively.
Literature	GENERIF_SUMMARY	together, our results suggest that c-myc, p53, C-erbB-2 and COX-2 proteins do not correlate with cancer stage or follow-up in Esophageal squamous cell carcinoma
Literature	GENERIF_SUMMARY	HER2-positive Barrett's adenocarcinomas are associated with p53 overexpression and lesion protrusion at the early disease stage.
Literature	GENERIF_SUMMARY	Amplification of the ERBB2 gene in breast tumors is potentially initiated by a complex region that has unusual genomic features.
Literature	GENERIF_SUMMARY	Primary human ovarian epithelial cancer cells broadly express HER2 at immunologically-detectable levels.

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Literature	GENERIF_SUMMARY	The level of HER-2/neu mRNA was higher and RECK mRNA was lower in breast carcinoma than homologous paraneoplastic breast tissue.
Literature	GENERIF_SUMMARY	Twenty-seven point five percent of oesophageal adenocarcinomas showed evidence of HER2 over expression.
Literature	GENERIF_SUMMARY	The present study revealed that ESCC of Brazilian patients do not present mutations in hot spots of EGFR, K-RAS and BRAF and only a minor proportion present overexpression of EGFR or HER2.
Literature	GENERIF_SUMMARY	LC combination therapy was effective and well-tolerated in patients with HER2+ MBC with BMs, who had progressive disease after trastuzumab-containing therapy.
Literature	GENERIF_SUMMARY	Double HER2 blockade is an effective therapy in disseminated CDC even in the presence of brain metastases.
Literature	GENERIF_SUMMARY	analysis of the GP2 peptide of HER2 by terahertz spectroscopy
Literature	GENERIF_SUMMARY	ERBB1 and ERBB2 have different roles in response to antineoplastic agents in colorectal cancer cells
Literature	GENERIF_SUMMARY	The results demonstrated that EGF bound to EGFR more stably in the cells co-expressing EGFR and HER2, and the binding enhancement in the presence of HER2 was inhibited by either Trastuzumab or Pertuzumab
Literature	GENERIF_SUMMARY	EGFR and Her2 were significantly upregulated in lymph node metastasized colorectal cancer (LNM CRC) compared to non-LNM CRC.
Literature	GENERIF_SUMMARY	Results suggest that patients with HER2 mutation-positive breast cancers could benefit from existing HER2-targeted drugs.
Literature	GENERIF_SUMMARY	HER2 examination on fluorescence in situ hybridization using fine-needle aspiration cytology samples of tumors in recurrent/metastatic sites or disseminated tumor cells in effusion is beneficial
Literature	GENERIF_SUMMARY	localizes in mitochondria of cancer cells and negatively regulates mitochondrial respiratory functions
Literature	GENERIF_SUMMARY	HER2 status is not an independent prognostic biomarker in early oesophago-gastric adenocarcinoma.
Literature	GENERIF_SUMMARY	The major effect of EBP1 on ErbB2 mRNA expression levels is at the transcriptional level.
Literature	GENERIF_SUMMARY	Findings suggest that autocrine motility factor (AMF)-HER2 interaction might be a novel target for therapeutic management of patients with breast cancer, whose disease is resistant to trastuzumab.
Literature	GENERIF_SUMMARY	HER-2/neu overexpression/amplification is associated with gastric cancer.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Silencing of HER2 caused a greater growth arrest and apoptosis in trastuzumab responding compared to the non-responding cell lines, indicating that the resistant cells are inherently less dependent on the HER2 pathway.
Literature	GENERIF_SUMMARY	antierbB2 scFvFcIL2 fusion protein significantly delayed HER2 overexpression of nonsmall cell lung cancer (Calu3) in human nonsmall cell lung cancer xenografts in nude mice
Literature	GENERIF_SUMMARY	HER2 induces transcriptional activation of leptin in MCF10A human breast epithelial cells.
Literature	GENERIF_SUMMARY	Fcab clones that retained their affinity to Her2/neu but exhibited a significantly increased conformational stability and resistance to aggregation could be evolved
Literature	GENERIF_SUMMARY	The proposed enhanced method substantially improves HER2 FISH assessment in breast cancer biopsy specimens because the evaluation of HER2/CEP17 copy numbers is more representative, easier, and faster than the conventional method.
Literature	GENERIF_SUMMARY	Expression levels of ERBB2, caveolin-1 and DNA-PKcs were increased.
Literature	GENERIF_SUMMARY	High HER2 expression is associated with gastric carcinomas with advanced stages.
Literature	GENERIF_SUMMARY	Newly developed ovarian serous adenocarcinoma cell line, TU-OS-4 overexpresses human her2 protein.
Literature	GENERIF_SUMMARY	There was no statistically significant association between histopathological parameters and Her-2 gene polymorphism/expression in patients with gastric cancer
Literature	GENERIF_SUMMARY	Her-2 expression in circulating tumor cells varied greatly within and among patients compared with Her-2 expression in leukocytes. Her-2 expression is heterogeneous among circulating tumor cells within each patient.
Literature	GENERIF_SUMMARY	an FRS2beta-CIN85/CD2AP-Cbl axis for downregulation of ErbB2 may regulate
Literature	GENERIF_SUMMARY	The positive rate of vasculogenic mimicry in invasive breast carcinoma correlated with the increased HER2 expression.
Literature	GENERIF_SUMMARY	HER-2 amplification is detectable in a relevant proportion (26.7%) of rectal cancer patients.
Literature	GENERIF_SUMMARY	we show that the expression of p95HER2 in breast epithelial cell lines results in either increased proliferation or senescence.
Literature	GENERIF_SUMMARY	Brk/PTK6 cooperates with HER2 and Src in regulating breast cancer cell survival and epithelial-to-mesenchymal transition.

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Literature	GENERIF_SUMMARY	HER-2 functions as an oncogene in SKOV3 human ovarian carcinoma cells.
Literature	GENERIF_SUMMARY	The results suggest that miR-199b-5p may have the potential to be a novel important alternative therapeutic target for HER2-positive breast cancer.
Literature	GENERIF_SUMMARY	WNT10B/beta-catenin signalling induces HMGA2 and proliferation in metastatic breast cancer tumours devoid of ERalpha, PR and HER2 expression.
Literature	GENERIF_SUMMARY	ATG12 is a novel determinant of breast cancer primary resistance to HER1/2-targeted therapeutics.
Literature	GENERIF_SUMMARY	irradiation induces HER-2/neu gene amplification and CEP17 polysomy thereby enhancing expression of this protein in breast cancer cell lines rendering them susceptible to treatment with trastuzumab
Literature	GENERIF_SUMMARY	Thirty of 344 patients (9%) whose tumors were initially HER2-negative were found by FISH to have supernumerary HER2 gene copies (up to 15 copies) after neoadjuvant chemotherapy.
Literature	GENERIF_SUMMARY	Results suggest that inhibition of PTPalpha can have a beneficial effect on HER2-positive breast cancers, but that inhibition of additional targets is needed to block breast tumorigenesis.
Literature	GENERIF_SUMMARY	these data show that ionizing radiation-mediated activation of ErbB2 and induction of 14-3-3zeta collaborate to regulate FoxM1 and promote invasion of breast cancer cells
Literature	GENERIF_SUMMARY	HER2 stabilizes survivin while concomitantly down-regulating survivin gene transcription by suppressing Notch cleavage.
Literature	GENERIF_SUMMARY	HER2 exon 20 mutation is associated with non-small-cell lung cancer.
Literature	GENERIF_SUMMARY	Polymorphic HER2 variant alanine allele carries higher frequency in Parkinson's disease patients, linking the Parkinson's and HER2 polymorphism and suggesting an effect of the C-terminal domain of HER2 in dopaminergic cell survival.
Literature	GENERIF_SUMMARY	The present data suggest that ErbB2, ErbB3, and ErbB4 genes may have a role in bladder cancer tumorigenesis.
Literature	GENERIF_SUMMARY	HER-2 positivity and gene amplification are fairly frequent and independently associated with poor survival in esophageal adenocarcinoma
Literature	GENERIF_SUMMARY	TFAP2C amplification and overexpression represents a genetic dependency in ERBB2+ve breast cancer.
Literature	GENERIF_SUMMARY	Report c-ERBB-2 mutations in pancreatic ductal carcinoma and their relationship to patient survival.
Literature	GENERIF_SUMMARY	Our preliminary data demonstrate score 3+ staining for ErbB2 in 10% of adamantinomatous

Main Category	Category	Terms
		craniopharyngiomas and different localization of 3+ ErbB2 cells and cells with nuclear beta-catenin immunoreactivity.
Literature	GENERIF_SUMMARY	HER2 gene copy number status is asociated with clinical efficacy in metastatic colorectal cancer
Literature	GENERIF_SUMMARY	Heterogeneous HER2 expression in primary gastric tumors and the variable lymph node metastasis status suggest evaluating multiple samples.
Literature	GENERIF_SUMMARY	These findings provide proof and the foundation for the molecular and biological relationships of ErbB-2 and TACC1 in breast cancer.
Literature	GENERIF_SUMMARY	In breast cancer, dual in situ hybrid-ization may underestimate the HER2/chromosome 17 ratio, or fluorescence in situ hybridization may overestimate this ratio.
Literature	GENERIF_SUMMARY	There was no statistically significant correlation between histopathologic grading of salivary gland mucoepidermoid carcinoma and c-erbB-2 expression.
Literature	GENERIF_SUMMARY	racially restricted contributions of GM and FcgammaR genotypes to humoral immunity to HER2
Literature	GENERIF_SUMMARY	Report nuclei-based classifier as a tool for automated quantitative HER2 FISH signals analysis in nuclei in breast cancer specimens.
Literature	GENERIF_SUMMARY	ErbB2-dependent chemotaxis requires microtubule capture and stabilization coordinated by distinct signaling pathways.
Literature	GENERIF_SUMMARY	Response to trastuzumab is correlated to HER2 mRNA levels only in ESR1-positive tumors. This study adds further evidence to the different biology of both subsets within the HER2-positive group.
Literature	GENERIF_SUMMARY	These findings are very important for understanding the occurrence, development, prognosis and treatment of breast cancer in Chinese population.
Literature	GENERIF_SUMMARY	Our study suggested that ER/PR status and histological grade are risk factors for the development of bone metastasis in breast cancer patients.
Literature	GENERIF_SUMMARY	Data suggest that ErbB receptors (ERBB2, ERBB3, and ERBB4) are expressed in most endometrial cancers; expression of ErbB receptors appears to be related to poor prognosis; dismal outcomes correlate with cancers positive for all three ErbB receptors.
Literature	GENERIF_SUMMARY	The rate of cases overexpressing Her2 in advanced gastric adenocarcinoma in our study (35.5%) is higher than that observed in most series in the literature
Literature	GENERIF_SUMMARY	our results suggest that the detection of ESR1/PGR/ERBB2 mRNA levels by RT-QPCR is a better approach for subtyping breast cancer and predicting the prognosis.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	HER2 is overexpressed in 13% to 20% of NSCLCs (with 3+ expression in only 2% to 6%). HER2 insertions have been found to induce constitutive activation of the HER2 kinase in a ligand-independent fashion.
Literature	GENERIF_SUMMARY	High HER2 expression is associated with lung adenocarcinoma-associated cytologically negative pleural effusions.
Literature	GENERIF_SUMMARY	Eag1 K+ channel and ErbB were expressed in all human pituitary adenomas examined while ErbB2 expression was more variable.
Literature	GENERIF_SUMMARY	Quantitative HER2 determination by Centaur did not define a new group of patients eligible for HER2-targeted therapy.
Literature	GENERIF_SUMMARY	FAS and HER2 overexpression are common in esophageal carcinomas
Literature	GENERIF_SUMMARY	our data indicate that the ERBB2 gene is a novel MBP-1 target, and analysis of primary tumors suggests that the concomitant high expression of MBP-1 and HDAC1 may be considered a diagnostic marker of cancer progression for breast IDC.
Literature	GENERIF_SUMMARY	Latest advances in HER2/neu status diagnostic assessment and the most relevant research emerging from ""Omics"" (genomics, epigenetics, transcriptomics, and proteomics) studies on HER2/neu-positive breast cancer. Review."
Literature	GENERIF_SUMMARY	These data suggest that HER2/Akt is an important negative regulator of p57 (Kip2), and that p57 restoration in HER2-overexpressing cells can reduce breast tumor growth.
Literature	GENERIF_SUMMARY	Simultaneous overexpression of CD44 and c-erb-B2 may be used as a marker of poor prognosis in gastric carcinoma.
Literature	GENERIF_SUMMARY	we show that HER2 is selectively expressed in and regulates self-renewal of the cancer stem cell (CSC) population in estrogen receptor-positive (ER(+)), HER2(-) luminal breast cancers.
Literature	GENERIF_SUMMARY	HER2 amplification was a rare event (1.9%) in squamous cell carcinomas of the vulva.
Literature	GENERIF_SUMMARY	interaction between LPP and alpha-actinin, an actin cross-linking protein, is necessary for TGFbeta-induced migration and invasion of ErbB2-expressing breast cancer cells.
Literature	GENERIF_SUMMARY	HER-2/neu induced the improvement of autocrine estrogen in endometrial carcinoma cell through triggering the COX-2/P450arom signal.
Literature	GENERIF_SUMMARY	HER2 overexpression was associated with an increasing risk of developing metastases.
Literature	GENERIF_SUMMARY	PTEN, VEGF, HER2, p53 expressions were compared

Main Category	Category	Terms
		with bevacizumab response.
Literature	GENERIF_SUMMARY	These results suggest that highly malignant differentiated thyroid cancer can be detected by monitoring HER2 status and telomere shortening, and that trastuzumab therapy may be effective for refractory thyroid cancer.
Literature	GENERIF_SUMMARY	The study aimed at examining the relationship between HER2 immunohistochemistry assessment scores in paired core needle biopsies and whole tissue sections of invasive ductal breast carcinoma.
Literature	GENERIF_SUMMARY	Data indicate that EGFR T790M as the most common mechanism of acquired resistance, whereas MET amplification, HER2 amplification, and small cell histologic transformation occur less frequently.
Literature	GENERIF_SUMMARY	The level of HER2/neu amplification tends to increase from the primary tumour to the paired metastases in a significant proportion of patients with HER2-positive metastatic breast neoplasms.
Literature	GENERIF_SUMMARY	Topo IIalpha, together with HER-2, might help to select for patients who could benefit from anthracycline-based neoadjuvant chemotherapy and identify non-complete responders at a higher risk of disease recurrence or death.
Literature	GENERIF_SUMMARY	results indicate that specific amplified and/ or overexpressed genes found in ERBB2-amplified breast cancer may mediate response to ERBB2-targeting agents
Literature	GENERIF_SUMMARY	Results indicate that neither total cfDNA nor HER2 gene copy number is elevated in primary breast cancer patients, and the level of both parameters increased during neoadjuvant chemotherapy, but without any relation to treatment effect.
Literature	GENERIF_SUMMARY	an NRG1-ERBB3-ERBB2 signaling unit operates in melanoma cells where it promotes growth and survival.
Literature	GENERIF_SUMMARY	Data indicate that monitoring tissue HER-2-positive breast cancer patients with serum HER-2 has a sufficient sensitivity to detect metastatic recurrence, while its use in monitoring of tissue HER-2-negative patients is unsatisfactory.
Literature	GENERIF_SUMMARY	Nucleophosmin/B23 is a favorable prognostic indicator for gastric cancer, but Her-2/neu has no relationship with prognosis.
Literature	GENERIF_SUMMARY	completely human, functionalized GrB construct can effectively target Her2/neu-expressing cells and displays impressive in vitro and in vivo activity.
Literature	GENERIF_SUMMARY	HER2-positive disseminated tumor cells in early breast cancer patients is associated with an increased risk of relapse.
Literature	GENERIF_SUMMARY	the suppressive role of Necl-2 in the HRG-induced

Main Category	Category	Terms
		ErbB2/ErbB3 signaling is regulated by miR-199a at least through the reduction of the ST6GAL1-catalyzed sialylation of Necl-2
Literature	GENERIF_SUMMARY	EpCAM expression was associated with an unfavourable prognosis in the basal-like and luminal B HER2(+) subtypes but associated with a favourable prognosis in the HER2 subtype breast cancer.
Literature	GENERIF_SUMMARY	Her-2/neu over-expression and parameters assessing metastatic burden in the SLN, particularly total metastatic area (TMA), predicted the presence of NSLN involvement and extranodal disease in patients with breast carcinoma and SLN metastases.
Literature	GENERIF_SUMMARY	A significant association between HER2 amplification and treatment with respect to both disease-free survival and overall survival in gastrointestinal adenocarcinoma.
Literature	GENERIF_SUMMARY	Immunostain of Her2/neu has been positive in 42.2% of cases, the poorly differentiated carcinomas presenting high scores regardless of tumor stage.
Literature	GENERIF_SUMMARY	The her2 receptor attachment function of measles virus hemagglutinin is not required for its fusion-helper function.
Literature	GENERIF_SUMMARY	these results suggest that coupling HDAC and HER2 inhibitory activities to an EGFR inhibitor may potentially be effective in overcoming drug resistance and preventing cancer cell migration
Literature	GENERIF_SUMMARY	HER2 mediates the effects of epidermal growth factor (EGF) on Snail, Slug and E-cadherin expression as well as on ovarian cancer cell invasion.
Literature	GENERIF_SUMMARY	Data suggest that p63/Her2/neu expressing precursor intermediate in comedo-DCIS may provide a cellular basis for emergence of p63+/Her2/neu- or p63+/Her2/neu+ basal-like breast cancer.
Literature	GENERIF_SUMMARY	BMs were detected very rarely in asymptomatic, non-metastatic HER2-positive breast cancer patients compared with asymptomatic, metastatic patients.
Literature	GENERIF_SUMMARY	A positive correlation was demonstrated between lymphocyte infiltration and HER2 gene amplification in breast cancer.
Literature	GENERIF_SUMMARY	newly identified HER2 epitopes seem to be crucial for an effective immune response against HER2-overexpressing breast cancers and might help discriminating between metastatic breast cancer and early breast cancer
Literature	GENERIF_SUMMARY	Studies indicate that epidermal growth factor-2 (HER2) overexpression occurs in many other tumor types, including esophageal cancer.
Literature	GENERIF_SUMMARY	Data indicate that genomic profiling of relapsed CDH1-mutated invasive lobular carcinoma (ILC) featured a high incidence of ERBB2 alterations.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Data indicate a signaling pathway from ErbB2 to p21-Activated kinase-1 (Pak1) to beta-catenin that is required for efficient transformation of mammary epithelial cells.
Literature	GENERIF_SUMMARY	Her2/neu alterations may occur early and increasingly during Barrett's malignant progression.
Literature	GENERIF_SUMMARY	DOCK1-Rac signaling as an HER2 effector pathway essential for HER2-mediated breast cancer progression to metastasis.
Literature	GENERIF_SUMMARY	HER2 655V polymorphism is potential risk factor for development of breast cancer.
Literature	GENERIF_SUMMARY	Report HER2 gene amplification and protein expression in Chinese patients with resectable gastric cancer and the association with clinicopathological characteristics and survival.
Literature	GENERIF_SUMMARY	This study demonistrated that Her2neu amplification associates with Co-deletion 1p/14q in recurrent meningiomas.
Literature	GENERIF_SUMMARY	This study of patients presenting with HER2-mutated non-small-cell lung cancer, were found to be mainly female, nonsmokers, and exclusively suffering from adenocarcinoma subtype disease.
Literature	GENERIF_SUMMARY	results indicate that Necl-4 serves as a tumor suppressor by inhibiting the ErbB2/ErbB3 signaling and hemidesmosome disassembly
Literature	GENERIF_SUMMARY	These results confirm that the outcome of HER2-positive patients treated with trastuzumab is significantly worse in patients with PIK3CA-mutated compared with wild-type tumours.
Literature	GENERIF_SUMMARY	the acquired substrate preference for GAB1 is critical for the ERBB2 mutant-induced oncogenesis.
Literature	GENERIF_SUMMARY	No statistically significant correlation between trastuzumab resistance and the expression of HER2 was noted in breast cancer.
Literature	GENERIF_SUMMARY	Metformin can inhibit the proliferation and promote apoptosis of HER2 positive breast cancer cells,which is maybe related to inhibition of HSP90.
Literature	GENERIF_SUMMARY	Report changes in HER2 expression in breast neoplasms after neoadjuvant dose-dense doxorubicin, cyclophosphamide followed by paclitaxel.
Literature	GENERIF_SUMMARY	Moderate overexpression of HER-2/neu was detected in a small proportion of colorectal cancer patients.
Literature	GENERIF_SUMMARY	Dysregulation of miR-125a-5p/125b and HER2 is an early event in the gastric (intestinal-type) and esophageal (Barrett) oncogenesis.
Literature	GENERIF_SUMMARY	The clinicopathological factors and incidence of HER2 expression of Mucinous epithelial ovarian cancer in a

Main Category	Category	Terms
		large Asian retrospective cohort from Singapore, is reported.
Literature	GENERIF_SUMMARY	a proportion of patients with breast cancer harbor tumor cells in blood and/or bone marrow, and the HER2 status of these residual tumor cells can be different from that of the primary tumor.
Literature	GENERIF_SUMMARY	MUC1 and HER2 might be closely associated with invasive phenotype of intraductal papillary mucinous neoplasm.
Literature	GENERIF_SUMMARY	Analysis of plasma DNA with digital PCR has the potential to screen for the acquisition of HER2 amplification in metastatic breast cancer.
Literature	GENERIF_SUMMARY	Approximately 10% of Chinese patients with primary gastric cancer (GC) and gastroesophageal junction (GEJ)adenocarcinoma were HER2-positive on immunohistochemistry.
Literature	GENERIF_SUMMARY	a positive correlation exists between flotillin2 and erbB2 expression levels
Literature	GENERIF_SUMMARY	TPD52 is a survival factor in ERBB2-amplified breast cancer.
Literature	GENERIF_SUMMARY	we found that HER2 or GCDFP15 protein overexpression is a sensitive and specific tool to differentiate MA from BL in the context of ER negative tumors.
Literature	GENERIF_SUMMARY	Two independent high-risk primary breast cancer subgroups for developing brain metastases were identified, represented by genetic alterations in either HER2 or EGFR/PTEN-driven pathways.
Literature	GENERIF_SUMMARY	Data suggest that up-regulation of signaling via ERBB2 and EGFR (epidermal growth factor receptor) by leptin contributes to vascular dysfunction (as seen in obesity, metabolic syndrome, and diabetes). [REVIEW]
Literature	GENERIF_SUMMARY	Serum HER 2 extracellular domain level is correlated with tissue HER 2 status in metastatic gastric or gastro-oesophageal junction adenocarcinoma
Literature	GENERIF_SUMMARY	Altogether the results indicate that pertuzumab can be a valuable therapeutic agent not only in cancers overexpressing ErbB2 but also in cancers co-expressing ErbB2 and ErbB3.
Literature	GENERIF_SUMMARY	overexpressed in a subset of pancreatic ductal adenocarcinomas, no association with survival
Literature	GENERIF_SUMMARY	HER2 protein expression and HER2 gene amplification are infrequent in small intestinal carcinomas.
Literature	GENERIF_SUMMARY	Findings suggest an association between lapatinib-induced autophagy and the disruption of Her2-Beclin-1 complex.
Literature	GENERIF_SUMMARY	HER2 inhibited calpain-1 activity through upregulating

Main Category	Category	Terms
		calpastatin, an endogenous calpain inhibitor.
Literature	GENERIF_SUMMARY	The patients homozygous for the Valine allele showed strong HER2 protein expression. The results showed that the valine allele may be an indicator of genetic susceptibility to ovarian carcinoma
Literature	GENERIF_SUMMARY	Data suggest that ssessment of Her-2 expression status may be valuable for the targeted therapeutic management of colorectal cancer (CRC).
Literature	GENERIF_SUMMARY	Data indicate that Lrig3 opposes Lrig1 negative regulatory action and enhances ErbB receptors ERBB2, ERBB3 and ERBB4 stability.
Literature	GENERIF_SUMMARY	This study confirms that discordance in ER and PR receptor expression between the primary breast tumor and the corresponding metastatic lesions is high, whereas HER2 status remains relatively constant.
Literature	GENERIF_SUMMARY	A downregulation of FASN effectively inhibits the activity of ""HER2-PI3K/Akt axis"" and alters the malignant phenotype in colorectal cancer cells."
Literature	GENERIF_SUMMARY	CDC25a is a target of HER2 signaling in human breast cancer.
Literature	GENERIF_SUMMARY	Review discusses the clinical significance of HER2 discordance in breast cancer.
Literature	GENERIF_SUMMARY	DNA from 4 HPV positive and 4 HPV negative fresh frozen primary HNSCC were subject to comprehensive genome-wide methylation profiling.Pathway analysis of 1168 methylated genes showed 8 signal transduction pathways (ERB2) of which 62% are hypermethylated.
Literature	GENERIF_SUMMARY	Studies suggest that the efficacy of HER2 targeting agents may be related to their ability to target the breast cancer stem cell (CSC) population.
Literature	GENERIF_SUMMARY	HER2 mutation was present in one male, non-smoking patient with low differentiated adenocarcinoma (0.67% of all patients with brain metastases from Non-small cell lung cancer and 1.5% of patients with adenocarcinoma).
Literature	GENERIF_SUMMARY	Data suggest that detection of ductal carcinoma in situ (DCIS) based on marker expression during breast conserving surgery should be possible with a panel of molecular imaging tracers targeting CD44v6, GLUT1, HER2, IGF1-R, and EGFR.
Literature	GENERIF_SUMMARY	HER2 over-expression in gastric adenocarcinoma of Chinese patients was significantly more common in proximal than in distal cancers.
Literature	GENERIF_SUMMARY	Lower expression level of HER2 and high expression levels of phosphorylated HER2 is associated with breast cancer.
Literature	GENERIF_SUMMARY	positive expression of phosphorylated pHER2 may yield additional information regarding the poor prognosis of

Main Category	Category	Terms
		breast cancer and could be used for pre-selection of patients with HER2-overexpressing breast cancer displaying resistance to trastuzumab treatment
Literature	GENERIF_SUMMARY	heavy alcohol use during the course of trastuzumab treatment for breast cancer and the HER2 Ile/Val genotype may constitute risk factors for cardiac toxicity
Literature	GENERIF_SUMMARY	Vimentin expression in epithelial cells of the surgically resected gastric adenocarcinoma tissue is an independent predictor of short survival, and Her-2 status shows a valuable correlation with clinical parameters.
Literature	GENERIF_SUMMARY	ER, PR, and HER2 expression in the biopsy and final pathology are still needed for selecting patients for postoperative adjuvant treatment in clinical practice.
Literature	GENERIF_SUMMARY	After 3.5 years follow-up, we failed to show that 6 months of treatment with trastuzumab was non-inferior to 12 months of trastuzumab. Despite the higher rates of cardiac events, 12 months of adjuvant trastuzmab should remain the standard of care.
Literature	GENERIF_SUMMARY	in this largest comprehensive study, 35% of endometrial serous carcinoma harbors HER2 protein overexpression and/or gene amplification.
Literature	GENERIF_SUMMARY	Androgen blockade decreases serum PSA by eliminating HER-2 negative prostate cancer cells.
Literature	GENERIF_SUMMARY	Dual staining technique which allows simultaneous detection of myoepithelial marker protein and cancerous HER2 gene is feasible.
Literature	GENERIF_SUMMARY	HER2-I655V polymorphism is associated to a left ventricular ejection fraction
Literature	GENERIF_SUMMARY	The CD151 complex has a critical regulatory role in ErbB2-dependent signaling.
Literature	GENERIF_SUMMARY	Increased HER2 expression is associated with cervical clear cell carcinoma.
Literature	GENERIF_SUMMARY	Lapatinib inhibited the proliferation and the rate of DNA synthesis in HER2-positive cells
Literature	GENERIF_SUMMARY	Assessed the HER2 status in gastric cancer tissue microarrays and corresponding whole sections using HercepTest immunohistochemistry (IHC), and to correlate it and to assess the concordance of HER2 IHC and fluorescence in situ hybridization in TMAs.
Literature	GENERIF_SUMMARY	Results show that methylation of histone methylation mark H3K9 can be induced and this epigenetic mark was instructive in promoting downregulation of HER2/neu expression.
Literature	GENERIF_SUMMARY	Report prognostic value of HER-2 determination in fine needle aspirates of breast tumors.
Literature	GENERIF_SUMMARY	Immunohistochemical expression of HER-2 was not

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		associated with characteristics of invasive breast cancers.
Literature	GENERIF_SUMMARY	HER-2 levels were altered after neoadjuvant chemotherapy in breast cancer.
Literature	GENERIF_SUMMARY	In gastric adenocarcinoma, Her-2/neu, as a marker of long time survival prediction, seems to be limited.
Literature	GENERIF_SUMMARY	findings show that a significant percentage of breast tumors coexpress MET and ERBB2; ERBB2(+) breast cancers with MET(+)/ERBB2(+) subpopulations may have an innate resistance to ERBB2 inhibition
Literature	GENERIF_SUMMARY	EGFR, but not ERBB2, is expressed in precursor lesions of squamous cervical neoplasia and is related to the neoplastic progression
Literature	GENERIF_SUMMARY	P130Cas overexpression synergizes with ErbB2 in mammary cell transformation and promotes ErbB2-dependent invasion.
Literature	GENERIF_SUMMARY	HER2 receptor expression is detectable by immunohistochemistry in approximately 20% of non-small cell lung cancers and in 2% of NSCLC using gene amplification by FISH. HER2 is activated by axon 20 in frame insertion mutations in 2% of lung adenocarcinomas
Literature	GENERIF_SUMMARY	literature review comprises immunohistochemical studies on the clinical significance of c-erbB2/HER2 overexpression in gliomas, medulloblastomas, and meningiomas
Literature	GENERIF_SUMMARY	Carboxyl group footprinting mass spectrometry and molecular dynamics identify key interactions in the HER2-HER3 receptor tyrosine kinase interface.
Literature	GENERIF_SUMMARY	Data show that the inactive conformation of ERBB2 does not interact with HSP90 chaperone.
Literature	GENERIF_SUMMARY	Results show that scFvs against HER-2/neu and carcinoembryonic antigen (CEA) were isolated.
Literature	GENERIF_SUMMARY	proteomic analysis of HER2 in breast cancer cells reveals proteomic changes triggered by the amplification of this single gene, and the downstream biological changes triggered by HER2 overexpression
Literature	GENERIF_SUMMARY	HER2 gene amplification and an increased EGFR gene copy number (with balanced chromosome 7 high-polysomy) were each detected in four of 28 (14.3%) salivary mucoepidermoid carcinomas.
Literature	GENERIF_SUMMARY	we demonstrated the prognostic value in the various disease subtypes, including HER-2-positive disease irrespective of therapy.
Literature	GENERIF_SUMMARY	Findings suggest that aluminum concentration does not affect genomic stability of ERBB2, C-MYC, and CCND1 genes in breast tissues.

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Literature	GENERIF_SUMMARY	Mediator of ERBB2-driven cell motility (MEMO) promotes extranuclear estrogen receptor signaling involving the growth factor receptors IGF1R and ERBB2.
Literature	GENERIF_SUMMARY	Expression of Delta-HER2 induces tumor formation and metastasis.
Literature	GENERIF_SUMMARY	This meta-analysis result suggested that EGFR or HER2 should have significant predictive ability for estimating overall survival in GC patients and may be useful for defining prognosis of GC patients.
Literature	GENERIF_SUMMARY	In HER2-positive gastric cancer.
Literature	GENERIF_SUMMARY	HER-2/neu overexpression was not associated with a significant impact on survival.
Literature	GENERIF_SUMMARY	The correlation of Her2 expression with the expression of Her2 chaperoning HSP90 may indicate a synergistic regulation.
Literature	GENERIF_SUMMARY	report results from long-timescale molecular dynamics simulations indicating that a single ligand is necessary and sufficient to stabilize the ectodomain interface of Her2 heterodimers, which assume an asymmetric conformation similar to dEGFR dimers
Literature	GENERIF_SUMMARY	Uterine serous carcinoma cell lines exhibiting HER2 gene amplification were significantly more sensitive to GDC-0980 exposure.
Literature	GENERIF_SUMMARY	Variability in HER2 staining in HER2-positive tumours has no role in clinical management with adjuvant trastuzumab
Literature	GENERIF_SUMMARY	Findings support the development of CDK and Raf co-targeting strategies in EGFR/HER-2-overexpressing or RAS/RAF mutant breast cancer (BC).
Literature	GENERIF_SUMMARY	Both FGF1 and ERBB2 significantly influenced overall survival in breast cancer patients, especially among women with low levels of Native American ancestry.
Literature	GENERIF_SUMMARY	Nuclear T-STAR protein expression correlates with HER2 status, hormone receptor negativity and prolonged recurrence free survival in primary breast cancer and decreased cancer cell growth in vitro.
Literature	GENERIF_SUMMARY	The frequencies of EGFR, KRAS, BRAF, and HER-2 mutations in BM-NSCLC were 2.6, 38.5, 0, and 0% respectively.
Literature	GENERIF_SUMMARY	Myocardial strain imaging is more sensitive than left ventricular ejection fraction for the early detection and intermediate term monitoring of LV systolic function following anthracycline chemotherapy in HER2/neu negative breast cancer patients.
Literature	GENERIF_SUMMARY	HER-2 alteration is not an independent prognostic factor for curatively resectable gastric cancer
Literature	GENERIF_SUMMARY	LOXL2 expression in normal epithelial cells can induce

Main Category	Category	Terms
		abnormal changes that resemble oncogenic transformation and cancer progression.
Literature	GENERIF_SUMMARY	Pertuzumab, a novel HER2 dimerization inhibitor, has been shown to be effective in the treatment of metastatic HER2-positive breast cancer--{REVIEW}
Literature	GENERIF_SUMMARY	constructed an integrated mathematical model of HER activation and trafficking. Studied HER phosphorylation and abundance data collected in a panel of human mammary epithelial cells expressing varying levels of EGFR/HER1, HER2 and HER3.
Literature	GENERIF_SUMMARY	Data indicate that all mesenchymal colony clusters displayed a predominant CD44+/CD24- phenotype with decreased HER-2 expression and elevated levels of a beta1-integrin isoform with a high degree of N-glycosylation.
Literature	GENERIF_SUMMARY	In tumours from postmenopausal women harbouring normal HER2 gene copy numbers, we found HER2 and HER4, but HER3 levels in particular, to be elevated compared to controls
Literature	GENERIF_SUMMARY	Patients with HER2 loss tend to have a higher risk of relapse compared with maintained HER2 positivity in breast cancer.
Literature	GENERIF_SUMMARY	all pairings of the EGF receptor, ErbB2, and ErbB3 form ligand-independent dimers/oligomers.
Literature	GENERIF_SUMMARY	HER2 as a promising target for cytotoxicity T cells in human melanoma therapy.
Literature	GENERIF_SUMMARY	HER2 expression was found in 10% of tumors and was related to only one histological type of tumor - salivary duct carcinoma.
Literature	GENERIF_SUMMARY	In contrast to the higher incidence reported at the population level, 20-30% HER2-positive cases for metastatic carcinomas, and only 11% of the screen-detected breast carcinomas displayed HER2/neu gene amplification
Literature	GENERIF_SUMMARY	The suitable cut-off value for p53 overexpression was determined to be 50%, and p53 overexpression appears to be a significant prognostic factor in patients with luminal/HER2-negative breast cancer
Literature	GENERIF_SUMMARY	CREB plays an important role in the HER-2/neu-mediated transformation by altering in vitro and in vivo growth characteristics.
Literature	GENERIF_SUMMARY	presence of cyclin E isoforms in p53 silenced clones also does not altered HER-2 expression
Literature	GENERIF_SUMMARY	Quantitative assessment of intratumoral HER2 heterogeneity may offer an opportunity to improve the identification of patients likely to respond to HER2-targeted therapies.
Literature	GENERIF_SUMMARY	HER2 overexpression is associated with an immune

Main Category	Category	Terms
		escape phenotype. [review]
Literature	GENERIF_SUMMARY	MTK1 was identified in the HER2/HER3-HRG mediated extracellular acidification and cell migration pathway in breast cancer cells.
Literature	GENERIF_SUMMARY	An heregulin-EGFR-HER3 autocrine signaling axis can mediate acquired lapatinib resistance in HER2+ breast cancer models.
Literature	GENERIF_SUMMARY	Data show that peptide-pulsed dendritic cells (DCs) prime naive CD8-positive (+) T cells to exert HER-2/neu peptide-specific cytotoxicity and CD4+ T cells to proliferate in a peptide-dependent manner.
Literature	GENERIF_SUMMARY	HER2 and MYC are frequently coamplified in breast cancer, associated with aggressive clinical behaviour and poor outcome.
Literature	GENERIF_SUMMARY	ErbB2 expression is inversely associated with recurrence in urothelial tumors of the upper urinary tract.
Literature	GENERIF_SUMMARY	An isogenic gene expression signature was identified between triple-negative and HER2-positive breast cancers.
Literature	GENERIF_SUMMARY	Specific cytolytic activity of retargeted T cells to kill HER2 positive SKBR3 cell.
Literature	GENERIF_SUMMARY	PTPRO methylation is associated with poor survival only in HER2-positive patients.
Literature	GENERIF_SUMMARY	Crystal structure of the human HER2 receptor in complex with a duplex DARPin inhibitor.
Literature	GENERIF_SUMMARY	Combined HER2-targeted therapy produced a numerically but insignificantly higher pathological complete response percentage than single-agent HER2-directed therapy.
Literature	GENERIF_SUMMARY	The specific binding of the RNA aptamer to the HER2 receptor.
Literature	GENERIF_SUMMARY	Breast cancer with a HER2 positive subtype is associated with the longest survival after diagnosis of distant metastases.
Literature	GENERIF_SUMMARY	In spite of a 52-week adjuvant trastuzumab treatment, classic poor prognostic factors for invasive EBC remained as such in patients with HER2-positive EBC.
Literature	GENERIF_SUMMARY	Consensus scoring of breast carcinomas for HER-2 immunohistochemistry by experienced pathologists is likely to decrease the use of fluorescent in situ hybridisation testing.
Literature	GENERIF_SUMMARY	HRG1 mediates resistance to lapatinib through HER3 and AKT activation, and this depends on residual HER2 activity.
Literature	GENERIF_SUMMARY	our data indicate that ErbB2 activation promotes GLS1

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		expression via a PI3K-Akt-independent NF-kappaB pathway in breast cancer cells, identifying another oncogenic signaling pathway which stimulates GLS1 expression
Literature	GENERIF_SUMMARY	In cases with cluster amplification, HER2 protein overexpression by immunohistochemistry closely correlated at the cellular level with HER2/NEU gene amplification.
Literature	GENERIF_SUMMARY	HSP90 inhibitor was more efficacious in HER2-positive tumors than in HER2-negative tumors.
Literature	GENERIF_SUMMARY	Data indicate that a mean HER2/CEP17 ratio of 4.7 was identified as the optimal cutoff value discriminating sensitive and refractory patients.
Literature	GENERIF_SUMMARY	Although up to 30% of non-small-cell lung cancer cases display HER2 overexpression, amplification or mutation, the prognostic significance of HER2 dysregulation remains undefined
Literature	GENERIF_SUMMARY	Ligand stimulation of non-overexpressed ERBB2 transiently removes UCP2 and paradoxically reduces the mitochondrial membrane potential, oxygen consumption, and OXPHOS on a signaling time scale.
Literature	GENERIF_SUMMARY	Investigated ERBB2 expression and invasiveness of glioma cells. ERBB2 shRNA transfection vector could effectively inhibit expression of ERBB2; while ERBB2 overexpressing vector transfection could significantly improve expression of ERBB2 in TJ905 cells.
Literature	GENERIF_SUMMARY	HER2 expression is not associated with a difference in the treatment outcome in patients with rectal adenocarcinoma.
Literature	GENERIF_SUMMARY	Seven novel microRNAs, direct HER2 regulators were identified. miR-342-5p was associated with better survival in breast cancer patients.
Literature	GENERIF_SUMMARY	The use of the 4B5 antibody on whole-tissue sections was the most accurate immunohistochemical method for evaluating HER2 expression in gastric adenocarcinoma.
Literature	GENERIF_SUMMARY	Data indicate that HER2 positive was an independent adverse prognostic factor in group B patients without trastuzumab and group C patients of HER2-negative without trastuzumab.
Literature	GENERIF_SUMMARY	siRNA cocktail, and co-downregulation of VEGF and HER2 result in significant inhibition of gastric cancer growth and migration in vitro
Literature	GENERIF_SUMMARY	marker of main subtypes of invasive breast cancer in China
Literature	GENERIF_SUMMARY	TFF3 expression at both mRNA and protein levels was significantly downregulated by ErbB-2 silencing in breast cancer cell lines

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Literature	GENERIF_SUMMARY	investigation of STAT3 tyrosine-705 phosphorylation and its correlation with its possible regulator HER2 in the prognosis of pancreatic ductal carcinoma
Literature	GENERIF_SUMMARY	Different HER2 mRNA profiles, possibly in relation to contact between tumor cells and lymphocytes, might help to explain the discrepancy between gene amplification and protein overexpression results in gastric cancers.
Literature	GENERIF_SUMMARY	ERBB2 amplification is more frequent in micropapillary urothelial carcinoma than typical urothelial carcinoma, and patients with micropapillary carcinoma who have ERBB2 amplification have worse cancer-specific survival than those who do not.
Literature	GENERIF_SUMMARY	Next generation sequencing of micropapillary urothelial carcinoma revealed a high incidence of mutation in the extracellular domain of ERBB2.
Literature	GENERIF_SUMMARY	Immunohistochemical staining for EGFR, HER2 and IGF-1R was undertaken in 31 ovarian adult granulosa cell tumors. Tumor DNA was also analysed for mutations in the tyrosine kinase domain of EGFR (exons 18-21).
Literature	GENERIF_SUMMARY	new role for PTPalpha in the regulation of motility of mammary epithelial cells in response to ErbB2 activation.
Literature	GENERIF_SUMMARY	4 genes (EGFR, ERBB2, PTK2, and RAF1) with five SNPs (rs11238349, rs17172438, rs984654, rs11773818, and rs17172432) might be pivotal factors in the development of prostate cancer.
Literature	GENERIF_SUMMARY	HER2 and IGF-IR may be clinically beneficial in minimizing the acquired resistance to trastuzumab therapy.
Literature	GENERIF_SUMMARY	HER2-mediated phosphorylation destabilizes PUMA protein.
Literature	GENERIF_SUMMARY	Her2 status is associated with key clinical features and pathological characteristics of epithelial ovarian tumors but not with survival.
Literature	GENERIF_SUMMARY	data help to understand the HER2 overexpression in oxidative signaling and may enable the development of new strategies for anti-HER2 therapy
Literature	GENERIF_SUMMARY	HER2 status in primary gastric cancer is a reliable basis for deciding whether to treat with anti-HER2 agents in patients with lymph node metastases.
Literature	GENERIF_SUMMARY	Studies indicate that changes in oestrogen receptor (ER), progesterone receptor (PgR) and epidermal growth factor receptor 2 (HER2) expression may occur during the natural history of breast cancer.
Literature	GENERIF_SUMMARY	In ER-positive cancers, high LI correlated with biologic parameters associated with poor prognosis, whereas in HER2 positive cancers, LI correlated with biologic

Main Category	Category	Terms
		parameters of good prognosis.
Literature	GENERIF_SUMMARY	Data suggest that trastuzumab may be effective in colorectal cancer, but only in tumors with the HER2 protein on their cell membrane, which accounts for approximately 5% of the patients.
Literature	GENERIF_SUMMARY	LKB1 expression in cancer cell resulted in dephosphorylation of several tumor-enhancing RTKs, including ErbB2, hepatocyte growth factor receptor (c-Met), EphA2, rearranged during transfection (RET), and insulin-like growth factor I receptor.
Literature	GENERIF_SUMMARY	Data show that S100A14 and HER2 are colocalized in plasma membrane of breast cancer tissue cells and breast cancer cell lines.
Literature	GENERIF_SUMMARY	miR-221 may promote trastuzumab resistance and metastasis of HER2-positive breast cancers by targeting PTEN, suggesting its role as a potential biomarker for progression and poor prognosis
Literature	GENERIF_SUMMARY	we found that there were no associations between genetic variants in the 17q21 ERBB2 amplicon and GC susceptibility.
Literature	GENERIF_SUMMARY	We evaluated HER2 and cMet protein expression, and FGFR2 gene amplification to assess their prognostic significance in patients with gastric adenocarcinoma
Literature	GENERIF_SUMMARY	Expression of ERBB2 is increased in nerve tissue from patients with neurofibromatosis type 2.
Literature	GENERIF_SUMMARY	High-level ERBB2 amplification is associated with shorter time-to-metastasis, but improved response to trastuzumab in metastatic breast cancer.
Literature	GENERIF_SUMMARY	Findings strongly suggest that mutations in the transmembrane domain of HER2 may be oncogenic, causing hereditary and sporadic lung adenocarcinomas.
Literature	GENERIF_SUMMARY	calretinin immunoexpression and the hormonal status for estrogen receptors (ER), progesterone receptors (PR), and ERRB2, in 33 cases of cutaneous metastases of ductal breast carcinoma, as well as in seven primary cutaneous apocrine tumors, were studied
Literature	GENERIF_SUMMARY	Data indicate that responses were noted in patients with HER2-amplified breast cancer resistant to trastuzumab, HER2-mutant non-small-cell lung cancer, and tumor types without identified mutations in the HER-PI3K-mTOR pathway.
Literature	GENERIF_SUMMARY	Capecitabine/cisplatin doublet has an encouraging antitumor activity with acceptable and manageable toxicity in anthracycline- and taxane-pretreated HER-2 negative metastatic breast carcinoma patients.
Literature	GENERIF_SUMMARY	A prognostic biomarker for trastuzumab treatment in HER2-positive MBCs.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Data demonstrate that the E2F transcription factors are integral to HER2+ tumor development and progression.
Literature	GENERIF_SUMMARY	ER and HER2 are useful markers in core needle biopsies, providing a reliable base for preoperative decisions in neoadjuvant chemotherapy, and may be omitted on surgical specimens with certain precautions and exceptions.
Literature	GENERIF_SUMMARY	High co-expression of HER-2, IL-6, TNF-alpha and TGF-beta1 is asosciated with breast cancer.
Literature	GENERIF_SUMMARY	ErbB2 overexpression controls HSF1 activity, with subsequent stabilization of tumor-promoting HSP90 target, MIF.
Literature	GENERIF_SUMMARY	High HER2 expression is associated with brain metastases arising from breast cancer with stereotactic radiosurgery.
Literature	GENERIF_SUMMARY	ALTTO ineligibility did not change when HER2 testing was performed ... a dual antibody ER assay had fewer false-negative test results than an assay with a single antibody.
Literature	GENERIF_SUMMARY	These results could justify more extensive research to assess p53 and HER-2/neu gene status as significant prognostic factors in oral cancers.
Literature	GENERIF_SUMMARY	Lapatinib plus vinorelbine offers an effective treatment option for patients with HER2-overexpressing MBC, having displayed comparable efficacy and tolerability rates to lapatinib plus capecitabine
Literature	GENERIF_SUMMARY	In HER2-positive breast cancers, HR expression is associated with a poor prognosis despite the hormone therapy. For this population, a personalized management may be required.
Literature	GENERIF_SUMMARY	ERBB2 is overexpressed in fibrolamellar hepatocellular carcinoma.
Literature	GENERIF_SUMMARY	Anti-huHER2 antibodies elicited in the tolerant host exert antitumor activity.
Literature	GENERIF_SUMMARY	onstitutively active ERBB2 sensitizes human breast epithelial cells to agents that induce endoplasmic reticulum stress
Literature	GENERIF_SUMMARY	EGFR and HER-2 may be promising prognostic and therapeutic targets in canine gastric epithelial neoplasms
Literature	GENERIF_SUMMARY	Patients with stages I-III (including inflammatory) HER2-positive breast cancer were randomised to receive epirubicin.
Literature	GENERIF_SUMMARY	[review] HER2 over-expression is significantly correlated with phosphatidylinositol (PI)3 kinase-Akt expression in gastric cancer (GC) tissues and plays an important role in HER2-positive GC.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Altogether, our study uncovered the existence of a novel Akt-HSF-1 signaling axis that leads to Slug upregulation and EMT, and potentially contributes to progression of HER2-positive breast cancer
Literature	GENERIF_SUMMARY	The efficacy and safety of trastuzumab combined with SP in patients with human epidermal growth factor receptor type 2 (HER2)-positive AGC.
Literature	GENERIF_SUMMARY	Intratumoral heterogeneity of ERBB2 amplification, defined as less than 50% of cancer cells positive for ERBB2 amplification, was found in 41% (21/51) of ERBB2-amplified gastric adenocarcinoma tumors
Literature	GENERIF_SUMMARY	Data suggest that epidermal growth factor receptor 2 discordance is mostly due to testing with only immunohistochemical (IHC) method.
Literature	GENERIF_SUMMARY	ERBB2-positive breast cancers are partially autophagy-deficient even in a wild-type BECN1 background
Literature	GENERIF_SUMMARY	The folding pf Her2 is driven by chain A, which has a poor capacity to self-recognize and a strong tendency to form the internal bridge between Cys565 and Cys574.
Literature	GENERIF_SUMMARY	Eevaluation of HER2 status is certainly an important tool during the diagnosis of gastric cancer patients
Literature	GENERIF_SUMMARY	Expression of beta-catenin, Ki67 and Her-2/Neu was studied in gastric cancer. The protein expression of beta-catenin, Ki67 and Her-2/Neu had close relationship with the clinical pathologic characteristics of gastric cancer.
Literature	GENERIF_SUMMARY	Positivity for HER2, MET and FGFR2 alternations was found in 23 (13.4%), 21 (12.2%) and 9 (5.2%) patients.
Literature	GENERIF_SUMMARY	Identify a molecular signature based on levels of pEGFR, pERBB2, and SMOX for the initiation of gastric carcinogenesis in helicobacter infected gastric epithelial cells.
Literature	GENERIF_SUMMARY	Suggest that HER2 positivity in upper urinary tract urothelial carcinoma is an independent predictive marker for early recurrence of urothelial carcinoma in the residual urinary bladder after surgery.
Literature	GENERIF_SUMMARY	The majority of SDCs express AR, HER2, and/or EGFR.
Literature	GENERIF_SUMMARY	A dual tyrosine kinase inhibitor (TKI) of epithelial growth factor receptor (EGFR)/HER2.
Literature	GENERIF_SUMMARY	Epigenetic silencing of miR-375 induces trastuzumab resistance in HER2-positive breast cancer by targeting IGF1R.
Literature	GENERIF_SUMMARY	Labeling anti-HER2 Nanobody 5F7GGC with *I-SGMIB yields a promising new conjugate for targeting HER2-expressing malignancies
Literature	GENERIF_SUMMARY	High epidermal growth factor receptor-2 is associated

Main Category	Category	Terms
		with locoregional recurrence in patients with T1-T2 breast cancer.
Literature	GENERIF_SUMMARY	Although there was no difference in survival, C-erbB-2 overexpression was observed only in the glioblastoma subtype of Astrocytic tumors.
Literature	GENERIF_SUMMARY	HER2 Ile655Val polymorphism affects the function of HER2 gene only restricted in HER2-positive breast cancers
Literature	GENERIF_SUMMARY	Overexpression of HER2 is associated with gastric adenocarcinoma.
Literature	GENERIF_SUMMARY	Our results demonstrate that both the wild-type and the mutant are 100% dimeric, suggesting that the Neu TM helix dimerizes more efficiently than other RTK TM domains in mammalian membranes
Literature	GENERIF_SUMMARY	the study showed a significant association between low pleural glucose level and overexpression of HER2 in lung cancer.
Literature	GENERIF_SUMMARY	HER2 was positive in nearly 52 % of the primary tumors, and these expressed HER2 in corresponding lymph node and liver metastases in 93 and 100 % of the cases, respectively
Literature	GENERIF_SUMMARY	our findings suggest miR-630 as a key regulator of cancer cell progression in HER2 over-expressing breast cancer, through targeting of IGF1R.
Literature	GENERIF_SUMMARY	). Our study demonstrates strong concordance between biopsy and resection specimens for HER2 overexpression in gastric cancer. Discordance was correlated with tumor heterogeneity
Literature	GENERIF_SUMMARY	Multiple HER2 gene copies lead to different protein expression patterns (score 1+ to 3+) but pure gene amplification is only a subset of them
Literature	GENERIF_SUMMARY	there was no correlation between the p95-HER2 expression and resistance to trastuzumab.
Literature	GENERIF_SUMMARY	An EGFR/HER2-Bispecific and enediyne-energized fusion protein shows high efficacy against esophageal cancer.
Literature	GENERIF_SUMMARY	CONCLUSION
Literature	GENERIF_SUMMARY	HER2 overexpressing tumours in older women showed relatively a less aggressive phenotype and did not show any inferior long-term clinical outcome despite not having received chemotherapy when compared with the younger patients.
Literature	GENERIF_SUMMARY	Data show that high P95HER2 (p95, a truncated form of the HER2) expression correlated with shorter progression-free survival and overall survival in the hormone receptor-positive subgroup of patients, but not in the hormone receptor-negative group.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	The evidence of HER2 positive metastatic lesion and primary colorectal cancer suggest that HER2 assessment might be considered in selected cases when this may help change the therapeutic decision.
Literature	GENERIF_SUMMARY	none of the associations were replicated whether analysed for leprosy per se, LL leprosy, TT leprosy, erythema nodosum leprosum or reversal reaction conditions.
Literature	GENERIF_SUMMARY	Affinity-purified bispecific Affibody molecules, targeting HER2 and HER3.
Literature	GENERIF_SUMMARY	Estradiol does not repress HER2 gene transcription but the repressive response can be restored by overexpression of ANCO1.
Literature	GENERIF_SUMMARY	No TOP2A amplification occurred without HER2 co-amplification in an analysis of breast cancer in tissue microarrays from a clinical cohort study.
Literature	GENERIF_SUMMARY	In conclusion HER2 and GATA4 are new molecular prognostic markers of GCT recurrence, which could be utilized to optimize the management and follow-up of patients with early-stage GCTs
Literature	GENERIF_SUMMARY	we propose that the HER2 gene copy number value should be included as an additional indicator for stratifying both the management and the follow-up of breast cancer patients.
Literature	GENERIF_SUMMARY	the ErbB3/4 ligand heregulin (HRG) reduced both ErbB2 and ErbB3 mRNA and protein levels in human breast cancer cell lines.
Literature	GENERIF_SUMMARY	Data indicate that signal transducer and activator of transcription 1 (STAT1) activation leads to HER2/ERBB2 downregulation in cancer cells.
Literature	GENERIF_SUMMARY	HER2 amplification is found in a subset of uterine papillary serous carcinomas. Screening with immunohistochemistry will overestimate the number of cases showing underlying HER2 gene amplification.
Literature	GENERIF_SUMMARY	We show for the first time the presence of HER2/HER3 heterodimers and the loss of p21 expression in HER2+ breast cancer predicts a significantly poorer outcome when submitted to adjuvant trastuzumab
Literature	GENERIF_SUMMARY	The role of targeted agents for metastatic HER2-positive breast cancer is under active clinical trial investigation and we await the maturation of trial results and long-term survival data
Literature	GENERIF_SUMMARY	Results suggest that FoxM1 and Her-2 are important diagnostic markers for gastric cancer.
Literature	GENERIF_SUMMARY	These findings deciphered the molecular mechanism of HBx-mediated HER2 upregulation in HBV-associated HCC.
Literature	GENERIF_SUMMARY	The expression of her-2 is closely related to

Main Category	Category	Terms
		differentiation degree, infiltration depth, lymph nodes' metastasis and pTNM stage of the esophageal cancer and to prognosis.
Literature	GENERIF_SUMMARY	Positive HER-2 status was significantly associated with poor 5-year disease-free survival and 5-year overall survival in patients with locally advanced rectal cancer.
Literature	GENERIF_SUMMARY	conclude that ARTN is one mediator of acquired resistance to trastuzumab in HER2-positive mammary carcinoma cells
Literature	GENERIF_SUMMARY	Delayed fixation impaired HER2 protein expression assessed using immunohistochemistry in 22% of 3+ carcinomas. HER2 amplification assessed using FISH may be less affected.
Literature	GENERIF_SUMMARY	HER2 overexpression drive the aggressive behavior of the estrogen receptor-positive breast tumors without downregulation of luminal proteins.
Literature	GENERIF_SUMMARY	Depletion of either flotillin-1 or flotillin-2 resulted in downregulation of ErbB3 and a selective reduction of ErbB2-ErbB3 receptor complexes.
Literature	GENERIF_SUMMARY	CD24 supports the expression of HER2 and the consequent activation of PI3K-Akt signaling in HER2-positive breast cancer cells.
Literature	GENERIF_SUMMARY	HER2 overexpression is not an independent prognostic marker of colorectal cancer, but DcR3 overexpression is highly correlated with lymph node metastasis and poor overall survival
Literature	GENERIF_SUMMARY	HOTAIR functions as a competing endogenous RNA to regulate HER2 expression by sponging miR-331-3p in gastric cancer.
Literature	GENERIF_SUMMARY	Results suggest that serum HER2 could be a useful real-time marker for tumor burden and recurrence in patients with HER2-positive disease.
Literature	GENERIF_SUMMARY	Multiple roles of perforin in hampering ERBB-2 (Her-2/neu) carcinogenesis in transgenic male mice.
Literature	GENERIF_SUMMARY	These data reveal the distinct roles of EGFR and HER2 on expression of matrix macromolecules in cancer cells.
Literature	GENERIF_SUMMARY	The ERBB2 mutation was revealed to be one of the most frequent alterations in small bowel adenocarcinoma with a distribution dependent on tumour location.
Literature	GENERIF_SUMMARY	new class of protein cancer biomarker candidates
Literature	GENERIF_SUMMARY	Results show that c-erbB2-induced epithelial-mesenchymal transition is not affected by ectopic expression of E-cadherin.
Literature	GENERIF_SUMMARY	Suggest prolidase as a high affinity ligand of the ErbB2 receptor which can regulate ErbB2 signaling and cell

Main Category	Category	Terms
		growth.
Literature	GENERIF_SUMMARY	interaction of a kinase of the Src family with ErbB2 C-terminal domain
Literature	GENERIF_SUMMARY	The HER2 oncogene is a bi-functional locus.
Literature	GENERIF_SUMMARY	These findings suggest cooperative roles for bile acid and erbB2 activation in epithelial cell proliferation; bile acid appears to accelerate erbB2-induced pro-tumorigenic activities in the biliary tract and skin.
Literature	GENERIF_SUMMARY	HPV 16 infection induce upregulation of HER2/neu, pStat3 and IFI16.
Literature	GENERIF_SUMMARY	Study addresses whether the ERBB3 mutations altering Val-104 were present in homologous amino acids of ERBB family genes EGFR, ERBB2 and ERBB4 in CRC.
Literature	GENERIF_SUMMARY	No significant differences according to the HER2 inhibition were found.
Literature	GENERIF_SUMMARY	HER2 gene amplification was identified in a minority of CRC patients.
Literature	GENERIF_SUMMARY	These results suggest that H. pylori CagA may induce overexpression of the HER-2 protein by increasing HER-2 DNA and mRNA copy number
Literature	GENERIF_SUMMARY	High levels of HER2 are associated with achievement of a pathologic complete response in the preoperative setting, while levels of phospho-HER2 were not predictive of response in breast cancer patients.
Literature	GENERIF_SUMMARY	Complete pharmacologic EGFR/HER2 inhibition is required to reverse Src-dependent resistance to lapatinib in breast cancer.
Literature	GENERIF_SUMMARY	these results suggest that there may be a subgroup of patients with an exceptional response to trastuzumab emtansine based on high HER2 mRNA expression levels.
Literature	GENERIF_SUMMARY	Studies suggest that C-erbB2 (HER-2/neu)overexpression is related to poor prognostic of osteosarcoma and can be a useful clinical prognostic factor for those patients.
Literature	GENERIF_SUMMARY	Regardless of several limitations, our study suggested that HER-2 overexpression probably had little impact on CRC survival.
Literature	GENERIF_SUMMARY	Identifying FGFR2 as a transcriptional regulator of rDNA in bone unexpectedly reveals a nucleolar route for FGF signaling that allows for independent regulation of osteoprogenitor cell proliferation and differentiation.
Literature	GENERIF_SUMMARY	These data support the concept that activation of ERBB family members is sufficient to bypass dependency on FGFR3 and suggest that concurrent inhibition of these two pathways may be desirable when targeting FGFR3-dependent cancers.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	These results suggest an important role of HER2 in the activation of G2/M checkpoint response following IR.
Literature	GENERIF_SUMMARY	Together, our results indicate that c-FLIP(L) promotes TRAIL resistance and inhibits caspase-3 and caspase-8 activation in HER2-positive breast cancer.
Literature	GENERIF_SUMMARY	HER2/neu protein is credibly overexpressed in colon and gastric adenocarcinomas in immunohistochemistry
Literature	GENERIF_SUMMARY	HER-2 was expressed in a proportion of osteosarcoma patients, and can serve as a target for antineoplastic therapy. (Review)
Literature	GENERIF_SUMMARY	Results show that Cholangiocarcinoma express differentially HER2 protein based on tumor location.
Literature	GENERIF_SUMMARY	ERBB2 collaborates with androgen signaling to promote prostate cancer metastasis.
Literature	GENERIF_SUMMARY	These data indicate that Pick1 is involved in regulating the cell-cell junction in epithelial cells.
Literature	GENERIF_SUMMARY	HER2 amplification characterizes one intrinsic subtype of gastric cancer associated with a specific phenotype, as it is most commonly found in proximal gastric cancers. [Review]
Literature	GENERIF_SUMMARY	Higher ADAM10 levels correlated with poorer relapse-free survival (p
Literature	GENERIF_SUMMARY	Increased ErbB2 mRNA expression in peripheral blood leukocytes from breast cancer patients.
Literature	GENERIF_SUMMARY	HER2 overexpression is more common in male patients with high grade urothelial carcinoma of the renal pelvis.
Literature	GENERIF_SUMMARY	studies suggest that p95HER2 constitutes a new surrogate marker for an aggressive HER2-positive breast cancer subtype with distinct clinical and biological features
Literature	GENERIF_SUMMARY	Alu and LINE-1 hypomethylation is associated with HER2 enriched subtype of breast cancer
Literature	GENERIF_SUMMARY	results demonstrate that HER2-amplified ductal carcinoma in situ constitute a clearly distinct subgroup which is characterized by histopathological features associated with poor prognosis
Literature	GENERIF_SUMMARY	High circulating epidermal growth factor receptor 2 is associated with hyperglycaemia and insulin resistance
Literature	GENERIF_SUMMARY	Study identified novel microRNAs that were differentially expressed according to the HER2 status across different tumor types.
Literature	GENERIF_SUMMARY	DFF has no statistically significant effect on the HER2/CEP17 ratio tested by Dual-ISH. However, significant artifacts could occur in samples fixed after 1 hour of DFF time.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Her2 expression in in locally advanced gastric cancer with extensive lymph node metastasis
Literature	GENERIF_SUMMARY	Provide additional prognostic information when evaluating clinical features of ER(+), HER2(-) early breast cancer.
Literature	GENERIF_SUMMARY	HER2 positivity was seen in 17.6% of all cases of breast cancer from a single center in Kenya.
Literature	GENERIF_SUMMARY	Besides tumour stage, grading and nodal status, the Her2-neu score 0 is able to function as a prognostic factor in patients with TNBC.
Literature	GENERIF_SUMMARY	In tDNA, 50 mutations (36 EGFR, 5 ERBB2, 4 KRAS, 3 BRAF, and 2 PIK3CA) were identified, of which 26 were detected in cfDNA. These data demonstrate the feasibility and potential utility of mutation screening in the detection of a range of tumor biomarkers
Literature	GENERIF_SUMMARY	Relationship between hormone receptors and Her-2/neu in proportionally inverse; when high hormone receptors negativity is present there is also a Her-2/neu highly overexpressed
Literature	GENERIF_SUMMARY	HER2/HER3 co-overexpression is significantly associated with impaired survival from diagnosis of metastatic disease in patients with HER2-positive metastatic breast cancer.
Literature	GENERIF_SUMMARY	HER-2/neu but not PTEN expression is an independent prognostic factor for type I endometrial carcinoma
Literature	GENERIF_SUMMARY	HER2 gene amplification was associated with breast cancer.
Literature	GENERIF_SUMMARY	HER2 protein overexpression, gene amplification, and gene mutations appeared to be uncommon in lung adenocarcinoma.
Literature	GENERIF_SUMMARY	Data suggest that expression of EGF, EGF receptor, and c-erb-B2/ERBB2 in gastric mucosa is up-regulated during (1) gastritis with Helicobacter pylori infection, (2) gastric cancer, and (3) gastric cancer with Helicobacter pylori infection.
Literature	GENERIF_SUMMARY	Report protective role of ErbB2 modulation in human-induced pluripotent stem cell-derived cardiomyocytes exposed to doxorubicin.
Literature	GENERIF_SUMMARY	Transcriptional analysis revealed 15 genes to be differentially expressed between the two groups highlighting that Her2 signaling in primary tumors could be associated with specific blood gene expression
Literature	GENERIF_SUMMARY	HER-2 overexpression is associated with colorectal cancer.
Literature	GENERIF_SUMMARY	PTPN18 regulates HER2-mediated cellular functions through defining both its phosphorylation and ubiquitination barcodes.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	The meta-analysis suggests that HER-2/neu overexpression might not be a significantly prognostic indicator for patients with CRC.
Literature	GENERIF_SUMMARY	High HER2 expression is associated with breast cancer.
Literature	GENERIF_SUMMARY	A mutually exclusive pattern of FGFR2, HER2, and KRAS gene amplification and survival trends in gastric cancer patients.
Literature	GENERIF_SUMMARY	TOP2A FISH and qPCR may not identify the same pathology on chr17q. Non-HER2 chr17 CN patterns may further predict outcome in breast cancer patients with known favorable and unfavorable prognosis.
Literature	GENERIF_SUMMARY	The expressions of HER-2/neu (c-erbB2), survivin and cycline D1 are positively correlated with the malignant potential of serous ovarian neoplasms.
Literature	GENERIF_SUMMARY	Overexpression and gene amplification of HER2 was found in biliary tract carcinomas.
Literature	GENERIF_SUMMARY	HER2 overexpression in ovarian primary mucinous adenocarcinomas is associated with an expansile, but not an infiltrative, invasion pattern and a favorable prognosis
Literature	GENERIF_SUMMARY	serum HER2 ECD are highly correlated with tissue HER2 status in metastatic gastric cancer
Literature	GENERIF_SUMMARY	Overexpression of HER-2 with high level of PSA are associated with prostatic adenocarcinoma.
Literature	GENERIF_SUMMARY	High HER2 expression is associated with gastric cancer.
Literature	GENERIF_SUMMARY	Results show that poly (ADP-ribose) polymerase (PARP) inhibition has antitumour effects and increases trastuzumab activity in oncogene protein HER2 overexpressing breast cancer.
Literature	GENERIF_SUMMARY	Complete HER2 concordance existed between IHC and FISH tests for the Her2 gene copies per tumor cell in mucinous epithelial ovarian cancers.
Literature	GENERIF_SUMMARY	This work reveals that DeltaNErbB2 strongly affects several major pro- and antiapoptotic pathways and provides mechanistic insight into the role of NHE1 in chemotherapy resistance
Literature	GENERIF_SUMMARY	Differentiation from primary sarcoma is of paramount importance. Skeletal muscle metastases usually indicate an advanced disease associated with poor prognosis. Treatment should be individualized depending on the patient's clinical condition.
Literature	GENERIF_SUMMARY	our results establish the d16HER2 signaling axis as a signature for decreased risk of relapse after trastuzumab treatment.
Literature	GENERIF_SUMMARY	In this study, PR negativity was a prognostic factor for early relapse in luminal B/HER2-negative breast cancer,

Main Category	Category	Terms
		while a high Ki-67 index suggested a higher risk of early relapse.
Literature	GENERIF_SUMMARY	in responding cells (LNCap and DU-145) HER2 expression decreased, accompanied by increased EGFR expression
Literature	GENERIF_SUMMARY	Four weeks of neoadjuvant lapatinib in patients with HER2-positive DCIS resulted in inhibition of HER2 and RAS/MAPK signaling pathway
Literature	GENERIF_SUMMARY	HER2 promotes tumor cell survival by inhibiting apoptosis i.e. by downregulating CASP-3 and CASP-8.
Literature	GENERIF_SUMMARY	Her2 and estrogen receptor status influence the prognostic significance of FOXP3 positive tumor infiltrating lymphocytes in breast cancer.
Literature	GENERIF_SUMMARY	Expression levels of ER, PR, and HER2 were analyzed to determine clinical significance of four molecular subtypes of endometrial cancer.
Literature	GENERIF_SUMMARY	Elevated expression of the ER-alpha36-EGFR/HER2 loops is one of the mechanisms by which ER-positive breast cancer cells escape tamoxifen therapy.
Literature	GENERIF_SUMMARY	HER2 positivity significantly correlated with higher stages and lymph node metastasis in colorectal carcinoma.
Literature	GENERIF_SUMMARY	There was no association between ErbB2-IR and the androgen receptor (AR)-IR in the tumours, but an interaction between the two parameters was seen with respect to their association with the tumour stage.
Literature	GENERIF_SUMMARY	Our findings show an evident change in ER, PR and HER2 between breast primary tumors and relapsing tumors. The unstable status for ER or HER2 in breast cancer seems to be clinically significant and to correlate with a worse prognosis.
Literature	GENERIF_SUMMARY	EndoPredict high-risk ER+/HER2- breast cancers are more likely to respond to anthracycline-based chemotherapy.
Literature	GENERIF_SUMMARY	Our study demonstrated that HER2 overexpression is an independent prognostic factor in patients with any stage of resectable gastric cancer
Literature	GENERIF_SUMMARY	Activation of PKC-alpha and -delta mediates a novel positive feedback loop by promoting ErbB2 entry into the endocytic recycling compartment.
Literature	GENERIF_SUMMARY	Additive impact of HER2-/PTK6-RNAi on interactions with HER3 or IGF-1R leads to reduced breast cancer progression
Literature	GENERIF_SUMMARY	The HER2Bi-armed activated T cells with anti-tumor effects may provide a promising immunotherapy for colorectal carcinoma in the future.
Literature	GENERIF_SUMMARY	NTS/NTSR1 complex enhances breast tumor

Main Category	Category	Terms
		aggressiveness via EGFR/HER2/HER3 pathway.
Literature	GENERIF_SUMMARY	Neurotensin and its receptor NTSR1 causes EGFR, HER2 and HER3 over-expression and their autocrine/ paracrine activation in lung adenocarcinomas, confirming responsiveness to erlotinib.
Literature	GENERIF_SUMMARY	Chemotherapy sensitizes p95HER2/611CTF-positive patient-derived xenograft tumors to trastuzumab. This sensitization may be related to HER2 stabilization induced by chemotherapy in p95HER2/611CTF-positive cells.
Literature	GENERIF_SUMMARY	This study has revealed a strong association between the expression of WLS and HER2 that has important biological and clinical implications.
Literature	GENERIF_SUMMARY	Patients with HER2neu-positive ductal carcinoma In Situ had greater extent of disease by imaging and were more likely to undergo mastectomy than HER2neu-negative.
Literature	GENERIF_SUMMARY	Erbin is an ErbB2 regulator for breast tumor formation and progression
Literature	GENERIF_SUMMARY	HER-2/neu overexpression is associated with gastric carcinoma.
Literature	GENERIF_SUMMARY	there is HER2 and EGFR, but not VEGFR expression in micropapillary urothelial carcinoma with invasive urothelial carcinoma
Literature	GENERIF_SUMMARY	Both HER2 and TACC1 expression were independent prognostic factors in patients with gastric carcinoma.
Literature	GENERIF_SUMMARY	Tumour SUVmax could be an imaging biomarker for predicting ALNM Tumour SUVmax predicting ALNM is effective in ER-positive/HER2-negative and HER2-positive subtypes
Literature	GENERIF_SUMMARY	HER2 interacts with beta-catenin, leading to dissolution of the adherens junction, decreased cell-cell adhesion, and disruption of the pulmonary epithelial barrier.
Literature	GENERIF_SUMMARY	Mutations in exon 15 and 20 were more potent than wild-type HER2 in associating with activating signal transducers and inducing survival, invasiveness, and tumorigenicity.
Literature	GENERIF_SUMMARY	Discordance between the results of immunohistochemistry/fluorescence in situ hybridization detection of HER2 in both ductal and lobular carcinomas is largely associated with chromosome 17 aneuploidy.
Literature	GENERIF_SUMMARY	This study contributes to elucidating the intrinsic relations among breast cancer subgroups defined by ER, PR and HER2 expression via integrating mRNA and miRNA expression
Literature	GENERIF_SUMMARY	HER2 overexpression is associated with ovarian mucinous carcinomas.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	The content of HER-2 protein and amplification of HER2 and CEP17 correlated with the clinical morphological parameters of patients with nonsmall cell lung cancer.
Literature	GENERIF_SUMMARY	Epidermal growth factor receptor overexpression, but not high EGFR copy number, is a poor prognostic factor in HER2-positive primary breast cancer.
Literature	GENERIF_SUMMARY	The HER-2 form of breast cancer is more aggressive and associated with worse survival outcomes compared with patients without HER-2 overexpression.
Literature	GENERIF_SUMMARY	These results indicate that pathologic activation of Src is a potential mechanism of acquired resistance to ERBB2 inhibition in ERBB2-amplified gastroesophageal cancer.
Literature	GENERIF_SUMMARY	Immunohistochemical profile of breast cancer with respect to estrogen receptor and HER2 status.
Literature	GENERIF_SUMMARY	Elevated circulating Her-2/neu extracellular domain is associated with aggressive breast cancer.
Literature	GENERIF_SUMMARY	This model demonstrates the role of ErbB receptors underlying prolactinoma tumorigenesis and the feasibility of targeting these receptors for translation to treatment of refractory prolactinomas.
Literature	GENERIF_SUMMARY	Combined inhibition of MKP1 and HER2 enhanced cell killing in breast cancer. Together, our findings identify a new mechanism of resistance in breast tumors and reveal MKP1 as a novel therapeutic target for radiosensitization.
Literature	GENERIF_SUMMARY	Circulating tumor cells expression of ER, BCL-2, HER2, and Ki67 can be reproducibly measured with high analytical validity using the CellSearch System.
Literature	GENERIF_SUMMARY	The ratios of IGF-I/IGFBP-3 were increased in postmenopausal Taiwanese women with breast cancer, irrespective of their ages, ER and PR statuses, and HER2 expression.
Literature	GENERIF_SUMMARY	suggest that accurate assessment of HER2 status, including HER2 heterogeneity
Literature	GENERIF_SUMMARY	it is essential to confirm any EGFR L858R-positive cases by molecular methods or at least discard the presence of HER2 overexpression/amplification before rendering a diagnosis
Literature	GENERIF_SUMMARY	Reduced tumor invasivenes was the most significant biologic effect in breast cancer cells when HER2 and IGF-1R were cotargeted.
Literature	GENERIF_SUMMARY	HER2 expression in fine needle aspirates of lymph nodes detected by preoperative axillary ultrasound in breast cancer patients.
Literature	GENERIF_SUMMARY	Data show that a phenotypical implication of 655Val could be related to a cardiotoxicity risk upon Trastuzumab treatment.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Her2/Her3 and Met/Her3 were verified in human GC.
Literature	GENERIF_SUMMARY	Study indicates that quantification of HER2 mRNA expression with the RT-qPCR could be an alternative method of conventional IHC/FISH methods.
Literature	GENERIF_SUMMARY	EGFR is commonly coamplified with ERBB2, and in the setting of ERBB2 amplification, higher EGFR expression appears to mark tumors with greater sensitivity to dual EGFR/ERBB2 kinase inhibitors.
Literature	GENERIF_SUMMARY	The study determined the prognostic significance of Her2 status in gastric cancer and identified clinicopathological factors predictive of Her2 positivity.
Literature	GENERIF_SUMMARY	LMP2A may suppress the HER2 expression through the TWIST/YB-1 axis in Epstein-Barr virus-associated gastric carcinoma
Literature	GENERIF_SUMMARY	The single nucleotide polymorphism +936 C/T VEGF is associated with human epidermal growth factor receptor 2 expression in Moroccan breast cancer women.
Literature	GENERIF_SUMMARY	Phosphorylated ErbB1-4 and heregulin contribute to poorer patient prognosis in CRC. This heregulin-ErbB family member autocrine loop may be a candidate for targeted treatment of CRC.
Literature	GENERIF_SUMMARY	Increased HER2 expression is associated with cancers of pancreas and biliary tract neoplasms.
Literature	GENERIF_SUMMARY	Doxorubicin and methotrexate combination therapy down regulates HER2 gene expression and improve clinical outcome is OSCCs.
Literature	GENERIF_SUMMARY	Data suggest a role for stem cells proliferation in cell surface receptor HER2-regulated breast cancer tissue.
Literature	GENERIF_SUMMARY	These results for the first time indicated that a cross talk in FASN and HER2 expressions might be associated with prognosis in malignant ovarian cancer.
Literature	GENERIF_SUMMARY	Data indicate that HER2-negative tumors with HER2-ICD positivity exhibit clinical behavior closer to that of HER2-positive tumors.
Literature	GENERIF_SUMMARY	HER2 over-expression rate is relatively low among stage I-III gastric cancer patients, and might be generally not associated with clinicopathological characteristics.
Literature	GENERIF_SUMMARY	Serum Her2 level combined with CA-15-3 were strong and independent indicators for a worse disease free survival in patients with early breast cancer.
Literature	GENERIF_SUMMARY	Patients with HER2-amplified tumors were more likely to present PIK3CA mutations compared to patients with triple negative tumors.
Literature	GENERIF_SUMMARY	our findings indicate that total levels of HER2 can predict benefit for the combination of L+T. The added value of HER2 quantification is indeed its predictive

Main Category	Category	Terms
		power.
Literature	GENERIF_SUMMARY	Active HER2 binds and regulates the NRF2-dependent transcriptional activation.
Literature	GENERIF_SUMMARY	HER2 mutations were observed in 3.6% of NSCLCs, Amongst the patients with HER2 mutations, HER2 amplification and phosphorylation were independent prognostic factors.
Literature	GENERIF_SUMMARY	HER2 gene amplifications, and wild type KRAS genes are common in eyelid sebaceous carcinomas
Literature	GENERIF_SUMMARY	analysis of activator-receiver preferences among ErbB proteins
Literature	GENERIF_SUMMARY	Data show that the combination of PI3 kinase inhibitors with ErbB receptors HER2/HER3 targeting antibodies may constitute a promising therapeutic strategy for prostate cancer.
Literature	GENERIF_SUMMARY	Study shows that although HER2 expression itself was not a prognostic factor, the combination of HER2 expression and loss of heterozygosity of PTEN exacerbates the malignant potential of gastric cancer through its proliferative function.
Literature	GENERIF_SUMMARY	the concordance rate in HER2 status between primary gastric carcinoma (GC) and synchronous lymphnode metastases, was investigated.
Literature	GENERIF_SUMMARY	Standard slide-based tests for overexpression or amplification of ERBB2 would fail to detect the majority of activating mutations that occur overwhelmingly in the absence of copy number changes
Literature	GENERIF_SUMMARY	study found that serum HER2 extracellular domain (ECD) level was correlated with a poor prognosis in primary breast cancer, and that patients with an elevated HER2 ECD level had worse survival rate than those with normal HER2 ECD levels.
Literature	GENERIF_SUMMARY	one can suppose that the structural features of DNA are involved in the formation of ERBB2-containing amplicon borders in breast cancer cells and the data are of importance for understanding the mechanisms of oncogene amplification.
Literature	GENERIF_SUMMARY	Case Report
Literature	GENERIF_SUMMARY	This study suggests that TFF3 is an independent indicator for survival in GC, while HER2 is not associated with the outcome. Patients with HER2-negative/TFF3-negative GC have the best outcome.
Literature	GENERIF_SUMMARY	we investigate whether inhibiting HUNK impairs in vivo tumor growth that is initiated by HER2 inhibitor-resistant breast cancer cells.
Literature	GENERIF_SUMMARY	Trastuzumab(monoclonal antibody targeting HER2)significantly improves the survival of women with HER2 breast cancer.IHC and FISH testing of HER2 must

Main Category	Category	Terms
		be accurate results to predicting patients' responses to HER2-targeted therapy.
Literature	GENERIF_SUMMARY	This meta-analysis suggests that HER-2-positive status is not an independent adverse prognostic factor for survival among Inflammatory breast cancer patient cases.
Literature	GENERIF_SUMMARY	The meta-analysis findings suggest that the HER2 Ile655Val polymorphism is marginally associated with breast cancer susceptibility in worldwide populations with additive and dominant models, but not a recessive model.
Literature	GENERIF_SUMMARY	The addition of CDK4/6 inhibitor palbociclib to letrozole in this phase 2 study significantly improved progression-free survival in women with advanced oestrogen receptor-positive and HER2-negative breast cancer.
Literature	GENERIF_SUMMARY	miR-7 regulated pathways, including EGFR and Src kinase, represent targets for the therapeutic intervention of refractory and metastatic HER2Delta16 driven breast cancer
Literature	GENERIF_SUMMARY	Activating the PI3K pathway is associated with low pathological complete response to trastuzumab-based treatment in HER2-positive breast cancer
Literature	GENERIF_SUMMARY	HER2 signalling mechanisms along with the repertoire of HER family and other targeted therapies, it is likely that the near future holds further dramatic improvements to the prognosis of women with HER2-positive breast cancer.
Literature	GENERIF_SUMMARY	HER2 positivity occurs in 15.6 % of invasive gastroesophageal adenocarcinoma in Western patients, of which the majority is esophageal primary tumors and of the intestinal tumor type.
Literature	GENERIF_SUMMARY	An increase in the TNM stage and pathological tumor size status correlated with an increase in EGFR and cerbB2 expression in larynx carcinoma.
Literature	GENERIF_SUMMARY	Report rapid fluorescence in situ hybridisation for HER2 assessment in breast and gastro-oesophageal cancers.
Literature	GENERIF_SUMMARY	roles of ESR1-mediated transactivation and transrepression in the regulation of invasiveness of ESR1+/HER2+ breast cancer cells by E2
Literature	GENERIF_SUMMARY	p27 accumulated in the cytoplasm of breast cancer cells exclusively in cell lines that are Her2+.
Literature	GENERIF_SUMMARY	ErbB2 and ErbB3 homodimers are endocytosis deficient owing to the lack of endocytic codes. Interestingly, EGFR-ErbB2 or EGFR-ErbB3 heterodimers are also endocytosis deficient
Literature	GENERIF_SUMMARY	The synergism between HER2 and EGFR in determining radiosensitivity.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	loss of Postn in an ErbB2/Neu/HER2 overexpression model results in apocrine-like tumors that activate an AR-dependent pathway.
Literature	GENERIF_SUMMARY	Data show that hypoxia promotes lapatinib resistance in ERBB2-positive breast cancer cells through activation of the MEK-ERK pathway a HIF-1-dependent manner via regulation of dual-specificity phosphatase 2 (DUSP2).
Literature	GENERIF_SUMMARY	PRP4K functions as a HER2-regulated modifier of taxane sensitivity that may have prognostic value as a marker of better overall survival in taxane-treated ovarian cancer patients.
Literature	GENERIF_SUMMARY	High grade endometrial cancer expresses higher levels of p95HER2.
Literature	GENERIF_SUMMARY	Suggest that TXNIP plays a critical role in anti-Her-1/Her-2 treatment and may be a potential prognostic marker in breast cancer.
Literature	GENERIF_SUMMARY	Gastric and esophageal neoplasms are characterized by wide HER2 genetic heterogeneity.
Literature	GENERIF_SUMMARY	The tripartite antibody mixture induced efficient HER2 internalization.
Literature	GENERIF_SUMMARY	Gastric cancer cells that express the HER-2 oncogene spur the activation of NF-kappaB that can upregulate the expression of ICAM-1 and induce the expression of MMP-9, which hydrolyzes ICAM-1 into s-ICAM-1 to promote tumor immune escape.
Literature	GENERIF_SUMMARY	Ad-HER2-ECD can efficiently transduce HER2-ECD into HER2-negative human cancer cells. PIT with trastuzumab-IR700 induced direct cell membrane destruction of Ad-HER2-ECD-transduced HER2-negative cancer cells
Literature	GENERIF_SUMMARY	Our study suggests addition of imetelstat to trastuzumab may enhance the effects of HER2 inhibition therapy, especially in the CSC population.
Literature	GENERIF_SUMMARY	Both HER2 and MMP-9 may be required to effectively block HER2 signaling in gastric cancer therapy.
Literature	GENERIF_SUMMARY	results show that there is no association between the genotyped SNPs and the disease (p>0.05) in this population.
Literature	GENERIF_SUMMARY	Missense mutations in the gene ERBB2 are exclusively present in bladder cancer patients responding to neoadjuvant chemotherapy.
Literature	GENERIF_SUMMARY	there is a correlation between plasma membrane associated beta-catenin in HER2-positive breast cancer and survival outcome
Literature	GENERIF_SUMMARY	In urothelial carcinoma overexpression of HER2 protein and gene was evident in all stages, regardless of schistosomiasis status.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	determination of HER2 copy number variations by SNP array-based genomic segmentation analysis is an effective supplement to IHC/FISH HER2 analysis.
Literature	GENERIF_SUMMARY	Symptom burden and quality of life among women with HER2(+) metastatic breast cancer.
Literature	GENERIF_SUMMARY	HER2-trogocytosis is visible evidence of tumor microenvironment interaction between cancer cells and immune cells.
Literature	GENERIF_SUMMARY	Dacomitinib growth inhibition was associated with a significant and dose-dependent decline in phosphorylated HER2/neu and S6 transcription factor and a dose-dependent and time-dependent cell cycle arrest
Literature	GENERIF_SUMMARY	The selective anticancer activity of hirsutine in HER2-positive breast cancer by inducing a DNA damage response.
Literature	GENERIF_SUMMARY	Our findings suggested that there was positive feedback regulation between FASN and HER2 expression and phosphorylation in osteosarcoma cells.
Literature	GENERIF_SUMMARY	suggest that ECM-detached ErbB2-expressing cells may be uniquely susceptible to targeted therapy against EGFR and that this sensitivity could be exploited for specific elimination of ECM-detached cancer cells
Literature	GENERIF_SUMMARY	Findings indicate a pathway involving protein-tyrosine phosphatase D2 PTPD2 (PTPN14) and the lipid second messenger phosphatidic acid that promotes ERBB2 receptor function.
Literature	GENERIF_SUMMARY	The expression of Bcl-2 was an independent, favorable prognostic factor in breast cancer patients with the HR+/HER2- subtype.
Literature	GENERIF_SUMMARY	HER2 positive status had no significant prognostic impact in terms of survival in advanced gastric cancer patients.
Literature	GENERIF_SUMMARY	c-MYC expression is up-regulated through the cross-talk between ER and HER2 in aromatase inhibitor resistant breast cancer cells.
Literature	GENERIF_SUMMARY	Loss of HER2 amplification is associated with melanoma.
Literature	GENERIF_SUMMARY	Lapatinib failed to show efficacy as either monotherapy or in combination with capecitabine in the treatment of HER2 amplified gastro-esophageal cancers.
Literature	GENERIF_SUMMARY	Expression of HER2 and pAKT define a population of patients with stage IV pulmonary adenocarcinoma with a distinct disease course.
Literature	GENERIF_SUMMARY	Studies suggest that observations may have significant therapeutic implications in cancers not traditionally thought to benefit from anti-HER-2 proto-oncogene protein) therapies.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	HER2 IHC status, HER2/CEP17 ratio, and HER2 GCN were correlated with clinical outcomes of trastuzumab-based treatment in HER2-positive advanced gastric cancer
Literature	GENERIF_SUMMARY	High Tau-protein and low PTEN expression showed a significant association with poor response to H + P chemotherapy in patients with HER2-positive advanced breast cancer.
Literature	GENERIF_SUMMARY	Report high expression of HER2 in gastric adenocarcinomas.
Literature	GENERIF_SUMMARY	HER2 positivity was increased in BRCA2 breast cancers compared to non-BRCA familial and BRCA1 breast cancers.
Literature	GENERIF_SUMMARY	HER3 overexpression strongly correlated with HER2 positivity in this cohort of patients
Literature	GENERIF_SUMMARY	CK5/6 have roles in biologically aggressive tumors likely to display resistance to trastuzumab ab initio in women diagnosed with HER2+ cancer
Literature	GENERIF_SUMMARY	Genetic induction of fibronectin by HER2 triggers cell adhesion and invasion capacities.
Literature	GENERIF_SUMMARY	HER2/CD3 BsAb efficiently inhibited the growth of breast cancer tissue by activating and inducing the proliferation of tumor tissue infiltrating lymphocytes.
Literature	GENERIF_SUMMARY	Study shows an inverse association between the level of HER2 amplification and survival time in a relatively large cohort of HER2-positive breast cancer patients where greater HER2 amplification level is correlated with worse outcome.
Literature	GENERIF_SUMMARY	Results show that Her2 gene is particularly amplified in 38% of breast tumors of patients exposed to ionizing radiation during there childhood.
Literature	GENERIF_SUMMARY	In this study, we demonstrate that 5a-induced cell cycle arrest and apoptosis by inhibiting EGFR and HER2 activity and downstream activation of PI3K/Akt and MEK/Erk pathways.
Literature	GENERIF_SUMMARY	A substantially higher frequency of HER2 mutations occur in pleomorphic invasive lobular carcinoma, compared with that in classic invasive lobular carcinoma.
Literature	GENERIF_SUMMARY	Progesterone receptor status exerts a significant impact on prognosis of estrogen receptor+/HER2- breast cancer.
Literature	GENERIF_SUMMARY	Neoadjuvant chemotherapy reduces HER2 expression in breast cancer patients.
Literature	GENERIF_SUMMARY	data conclude that the QPA is able to determine ERBB2 gene status with high accuracy, while also overcoming the limitations of conventional techniques and providing better cost-effectiveness

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Human epidermal growth factor receptor 2 (HER2)-activating mutations are present in 1% to 2% of lung adenocarcinomas, with predominance in never smokers, association with gene amplification and overexpression
Literature	GENERIF_SUMMARY	HER2 status was highly concordant between the paired primary and brain tumors, whereas changes of ER and PR status occurred in a substantial proportion of the patients
Literature	GENERIF_SUMMARY	overexpression of HER2 exhibited stronger negative effect on the prognosis of ILC patients than that in IDC patients, suggesting that treatment targeting HER2 is crucial for this BC subgroup
Literature	GENERIF_SUMMARY	HER2 amplified bladder cancer feature specific morphological characteristics.
Literature	GENERIF_SUMMARY	in breast carcinoma, HER2 status in the presence of polysomy 17 may vary with the scoring criteria used
Literature	GENERIF_SUMMARY	HER2 and EGFR)on the breast cancer xenografts in a mouse orthotopic model were successfully detected in a multiplexed way, illustrating the potential of FRES as a molecular diagnostic instrument
Literature	GENERIF_SUMMARY	EGFR, ErbB2 and MET genes are frequently amplified in gastric carcinoma.
Literature	GENERIF_SUMMARY	Report shows that HER2 positive breast cancers exhibit increased expression of PARP1 and phospho-p65.
Literature	GENERIF_SUMMARY	Data show that proto-oncogene protein HER-2 is directly regulated by microRNAs miR-125a and miR-125b in small cell lung cancer (SCLC).
Literature	GENERIF_SUMMARY	the tumor HER2 status determined in the biopsy specimen significantly reflects the molecular biological properties of gastric cancer and may be clinically used to determine indications for the use of targeted drugs
Literature	GENERIF_SUMMARY	HER2 expression in gastric/GEJ adenocarcinomas shows some differences compared to what is observed in breast adenocarcinomas, most notably basolateral or lateral membranous positivity and increased heterogeneity. [review]
Literature	GENERIF_SUMMARY	negative-ER ,negative-PR ,positive-p53 and higher Ki-67 expressions were related to a higher mtDNA copy ratio. In addition, advanced T-status (p=0.019) and negative-HER-2/neu expression (p=0.061) were associated with mtDNA D310 mutations
Literature	GENERIF_SUMMARY	Presence of circulating Her2-reactive CD8 + T-cells is associated with lower frequencies of myeloid-derived suppressor cells and regulatory T cells, and better survival in older breast cancer patients.
Literature	GENERIF_SUMMARY	Considering simultaneously tumor pathophenotypes and several molecular levels, we show the heterogeneous behavior of ERBB2-positive breast

Main Category	Category	Terms
		cancer in terms of disease progression
Literature	GENERIF_SUMMARY	The tumorigenic function of the HER2 H878Y mutation.
Literature	GENERIF_SUMMARY	HER2 expression may have potential as a prognostic factor for intestinal cancer types
Literature	GENERIF_SUMMARY	using dual paraffin blocks to assess Her2/neu expression of gastric cancer may help identify more patients with Her2/neu-positive gastric cancer
Literature	GENERIF_SUMMARY	We did not find that c-erbB-2 expression was associated with the prognosis of gastric cancer. [Meta-analysis]
Literature	GENERIF_SUMMARY	explore the effects of the HER2 gene and Herceptin on circulating tumor cells.
Literature	GENERIF_SUMMARY	Her2/neu single nucleotide polymorphisms may be useful in conjunction with other biomarkers to risk stratify patients in order to optimize clinical management.
Literature	GENERIF_SUMMARY	Quantitative HER2 measurement by HERmark is highly sensitive, accurately quantifies HER2 protein expression and correlates well with routine HER2 testing.
Literature	GENERIF_SUMMARY	Essential activating sequence exists in the extracellular domain of HER2. Disruption of this sequence disables HER2 dimerization loop, blocks subsequent activation of HER2-driven oncogenic signaling, and generates a dominant-negative form of HER2.
Literature	GENERIF_SUMMARY	results provide the molecular basis of a novel resistance mechanism to trastuzumab-based anti-HER2 therapy, supporting additional molecule stratification within HER2-positive GC patients for more effective therapy options
Literature	GENERIF_SUMMARY	Data (including data from SZ95 sebocyte cell line) suggest sebocytes abundantly express EGFR (epidermal growth factor receptor) with lower levels of ERBB2 and ERBB3; expression of these receptors appears coordinated with lipogenesis in specific ways.
Literature	GENERIF_SUMMARY	ErbB2 modulates gemcitabine and irinotecan/SN-38 chemoresistance of human pancreatic cancer cells via hCNT1 transporter and multidrug-resistance associated protein MRP-2.
Literature	GENERIF_SUMMARY	HER2 mutation is associated with lung cancers.
Literature	GENERIF_SUMMARY	Regardless of methodology, basal-like cases seldom expressed ER/ESR1 or PR/PGR and were associated with the lowest expression level of HER2/ERBB2 relative to other subtypes.
Literature	GENERIF_SUMMARY	HER2 overexpression was seen in one-third of primary gastric cancers with lymph node metastases. As the receptor expression may lose/gain in the metastases assessment of the receptor status in metastatic lesions is encouraged.

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Literature	GENERIF_SUMMARY	HER2 detection in mixed gastric carcinoma displays high heterogeneity.
Literature	GENERIF_SUMMARY	Data show that proto-oncogene protein HER-2 overexpression occurs in only 8% of patients with colorectal cancer (CRLM) but with 75% of cases proto-oncogene protein HER-3 is frequently overexpressed in CRLM.
Literature	GENERIF_SUMMARY	Our data suggest that physical activity reduces risk of invasive breast cancers that lack HER2 overexpression, increasing our understanding of the biological mechanisms by which physical activity acts.
Literature	GENERIF_SUMMARY	HER2 2+ status is an unfavorable prognostic factor regarding the OS of breast cancer patients with intermediate grade of differentiation and could be used to identify patients, who may benefit from adjuvant therapy.
Literature	GENERIF_SUMMARY	Exogenously expressed HER2 failed to associate with EGFR or HER3 in MDA-MB-231 cells, while overexpression of HER2 enhanced HER family dimerization in MCF-10A and MCF-7 cells
Literature	GENERIF_SUMMARY	Metformin increases survival in hormone receptor-positive, HER2-positive breast cancer patients with diabetes
Literature	GENERIF_SUMMARY	Positive stromal PAI-1 protein expression may identify a subgroup of HER2-negative advanced breast cancer patients.
Literature	GENERIF_SUMMARY	mir-4728, an intron of the ErbB2 gene, is a negative regulator of MAPK signaling through directly targeting the ERK upstream kinase MST4 and exerts numerous tumor-suppressive properties
Literature	GENERIF_SUMMARY	SATB1 and HER2 expression correlated with poorly differentiated breast cancer and indicated an unfavorable prognosis.
Literature	GENERIF_SUMMARY	Gene amplification and immunohistochemical overexpression of ERBB2 occurred in 13% of all ampullary carcinomas, therefore providing a potential target for anti-HER2 therapy in these tumors.
Literature	GENERIF_SUMMARY	Measurements of HER2 experession in intracellular domain (ICD) and extracellular domain (ECD) in breast cancer suggest a subclassification of HER2-positive tumors
Literature	GENERIF_SUMMARY	This review summarizes the molecular characteristics, etiology, preclinical tools and therapeutic options for hormone receptors+ (i.e., estrogen receptor and progesterone receptor) HER2+ breast cancer. [review]
Literature	GENERIF_SUMMARY	Downregulation of the expression of HER2 by small interfering RNA in human meningioma cells significantly inhibited cell motility and proliferation, led to cell cycle arrest at the G0/G1phase and increased early apoptosis.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	HER2 role in gastric tumor metabolism and its prognostic implications
Literature	GENERIF_SUMMARY	HER2-mTOR signaling-driven breast cancer cells require ER-associated degradation to survive.
Literature	GENERIF_SUMMARY	HER2 inhibition can enhance or restore ER expression with parallel Bcl2 upregulation, representing an ER-dependent survival mechanism potentially leading to anti-HER2 resistance.
Literature	GENERIF_SUMMARY	HER2 overexpression or intratumoural heterogeneity was not prognostic, but primary-metastatic conversion was an independent predictor of a shorter overall survival (hazard ratio = 4.93).
Literature	GENERIF_SUMMARY	The aim of the study was to evaluate the prognostic ability of the transcriptional profiling of the HER family genes in early breast cancer.
Literature	GENERIF_SUMMARY	Report allelic imbalance at the HER2/TOP2A locus in breast cancer.
Literature	GENERIF_SUMMARY	In non-small cell lung cancer there is an elevated HER2 protein level and, well less frequently, altered activity in the HER2 gene.
Literature	GENERIF_SUMMARY	COX-2 and HER-2 are important markers for invasion and metastasis of colorectal cancer, and they act together to regulate the invasion and metastasis of colorectal cancer.
Literature	GENERIF_SUMMARY	These results support the role of the HER2 655 A>G polymorphism as a genetic marker of trastuzumab-induced cardiotoxicity in HER2-positive breast cancer patients.
Literature	GENERIF_SUMMARY	We also established the rarity of HER2 mutation in TNBC.
Literature	GENERIF_SUMMARY	Improved prognostic tools for DCIS patients are warranted to tailor adjuvant therapy. Here, we demonstrate that HER2 positive disease in the primary DCIS is associated with lower risk of recurrent invasive breast cancer.
Literature	GENERIF_SUMMARY	Ezrin and HER2/neu are overexpressed and coexpressed in osteosarcoma with adverse prognostic features such as high grade. Therefore, ezrin and HER2/neu could be potential prognostic markers and treatment targets for osteosarcoma
Literature	GENERIF_SUMMARY	ERBB2 gene amplification is detected in approximately 40% of isolated HLA-G(+) trophoblasts.
Literature	GENERIF_SUMMARY	loss of HER2 protein expression due to loss of DMs containing HER2 is not a likely mechanism of resistance to anti-HER2 therapies.
Literature	GENERIF_SUMMARY	AR is not associated with the proliferation marker Ki-67, but is associated with lower 21-gene recurrence score in estrogen receptor positive/HER2 negative breast

Main Category	Category	Terms
		cancer.
Literature	GENERIF_SUMMARY	GPNMB might be a surrogate marker for BC and may cross talk with the HER2 signal pathway.
Literature	GENERIF_SUMMARY	Study shows the phosphorylation of EGFR and HER2 after MEK inhibition in a MEK1-mutated gastric cancer cell line suggesting that EGFR and HER2 activation may serve as a potential mechanism responsible for the resistance to a MEK inhibitor.
Literature	GENERIF_SUMMARY	High co-expression of Sp1 and HER-2 is correlated with gastric cancer.
Literature	GENERIF_SUMMARY	The expressions of VEGF, HER-2, and EGFR mRNA were related to the lymph node metastasis in ESCC and pathologic differentiation degree.
Literature	GENERIF_SUMMARY	Results show that ESR1 and ERBB2 were found to modulate the most numbers of gene pairs in breast neoplasm and they share complex interplay between each other suggesting a role in breast cancer pathogenesis through gene regulation.
Literature	GENERIF_SUMMARY	Given the high prevalence of lung cancer and the availability of targeted therapy, Chinese lung ADC patients without EGFR and KRAS mutations are recommended for HER2 and BRAF mutations detection, especially for those never smokers
Literature	GENERIF_SUMMARY	FASN and HER2 have roles in breast cancer and are blocked by EGCG with pertuzumab or temsirolimus
Literature	GENERIF_SUMMARY	Our results indicate that the transactivation of HER2 and EGFR by the pro-inflammatory cytokine/ neuropeptide SP in BC cells is a c-Src and MMP-dependent process.
Literature	GENERIF_SUMMARY	C-erbB-2 was expressed in gastric carcinoma, gastric mucosal atypical hyperplasia and gastritis. Expression correlated positively with degree of cell differentiation. C-erbB-2 played an important role in the pathogenic mechanism of gastric carcinoma.
Literature	GENERIF_SUMMARY	VEGFC and VEGFR1 mRNA overexpression is of prognostic value, dependent on HER2 status.
Literature	GENERIF_SUMMARY	In vitro affinities of the panel of evolved DARPins directly correlated with the fluorescence intensities of evolved DARPins bound to HER2 on a breast cancer cell line
Literature	GENERIF_SUMMARY	Expression of total HER2, total HER3 were significantly elevated in HPV(+) HNSCC.
Literature	GENERIF_SUMMARY	Increased gene copy number of HER2 and concordant protein overexpression was found in a subset of eyelid sebaceous gland carcinoma.
Literature	GENERIF_SUMMARY	Data suggest that AMP-activated protein kinase (AMPK) regulates HER-2 proto-oncogene protein and epidermal growth factor receptor (EGFR) in HER2-

Main Category	Category	Terms
		amplified breast cancer cells and activation of AMPK might provide therapeutic benefit in such cancers.
Literature	GENERIF_SUMMARY	MLK3 represents a newly recognized integral component of HER2 biology in HER2+ breast tumors.
Literature	GENERIF_SUMMARY	8.7% of this Pakistani breast cancer cohort were HER2 overexpressing.
Literature	GENERIF_SUMMARY	ErbB2 receptor tyrosine kinase in AECs was activated by annexin V, and blockade reduced the fibrotic potential of annexin V-treated AEC-conditioned medium.
Literature	GENERIF_SUMMARY	Promote the internalization of T-DM1 into HER-2 positive cancer cells.
Literature	GENERIF_SUMMARY	Helicobacter pylori-positive gastric cancer showed more Her2/neu overexpression.
Literature	GENERIF_SUMMARY	HOXB7 Is an ERalpha Cofactor in the Activation of HER2 and Multiple ER Target Genes Leading to Endocrine Resistance
Literature	GENERIF_SUMMARY	In this study, we report that miR-205-5p is highly expressed in BCSCs and represses directly ERBB2 and indirectly EGFR leading to resistance to targeted therapy.
Literature	GENERIF_SUMMARY	The presence of tumor infiltrating lymphocytes at diagnosis is an independent, positive, prognostic marker in HER2-positive early breast cancer treated with neoadjuvant anti-HER2 agents.
Literature	GENERIF_SUMMARY	Her-2 overexpression results in ICA were similar to previous reports, the finding of 28% in HGD was unexpected and may have clinical implications. Positive Her-2 DISH in 6% of LGD is novel, suggesting a role of Her-2 during BE progression
Literature	GENERIF_SUMMARY	High HER2 expression is associated with breast cancer.
Literature	GENERIF_SUMMARY	HER2 and PTEN expression appears as early as the 9-12 weeks period in the digestive tract, but HER2 expression decreases in the 21-24 weeks period and then disappears. EGFR appears only during the 13-16 weeks period and p53 is strong until week 21.
Literature	GENERIF_SUMMARY	these studies support the rationale to target the NRG1-ErbB3-ErbB2 axis as a novel treatment strategy in a subset of cutaneous melanomas.
Literature	GENERIF_SUMMARY	SNP in HER2 codon 655 and investigation of HER2 and HER3 expression could be helpful to outline the prognosis for patients with lung adenocarcinoma.
Literature	GENERIF_SUMMARY	Supporting the clinical relevance of our results, we found an inverse correlation between ErbB-2/Stat3 nuclear co-expression and PDCD4 expression in ErbB-2-positive primary invasive breast cancer
Literature	GENERIF_SUMMARY	HER2-positive early breast cancer patients receiving

Main Category	Category	Terms
		adjuvant trastuzumab with high pretreatment neutrophil/lymphocyte ratio had shorter disease-free survival.
Literature	GENERIF_SUMMARY	Suggest that clinical course of KI67/Her2 positive breast cancer was determined by hormone receptor status rather than Ki67 expression.
Literature	GENERIF_SUMMARY	The combined impacts of alphav-integrin and HER2 influence the invasive phenotype of breast cancer cells.
Literature	GENERIF_SUMMARY	Our results suggest that it may be valuable to add agents targeting the STAT3 pathway to trastuzumab for treatment of HER2-positive breast cancer.
Literature	GENERIF_SUMMARY	It is suggested that tumor substaging and HER2 status are independent predictive markers for tumor progression or recurrence, and thus could be included in diagnostic and therapeutic management for NMIBC.
Literature	GENERIF_SUMMARY	ARPE-19 cells have low ErbB2 expression, and overexpression and knockdown experiments revealed that robust ErbB2 expression bypassed the requirement for clathrin for EGF-stimulated Akt phosphorylation.
Literature	GENERIF_SUMMARY	Suggest that androgen receptor and PTEN status might be considered as indicators of trastuzumab sensitivity in HER2-positive breast cancer patients.
Literature	GENERIF_SUMMARY	results suggest ErbB-2 is a prominent player in mediating the ligand-dependent and -independent activation of androgen receptor in androgen-sensitive and androgen-independent/castration-resistant PCa cells respectively for PCa progression and survival.
Literature	GENERIF_SUMMARY	Our results establish HER2Ab-NSCs as a novel, nontoxic, and rational therapeutic approach for the successful treatment of HER2 overexpressing BCBM, which now warrants further preclinical and clinical investigation.
Literature	GENERIF_SUMMARY	Data indicate that increased expression of erbB3 plays a pivotal role in activating downstream PI-3K/Akt pathway and promoting erbB2-driven mammary/breast tumorigenesis.
Literature	GENERIF_SUMMARY	High HER2 expression is associated with gastric cancer.
Literature	GENERIF_SUMMARY	The finding from this present meta-analysis suggested that HER-2 overexpression was not related to clinicopathological characteristics and poor prognostic of colorectal cancer patients.
Literature	GENERIF_SUMMARY	p95HER2-positive CTCs can be detected in both early and metastatic breast cancer patients.
Literature	GENERIF_SUMMARY	Differential Gene Expression of BRCA1,ERBB2 and TP53 biomarkers between Human Breast Tissue and Peripheral Blood Samples of Breast Cancer
Literature	GENERIF_SUMMARY	findings support the use of in situ single cell-based

Main Category	Category	Terms
		methods in cancer genomics and imply that chemotherapy before HER2-targeted therapy may promote treatment resistance
Literature	GENERIF_SUMMARY	Preclinical data suggest that the combination of multi-points Neu protein (HER-2) targeted therapy may break the drug resistance.
Literature	GENERIF_SUMMARY	The proportion of a positive HER2 status in patients with gastro-oesophageal adenocarcinoma and brain metastasis involvement was higher than expected.
Literature	GENERIF_SUMMARY	Her2 and Ki67 expression predicts recurrence of ductal carcinoma in situ.
Literature	GENERIF_SUMMARY	factor
Literature	GENERIF_SUMMARY	Three factor model revealed significant gene-gene interaction for PDGFB +286A>G, PDGFB +1135A>C and HER2 Ile165Val SNPs with GBC. Protein-protein interaction showed significant association of PDGFB and HER2 with EGFR receptor signaling pathway.
Literature	GENERIF_SUMMARY	Salivary Her2/neu levels were found to be significantly high in oral cancer.
Literature	GENERIF_SUMMARY	HER2 mutations were identified as the only driver mutation in a significant proportion of solid cancers. Evaluation of anti-HER2 therapies in nonamplified, HER2-mutated cancers is warranted.
Literature	GENERIF_SUMMARY	HER2 rs1058808 and rs2517956 polymorphisms are associated with its protein expression in breast cancer
Literature	GENERIF_SUMMARY	p-rpS6 is a robust post-treatment indicator of HER2 pathway-targeted therapy resistance
Literature	GENERIF_SUMMARY	ErbB2 and EGFR have previously been considered as undruggable targets for CART cells.
Literature	GENERIF_SUMMARY	Data show that dual epidermal growth factor receptor 2 HER2/phosphatidylinositol 3-kinase PIK3CA blockade may be a drug therapy for uterine serous carcinoma (USC) with HER2 gene amplification and mutated wild-type PIK3CA resistant to chemotherapy.
Literature	GENERIF_SUMMARY	Data indicate that high proto-oncogene protein HER-2 (HER2) and low estrogen receptor alpha (ESR1) mRNA levels are required for achieving trastuzumab response.
Literature	GENERIF_SUMMARY	findings demonstrate that Her2+/+ mice, similar to Pds5b-/- mice, are embryonic lethal and confirm the necessity for Pds5b in embryonic development
Literature	GENERIF_SUMMARY	Data showthat erbB-2 receptor protein-tyrosine kinase (HER2) represents an oncogene in ovarian cancer, and suggest that targeting this tyrosine kinase with T-DM1 may be therapeutically effective, especially in ovarian tumors with high content of HER2.
Literature	GENERIF_SUMMARY	18F-fluorodeoxyglucose (FDG) PET/CT after two cycles of neoadjuvant therapy may predict response in HER2-

Main Category	Category	Terms
		negative, but not in HER2-positive breast cancer.
Literature	GENERIF_SUMMARY	ER negative breast cancer showed different expression of sarcosine metabolism related proteins according to AR and HER-2 status.
Literature	GENERIF_SUMMARY	HER2 are highly expressed, and HER2 positivity could be considered as a significant biomarker of poor prognosis in molecular apocrine breast cancer.
Literature	GENERIF_SUMMARY	Data suggest that the association between ER, PR and HER-2/neu is age-related.
Literature	GENERIF_SUMMARY	HER2 and EGFR gene copy number gains may play an important role in the progression of carcinoma ex pleomorphic adenoma, in particular ductal-type CXPAs.
Literature	GENERIF_SUMMARY	Automated HER2 testing of gastric carcinoma biopsies using the Leica Bond system was useful and efficient. This method allowed us to improve normal routine procedures, minimizing time and costs as well as handling and observation errors
Literature	GENERIF_SUMMARY	Our results showed that HER2 immunohistochemistry (IHC) was well concordant with in situ hybridization (ISH) in HER2 IHC score 0/1+ or 3+. Although this meta-analysis showed higher diagnostic accuracy of HER2 IHC, more detailed criteria for HER2 IHC score 2+ cases will be required to predict HER2 status
Literature	GENERIF_SUMMARY	PKCalpha Attenuates Jagged-1-Mediated Notch Signaling in ErbB-2-Positive Breast Cancer to Reverse Trastuzumab Resistance
Literature	GENERIF_SUMMARY	demonstrate that HER2 is captured via a site, comprised of amino acids 210-240, in the extracellular domain of human Sdc1, and EGFR is captured via an extracellular site comprised of amino acids 87-131 in human Sdc4
Literature	GENERIF_SUMMARY	SRSF3 and hnRNP H1 are the first splicing factors identified which regulate the production of these functionally distinct HER2 splice variants and therefore maybe important for the regulation of HER2 signaling.
Literature	GENERIF_SUMMARY	Results demonstrate that acquired resistance of colorectal tumour cells to treatment with anti-EGFR mAb ICR62 is accompanied by an increased level of cell surface EGFR, and upregulation of phosphorylated HER-2 and HER-3 proteins.
Literature	GENERIF_SUMMARY	Tumor driven by gain-of-function HER2 H878Y mutant is highly sensitive to HER2 inhibitor.
Literature	GENERIF_SUMMARY	Her2 mutation is associated with response to therapy in lung cancer.
Literature	GENERIF_SUMMARY	HER2 mutations are associated with refractory metastatic breast cancers.
Literature	GENERIF_SUMMARY	CORRELATION BETWEEN CLINICAL PATHOLOGY OF LUMINAL B BREAST CANCER AND DETERMINATION

Main Category	Category	Terms
		OF ESTROGEN RECEPTOR, PROGESTERONE RECEPTOR AND HER2 EXPRESSION COMBINED WITH NUCLEAR MORPHOLOGY.
Literature	GENERIF_SUMMARY	Quantitative HER2 measurement by SRM-MS is superior to IHC and ISH in predicting outcome after treatment with anti-HER2 therapy
Literature	GENERIF_SUMMARY	HER2 expression rate increased in patients diagnosed with esophageal carcinoma.No correlation was found in patients survival.
Literature	GENERIF_SUMMARY	Uni- and multivariate analyses included age, tumor size, grade, ER, and Her2; in multivariate analysis, MammaPrint reached near-significance (HR 3.01; p 0.08).
Literature	GENERIF_SUMMARY	Positive HER2 expression is associated with trastuzumab benefit in breast cancer.
Literature	GENERIF_SUMMARY	These results are encouraging for further development of these bsRICs for dual-receptor-targeted radioimmunotherapy of breast cancer coexpressing HER2 and EGFR, including trastuzumab-resistant tumors
Literature	GENERIF_SUMMARY	It blocked activation of EGFR, HER2 and downstream Akt and Erk.
Literature	GENERIF_SUMMARY	Targeting this interaction of membrane surface glycoprotein and its animal lectin in Her-2 negative breast cancer may be of therapeutic value
Literature	GENERIF_SUMMARY	critical pathophysiologic event in the evolution of HER2-amplified cancers is the loss of the input signals that normally drive TORC2 signaling, repositioning it under Akt dependency, and fundamentally altering the role of HER3
Literature	GENERIF_SUMMARY	Structural analysis of kinase-inhibitor complexes unraveled that HER2 T798M mutation induces significant steric hindrance to class I inhibitors, but can establish additional nonbonded interactions for class III inhibitors.
Literature	GENERIF_SUMMARY	loss of FOXO1 promotes GC growth and metastasis by upregulating HER2 expression.
Literature	GENERIF_SUMMARY	clinically sizeable 5% fraction of KRAS wild-type colorectal cancer patients was found to be ERBB2-positive according to the colorectal cancer-specific ERBB2 scoring criteria
Literature	GENERIF_SUMMARY	FcgammaRIIIA/FcgammaRIIA gene polymorphisms and HER-2 may have a role in antibody-dependent cellular cytotoxicity and clinical response to trastuzumab in breast cancer
Literature	GENERIF_SUMMARY	MicroRNA-21 links epithelial-to-mesenchymal transition and inflammatory signals to confer resistance to neoadjuvant trastuzumab and chemotherapy in HER2-positive breast cancer patients.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	we provide a wide-ranging overview of potential biomarkers capable of stratifying patients regarding their response to trastuzumab includING HER2 amplification--{REVIEW}
Literature	GENERIF_SUMMARY	To predict HER2 status precisely in gastric carcinoma resection specimens, at least 4 biopsy fragments containing tumor cells are required.
Literature	GENERIF_SUMMARY	HER2 gene rs1136201 and rs1058808 polymorphisms and haplotype C-T-G-G may be related to osteosarcoma susceptibility in Chinese Han population
Literature	GENERIF_SUMMARY	High levels of stromal tumor infiltrating lymphocytes were associated with lack of trastuzumab therapy benefit in early stage HER2-positive breast cancer.
Literature	GENERIF_SUMMARY	Results suggest reclassification of HER-2 proto-oncogene protein (cHER2+) tumors based on the spectrum of molecular breast cancer subtypes.
Literature	GENERIF_SUMMARY	Combined treatment with GEM and T-DM1 might confer a potent therapeutic modality against PDA as a result of GEM-mediated HER2 up-regulation.
Literature	GENERIF_SUMMARY	The KIT and PDGFRB mutations were predicted to be pathogenic using in silico analysis, whereas the ERBB2 mutation was predicted to be benign...the patient was treated with pazopanib and achieved a partial response that lasted for 7.5 months
Literature	GENERIF_SUMMARY	HER2 genetic alteration in non-small cell lung cancer is associated with the treatment outcome with tyrosine kinase inhibitors. [review]
Literature	GENERIF_SUMMARY	The reduction in Phosphorylated-HER2 (P-HER2) and P-EGFR in the ex vivo-treated biopsy demonstrated good correlation with the in vivo response at day 10.
Literature	GENERIF_SUMMARY	findings newly identify a physiological function of the NRG1-ErbB2-ErbB3 axis in trophoblast survival during human placental development.
Literature	GENERIF_SUMMARY	Results indicate that p110alpha primarily mediates PI3K/AKT signaling in HER2+ PTEN deficient breast tumors whereas p110beta conditionally mediates HER2/PI3K signaling in the absence of p110alpha.
Literature	GENERIF_SUMMARY	Our study indicates that apigenin induces apoptosis through inhibition of STAT3 signalling and could serve as a useful compound to prevent or treat HER2-overexpressing breast cancer.
Literature	GENERIF_SUMMARY	only lapatinib enhanced trastuzumab-mediated ADCC via the upregulation of HER2 expression in MPM cells, suggesting that sequential combination of lapatinib and trastuzumab may be a promising strategy for MPM treatment.
Literature	GENERIF_SUMMARY	HER2 Codon 655 (Ile/Val) Polymorphism is not associated with Breast Cancer.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	HER2 missense mutations have distinct effects on oncogenic signaling and migration.
Literature	GENERIF_SUMMARY	E2F3 silencing decreases Her2-positive mammary tumor growth by reducing percentage of cells undergoing mitosis.
Literature	GENERIF_SUMMARY	Data indicate that the combination with the HER-2 proto-oncogene protein (HER2) inhibitor lapatinib and the inhibition of mTOR serine-threonine kinase (mTor) affords the highest degree of synergy.
Literature	GENERIF_SUMMARY	In addition HER2 discordance was more likely amongst different tumour foci if these arose in distinct and separate areas of DCIS
Literature	GENERIF_SUMMARY	series confirms a high concordance rate of 95%, supporting that testing of primary tumours and metastases is equally valid and providing clinical rationale for the addition of anti-HER2 therapy in HER2+ disseminated disease
Literature	GENERIF_SUMMARY	Continued use of trastuzumab in HER2(+) PTEN(-) cells increased the frequency of cancer stem cells.
Literature	GENERIF_SUMMARY	indicate that gains of DLX4 and ERBB2 occur in South African breast cancer patients irrespectively of their race and factors known to influence prognosis
Literature	GENERIF_SUMMARY	Our data suggest that b2-AR may be a new useful biomarker for predicting prognosis in Her2-positive breast cancer and may also be a promising selective therapeutic target for the aggressive subtype of breast cancer.
Literature	GENERIF_SUMMARY	Our study is the first to demonstrate the associations of high nuclear Gli1 expression with resistance to trastuzumab-based NAT and subsequent worse prognosis in HER2-positive disease. These findings suggest that the nuclear Gli1 protein may be a novel target of NAT for HER2-positive breast cancer.
Literature	GENERIF_SUMMARY	For key genes ERBB2 and ESR1, we demonstrate that overexpression is linked to the production of highly modified and truncated splice variants in tumours, but not controls, correlated with tumour subtype
Literature	GENERIF_SUMMARY	Study demonstrated that NAMPT, VEGF and HER2 were overexpressed in breast tumors as well as in the serum and their circulating levels declined after tumor removal, suggesting a clinical application of this triplet as biomarker for breast cancer diagnosis.
Literature	GENERIF_SUMMARY	Data suggest that NDRG1 down-regulates expression and activation of HER1/EGFR, HER2/ERBB2, and HER3/ERBB3 in response to epidermal growth factor (EGF) ligand in pancreatic/colonic neoplasm cells.
Literature	GENERIF_SUMMARY	the consistency test results indicated the inter-observer agreement was more robust in MS images for HER2 , CK5/6 and ER than that in RGB images for HER2 , CK5/6 and ER .

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Left ventricular dysfunction was low in node negative ERBB2 breast cancer patients receiving weekly paclitaxel/trastuzumab.
Literature	GENERIF_SUMMARY	Data indicate that co-inhibition of FGFR1 and HER2 or PDGFRalpha led to enhanced drug responses.
Literature	GENERIF_SUMMARY	Estrogen receptor, progesterone receptor or human epidermal growth factor receptor 2 samples with high-methylation risk score had the worst 5-year survival rate and overall survival time.
Literature	GENERIF_SUMMARY	Analysis of patients with ER-/PgR+/HER2- breast cancers showed that this subtype exhibited an aggressive behavior compared to triple negative breast cancers.
Literature	GENERIF_SUMMARY	High expression of HER2 is associated with gastric cancer.
Literature	GENERIF_SUMMARY	The authors identified an alternative HER2 transmembrane domain dimerization coupled with self-association of membrane-embedded cytoplasmic juxtamembrane region, effectively inhibiting the receptor kinase activity.
Literature	GENERIF_SUMMARY	Report use of protein kinase inhibitors in treating HER2-positive breast cancer with progressive brain metastases.
Literature	GENERIF_SUMMARY	Pretreatment imaging of HER2 targeting, combined with early metabolic response assessment holds great promise for improving the understanding of tumor heterogeneity in mBC and for selecting patients who will/will not benefit from T-DM1.
Literature	GENERIF_SUMMARY	Cigarette smoke activates pulmonary epithelial HER2.
Literature	GENERIF_SUMMARY	we show an association of HER2-overexpression and PGC-1beta. PGC-1beta knockdown impairs HER2-overexpressing cells proliferation acting on ERRalpha signaling, metabolism, and redox balance.
Literature	GENERIF_SUMMARY	Our discovery of HER2 expression in RB may lead to innovative and targeted drug treatment options designed to spare the eye and preserve vision in RB patients.
Literature	GENERIF_SUMMARY	Report quantitative real-time polymerase chain reaction is an alternative method for the detection of HER-2 amplification in formalin-fixed paraffin-embedded breast cancer samples.
Literature	GENERIF_SUMMARY	Data show the clinical efficacy of lapatinib, a small-molecule receptor tyrosine-protein kinase erbB-2 (HER2) inhibitor, in HER2-positive breast cancer patients.
Literature	GENERIF_SUMMARY	Metformin inhibits breast tumor angiogenesis and suggest role of HIF-1alpha-VEGF signaling axis in mediating HER2-induced tumor angiogenesis.

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Literature	GENERIF_SUMMARY	In patients with advanced cancer of gastric carcinomas and gastroesophageal junction, HER2-neu overexpression was more associated with the intestinal cancer subtype.
Literature	GENERIF_SUMMARY	Data show that droplet digital PCR (ddPCR) has the advantage of automation and allows levels of epidermal growth factor receptor 2 (HER2) amplification to be easily evaluated in large numbers of samples to define HER2 status.
Literature	GENERIF_SUMMARY	Incidence of cardiotoxicity among cohort of Italian women treated with trastuzumab for HER2-positive MBC appeared higher than that reported in RCTs, particularly in elder patients.
Literature	GENERIF_SUMMARY	Blocked autophagy in response to inhibitors of HER2.
Literature	GENERIF_SUMMARY	variation within the codon 655 of HER2 does not alter the Breast Cancer risk in Caucasian Austrian women
Literature	GENERIF_SUMMARY	HER2 mutations are rare in HER2-positive Japanese breast cancer patients.
Literature	GENERIF_SUMMARY	MSM decreased the ability of STAT5b to bind the promoter of the HER2 gene.
Literature	GENERIF_SUMMARY	NRG1 promotes the proliferation and migration of coronary artery smooth muscle cells by enhancing the phosphorylation of ErbB2, ErbB3 and ErbB4.
Literature	GENERIF_SUMMARY	positive expression of p53 and Bcl-2 and amplification of c-erb-B2 gene were detected in transitional cell carcinoma and squamous cell carcinoma and they were correlated with urinary bladder cancer advanced tumor grades
Literature	GENERIF_SUMMARY	HER2 genomic amplification is associated with cetuximab-resistant colorectal cancer.
Literature	GENERIF_SUMMARY	This suggests that a high HER2 FISH ratio is a potential indicator for a high pathologic complete response rate and a better prognosis when patients are treated with neoadjuvant systemic treatment-trastuzumab .
Literature	GENERIF_SUMMARY	There was a non-significant trend toward increased ovarian cancer risk in Val/Val homozygosity of the codon 655 (Ile655Val) polymorphism(rs1136201), especially in women aged above 50 years
Literature	GENERIF_SUMMARY	PKG II inhibits EGFinduced activation of HER2 through binding with and causing threonine 686 phosphorylation of this oncogenic protein.
Literature	GENERIF_SUMMARY	Increased expression of HER2 was linked to poor outcome in lung cancer patients.
Literature	GENERIF_SUMMARY	Patients aged
Literature	GENERIF_SUMMARY	in HER2-overexpressing D492 cells, EGFR can behave as a tumor suppressor, by pushing the cells towards epithelial differentiation

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	A higher proportion of HER2-overexpressing cases experienced recurrence, but the difference was not significant
Literature	GENERIF_SUMMARY	loss of HER2-positive status can occur after neoadjuvant treatment in patients with primary HER2-positive breast cancer.
Literature	GENERIF_SUMMARY	obesity-associated systemic factors suppress ERbeta expression in breast cancer cells via a HER2-mediated pathway
Literature	GENERIF_SUMMARY	HER2 was overexpressed in colorectal cancer cell lines. Its expression level was negatively regulated by miR331-3p. HER2 knockdown inhibited cell proliferation and phosphorylation of Akt and ERK1/2.
Literature	GENERIF_SUMMARY	HER2 mutations are not associated with HER2 amplification, thus suggesting a distinct entity and therapeutic target. HER2-positive lung cancer may not be an adequate term, and patient cohorts for the study of HER2-targeted agents should be defined by the specific HER2 alteration present.
Literature	GENERIF_SUMMARY	an important role for MUCL1 in the proliferation of breast cancer cells
Literature	GENERIF_SUMMARY	PMCA2 interacts with HER2 in specific actin-rich membrane domains.
Literature	GENERIF_SUMMARY	HER2 gene amplification is associated with invasive breast carcinoma.
Literature	GENERIF_SUMMARY	High HER2 expression is associated with breast cancer.
Literature	GENERIF_SUMMARY	Has moderate activity in HER2-overexpressing tumors.
Literature	GENERIF_SUMMARY	Assessment of the HER2 status of biopsy tissues could predict the status of the whole tumor; however, a proportion of these cases may be discordant because of intratumoral heterogeneity
Literature	GENERIF_SUMMARY	High HER2 expression is associated with gastric cancer.
Literature	GENERIF_SUMMARY	The combination of MM-302 and trastuzumab induced synergistic antitumor activity in HER2-overexpressing xenograft models of breast and gastric cancer.
Literature	GENERIF_SUMMARY	Studies indicate that overexpression of the HER2 proto-oncogene protein (HER2) may be found without the amplification of the HER2 gene, and HER2 gene amplification can occur without overexpression of the HER2 protein.
Literature	GENERIF_SUMMARY	NRF2 regulates both HER2 and HER3 receptors to influence cellular responses to HER2 targeting.
Literature	GENERIF_SUMMARY	Report link between the Ala1170Pro SNP and HER2 positivity and between HER2 overexpression and loss of heterozygosity in breast cancer.

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Literature	GENERIF_SUMMARY	FGFR1 protein expression may be a biomarker of ER-positive/HER2-negative primary breast cancer with possible resistance to standard treatment, and may be a useful tool to identify more specific patients who would benefit from FGFR-1 targeted therapy.
Literature	GENERIF_SUMMARY	we show that WASF3 is present in the HER2 immunocomplex and suppression of WASF3 function leads to suppression of invasion even in the presence of HER2 expression.The engagement of WASF3 with the HER2/HER3 complex facilitates its phospho-activation and transcriptional upregulation, which is facilitated by HER2/HER3 activation of JAK/STAT signaling.
Literature	GENERIF_SUMMARY	demonstrated ERBB2 was a direct target of miR-375, over-expression of miR-375 decrease the both mRNA and protein expression of ERBB2 in human papillary thyroid carcinoma cells
Literature	GENERIF_SUMMARY	High HER2 amplification and expression is associated with breast cancer.
Literature	GENERIF_SUMMARY	Photoimmunotherapy in combination with 5-FU resulted in enhanced antitumor effects compared to PIT alone for HER2-expressing human gastric cancer in vitro and in vivo
Literature	GENERIF_SUMMARY	TargetPrint ER had a positive agreement of 98 %, and a negative agreement of 95 % with central pathology. Corresponding figures for PgR were 85 and 94 % and for HER-2 72 and 99 %. Agreement of mRNA versus central protein was not different when the same or a different portion of the tumor tissue was analyzed or when DCIS and/or normal tissue was included in the sample subjected to mRNA assays
Literature	GENERIF_SUMMARY	There was no association noted between the histologic grade or pathologic stage and positivity for either p53 or HER2/neu
Literature	GENERIF_SUMMARY	the MapQuanttrade mark assay, based on mRNA expression assay, provides an objective and quantitative assessment of Estrogen receptor, Progesterone receptor and HER2 status in invasive breast cancer.
Literature	GENERIF_SUMMARY	HER2 signaling promotes breast cancer cell proliferation through regulation of E2F1-driven DNA metabolism.
Literature	GENERIF_SUMMARY	The highest rates of erb-B2 expression in peripheral blood or tumor tissue were in the ER and PR negative and HER-2 positive group.
Literature	GENERIF_SUMMARY	HER2-neu is risk factor that can be a good prognostic and also predictive factor.
Literature	GENERIF_SUMMARY	In saliva the correlation between autoantibodies anti-MUC1 and anti-HER2 more important than in serum.
Literature	GENERIF_SUMMARY	Data provided evidences that HER2 overexpression is

Main Category	Category	Terms
		required for epithelial-mesenchymal transition and cell invasiveness in cisplatin-resistant gastric cancer cells.
Literature	GENERIF_SUMMARY	HER2 gene amplification is associated with breast cancer.
Literature	GENERIF_SUMMARY	We report that targeting HER3 and b1 integrin provides a particularly effective combination therapy approach for HER2-amplified cancers, surpassing the combination of HER2 and b1 integrin targeting, and evading some of the safety concerns associated with direct HER2-targeting.
Literature	GENERIF_SUMMARY	High HER2 copy number status is associated with gastric cancer.
Literature	GENERIF_SUMMARY	TIRF high-content imaging system simultaneously showed the expression pattern of EGFRs and EC50 value for CPT-induced apoptosis and necrosis in MCF-7, SK-BR-3 and JIMT-1 cancer cell lines
Literature	GENERIF_SUMMARY	while transcript and protein levels of EGFR and ErbB2 were up-regulated or unaffected, respectively, hepatitis c virus induced a substantial reduction of ErbB3 and ErbB4 expression.
Literature	GENERIF_SUMMARY	The study concluded that true HER2-amplified breast cancers are rare amongst BRCA1 mutation carriers and are less common in BRCA2 than background rates
Literature	GENERIF_SUMMARY	In this phase I/II study, 14 high-risk disease-free ovarian (OC) and breast cancer (BC) patients after completion of standard therapies were vaccinated with MUC1, ErbB2 and carcinoembryonic antigen (CEA) HLA-A2+-restricted peptides and Montanide.
Literature	GENERIF_SUMMARY	Our study has shown that implementation of the 2013 ASCO/CAP updated guideline has significant impact on HER2 classification for breast cancers with an equivocal HER2 IHC result and therefore increased the use of HER2-targeted therapy
Literature	GENERIF_SUMMARY	HER2 amplification is not associated with pancreatic ductal adenocarcinoma.
Literature	GENERIF_SUMMARY	TOM1L1 enhances ERBB2-induced cell invasiveness by promoting invadopodia formation and MT1-MMP trafficking to the plasma membrane.
Literature	GENERIF_SUMMARY	this is the first study showing an association of EGFR and ErbB2 serum levels with glioma more than a decade before diagnosis
Literature	GENERIF_SUMMARY	The TN and HER2(T-) subtypes predict higher rates of ipsilateral breast tumor recurrence(IBTR) and locoregional recurrence (LRR)after neoadjuvant chemotherapy (NAC) and breast-conserving therapy . A pCR is predictive of improved IBTR or LRR in TN subtype.
Literature	GENERIF_SUMMARY	In this trial, maintenance therapy with E + BEV in ER-positive, HER2-negative MBC patients with no evidence

Main Category	Category	Terms
		of progression after first-line T + BEV did not achieve longer PFS compared with continuation of T + BEV
Literature	GENERIF_SUMMARY	Our data demonstrate a critical role of Mcl-1 in Her2-positive BC cell survival under hypoxic conditions and provide the preclinical framework for the therapeutic use of novel Mcl-1- targeting agents to improve patient outcome in breast cancer .
Literature	GENERIF_SUMMARY	no clinical significance of serum levels in ovarian cancer
Literature	GENERIF_SUMMARY	In a transgenic model of breast cancer, Pten suppression using a tetracycline-regulatable shRNA cooperates with HER2/neu, leading to aggressive and metastatic disease with elevated signaling through PI3K and, surprisingly, the MAPK pathway.
Literature	GENERIF_SUMMARY	FASN, ErbB2-high-expressing SK-BR-3 cells displayed higher levels of glycolysis and migration than FASN, ErbB2-low-expressing MCF7 cells.
Literature	GENERIF_SUMMARY	ErbB2 signaling is elevated in a subset of patients with abiraterone-resistant prostate cancer.
Literature	GENERIF_SUMMARY	This study showed a significant grade of discordance of the HER2 status between primary tumors and disseminated tumor cells in the bone marrow of a relevant subgroup of patients.
Literature	GENERIF_SUMMARY	Results suggest that APIP stimulates cell proliferation via ERBB3, acting in conjunction with ERBB2, in gastric cancer cells.
Literature	GENERIF_SUMMARY	Fulvestrant plus palbociclib was associated with significant and consistent improvement in progression-free survival compared with fulvestrant plus placebo, irrespective of the degree of endocrine resistance, hormone-receptor expression level, and PIK3CA mutational status. The combination could be considered as a therapeutic option for patients with HER2-negative breast cancer
Literature	GENERIF_SUMMARY	In trials of HER2-targeted agents in HER2+ metastatic breast cancer , Progression-free survival (PFS) moderately correlates with OS at the individual level and treatment effects on PFS correlate moderately with those on overall mortality, providing only modest support for considering PFS as a surrogate. PFS does not completely substitute for OS in this setting
Literature	GENERIF_SUMMARY	ERBB2 Exon 20 Insertion-Mutation is associated with response to chemotherapy in Lung Adenocarcinomas.
Literature	GENERIF_SUMMARY	(18)F-FDG PET/CT may be useful for predicting the HER2 status of gastric cancer and for determining the therapeutic strategy
Literature	GENERIF_SUMMARY	HSP70 affects ERBB2 expression and ERBB2-mediated drug-resistance in breast cancer cells.
Literature	GENERIF_SUMMARY	The use of HER2 gene copy number, and the so-called

Main Category	Category	Terms
		chromosome 17 polysomy was ascertained to cause a statistically significant increase in HER2-positive cases from 25 to 42.
Literature	GENERIF_SUMMARY	The findings in the current study also call into question the role of intensive cardiac monitoring among patients treated with anti-HER2 therapy in the absence of anthracyclines
Literature	GENERIF_SUMMARY	MIF and cerbB2 were correlated with the occurrence and the development of endometrial cancer
Literature	GENERIF_SUMMARY	The AIC value for this model was 3333.06, while for a model using 10% or higher as the cutoff for ER positivity, it was 3333.38, indicating that the model fits were nearly identical. The improvement in fit of the model when ERBB2 status was added was highly significant, with 5-year disease-specific survival estimates ranging from 48% to 99% (P
Literature	GENERIF_SUMMARY	High HER2 expression is associated with non-small cell lung cancer.
Literature	GENERIF_SUMMARY	beta3-alphaC deletions are activating mutations in HER2. HER2 deltaLRENT is an activating deletion primarily through increased phosphorylation of its dimerization partner. These deletions are resistant to the alphaC 'Out' inhibitor lapatinib.
Literature	GENERIF_SUMMARY	The Her2-neu score 0 might be considered as an innovative prognostic factor for patients with TNBC indicating poor clinical outcome
Literature	GENERIF_SUMMARY	HER2 insertion could define a distinct subset of NSCLC, which had a higher prevalence among females, nonsmokers, and adenocarcinoma. HER2 should be in the clinical genotyping of lung cancer, so patients may benefit from HER2-targeted therapy.
Literature	GENERIF_SUMMARY	Trastuzumab treatment improves patients' survival regardless of HR status and should be offered to all HER2+ patients.
Literature	GENERIF_SUMMARY	We investigate the impact of central histology (n = 772), immunohistochemistry for intrinsic subtyping and IHC4, and dichotomous (GG) or continuous (GGI) genomic grade (n = 472) on patient outcome and benefit from taxane chemotherapy, focusing on HR+/HER2(HR
Literature	GENERIF_SUMMARY	High HER2 expression is associated with gastric cancer.
Literature	GENERIF_SUMMARY	The overall intention-to-treat analysis demonstrates no benefit to vaccination. However, the results confirm that the vaccine is safe and suggest that vaccination may have clinical benefit in patients with low HER2-expressing tumors, specifically TNBC.
Literature	GENERIF_SUMMARY	Our results suggest that the addition of bevacizumab to neoadjuvant and adjuvant chemotherapy does not provide clinical benefit to patients with non-metastatic HER2-negative inflammatory breast cancer.

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Literature	GENERIF_SUMMARY	measurements comprising the receptors EGFR, ERBB3 (HER3), and the cyclin-dependent kinase inhibitor p27 (CDKN1B) was found to accurately predict dependence on PI3K/AKT vs. MAPK/ERK signaling axes in HER2+ cancers
Literature	GENERIF_SUMMARY	HER2-positive tumors in our study had a lower Ki-67 index and were p53-positive.
Literature	GENERIF_SUMMARY	Our findings define 3-OST3A as a novel regulator of breast cancer pathogenicity, displaying tumor-suppressive or oncogenic activities in a cell- and tumor-dependent context, and demonstrate the clinical value of the HS-O-sulfotransferase 3-OST3A as a prognostic marker in HER2+ patients.
Literature	GENERIF_SUMMARY	septins protect ErbB2 from ubiquitylation, endocytosis and lysosomal degradation. Septin oligomerization regulates persistent expression of ErbB2/HER2 in gastric cancer cells.
Literature	GENERIF_SUMMARY	Findings show for this first time that ATG9A loss in trastuzumab resistant cells allowed Her2 to escape from lysosomal targeted degradation through K63 poly-ubiquitination via c-Cbl.
Literature	GENERIF_SUMMARY	This phase Ib/IIa study assessed the feasibility of T-DM1 + docetaxel in patients with human epidermal growth factor receptor 2 (HER2)-positive metastatic breast cancer (MBC) and T-DM1 + docetaxel +/- pertuzumab in patients with HER2-positive locally advanced breast cancer (LABC).
Literature	GENERIF_SUMMARY	HER2 amplification, age, and grade were not significantly different between ER-/PR+ and double receptor negative groups
Literature	GENERIF_SUMMARY	our findings reveal an intriguing miR-155-ErbB2 context in regulating the malignant transformation of breast epithelial cells, and thereby indicate a novel mode of action for miR-155 in ErbB2-positive breast cancer.
Literature	GENERIF_SUMMARY	Association of HER2 single nucleotide polymorphisms with the breast cancer risk in the Bangladeshi population
Literature	GENERIF_SUMMARY	HER Family Protein Expression in a Greek Population with Gastric Cancer. A Retrospective Hellenic Cooperative Oncology Group Study
Literature	GENERIF_SUMMARY	Expression and Prognostic Significance of Human Epidermal Growth Factor Receptors 1, 2 and 3 in Periampullary Adenocarcinoma
Literature	GENERIF_SUMMARY	Our meta-analysis study strongly suggests that HER-2 expression levels are clearly correlated with the clinicopathological features in colorectal cancers(CRC) therefore, HER-2 may be a potential biomarker for diagnosis and prognosis of CRC.
Literature	GENERIF_SUMMARY	study provides evidence that working at night may increase risk of ER, PR and HER2-positive subtypes of

Main Category	Category	Terms
		breast cancer particularly among premenopausal women
Literature	GENERIF_SUMMARY	A low uEGF/Cr ratio is associated with rapid decline in renal function and incident chronic kidney disease in diabetic patients.
Literature	GENERIF_SUMMARY	increased expression of stromal colXalpha1 and low TILs correlate with poor pathologic response in ER+/HER2+ breast tumors. Further studies are needed to confirm their predictive value and impact on long-term outcomes, and to determine whether this collagen exerts a protective effect on the cancer cells or simply reflects other factors within the tumor microenvironment
Literature	GENERIF_SUMMARY	the proportion of HER2 positive cases is lower than that reported in older literature and that pathological characteristics differ between HER2 3+ and HER2 2+/Her2 amplified cases
Literature	GENERIF_SUMMARY	we have identified a novel mechanism of ErbB2-mediated mechanism of anoikis resistance of ErbB2-overproducing breast epithelial cells. This mechanism allows such cells to grow without adhesion to the ECM and is driven by ErbB2-induced activation of Mek, subsequent EGFR upregulation and further EGFR-dependent Perp loss.
Literature	GENERIF_SUMMARY	Report localized therapy of ductal carcinoma in situ in mice using the immunoconjugate 225Ac linked-trastuzumab, which targets Her2, delivered through the intraductal route.
Literature	GENERIF_SUMMARY	Anti-HER-2 and hormonal therapy, except tamoxifen, have been underutilized in Korean male breast cancer patients compared to female breast cancer patients.
Literature	GENERIF_SUMMARY	The T-allele of ABCB1-C3435T gene polymorphism has demonstrated to be a good prognosis factor for resistance, progression free survival and overall survival in HER2-positive Breast Cancer patients.
Literature	GENERIF_SUMMARY	Results suggest that our approach could provide a platform to develop nano-diagnostic agents or nano-therapeutic agents for early detection and treatment of HER-2-positive cancers.
Literature	GENERIF_SUMMARY	the ErbB2DeltaEx16 tumors exhibit distinct signaling and gene expression profiles that correlate with activation of number of key transcription factors implicated in breast cancer metastasis and cancer stem cell renewal.
Literature	GENERIF_SUMMARY	Our findings underline the survival benefit achieved by eribulin used according to European Union label in the overall metastatic breast cancer population and in various subgroups of interest, including patients with HER2-negative and triple-negative disease
Literature	GENERIF_SUMMARY	Overall PIK3CA mutant/HER2+ tumours had significantly lower pCR rates compared with wild-type

Main Category	Category	Terms
		tumours, however mainly confined to the HR+/PIK3CA mutant population. No definite conclusions can be drawn regarding survival
Literature	GENERIF_SUMMARY	The predictive and prognostic impact of TILs in HER2-positive breast cancer.
Literature	GENERIF_SUMMARY	MiR-193a-5p/ERBB2 have roles in response to chemoradiation therapy of esophageal squamous cell carcinoma
Literature	GENERIF_SUMMARY	Disufiram/Cu induces apoptosis and eliminates cancer stem-like cells via the suppression of HER2/Akt signaling.
Literature	GENERIF_SUMMARY	Infiltrating lobular carcinoma (ILC)of the breast represents a biologically distinct subset of breast cancer, a biology defined by specific genetic aberrations in E-cadherin, the high prevalence of ER-positive disease, the relatively low frequency of HER2-positive disease, and specific mutational events revealed by deep sequencing of ILC genomes
Literature	GENERIF_SUMMARY	In this update, we focus on the current genomic data, targeted therapies including immune system modulators, and expand on HER2/neu testing and the use of agents against this target. Several other facets of GAC and its therapy are not to be included in this review but have been discussed elsewhere.
Literature	GENERIF_SUMMARY	or patients with non-small cell lung cancer with a EGFR mutation and an anaplastic lymphoma kinase mutation(ALK) with brain metastases, several agents have shown intracranial activity. Lapatinib has been tested for patients with brain metastases from breast cancer harboring a HER2 mutation. molecular targeted therapies have shown efficacy in BRAF-positive melanoma brain metastasis
Literature	GENERIF_SUMMARY	Last but not least, we briefly review a new paradigm of emerging approaches that involve the host immune response, HER2 breast cancer vaccines, and other immune strategies, including immune checkpoint inhibition.
Literature	GENERIF_SUMMARY	Increased copy number of HER2 may confer an increased risk of death from breast cancer despite negative HER2 status by ratio
Literature	GENERIF_SUMMARY	QSAR models have been built with activity data and binding conformations to begin our work in this paper as well as to provide a reliable tool for reasonable design of EGFR/HER-2 inhibitors in future.
Literature	GENERIF_SUMMARY	p95HER2(truncated HER2) was correlated with trastuzumab resistance, but it was not an independent factor of trastuzumab resistance.
Literature	GENERIF_SUMMARY	our results demonstrate for the first time that PA is a novel inhibitor of HER2 and CXCR4 expression via kinase pathways and may play a critical role in determining the metastatic potential of breast cancer

Main Category	Category	Terms
		cells
Literature	GENERIF_SUMMARY	No difference was observed in the HER-2-overexpression meningioma cells with the inhibition of JNK.
Literature	GENERIF_SUMMARY	HER2 mutation is associated with recurrent and metastatic breast cancer.
Literature	GENERIF_SUMMARY	HER2 reduces the radiosensitivity of breast cancer by activating Fak in vitro and in vivo.
Literature	GENERIF_SUMMARY	hPEPD-G278D is a dual inhibitor of ErbB1 and ErbB2 and selectively targets cancer cells overexpressing ErbB1 and/or ErbB2
Literature	GENERIF_SUMMARY	HER2 status is in good overall concordance between primary tumors and the corresponding metastatic sites. As there was a discrepancy in a minority of cases, HER2 status should be evaluated at both the primary and the metastatic sites, whenever possible.
Literature	GENERIF_SUMMARY	The current study reveals a significant association between cancer susceptibility and the HER-2 Ile655Val polymorphism in all genetic models. However, in Asian ethnic group, Ile655Val polymorphism lacked a significant association with cancer. [review, meta-analysis]
Literature	GENERIF_SUMMARY	Alteration of HER-2 was not detected in ovarian serous neoplasms; however, in mucinous carcinoma, HER-2 amplification and overexpression occur.
Literature	GENERIF_SUMMARY	Targeting both HER2 and HER3 resulted in an improved treatment effects on HER2-positive gastric cancer.
Literature	GENERIF_SUMMARY	results from this study demonstrate that MUC13 functionally interacts and activates HER2 at p1248 in PDAC cells, leading to stimulation of HER2 signaling cascade, including ERK1/2, FAK, AKT and PAK1 as well as regulation of the growth, cytoskeleton remodeling and motility, invasion of PDAC cells-all collectively contributing to PDAC progression.
Literature	GENERIF_SUMMARY	The endometrial tissue showed significantly lower levels of miR-200b, miR-15a-5p, miR-19b-1-5p, miR-146a-5p, and miR-200c, and higher levels of miR-16-5p, miR-106b-5p, and miR-145-5p. VEGFA was significantly upregulated, whereas EGFR2, PTEN, and CXCR4 were markedly downregulated, in the endometriotic tissues compared to that in the normal endometrial tissues
Literature	GENERIF_SUMMARY	This study used complementary quantitative gel-based and gel-free proteomic approaches to find a panel of urinary protein markers that could discriminate HER2 enriched (HE) subtype breast cancer from the healthy controls.
Literature	GENERIF_SUMMARY	High HER2 expression is associated with liver metastasis in gastric cancer.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Oncogenic mutations in Her-2/neu or k-ras had no association with the severity of endometrial cancer.
Literature	GENERIF_SUMMARY	Results provide evidence that PTPRO inhibits ERBB2-driven breast cancer through dephosphorylation leading to dual effects of ERBB2 signaling suppression and endosomal internalization of ERBB2.
Literature	GENERIF_SUMMARY	In breast cancer patients treated with trastuzumab, the germline Ala1170Pro homozygous ERBB2 genotype was associated with a lower likelihood of the occurrence of a cardiac event compared with Pro/Pro + Ala/Pro genotypes.
Literature	GENERIF_SUMMARY	Breast cancer patients with HER2 equivocal status have no significantly higher risk of recurrence.
Literature	GENERIF_SUMMARY	Cetuximab treatment induced HER3 activation and HER2/HER3 dimerization in head and neck squamous cell carcinoma cell lines.
Literature	GENERIF_SUMMARY	Importantly, TPL exhibited a far greater antitumor effect on ErbB2-overexpressing gastric cancer cell line than trastuzumab plus pertuzumab, suggesting that it may be a promising agent for the treatment of gastric cancer.
Literature	GENERIF_SUMMARY	6 patients with metastatic HER2 positive lesions were found HER2 negative in primary cancers.
Literature	GENERIF_SUMMARY	Kataegis loci are enriched in high-grade, HER2(+) tumors in patients diagnosed with breast cancer.
Literature	GENERIF_SUMMARY	An interdisciplinary group of German experts has taken on the challenges of HER2 testing in gastric cancer as an opportunity to address essential aspects and questions on the practical use of HER2 testing in this indication from the perspective of pathologists and therapists.
Literature	GENERIF_SUMMARY	showed a high concordance with immunohistochemistry/in situ hybridization of 95 % (kappa 0.81) for estrogen receptor, 81 % (kappa 0.56) for progesterone receptor, and 94 % (kappa 0.76) for HER2
Literature	GENERIF_SUMMARY	FcgammaRIIIa V allele-restricted pathological complete response benefit from neoadjuvant trastuzumab plus lapatinib in HER2+ breast cancer.
Literature	GENERIF_SUMMARY	characterized the specific interactome of the epidermal growth factor receptor (EGFR) family member ERBB2 when in the form of a homodimer or when in the form of a heterodimer with either EGFR or ERBB3.
Literature	GENERIF_SUMMARY	Human epidermal growth factor receptor 2 amplification is significantly associated with cell differentiation of the malignant component in ICXPA [salivary invasive carcinoma ex pleomorphic adenoma] and it implies an unfavourable prognosis.
Literature	GENERIF_SUMMARY	Review of the role of HER-2 mutation in non-small cell

Main Category	Category	Terms
		lung carcinoma development, diagnosis, and treatment.
Literature	GENERIF_SUMMARY	this population-based case-control study verified association of the C allele of rs2910164.
Literature	GENERIF_SUMMARY	Data show that inhibition of S100 calcium binding protein P (S100P) results in reversing trastuzumab resistance (TzR) in proto-oncogene protein HER-2 (HER2)-driven breast cancers.
Literature	GENERIF_SUMMARY	Based on these findings, HER2 screening using qRT-PCR technique should also be incorporated for molecular subtyping of breast cancer. Screening of HER2 along with other biomarkers will impart a significant role towards disease prognosis.
Literature	GENERIF_SUMMARY	Although, we obtained a similar effect for high HER2 gene copy numbers, this provides only an indirect speculation and not a proof that high HER2/CEP17 ratios may induce HER2 resistance
Literature	GENERIF_SUMMARY	HER2 and NSE serum levels were independent prognostic factors in patients with brain metastases in breast cancer. Serum HER2 and MMP-9 appear to be associated with BM in metastatic BC patients. Their predictive value for BM still needs to be evaluated in further prospective studies.
Literature	GENERIF_SUMMARY	This concise handbook provides oncologists and other healthcare providers with crucial updates in the field, including an updated review of the current understanding of the biology of the HER2 pathway, an overview of HER2-testing, and evidence-based discussions of available and emerging HER2-targeted treatment options
Literature	GENERIF_SUMMARY	Expression of p16 may be used as a prognostic factor of overall survival and stage of ovarian carcinoma, while HER2 expression may not be used as a prognostic factor of overall survival.
Literature	GENERIF_SUMMARY	Dual targeting and enhanced cytotoxicity to HER2-overexpressing breast and stomach tumors by immunoapoptotin-armored mesenchymal stem cells.
Literature	GENERIF_SUMMARY	Delta16HER2 splice variant is commonly coexpressed with HER2 and identified as a clinically important HER2 molecular alteration promoting aggressive metastatic breast cancer. Dacomitinib, an irreversible small molecule pan-HER inhibitor, was able to completely suppress Delta16HER2-driven breast carcinogenesis.
Literature	GENERIF_SUMMARY	HER2 is over expressed in micropapillary urothelial carcinoma of the bladder.
Literature	GENERIF_SUMMARY	In this study, both tumor-related and immune-related features contribute to the modulation of pCR after neoadjuvant chemotherapy plus anti-HER2 agents. Immune signatures rather than TILs added significant prediction of pCR beyond PAM50 intrinsic subtypes
Literature	GENERIF_SUMMARY	this mulinational study describes HER2 status in gastric

Main Category	Category	Terms
		and gastroesophageal junction cancer
Literature	GENERIF_SUMMARY	When neoadjuvant therapy with letrozole and lapatinib is given for Asian postmenopausal ER+HER2+ breast cancer, tumor-infiltrating lymphocytes, change of estrogen receptor expression is to be considered potential biomarkers in these patients
Literature	GENERIF_SUMMARY	Data show that epidermal growth factor receptor 2 (HER2) single-point substations such as D769H and D769Y as well as the gatekeeper mutation T798 M were predicted to cause strong resistance for an array of tyrosine kinase inhibitors (TKIs) by reshaping the geometric feature and physiochemical property of the active site.
Literature	GENERIF_SUMMARY	relevant genomic aberrations such as mutations in the hotspot regions of exon 9 and 20 of the PIK3CA gene can be detected in single circulating tumor cells and might provide insights into mechanisms of resistance to HER2-targeted therapies.
Literature	GENERIF_SUMMARY	c-Met overexpression, HER-2 gene amplification, and SPTBN1-ALK gene fusion can coexist in lung adenocarcinoma and may become a potential biomarker of cancer refractory to crizotinib, chemotherapy, and radiotherapy as well as of a relatively poor prognosis.
Literature	GENERIF_SUMMARY	The conjugation of benzylguanine-modified auristatin F to EGFR-specific 425(scFv)-SNAP and HER2-specific alphaHER2(scFv)-SNAP resulted in two potent recombinant antibody-drug conjugates that specifically killed breast cancer cell lines by inducing apoptosis when applied at nanomolar concentrations.
Literature	GENERIF_SUMMARY	This trial did not clearly demonstrate that TC is non-inferior to dose-dense FEC --> D. However, 3-year DFS rates were excellent in both arms for women with node-positive, HER2-negative early breast cancer.
Literature	GENERIF_SUMMARY	Innon-muscle-invasive bladder cancer expression of ESR1, ERBB2 and MKI67 are significantly different between stage and grade
Literature	GENERIF_SUMMARY	EHF is a novel functional oncogene in thyroid cancer by transcriptionally regulating HER2 and HER3, and may represent a potential therapeutic target for this cancer
Literature	GENERIF_SUMMARY	The main significance of this work is the discovery of EPO as a novel ligand for the HER2 receptor. Following HER2 activation, EPO induces activation of FAK and subsequent activation of beta1-integrin, via inside-out signaling. This complex results in downstream activation of ERK1/2 and a sustained up regulation of both MUC4 and the HER2 receptor
Literature	GENERIF_SUMMARY	Her2 oncogene transformation has a role in enhancing 5-aminolevulinic acid-mediated protoporphyrin IX production and photodynamic therapy response
Literature	GENERIF_SUMMARY	An interdisciplinary German expert group took the

Main Category	Category	Terms
		challenges of HER2 testing in gastric cancer as an opportunity to address essential aspects and questions for the practical use of HER2 testing in this indication from the perspective of pathologists and therapists.
Literature	GENERIF_SUMMARY	Phos-tag Western blot analysis revealed that TPA induced phosphorylation of ErbB2 in an ERK-dependent manner.
Literature	GENERIF_SUMMARY	Study showed that 25 % of patients with bladder transitional cell carcinoma has Her2/neu over-expression. This Her2/neu (over-expression/ amplification) status was significantly associated with disease aggressiveness and poor outcome.
Literature	GENERIF_SUMMARY	the antibody-sialidase conjugate desialylated tumor cells in a HER2-dependent manner, reduced binding by natural killer (NK) cell inhibitory sialic acid-binding Ig-like lectin (Siglec) receptors, and enhanced binding to the NK-activating receptor natural killer group 2D (NKG2D).
Literature	GENERIF_SUMMARY	HER2 loss was identified as a mechanism of acquired resistance to trastuzumab and chemotherapy in advanced HER2-positive gastric or GEJ cancer.
Literature	GENERIF_SUMMARY	In the CNIO-BR-003/GEICAM/2010-10 study, we were able to show a predictive role of 18F-FMISO-PET in HER2-negative early breast cancer treated with nintedanib, detecting the patients that do not experience benefit from it based on baseline tumor oxygenation levels
Literature	GENERIF_SUMMARY	Neratinib treatment leads to HSP90 release from ErbB2 and its subsequent ubiquitylation and endocytic degradation in breast cancer cells.
Literature	GENERIF_SUMMARY	the optimal discrimination cut-off for each cytokine
Literature	GENERIF_SUMMARY	These findings suggest that early-stage morphological alterations of HER2-positive BC cells during cancer progression can occur in a physical and signalling-independent manner.
Literature	GENERIF_SUMMARY	Survival analyses showed no significant prognostic impact of ERBB2 mutations. Our study demonstrates that high grade ERBB2-negative ILC display a high frequency of ERBB2 mutations, and should be subjected to systematic genetic screening.
Literature	GENERIF_SUMMARY	High HER2 expression is associated with drug resistance in metastatic breast cancer.
Literature	GENERIF_SUMMARY	this is the first work to demonstrate 3'UTR-mediated NBCn1 regulation, shows that p95HER2 regulates NBCn1 expression at multiple levels, and substantiates the central position of p95HER2-NBCn1 signaling in breast cancer.
Literature	GENERIF_SUMMARY	Data indicate high TP53 mutation rates in triple negative breast cancer (TNBC) and HER2-positive breast cancer, but Mutations did not predict the

Main Category	Category	Terms
		response to an intense neoadjuvant chemotherapy in these two molecular breast cancer subtypes.
Literature	GENERIF_SUMMARY	Report xenograft tumor samples as an option for quality control in HER2 detection using fluorescence in situ hybridization and immunohistochemistry.
Literature	GENERIF_SUMMARY	Expression of HIF-1alpha was more often noted in invasive HER2+ cancers characterized by high degree of aggressiveness and poorer prognosis
Literature	GENERIF_SUMMARY	Her-2, N-ras and Nf1 have roles in brain onocogenesis
Literature	GENERIF_SUMMARY	HER2 amplification in breast cancer can be detected by tissue microarray and FISH in core biopsies
Literature	GENERIF_SUMMARY	HER2 overexpression or activation was detected by the PRO Onc assay in 22 % of HER2-negative patients with CTCs. However, HER2-targeted therapy was not effective in such patients. FISH and IHC staining remain the standards for HER2 determination.
Literature	GENERIF_SUMMARY	Following specific binding to and internalization into HER2-overexpressing tumor cells, the e23sFv-Fdt-casp6 protein induced tumor cell apoptosis and inhibited the proliferation of HER2-overexpressing A172 and U251MG cells in vitro, but not in U87MG cells with undetectable HER2
Literature	GENERIF_SUMMARY	the phosphorylation levels of ErbB2, ErbB3 and Akt and the protein levels of cyclin D1 were decreased by lapatinib treatment of HSC3, HSC4 and Ca9-22 cells
Literature	GENERIF_SUMMARY	our studies directed toward understanding the role of the d16HER2 variant in development and tumorigenesis of HER2-positive BC shed preliminary insights into the importance of this isoform in governing aggressiveness/progression of HER2-positive BC.
Literature	GENERIF_SUMMARY	A role for Her2/neu in identifying high-risk patients for malignant transformation.
Literature	GENERIF_SUMMARY	A trend of decreasing Her2/neu expression with increasing grade of tumor was observed.
Literature	GENERIF_SUMMARY	analysis of HER2 aberrations and heterogeneity in cancers of the digestive system [review]
Literature	GENERIF_SUMMARY	Findings suggest a starting point towards the development of epidermal growth factor receptor 2 (HER2) inhibitors using well-validated computational techniques.
Literature	GENERIF_SUMMARY	We aimed to investigate the prevalence, clinicopathologic characteristics, and functional relevance of novel HER2 mutations in breast cancer
Literature	GENERIF_SUMMARY	HER2/ECD is a prognosis factor that is valuable in evaluating the neoadjuvant treatment efficiency. HER2/ECD also appears to be a helpful surveillance biomarker for the early diagnosis of relapses and to predict the fate of metastases. This study brings evidences to

Main Category	Category	Terms
		support the use of HER2/ECD in the management of HER2-positive breast cancer.
Literature	GENERIF_SUMMARY	Biosensing techniques identify a subset of breast cancers with normal HER2 receptor levels with abnormally high levels of HER2 signaling.
Literature	GENERIF_SUMMARY	HER2-FASN lipogenic axis delineates a group of breast cancer patients that might benefit from treatment with therapeutic regimens containing FASN inhibitors
Literature	GENERIF_SUMMARY	Molecular inversion probe array analysis facilitates precise ERBB2 copy number detection in breast cancer tissues.
Literature	GENERIF_SUMMARY	ERCC1-SNP in combination with mRNA ERCC1, DPYD, and ERBB2 from pretherapeutic endoscopic biopsies can predict minor response to chemoradiation, as a basis for individualized therapy of advanced esophageal cancer.
Literature	GENERIF_SUMMARY	The lymph node status was associated with clinical stage, tumour grade, tumour histology and HER2/neu status.
Literature	GENERIF_SUMMARY	The results suggest that elevated serum HER2 level was associated with a clinicopathological aggressive phenotype of breast carcinoma and was related to tissue HER2 overexpression. Therefore, serum HER2 may be useful for monitoring the course of the disease and response to treatment.
Literature	GENERIF_SUMMARY	HER2 expression is associated with a significant rise of metastatic potential of breast cancer cells and could be a strong indicator of regional and distant metastasis of breast cancer.
Literature	GENERIF_SUMMARY	AZD9291 is the first and only FDA approved third-generation epidermal growth factor receptor (EGFR) tyrosine kinase inhibitors for T790M-positive advanced non-small cell patients. The resistance to AZD9291 arises after therapy. The mechanisms of resistance to third-generation inhibitors reported to date include the EGFR C797S mutation, EGFR L718Q mutation, and amplifications of HER-2, MET, or ERBB2. [review]
Literature	GENERIF_SUMMARY	Recommend performing the HER2 tests on gynecologic carcinomas of clear cell, serous, and undifferentiated types particularly when they arise in the endometrium.
Literature	GENERIF_SUMMARY	The dipole potential exerts significant effects on the ligand binding, clustering and signaling of ErbB2 protein.
Literature	GENERIF_SUMMARY	Findings suggest that the association of HER2 amplification with BRAF(V600E) mutation and telomere shortening may represent a marker of tumor aggressiveness, and, in refractory thyroid cancer, may warrant exploration as a site for targeted therapy.
Literature	GENERIF_SUMMARY	RAS mutations and HER2/MET amplification were the most frequently detected resistance mechanisms in

Main Category	Category	Terms
		both tissue and blood sample analysis. On the other hand, BRAF and EGFR ectodomain mutations were much rarer.The combined analysis of tissue and blood (ctDNA) results highlights the complexity of clonal evolution triggered by EGFR blockade
Literature	GENERIF_SUMMARY	describe a methodology that allows the analysis of ErbB-2-mediated plexin phosphorylation and signaling
Literature	GENERIF_SUMMARY	There were significant differences in the distribution of Ile655Val (p = 0.03) and Ala1170Pro (p = 0.01) genotypes between normal breast and breast tumor tissues.
Literature	GENERIF_SUMMARY	Mutations in HER2 gene is associated with breast cancer.
Literature	GENERIF_SUMMARY	this study showed that HER-2 mutation was not correlated to HER-2 protein overexpression, and existing follow-up data did not show a correlation between HER-2 mutation with DFS or OS.
Literature	GENERIF_SUMMARY	High HER2 expression is associated with Lung Cancer.
Literature	GENERIF_SUMMARY	our results indicate that EGFR promotes survival of prostate tumor-initiating cells and circulating tumor cells that metastasize to bone, whereas HER2 supports the growth of prostate cancer cells once they are established at metastatic sites
Literature	GENERIF_SUMMARY	Very low density (VLD) and HER2 positivity were prognostically significant factors independent of the NPI. Furthermore, the incorporation of VLD and HER2 to the NPI served to enhance its accuracy, thus offering a readily available and more accurate method for the evaluation of patient prognosis.
Literature	GENERIF_SUMMARY	HER2 is commonly amplified in UGI adenocarcinomas with amplification highly correlated to overexpression, and HER2 amplification levels correlated with PFS.
Literature	GENERIF_SUMMARY	Intralobular breast carcinoma patients with tumors carrying both CDH1 and ERBB2 mutations have a worse prognosis.
Literature	GENERIF_SUMMARY	miR-34a, like ErbB2, might be a diagnostic target in breast cancer.
Literature	GENERIF_SUMMARY	CRISPR-mediated targeting of HER2 inhibits breast cancer cell proliferation through a dominant negative mutation.
Literature	GENERIF_SUMMARY	K17 is expressed in a subset of triple-negative breast cancers, and is a marker of poor prognosis in patients with advanced stage and ER(-)/HER2(-) breast cancer.
Literature	GENERIF_SUMMARY	Parity and early [not later] age at first live birth (AFLB) showed a protective association with ER and the human epidermal growth factor receptor-2 (ER+/-/HER2+/-) and risk association with ER-/HER2- cancers. Associations of reproductive history with breast cancer risk varied among Danish women by ER+/- and ER+/-/

Main Category	Category	Terms
		HER2+/- expression and age-at-diagnosis, consistent with receptor-specific and age-related etiological het...
Literature	GENERIF_SUMMARY	tumor-promoting role for CCL2 acting through CCR2 on the tumor microenvironment
Literature	GENERIF_SUMMARY	Our findings indicate a significant strong association of cytoplasmic Her-2/neu expression with low grades and membranous Her-2/neu expression with high grades of colorectal cancer.
Literature	GENERIF_SUMMARY	HER2 mutations represent a distinct subset of NSCLC. NGS showed that HER2 mutations commonly co-existed with other driver genes.
Literature	GENERIF_SUMMARY	A total of 160 cases of mammary carcinoma were classified broadly according to their ER, PR, and HER2 expression into luminal.
Literature	GENERIF_SUMMARY	These results suggest that ANO1 Cl(-) channels may function as a transcriptional regulator of HER2, and ANO1 inhibitors have potential in the treatment of BCA patients with resistance to HER2-targeted therapy.
Literature	GENERIF_SUMMARY	High mRNA expression levels of delta-HER2 were associated with pSRC and increased proliferation of tumor cells. A poor prognosis may be predicted by the increase in delta/wt ratio in HER2-overexpressing breast cancer.
Literature	GENERIF_SUMMARY	Use molecular dynamics and pharmacophore-based screening to identify HER2 inhibitors active against gastric cancer derived from Syzygium alternifolium.
Literature	GENERIF_SUMMARY	HSP90 inhibition fosters ERBB2 polyubiquitination and proteasome-dependent generation of a non-ubiquitinated and inactive p116-ERBB2 form in endosomes which is trafficked to lysosomes.
Literature	GENERIF_SUMMARY	NRG1/ErbB2/Src/PTK2 signaling pathway may be a novel regulator of keloid fibroblast migration.
Literature	GENERIF_SUMMARY	Our study has provided new evidence that anti-HER2 treatment has a significant impact on HER2 loss. Far more importantly, the loss of HER2 amplification could identify non-pCR patients with high risk of disease relapse, which might help in tailoring following systemic treatment.
Literature	GENERIF_SUMMARY	These results provide a novel mechanism underlying the simvastatin-induced inhibition of tumor angiogenesis through regulating HER2/VEGF axis.
Literature	GENERIF_SUMMARY	Data indicate that the whole routine of epidermal growth factor receptor 2 (HER2) gene microfluidics-assisted FISH (MA-FISH) analysis on clinical breast cancer specimens could be performed in 1 day instead of the standard 2 days, thus improving the throughput and the screening capabilities.
Literature	GENERIF_SUMMARY	These findings rule out a significant deleterious effect of equivocal HER2 results. Alternative Ch17Ps may

Main Category	Category	Terms
		erroneously upgrade the HER2 status; therefore, they cannot be considered reliable in clinical practice.
Literature	GENERIF_SUMMARY	JNK and AKT, downstream effectors of HER2, co-operatively contribute to the metastatic potential of HER2-positive gastric cancer cells.
Literature	GENERIF_SUMMARY	HER2 Transmembrane Domain Mutations (V659/G660) That Stabilize Homo- and Heterodimerization Are Rare Oncogenic Drivers that are associated with Lung Adenocarcinoma.
Literature	GENERIF_SUMMARY	Distinct Receptor Tyrosine Kinase Subsets Mediate Anti-HER2 Drug Resistance in Breast Cancer
Literature	GENERIF_SUMMARY	Studies indicate that drugs targeting erb-b2 receptor tyrosine kinase 2 (c-erb B2/neu), vascular endothelial growth factor (VEGF) and EGF receptor (EGFR) under development for Inflammatory breast cancer (IBC).
Literature	GENERIF_SUMMARY	Breast cancer brain metastases commonly acquire alterations in clinically actionable genes, with metastasis-acquired ERBB2/HER2 alterations in approximately 20% of ERBB2/HER2-negative cases.
Literature	GENERIF_SUMMARY	Data suggest that HER2 expression may represent an additional aid to identify a subset of patients who are characterized by a worse prognosis and are potentially eligible for targeted therapy.
Literature	GENERIF_SUMMARY	HRG/HER2/HER3 signaling promotes AhR-mediated Memo-1 expression and migration in colorectal cancer
Literature	GENERIF_SUMMARY	The correlation between HER2 protein overexpression and some poor prognosis factors in patients affected by the breast ductal carcinoma in situ.
Literature	GENERIF_SUMMARY	Anti-HER2 DC1 vaccination is a safe and immunogenic treatment to induce tumor-specific T-cell responses in HER2(pos) patients; immune and clinical responses were similar independent of vaccination route. The immune response in the sentinel lymph nodes, rather than in the peripheral blood, may serve as an endpoint more reflective of antitumor activity
Literature	GENERIF_SUMMARY	The tumor model system we employed to test ACTIV therapy against large established tumors involved injection of tumor cells, gene modified to express human Her2, into transgenic mice, which expressed human Her2 in normal breast and brain
Literature	GENERIF_SUMMARY	Data show that added synthetic long peptides derived from widely expressed oncoproteins MUC1 and HER2/neu were reliably presented to CD4+ T-cells and cross-presented to CD8+ T-cells.
Literature	GENERIF_SUMMARY	ERBB2 may have a role in progression of non-muscle-invasive bladder cancers
Literature	GENERIF_SUMMARY	VEGF overexpression and the presence of vascular invasion were independent poor prognostic factors for HER-2 positive gastric cancers.

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Literature	GENERIF_SUMMARY	In HER2+ early breast cancer, a dual HER2-targeted combination of pertuzumab and trastuzumab, together with taxane-epirubicin-cyclophosphamide neoadjuvant chemotherapy, achieved high rates of pCR.
Literature	GENERIF_SUMMARY	DCC/18q and ERBB2/17q gene copy number variations are associated with disease-free survival in microsatellite stable colon cancer
Literature	GENERIF_SUMMARY	KRT19 intracellularly binds to HER2, playing a critical role in HER2 activation
Literature	GENERIF_SUMMARY	This study reports the novel findings that HER2 increases PARP1 protein via suppression of the let-7a miRNA, which regulates the PARP1 3'-UTR. Moreover, HER2 status correlates with high PARP1 and low let-7a in breast cancer clinical specimens.
Literature	GENERIF_SUMMARY	ONT-380 had a lower incidence and severity of diarrhea and rash than that typically associated with current dual HER2/EGFR inhibitors and showed notable antitumor activity in heavily pretreated HER2(+) Metastatic Breast Cancer patients, supporting its continued development
Literature	GENERIF_SUMMARY	we found an association of PD-L1 with luminal cancers in our cohort. Despite its association with less aggressive cancer subtype, an independent poor prognostic impact of PD-L1 in HER2-positive breast cancers was observed.
Literature	GENERIF_SUMMARY	HER2 mutation is associated with response to chemotherapy in lung adenocarcinoma.
Literature	GENERIF_SUMMARY	investigations revealed that NUMB and NUMBL interacted with small GTPase Rab7 to transition ERBB2 from early to late endosome for degradation.
Literature	GENERIF_SUMMARY	HER2 amplification, but not TOP2A amplification, is a predictor of unfavorable prognosis in breast cancer.
Literature	GENERIF_SUMMARY	Trastuzumab acidic variants had lesser binding to oncogene protein HER-2 (HER2) in comparison to the basic variants, and both acidic and basic variant showed no significant changes in their binding to soluble CD16a receptors.
Literature	GENERIF_SUMMARY	the expression of miR375 in human NSCLC cells was significantly downregulated and induction of miR375 may inhibit the proliferation of human NSCLC cells by inducing apoptosis. human epidermal growth factor receptor 2 (HER2) was a direct target gene of miR375.
Literature	GENERIF_SUMMARY	miR-133a inhibits proliferation and induces apoptosis of gastric cancer cells by downregulating the expression of ERBB2 and its downstream signaling molecules p-ERK1/2 and p-AKT
Literature	GENERIF_SUMMARY	UMAB36 could provide as an alternative highly specific IHC reagent for testing HER2 amplification in gastric cancer populations
Literature	GENERIF_SUMMARY	HER2 overexpression was evident in nearly 25% of the

Main Category	Category	Terms
		Malaysian patients with locally advanced or metastatic gastric cancer. The overexpression correlated significantly with male gender and diffuse-type tumors.
Literature	GENERIF_SUMMARY	Data indicate consistent results between molecular inversion probe (MIP) microarray and next-generation sequencing (NGS) in detecting HER2 protein and fibroblast growth factor receptor 1 (FGFR1) amplification in breast cancer.
Literature	GENERIF_SUMMARY	An HER2/CEP17 ratio of >6 in the pretherapeutic tumor biopsy is associated with a significantly higher pCR rate, particularly in HER2/HR copositive tumors, and can be used as a biomarker to predict response before neoadjuvant trastuzumab is initiated
Literature	GENERIF_SUMMARY	Our study suggests that HER2 overexpression or amplification is a poor prognostic factor in lung adenocarcinoma, although the frequency of such events is low.
Literature	GENERIF_SUMMARY	Data suggest that three new candidate genes involved in the development of rheumatoid arthritis (RA)
Literature	GENERIF_SUMMARY	ET-1 increased NSCLC clonal growth, whereas BQ123, ZD4054, lapatinib or gefitinib inhibited proliferation. The results indicate that ET-1 may regulate NSCLC cellular proliferation in an EGFR- and HER2-dependent manner
Literature	GENERIF_SUMMARY	miR-3622b-5p is involved in the proliferation and apoptosis of human ERBB2-positive cancer cells via targeting ERBB2/mTORC1 signaling pathway.
Literature	GENERIF_SUMMARY	Data suggest that HER2-negative patients with a HER2 somatic mutation are potentially good candidates for HER2-targeted therapy.
Literature	GENERIF_SUMMARY	Patients with triple-negative breast cancer have an increased serum HER2 concentration, and serum HER2 may be a valuable, novel biomarker for recurrence and metastasis in TNBC.
Literature	GENERIF_SUMMARY	Overexpression of HER-2 is a simple and reliable predictor for increased recurrence and poorer survival in patients with T3-gastric adenocarcinoma following surgical resection.
Literature	GENERIF_SUMMARY	Erb-b2 receptor tyrosine kinase 2 gene (ERBB2/HER2) copy number gain was identified as a putative resistance mechanism to afatinib.
Literature	GENERIF_SUMMARY	study identified proteomic changes associated with ErbB2 overexpression in a relevant mammary luminal Epithelial cells model and characterized in parallel the early phosphorylation events triggered by ErbB receptor-specific ligands in the context of ErbB2 overexpression.
Literature	GENERIF_SUMMARY	our data show that conventional assessment of HER2 by IHC or FISH should continue at this time, and that HER2 remains the only biomarker suitable for patient and/or regimen selection in this population.

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Literature	GENERIF_SUMMARY	possible role of hyperinsulinemia in the Barrett's Esophagus metaplasia-dysplasia-adenocarcinoma sequence through HER2 activation in esophageal epithelial cells
Literature	GENERIF_SUMMARY	EGFR and/or HER2 transactivation is involved in cell adhesion, cell migration and VEGF secretion produced by GHRH.
Literature	GENERIF_SUMMARY	In uterine serous carcinoma, 8% of patients with high HER2 intracellular domain (ICD) had low extracellular domain (ECD) levels (6/75 patients). In ovarian serous carcinoma, 42% of patients with high HER2 ICD had low ECD levels (29/69 patients).
Literature	GENERIF_SUMMARY	Studies indicate that dual anti-HER2 blockade treatment significantly increased the risk of developing severe diarrhea and treatment discontinuation in comparison with anti-HER2 monotherapy.
Literature	GENERIF_SUMMARY	ERBB2 oncogene at 17q12 is susceptible to palindromic gene amplification in HER2-positive breast tumors.
Literature	GENERIF_SUMMARY	There is a small, distinct category of HER2-positive CRC patients who are candidates for HER2-targeted therapy. CGS has the same sensitivity as IHC and FISH for detecting HER2-positive patients.
Literature	GENERIF_SUMMARY	NHERF1 acts with PMCA2 to regulate HER2 signaling and membrane retention in breast cancers
Literature	GENERIF_SUMMARY	We observed correlations between time to tumor relapse and the magnitude of changes of proliferation, luminal B, or HER2-E signatures in metastatic versus primary disease. Although the intrinsic subtype was largely maintained during metastatic progression, luminal/HER2-negative tumors acquired a luminal B or HER2-E profile during metastatic progression, likely reflecting tumor evolution or acquisition of estrogen independ
Literature	GENERIF_SUMMARY	Her-2-positive metastatic breast cancers, PTEN positivity was significantly associated with progressive disease, but not with PFS or OS
Literature	GENERIF_SUMMARY	The functionalization efficiency of Trastuzumab was higher than 70% and the antibody retained HER2 binding activity.
Literature	GENERIF_SUMMARY	We found an acquired HER2 gatekeeper mutation in a patient with HER2-mutant breast cancer upon clinical progression on neratinib. We speculate that HER2T798I may arise as a secondary mutation following response to effective HER2 tyrosine kinase inhibitors (TKI) in other cancers with HER2-activating mutations. This resistance may be overcome by other irreversible HER2 TKIs, such as afatinib.
Literature	GENERIF_SUMMARY	Her2, cMet and FGFR2 statuses were profiled in gastric cancer (GC) patients and the -derived tumor xenograft(PDX) models.

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Literature	GENERIF_SUMMARY	Data indicate that the major mechanism is the ability of p140Cap to interfere with ERBB2-dependent activation of Rac GTPase-controlled circuitries.
Literature	GENERIF_SUMMARY	Increased ezrin and HER2 expression in patients with salivary gland carcinomas represents a high-grade histopathological subtype.
Literature	GENERIF_SUMMARY	Measurement of HER2 signaling activity in the tumor cells of breast cancer patients is a feasible approach to explore as a biomarker to identify HER2-driven cancers not currently diagnosable with genomic techniques.
Literature	GENERIF_SUMMARY	Human epidermal growth receptor 2 (HER2) targeted therapies have survival benefit in adjuvant and metastatic HER2 positive breast cancer but are associated with cardiac dysfunction.
Literature	GENERIF_SUMMARY	From the significant impact of molecular profile on distant metastasis, subtype-specific individualization of systemic treatment and close surveillance are suggested. The preferential and long-term risk of brain metastasis in the HER2 subtype underlines the importance of alternative anti-HER2 therapies
Literature	GENERIF_SUMMARY	HER2 clusters together with HER3 in response to NRG, but their signaling relies on asymmetric kinase dimerization.
Literature	GENERIF_SUMMARY	Translational research for prediction of treatment benefit was performed through serial serum HER2-shed antigen assessments.
Literature	GENERIF_SUMMARY	E-cadherin expression in breast cancers is highly comparable between briefly fixed specimens and the corresponding traditionally fixed resection specimens.
Literature	GENERIF_SUMMARY	Combining the targets E-cadherin, epithelial membrane antigen (EMA), human epidermal growth receptor type 2 (Her2/neu), carcinoembryonic antigen (CEA) resulted in nearly 100% detection of ductal ovarian metastases, whereas the combination of EMA, Her2/neu and epithelial cell adhesion molecule (EpCAM) was most suitable to detect lobular ovarian metastases.
Literature	GENERIF_SUMMARY	Results indicate a mechanism of action for neuromedin U (NmU) in HER2-overexpressing breast cancer that enhances resistance to HER2-targeted drugs through conferring cancer stem cell (CSC) characteristics and expansion of the CSC phenotype.
Literature	GENERIF_SUMMARY	Combination of pegylated liposomal doxorubicin and lapatinib is feasible and potentially active in pretreated HER2-positive advanced breast cancer patients
Literature	GENERIF_SUMMARY	FOXO1 serves as an important linker between HER2 and MET signaling pathways through negative crosstalks and is a key regulator of the acquired lapatinib resistance in HER2-positive GC cells.
Literature	GENERIF_SUMMARY	HER2-Positive Circulating Tumor Cells are associated with Breast Cancer .

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Literature	GENERIF_SUMMARY	High serum HER2 extracellular domain (ECD) had a negative impact on overall survival in patients with gastric cancer. Serum HER2 ECD could be a highly specific surrogate biomarker for tissue HER2 status in gastric cancer.
Literature	GENERIF_SUMMARY	Elevated levels of ErbB2 are associated with Alzheimer's disease.
Literature	GENERIF_SUMMARY	Findings suggest that the modulation of aberrant expression of miR-9, which in turn induces oncogenic Her-2 protein activity, might hold promise for preventive and therapeutic management of breast cancer (BC).
Literature	GENERIF_SUMMARY	our findings identified a subset of EGFR and HER2 exon 20 insertion mutations that are sensitive to existing covalent quinazoline-based EGFR/HER2 inhibitors, with implications for current clinical treatment and next-generation small-molecule inhibitors.
Literature	GENERIF_SUMMARY	Results demonstrate that HER2 regulates the formation and/or maintenance of membrane protrusions and appears to contribute to its own membrane stability by preventing ubiquitination and internalization. Our findings suggest that HER2 signaling mediates these functions, in part, by regulating PMCA2 expression and maintaining low intracellular calcium levels.
Literature	GENERIF_SUMMARY	We highlight how Trastuzumab treatment changed the natural history of the HER2 receptor-positive breast cancer, demonstrating improved efficacy in changing the outcome of hormone receptor-positive patients.
Literature	GENERIF_SUMMARY	FGFR1 and/or FGF3 gene amplification correlated with a lower pathologic complete response in patients with HER2(+) early breast cancer treated with neoadjuvant anti-HER2 therapy.
Literature	GENERIF_SUMMARY	clinical trials using ERBB2 (HER2) or EGFR as targets have not fully appreciated the molecular heterogeneity in which activated ERBB2 and EGFR systems operate. Proper tumor classification is likely to be critical for arriving at thorough conclusions regarding new HER2 and EGFR based treatment regimes.
Literature	GENERIF_SUMMARY	A new prognostic model for HR-/HER2+ breast cancer based on the expression of MMP11 and CD2 was developed and the distant metastasis-free survival for patients in the high-risk group according to our model was significantly lower than that for those in the low-risk group.
Literature	GENERIF_SUMMARY	Evaluation of progesterone expression in axillary lymph node metastasis of ER-positive, HER2-negative breast cancer may enable prediction of patients who are less likely to benefit from adjuvant tamoxifen.
Literature	GENERIF_SUMMARY	Our data showed that the rs61552325 major variant decreases PC3 cell migration and its minor variant depresses DU145 cell invasion, suggesting that rs61552325 is likely an important change during prostate cancer invasion.

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Literature	GENERIF_SUMMARY	Data show that combined therapy using HER2 inhibitor and BRAF/MEK inhibitor presented more significant redifferentiation effect on papillary thyroid cancer cells harboring BRAFV600E than BRAF/MEK inhibitor alone.
Literature	GENERIF_SUMMARY	T-DM1 was well tolerated in Japanese patients with HER2-positive advanced breast cancer.
Literature	GENERIF_SUMMARY	Given an aberrant high level of sPLA2IIa in the tumor microenvironment that should be much higher than that in the blood, our findings support the notion that sPLA2IIa functions as a ligand for EGFR family receptors and supports CSC properties via HER/ERBB-elicited signaling, which may contribute to resistance to therapy and cancer progression
Literature	GENERIF_SUMMARY	The authors herein prove, for the first time, that the transcriptional repressor Blimp1 is a novel mediator of p130Cas/ErbB2-mediated invasiveness. Indeed, high Blimp1 expression levels are detected in invasive p130Cas/ErbB2 cells and correlate with metastatic status in human breast cancer patients.
Literature	GENERIF_SUMMARY	our findings suggest that heterodimerization of Her2 and TrkB receptors gives breast cancer cells a survival advantage in the brain and that dual inhibition of these receptors may hold therapeutic potential
Literature	GENERIF_SUMMARY	Our study showed that the expressions of HER2, STAT3, and SOCS3 are associated with the progression of ovarian cancer (OC), and higher expressions of HER2 and STAT3 and lower expression of SOCS3 predict poor prognosis of OC.
Literature	GENERIF_SUMMARY	Simultaneous suppression of ErbB kinases and androgen signaling by Celecoxib represents a novel strategy to interrupt the vicious cycle of AR/ErbB cross-talk with the primary purpose of undermining their resilient signaling in prostate cancer progression.
Literature	GENERIF_SUMMARY	Using concurrent ipsilateral samples HER2 discordance is noted in grade 2 to 3 ductal breast tumors.
Literature	GENERIF_SUMMARY	Real-time quantitative polymerase-chain-reaction may outperform fluorescent-in situ hybridization in identifying patients overexpressing HER2 protein.
Literature	GENERIF_SUMMARY	This is the first report to suggest that the reduction in risk occurs for low-dose aspirin and not for regular-dose aspirin and only among women with the hormone receptor-positive/HER2-negative subtype. This preliminary study builds on previous knowledge and further supports the need for formal cancer chemoprevention studies of low-dose aspirin.
Literature	GENERIF_SUMMARY	The RP2D of the combined foretinib and lapatinib is 45 mg and 1000 mg PO OD, respectively. Limited activity was seen with this combination in a predominantly unselected cohort of HER-2-positive patients with metastatic breast cancer .
Literature	GENERIF_SUMMARY	HER2(+) and triple-negative breast cancer cell lines

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		were treated with AR antagonist plus anti-HER2 mAb trastuzumab or mTOR inhibitor everolimus..AR antagonists synergize with FDA-approved breast cancer therapies such as everolimus and trastuzumab through distinct mechanisms. Treatment combinations are effective in trastuzumab-resistant HER2(+) breast cancer cells in vivo
Literature	GENERIF_SUMMARY	we demonstrate that oncogenic p110alphaH1047R, K-RasG12V, H-RasG12V, and Her2 use a common Akt1-FOXO3a pathway in suppression of DeltaNp63alpha expression and, consequently, promote cell migration/invasion and tumor metastasis. Thus, DeltaNp63alpha may function as a critical integrator of oncogenic signals in cell motility and cancer metastasis.
Literature	GENERIF_SUMMARY	Because the concordance rates of HER2 IHC score 2/3+ cases were lower than that of HER2 IHC score 0/1+ cases, further studies for detailed analysis criteria for HER2 IHC score 2+ or 3+ are required.
Literature	GENERIF_SUMMARY	HER2 reactivation through acquisition of the HER2L755S mutation was identified as a mechanism of acquired resistance to lapatinib-containing HER2-targeted therapy in preclinical HER2-amplified breast cancer models, which can be overcome by irreversible HER1/2 inhibitors
Literature	GENERIF_SUMMARY	Ten international pathology institutions participated in this study and determined messenger RNA expression levels of ERBB2, ESR1, PGR, and MKI67 in both centrally and locally extracted RNA from formalin-fixed, paraffin-embedded breast cancer specimens with the MammaTyper(R) test. Samples were measured repeatedly on different days within the local laboratories, and reproducibility was assessed by means of variance comp...
Literature	GENERIF_SUMMARY	Our cell and molecular biology experiments demonstrate that SHC and CTNND1 interact in an EGF- and HER2-dependent manner and that the cell junctions are phenotypically affected by HER2, breaking down in response to EGF and yet avoiding apoptosis as a result of cell junction loss.
Literature	GENERIF_SUMMARY	Combining ER and IGF1R targeting with HER2 targeted therapies may be an alternative to HER2 targeted therapy and chemotherapy for patients with HER2/ER/IGF1R positive breast cancer.
Literature	GENERIF_SUMMARY	No significant association could be seen between the level of macrophage and HER-2, ERbeta1 or MMP-9 expression
Literature	GENERIF_SUMMARY	These results suggested that SUVmax of 18F-FDG PET/CT could be used to predict and evaluate Her-2 or P53 related chemotherapy resistance of gastric adenocarcinoma patients.
Literature	GENERIF_SUMMARY	This article reviews aberrations in epidermal growth factor receptor, MET, and ERBB2, their therapeutic

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		implications, and future directions in targeting these pathways.
Literature	GENERIF_SUMMARY	findings present a probable mechanism of resistance against paclitaxel chemotherapy in HER2-positive breast cancers.
Literature	GENERIF_SUMMARY	The co-crystal structure shows that the Fc protein interacts with two HER2 molecules.
Literature	GENERIF_SUMMARY	Evidence on the efficacy of palliative trastuzumab-based therapy after failure of trastuzumab in the adjuvant setting is limited because of a minority of patients treated with adjuvant trastuzumab in clinical trials. The impact of trastuzumab resistance on the response on dual HER2 blockade with trastuzumab and pertuzumab and how resistance mechanisms can be used in the optimization of HER2-targeted treatment
Literature	GENERIF_SUMMARY	The number of tumor-containing fragments influences HER2 biopsy immunohistochemistry positivity in gastric cancer.
Literature	GENERIF_SUMMARY	Elevated levels of sHER2 are associated with worse survival, irrespective of the HER2 status of the tumor, but do not provide additional prognostic information in patients with known circulating tumor cell status.
Literature	GENERIF_SUMMARY	Report concordance between immunohistochemistry and fluorescence I situ hybridization for HER2 on tissue microarray sections.
Literature	GENERIF_SUMMARY	This article also reviews how to approach patients with advanced HER2-positive disease and discusses promising new therapies that are in development
Literature	GENERIF_SUMMARY	similarities in HER2 amplification status between the in situ and invasive component of admixed tumors hint at a common biological pathway for both components
Literature	GENERIF_SUMMARY	The results indicate significant contribution of IL8 on survival of hormonal dependent early-stage breast cancer patients and association with established parameters such as estrogen receptors/progesterone receptor and HER2.
Literature	GENERIF_SUMMARY	correlation was found between the CEUS and ARFI characteristics of breast cancer and HER-2 expression levels. This correlation was principally reflected in perfusion defects, perforator blood flow, PI, PT, K and VTQ
Literature	GENERIF_SUMMARY	These data, together with those recently published, indicate that HER-2 Subclonal Amplification has an impact on prognosis and should be taken into consideration to correctly plan the treatment of breast cancer patients
Literature	GENERIF_SUMMARY	This study demonstrated that discordance between the HER2 expression of circulating tumor cells (CTCs) and that of the primary tumor frequently occurs in early breast cancer. Future follow-up evaluation will assess

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		whether this discrepancy may contribute to trastuzumab resistance.
Literature	GENERIF_SUMMARY	Gastric cancer HER2 positivity was associated with lower median overall survival in Sri Lankan patients.
Literature	GENERIF_SUMMARY	Data suggest that NDRG1 attenuates oncogenic signaling by inhibiting formation of EGFR/HER2 and HER2/HER3 heterodimers and by down-regulating EGFR via a mechanism involving its degradation. (NDRG1 = N-myc downstream regulated gene 1 protein; EGFR = epidermal growth factor receptor; HER = human epidermal growth factor receptor) [REVIEW]
Literature	GENERIF_SUMMARY	Data suggest that ERBB2 is post-transcriptionally up-regulated in tamoxifen-resistant estrogen receptor-positive breast cancer cells; ERBB2 expression is decreased post-transcriptionally by miRNA26a/b binding to its 3prime- untranslated region; ERBB2 expression is increased post-transcriptionally by HuR binding to its 3prime-untranslated region. (ERBB2 = epidermal growth factor receptor 2; HuR = Hu antigen R)
Literature	GENERIF_SUMMARY	SMYD3-mediated methylation of HER2 at Lysine 175 may regulate the formation of HER2 homodimer and subsequent autophosphorylation and suggest that the SMYD3-mediated methylation pathway seems to be a good target for development of novel anti-cancer therapy.
Literature	GENERIF_SUMMARY	Study recommend that measuring RANKL together with HER2 in blood samples can be routinely applied to allow early detection of bone metastases in patients with breast cancer.
Literature	GENERIF_SUMMARY	human epidermal growth factor receptor 2 (HER-2) levels, were correlated well with TSP50/p-Samd2/3 and TSP50/p27 expression status. Thus, our studies revealed a novel regulatory mechanism underlying TSP50-induced cell proliferation and provided a new favorable intervention target for the treatment of breast cancer
Literature	GENERIF_SUMMARY	There were significant differences in human epidermal growth factor receptor 2 and human epidermal growth factor receptor 3 serum levels between the patients with tissue human epidermal growth factor receptor 2-positive and tissue human epidermal growth factor receptor 2-negative (p = 0.0026, p = 0.000011) tumors, but not in the serum levels of human epidermal growth factor receptor 4 (p = 0.054).
Literature	GENERIF_SUMMARY	In this study, snaR was identified as upregulated and to play a role in cancer progression of HER2-positive breast cancer cells. These results suggest snaR as a potential biomarker for HER2-positive breast cancer
Literature	GENERIF_SUMMARY	the present study suggests that HER2 has an important role in the regulation of the cancer stem-like cells phenotype in ALK translocated lung cancers that is

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		mainly orchestrated by HER2/HER3 heterodimers.
Literature	GENERIF_SUMMARY	The study suggests that high expression of HER-2 promoted sentinel lymph node metastasis.
Literature	GENERIF_SUMMARY	FSIP1 binds HER2 directly, which regulates breast cancer growth and invasiveness
Literature	GENERIF_SUMMARY	Neratinib is active in HER2(mut), nonamplified MBC. ctDNA sequencing offers a noninvasive strategy to identify patients with HER2(mut) cancers for clinical trial participation
Literature	GENERIF_SUMMARY	SYD985 may represent a novel and highly effective ADC against HER2-expressing Carcinosarcomas (CS). Clinical studies with SYD985 in patients harboring chemotherapy-resistant CS with low/moderate and high HER2 expression are warranted
Literature	GENERIF_SUMMARY	Although tertiary lymphoid structures (TLS), showed similar relationships with clinicopathologic features and biomarkers as tumor infiltrating lymphocytes , the prognostic value of TLS, particularly in HER2 positive cancers, was independent of TIL.
Literature	GENERIF_SUMMARY	HER2 amplification in the invasive carcinomas was negative in 87% of the cases and positive in 13% of the cases
Literature	GENERIF_SUMMARY	Study provides evidence that the hostile environment developed in spheroids has a key role in the acquisition of resistance to Trastuzumab and is associated with an increase in the number of breast cancer stem cells as well as a modulation in HER2 expression.
Literature	GENERIF_SUMMARY	ERBB2 methylation status in myocardial tissue contributes to a fraction of the variability in cardiac ERBB2 expression. This study provides a foundation to test whether differential DNA methylation status in the proximal promoter region of ERBB2 impacts on the cardiotoxic potential of trastuzumab by modifying the intracellular dynamics of ERBB2.
Literature	GENERIF_SUMMARY	Safety of pertuzumab supplementation in patients with HER2-positive metastatic breast cancer.
Literature	GENERIF_SUMMARY	Patients with activating exon 20 HER2 mutations appear to have worse survival out- comes compared with other patients with lung adenocar- cinomas with oncogenic drivers.
Literature	GENERIF_SUMMARY	A low level of cytotoxic T-cell infiltration in HER2 positive breast tumor's stroma was predictive of which women receive the most benefit fromtrastuzumab therapy compared with treatment with lapatinib.
Literature	GENERIF_SUMMARY	PIK3CA and/or ERBB family mutations were detected in 23 (31.1%) tumour samples tested, whereas PTEN expression was low in 31.1% of cases tested
Literature	GENERIF_SUMMARY	An initial scan using low radioactive [*I]SGMIB-2Rs15d allows patient selection and dosimetry calculations for

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		subsequent therapeutic [(131)I]SGMIB-2Rs15d and could thereby impact therapy outcome on HER2(+) breast cancer patients
Literature	GENERIF_SUMMARY	High HER2 expression and Gene Amplification is associated with Upper Tract Urothelial Carcinomas.
Literature	GENERIF_SUMMARY	Results indicate the importance of the LDL receptor (LDLR) in the growth of triple-negative and HER2-overexpressing breast cancers in the setting of elevated circulating LDL cholesterol (LDL-C).
Literature	GENERIF_SUMMARY	Data have found that the prevalence of HER2 mutations is about 3%. These genetic alterations are independently associated with HER2 amplification status, occurring in both ER-positive/HER2-negative diseases or HER2-enriched cancers. [review]
Literature	GENERIF_SUMMARY	The purpose of this study was to explore gene copy number (GCN) variation of EGFR, HER2, c-MYC, and MET in patients with primary colorectal cancer
Literature	GENERIF_SUMMARY	The RP2D of ganetespib is 150 mg/m(2) in combination with weekly paclitaxel plus trastuzumab. The combination was safe and well tolerated. Despite prior taxanes, pertuzumab, and T-DM1, clinical activity of this triplet regimen in this heavily pretreated cohort is promising and warrants further study in HER2-positive metastatic breast cancer
Literature	GENERIF_SUMMARY	auranofin could regulate the Her2/Akt/FOXO3 signaling pathway in SKOV3 cells and be used as a potential antitumor agent considering the expression of MUC4 in ovarian cancer patients.
Literature	GENERIF_SUMMARY	Ongoing characterization of the relationship between immune response and high-risk HER2pos Ductal carcinoma in situ will drive a tailored approach to therapy and should improve outcomes in patients diagnosed with these lesions
Literature	GENERIF_SUMMARY	This is a review of HER2 expression and function in gastric and colorectal adenocarcinomas. [Review Article]
Literature	GENERIF_SUMMARY	Here we show that, though OPCML interacts only with HER2 and not with EGFR, the interaction of OPCML with HER2 disrupts the formation of the HER2-EGFR heterodimer, and this translates into a better response to both lapatinib and erlotinib in HER2-expressing ovarian and breast cancer cell lines
Literature	GENERIF_SUMMARY	Patients with ER(+)/HER2(-) breast cancer were consecutively included for retrospective analysis. The contralateral parenchyma was segmented automatically on dynamic contrast-enhanced magnetic resonance imaging (DCE-MRI), where upon the average of the top-10% late enhancement was calculated
Literature	GENERIF_SUMMARY	CAR-engineered T (CAR-T) cells utilizing Her2-targeting DARPins G3 and 929 can target human epidermal growth factor receptor 2 (Her2)-overexpressing cancer

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		cells.
Literature	GENERIF_SUMMARY	Although trastuzumab-based therapy improved survival, the actual use of trastuzumab in the early stage of breast cancer may influence the subsequent therapeutic effect after metastasis. These findings from real-world research could help to optimize HER2 therapy after metastasis, especially in regions with limited access to these expensive targeted drugs.
Literature	GENERIF_SUMMARY	HER-2 and TOP2A genes amplification are 2 separate genetic yet closely related events in breast cancer. Polysomy of chromosome 17 is proposed to be an early event in occurrence of TOP2A gene amplification.
Literature	GENERIF_SUMMARY	In this review the information regarding various epigenetic markers implicated in HER2 positive breast cancer susceptibility and therapeutic-strategies has been compiled. [review]
Literature	GENERIF_SUMMARY	Electrochemical signals are significantly enhanced through the layer-by-layer assembly of pSC4-AuNPs due to its high conductivity and high effective area. With this innovative method, by taking the assay of a tumor marker as an example, human epidermal growth factor receptor 2 (ErbB2) was successfully measured with a detection limit of 0.5ng/mL
Literature	GENERIF_SUMMARY	Both ERBB2 and ERBB3 are essential for normal proliferation of skin keratinocytes, but in contrast to ERBB3, ERBB2 is essential for migration of human keratinocytes.
Literature	GENERIF_SUMMARY	The results suggested a possible link between tRNALeu overexpression and RSK1/MSK2 activation and ErbB2/ ErbB3 signaling, especially in breast cancer.
Literature	GENERIF_SUMMARY	Clear genetic patterns differentiating gastric papillary adenocarcinoma (GPA) and gastric tubular adenocarcinoma (GTA) could not be identified with a hierarchical cluster analysis, but microsatellite stability was linked with TP53 and Her2 overexpression.
Literature	GENERIF_SUMMARY	Patients in their 20s with breast cancer had unfavorable characteristics and worse prognosis than patients in their 30s and 40s. When stratified by tumor subtype, patients in their 20s with luminal subtype of breast cancer showed worse prognosis than older patients, whereas HER-2 and triple-negative breast cancer subtypes showed no significant differences.
Literature	GENERIF_SUMMARY	The SELECT BC study was performed in patients with HER2-negative metastatic breast cancer.
Literature	GENERIF_SUMMARY	Our findings suggest the possibility for changing the treatment paradigm of patients with ER+HER2-MBC, so a randomized prospective study is warranted to determine the optimum sequence of systemic therapies
Literature	GENERIF_SUMMARY	Relapse-free survival (RFS) was significantly shorter in patients who were double positive for EGFR and HER2. Our results suggest that EGFR and HER2 are potential

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		therapeutic targets and that their co-expression is a prognostic factor for cervical adenocarcinoma.
Literature	GENERIF_SUMMARY	New guidelines classify more HER2 FISH equivocal cases. Repeat HER2 testing clarifies HER2 status in about 50% of initial HER2 FISH equivocal cases. In addition, HER2 equivocal cases merit further study as there is limited information about prognosis and optimal treatment strategy for this population.
Literature	GENERIF_SUMMARY	ERBB2 amplification is driving resistance to erlotinib in lung adenocarcinoma.
Literature	GENERIF_SUMMARY	the estrogen- (ER), progesterone- (PgR) and HER2/neu receptor status of the primary tumor with brain metastases in a series of 24 consecutive breast cancer patients, is reported.
Literature	GENERIF_SUMMARY	ddPCR was able to identify HER2 amplification status in breast and gastric cancers with precise correlation with Immunohistochemistry and FISH results. This method might become a standard method for testing formalin-fixed paraffin-embedded samples.
Literature	GENERIF_SUMMARY	Study showed for the first time that co-occuring oncogenic HER2 and mutant PIK3CA could induce replication stress in mammary epithelial cells and drives breast cancer progression.
Literature	GENERIF_SUMMARY	In gastric cancer, PTEN and HER2 are potential candidate biomarkers for developing human antibodies that block PD-L1.
Literature	GENERIF_SUMMARY	These data demonstrate that the formation of this MPZL1-PTPN11-GRB2 complex is triggered by the attachment of HER2+ breast cancer cells to fibronectin.
Literature	GENERIF_SUMMARY	Controlled activation of specific EGFR, HER2 and HER3 homodimers and heterodimers caused oesophageal squamous epithelial cell migration and/or invasion.
Literature	GENERIF_SUMMARY	Concomitant high expression of ERalpha36, EGFR and HER2 was strongly associated with aggressive behaviors including extrathyroidal extension, lymph node metastasis and high TNM stage in papillary thyroid carcinomas.
Literature	GENERIF_SUMMARY	Focus on HER2 targeted therapies.
Literature	GENERIF_SUMMARY	Her2 expression is significantly correlated with tumor grade, depth of invasion and stage.
Literature	GENERIF_SUMMARY	Our study provides novel evidence of Endo II function in HER2+ cancer cell motility and trafficking of HER2 that relates to effective treatments with trastuzumab or T-DM1. Thus, differential expression of Endo II may relate to sensitivity or resistance to trastuzumab-based therapies for HER2+ cancers
Literature	GENERIF_SUMMARY	A major finding of our study is that one out of five (20%) patients with breast cancer BM had a receptor discrepancy between the primary tumor and the

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		subsequent BM, with loss of hormone receptors (ER and/or PR) expression, and gain of HER2 overexpression as the most commonly observed changes
Literature	GENERIF_SUMMARY	Vinorelbine/gemcitabine appeared to be a more valuable first-line treatment regimen than vinorelbine/ cisplatin in anthracycline- and taxane-pretreated patients with HER2-negative advanced breast cancer.
Literature	GENERIF_SUMMARY	In the largest series reported to date, patients with HER2-amplified m17 cancers treated with trastuzumab have outcomes comparable to patients from the large phase III adjuvant trastuzumab trials who were HER2-positive, supporting the critical role of HER2-directed therapy in this patient population.
Literature	GENERIF_SUMMARY	Her-2/neu protein overexpression was observed in 21 (37.5%) patients. No statistically significant association was seen between age and Her-2/neu protein overexpression.
Literature	GENERIF_SUMMARY	To avoid to offer patients an ineffective therapy, HER2 status should be studied more thoroughly in low amplified and equivocal cases which can have lower response rates and shorter time to progression to trastuzumab.
Literature	GENERIF_SUMMARY	the HER2 overexpression is associated with the prognostically unfavorable traditional clinicopathological features in breast cancers with maximum diameter
Literature	GENERIF_SUMMARY	Study discovered a novel enhancer HER2 gene body enhancer (HGE) in the 3' gene body of HER2. The HGE activates promoters 1 and 2 in trans., and hence the TFAP2C-mediated transcriptional induction of HER2 expression in breast cancer samples.
Literature	GENERIF_SUMMARY	Detection of a single ERBB2-amplified DCC was the most important risk factor for death from esophageal cancer (relative risk = 4.22; 95% CI = 1.91-9.32; p
Literature	GENERIF_SUMMARY	NRG1-dependent activation of HER3 induces primary resistance to trastuzumab in HER2-overexpressing breast cancer cells. HER3 monoclonal antibody combined with trastuzumab may serve as a treatment choice for patients with primary resistance to trastuzumab.
Literature	GENERIF_SUMMARY	Interaction between RhoGDI1 and ephrinB1 promotes cancer cell migration and invasion through RhoA activation.
Literature	GENERIF_SUMMARY	Breast cancer patients with HER2 exon 27 mutations have a worse survival, especially in HER2-negative patients.
Literature	GENERIF_SUMMARY	Asbestos-exposed patients displayed a significantly lower rate of EGFR mutations and a higher rate of HER2 mutations. Chrome-exposed patients exhibited enhanced HER2 and PIK3 mutation frequency. In

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		particular, asbestos-exposed patients have a lower chance of EGFR mutations.
Literature	GENERIF_SUMMARY	In total 1576 CTCs were determined HER-2 positive. We conclude that the use of image analysis enables a more reproducible quantification of treatment targets on CTCs and leads the way to fully automated and reproducible approaches
Literature	GENERIF_SUMMARY	Tumor-linked HER2 levels can be altered by several factors such as obesity, leptin and fatty acids (review).
Literature	GENERIF_SUMMARY	High HER2 expression promotes gastric cancer invasion and metastasis.
Literature	GENERIF_SUMMARY	The detection of HER2 in gastric cancer is highly heterogeneity, and the combined detection of HER2 protein expression, HER2 gene amplification and chemosensitivity can provide important reference markers for the benefit of antitumor drugs.
Literature	GENERIF_SUMMARY	HER2 expression is increased in oesophageal and gastric adenocarcinomas after neoadjuvant therapy but was not prognostic.
Literature	GENERIF_SUMMARY	Study from South Korean patients with stage II-III gastric cancer shows that HER2-positive patients had inferior overall survival and recurrence-free survival.
Literature	GENERIF_SUMMARY	Tumor infiltrating lymphocytes cannot be used as a prognostic biomarker in pT1a-b HER2-positive breast cancer.
Literature	GENERIF_SUMMARY	Tumor cell content was not associated with mutational rate for EGFR, BRAF and HER2 mutations. DNA quantity was not associated with mutational rate for EGFR, KRAS, BRAF and HER2
Literature	GENERIF_SUMMARY	High HER2 expression is associated with metastasis in breast Cancer.
Literature	GENERIF_SUMMARY	Data suggest ESE-1 silencing as a mean to treat HER2(+) patients who show resistance to anti-HER2 therapy.
Literature	GENERIF_SUMMARY	Data show that HER2 breast cancer patients with good performance status, controlled extracranial (EC) metastasis and single brain metastasis (BM) had better outcome.
Literature	GENERIF_SUMMARY	data show that ErbB2 and Akt are important for CNC migration and implicate other ErbB receptors and Akt-independent signaling pathways. Our findings provide the first example of a functional interaction between the extracellular domain of a type II classical cadherin and growth factor receptors
Literature	GENERIF_SUMMARY	Data indicate that the changes of proto-oncogene protein HER-2 (HER2) copy number in ctDNA could efficiently monitor trastuzumab efficacy.
Literature	GENERIF_SUMMARY	The interplay of dual MET/HER2 overexpression in the

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		AKT and ERK pathways for esophageal cancer is described. Therefore, combination therapy could be a novel strategy for EAC with amplification of both MET and HER2.
Literature	GENERIF_SUMMARY	Studied the prognostic value of HER2-HER3 and p-mTOR expression in gastric cancer tissue. HER3 overexpression was associated with depth of invasion; HER2-HER3 overexpression corresponded to gradually shortened 5-year overall survival; p-mTOR overexpression was found to be associated with patient age, tumor location, depth of invasion.
Literature	GENERIF_SUMMARY	Using FISH, amplification of the HER-2/neu gene was found in 8.9% of our samples.
Literature	GENERIF_SUMMARY	Results show that Her2 nuclear transport is directly regulated with Kpnbeta1 which interacts with Her2 protein .
Literature	GENERIF_SUMMARY	MiR-129-5p sensitized Her-2-positive breast cancer to trastuzumab by downregulating rpS6.
Literature	GENERIF_SUMMARY	The 12.5% overall overexpression/amplification or HER-2 in gastric and gastroesophageal junction adenocarcinomas is in concordance with previous reports. No correlation was found between tumor stage and HER2 overexpression/amplification. Determination of HER2 in limited tissue samples benefits from combinational immunohistochemistry and dual in situ hybridization testing.
Literature	GENERIF_SUMMARY	The good consistency of the results with those obtained by the enzyme-linked immunosorbent assay (ELISA) conventional method (p-value of
Literature	GENERIF_SUMMARY	The results show that Somatic ERBB2 mutations were detected in 3.15% patients. The ERBB2 mutation rate differed amongst different histological types. Patients carrying ERBB2 mutations were found to have worse prognosis than those with wild-type or PIK3CA-mutated ICCs but a better prognosis than those with KRAS-mutated ICCs.
Literature	GENERIF_SUMMARY	Asymmetric heterodimers of HER1/HER2 complexes were analyzed using molecular dynamics simulations. HER2 kinase domain prefers to serve as the receiver rather than the activator. Key binding residues of this dimer complex at N lobe of HER2 is ALA683 and at C lobe of HER1 are GLU914, GLU917, and ASP930.
Literature	GENERIF_SUMMARY	Trastuzumab in combination with docetaxel and S-1 showed effective antitumor activity and manageable toxicities as first-line treatment for patients with HER2-positive gastric cancer
Literature	GENERIF_SUMMARY	epicardial highly proliferative cells with higher cell surface expression of ErbB2 reproduced the phenotype of endothelial cells and were characterized by endothelial cell-like functional properties.
Literature	GENERIF_SUMMARY	The aim of this study was to improve knowledge of the

Main Category	Category	Terms
		HER2 and the E-cadherin alterations in the context of gastric cancer to characterize subtypes of patients that could better benefit from targeted therapy.
Literature	GENERIF_SUMMARY	Study showed the expression levels of Gli1 and HER2 were significantly higher in gastric cancer, and they are positively related. HER2 may regulate Gli1 by Akt-mTOR-p70S6K pathway.
Literature	GENERIF_SUMMARY	In patients with HER2-positive advanced breast cancer who have been heavily pretreated with anti-HER2 agents and cytotoxic chemotherapy, trastuzumab emtansine (T-DM1) is well tolerated and provided a meaningful progression-free survival of 6 months and an overall survival that has not been reached.
Literature	GENERIF_SUMMARY	Although new treatments have been approved for patients with ER+/HER2- advanced or MBC in the past decade, neither survival nor QoL appear to have improved significantly.
Literature	GENERIF_SUMMARY	Structural analysis demonstrated that the coupling of synthetic, natural or newly designed compounds impacts the conformational space of EGFR and HER2 differently.
Literature	GENERIF_SUMMARY	as proof of concept, the electrochemical sensor was successfully applied for detection of clinically relevant breast cancer biomarker Human Epidermal Growth Factor Receptor 2 (HER-2)
Literature	GENERIF_SUMMARY	Mutations in ERBB2 gene are associated with colorectal cancer.
Literature	GENERIF_SUMMARY	Higher RDI of chemotherapy is associated with better survival for ER+/PR+, HER2- patients and triple-negative breast cancer(TNBC) patients. To optimize survival benefits, RDI should be maintained >/= 85% in ER+/PR+, HER2- patients, and >/= 75% in TNBC patients.
Literature	GENERIF_SUMMARY	HER2-CAR incorporates humanized anti-HER2 scFv. Humanized anti-HER2-scFv is composed of the variable region of an immunoglobulin heavy chain and the variable region of the light chain linked through a polypeptide linker.
Literature	GENERIF_SUMMARY	Study revealed that PDGFRalpha/HER2 and PDGFRalpha/p53 co-expression exists in poorly differentiated oral squamous cell carcinoma cells, suggesting that cooperation between these proteins might enhance aggressive behavior of tumor.
Literature	GENERIF_SUMMARY	HER2 amplification is a distinct driver event seen in all breast cancer subtypes, rather than a subtype marker, with major characteristics restricted to amplification and overexpression of HER2 and neighboring genes. The HER2E subtype has a distinctive transcriptional landscape independent of HER2A that reflects androgen receptor signaling as replacement for estrogen receptor (ER)-driven tumorigenesis.
Literature	GENERIF_SUMMARY	signaling driven by HER2 promotes chronic

Main Category	Category	Terms
		inflammation needed to support cancer stem-like cell maintenance in HER2-positive breast cancers
Literature	GENERIF_SUMMARY	Study revealed that HER-2 mutant K753E showed some contrasting behaviour as compared to wild, L768S and V773L HER-2 in complex with lapatinib while similar to previously known lapatinib resistant L755S HER-2 mutant. Lapatinib showed stable but reverse orientation in binding site of K753E and the highest binding energy among studied HER2-lapatinib complexes but slightly lesser than L755S mutant.
Literature	GENERIF_SUMMARY	NVP-AUY922 displays potent anti-cancer activity in HER2-positive and trastuzumab-resistant breast cancer cells.
Literature	GENERIF_SUMMARY	The expression of C-Met and HER2 protein in lung adenocarcinoma is highly correlated, and whether it is synergistic in the targeted therapy of lung adenocarcinoma deserves further study.
Literature	GENERIF_SUMMARY	Relative to ER, PR and HER2 from medical records, central IHC staining and the addition of Ki67 or combined tumor grade improved accuracy for classifying PAM50-based luminal subtypes.
Literature	GENERIF_SUMMARY	he advantages offered by this platform in terms of high sensitivity, broad dynamic range and low-cost demonstrates great potential for improved monitoring of human epidermal growth factor receptor 2(HER2) levels for the management of breast and other cancers.
Literature	GENERIF_SUMMARY	IGF-1R-targeting therapy can overcome drug insensitiveness in HER2-amplified gastric cancer.
Literature	GENERIF_SUMMARY	Up-regulation of NDRG1 by DpdtC is a promising therapeutic approach in HER2-overexpressed cancers.
Literature	GENERIF_SUMMARY	HER2 and HER3 expression was detected in 22.2% and 86.1% of samples, respectively. The frequency of EGFR mutation was 45.7% and was not significantly different between stage 0 and IA1 (40.0% and 48.0%, respectively), suggesting that EGFR mutation does not correlate with cancer progression from stage 0 to IA1.
Literature	GENERIF_SUMMARY	Studied association of coamplification of microRNA 4728 and HER2 inhibitor resistance in HER2 amplified breast cancers.
Literature	GENERIF_SUMMARY	Aggressive adjuvant systemic treatments should be considered for patients with HER2-rich and triple-negative subtype who exhibit tumor shrinkage in NAC but still have high levels of Ki67.
Literature	GENERIF_SUMMARY	AhR is an important factor for the malignancy in HER2 overexpressing breast cancers.
Literature	GENERIF_SUMMARY	Performed in mice bearing subcutaneous HER2-overexpressing and low HER2-expressing tumors.
Literature	GENERIF_SUMMARY	Mendelian randomization identified HER2 as a novel, causal mediator of CKD. Blood HER2 concentration

Main Category	Category	Terms
		associated with CKD events in prediabetic ORIGIN trial participants. ACE inversely regulated HER2 levels.
Literature	GENERIF_SUMMARY	Increased platelet-to-lymphocyte ratio was a poor prognostic factor for both overall survival (OS) (primary endpoint) and failure-free survival (FFS) and warrants further investigation into its prognostic ability amongst patients with HER2-positive metastatic breast cancer
Literature	GENERIF_SUMMARY	In this study, authors analyzed the immunoexpression of EGFR, HER2 (EGFR2) and HER3 (EGFR3) in 41 cases of serous borderline ovarian tumors and carcinomas, in relation to the degree of differentiation and tumor stage.
Literature	GENERIF_SUMMARY	we demonstrated that the majority of BCs with equivocal HER2 FISH results remain equivocal after IHC testing. Alternative FISH testing of HER2 amplification in equivocal cases provides a standardized approach resulting in classifying almost half the cases as amplified.
Literature	GENERIF_SUMMARY	The combination of immunohistochemical expression of BRCA1, ER, PR, and HER-2/neu and clinicopathological details may be helpful in predicting the individuals more likely to carry BRCA1 mutations and thus selecting the candidate and family members for genetic screening for BRCA1 mutations.
Literature	GENERIF_SUMMARY	In early breast cancer, PIK3CA mutations seem to identify HER2+ patients who are less likely to reach pCR. The clinical implications of PIK3CA mutations tend to vary between exon 9 and exon 20. This mechanism should be explored in further studies.
Literature	GENERIF_SUMMARY	identify the presence of intratumoral heterogeneity of ERBB2 amplification and HER2 expression in Micropapillary urothelial carcinoma
Literature	GENERIF_SUMMARY	HER2 interacts with Beclin 1 in breast cancer cells and inhibits autophagy. Mice with increased basal autophagy due to a genetically engineered mutation in Becn1 are protected from human HER2-driven mammary tumorigenesis. HER2-mediated inhibition of Beclin 1 and autophagy likely contributes to HER2-mediated tumorigenesis.
Literature	GENERIF_SUMMARY	The growth and chemo-resistance of Prostate cancer cells may be suppressed via re-expression of miR-4319 that inhibits Her-2 and its signaling.
Literature	GENERIF_SUMMARY	Dual EGFR/HER2 knockdown by siRNAs further decreased the growth of Gastric Cancer cells treated with gefitinib alone, confirming that single-agent drug targeting does not achieve a maximal biological effect.
Literature	GENERIF_SUMMARY	we further dissected the diarrhea profile of the various patient dose cohorts and carried out in vitro investigations in human colon cell lines and explants to decipher the contribution and the mechanism of anti-HER2/3 therapeutic antibodies to intestinal epithelium malfunction

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Literature	GENERIF_SUMMARY	with this unbiased study design and representative group of patients with Ductal carcinoma in situ(DCIS) treated by breast-conserving surgery alone, COX-2, HER2, and periductal fibrosis were revealed as promising markers predicting progression of DCIS into ipsilateral invasive breast cancer.
Literature	GENERIF_SUMMARY	identified a novel interaction of RUNX1 and the ErbB2/HER2 signaling pathway in gastric cancer, which can potentially be exploited in the management of this malignancy
Literature	GENERIF_SUMMARY	We demonstrate that decreased HER2 and upregulation of MDR1 contribute to T-DM1 resistance in KPL-4 T-DM1-resistant cells.
Literature	GENERIF_SUMMARY	study highlights the role of tissue differentiation on pathological response to neoadjuvant chemotherapy in gastric cancer and shows no impact between FOXP3, HER2 and MET expression in terms of tumor regression grading
Literature	GENERIF_SUMMARY	HER2 gene amplification in circulating tumor DNA predicts HER2-positive breast neoplasms resistance to trastuzumab emtansine.
Literature	GENERIF_SUMMARY	Alcohol is associated with lower risk of H2E breast cancer relative to ER+ breast cancer.
Literature	GENERIF_SUMMARY	pharmacological inhibition of AXL specifically decreased the metastatic burden of mice developing HER2(+) breast cancer.
Literature	GENERIF_SUMMARY	results simultaneously showed the association between HER2 gene amplification and the expression levels of MAPK signaling pathway proteins and clinicopathologic characteristics in gastric cancer.
Literature	GENERIF_SUMMARY	Low estrogen receptor (ER)+ breast cancers are biologically similar to ER- tumors. Semiquantitative ER H-score is an important determinant of response to neoadjuvant chemotherapy.
Literature	GENERIF_SUMMARY	Her2 (Ile)655(Val) polymorphism is associated with breast cancer risk in overall
Literature	GENERIF_SUMMARY	Notch-1 protein expression predicts poorer survival in women with HER2+ breast cancer.
Literature	GENERIF_SUMMARY	Results showed that combining the results of IHC and FISH according to the HER2 testing algorithm is a useful method for accurately evaluating HER2-positive EMPD.
Literature	GENERIF_SUMMARY	By genetic elimination of HER3 or shRNA knockdown of HER3 in HER2-amplified cancer cells, study finds residual HER2-driven activation of PI3K/Akt pathway signaling that is driven by HER2 through direct and indirect mechanisms. Indirect mechanisms involved second messenger pathways, including Ras or Grb2. Mutation of Y1139 impaired the tumorigenic competency of HER2.

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Literature	GENERIF_SUMMARY	There was a statistically significant association between positive p95-HER2 expression and negative hormonal receptors expression (p=0.004), high Ki-67 expression (p
Literature	GENERIF_SUMMARY	It was concluded that miR494 inhibited the cancer initiating cells phenotype and reversed resistance to lapatinib by inhibiting FGFR2 in HER2positive gastric cancer.
Literature	GENERIF_SUMMARY	High HER2 expression is associated with Gastric Adenocarcinoma.
Literature	GENERIF_SUMMARY	ctDNA gene mutation profiles differed among HR/HER2 subtypes of metastatic breast cancer (MBC) patients. By identifying mutations associated with treatment resistance, we hope to improve therapy selection for MBC patients who received multiline treatment.
Literature	GENERIF_SUMMARY	Antitumor effect of lapatinib and cytotoxic agents by suppression of E2F1 in HER2positive breast cancer.
Literature	GENERIF_SUMMARY	Cholangiocarcinoma is a genetically diverse cancer. Alterations in CDKN2A/B and ERBB2 are associated with negative prognostic implications in patients with advanced disease. Somatic alterations with therapeutic implications were identified in almost half of patients.
Literature	GENERIF_SUMMARY	HER2 amplification was predictive of anti-EGFR therapy response and appeared to be prognostic in metastatic colorectal cancer patients.
Literature	GENERIF_SUMMARY	Authors showed that mRNA and protein levels of COX2 and HER2 were upregulated in CRC compared with the adjacent tissues. COX2 protein levels and nuclear COX2 expression were correlated with a poor prognosis of CRC patients. COX2 expression was positively associated with HER2 expression.
Literature	GENERIF_SUMMARY	Our data support the combination of therapies targeting ERBB2/3 and mTORC1 signalling, together with fulvestrant, in patients who relapse on endocrine therapy and retain a functional ER.
Literature	GENERIF_SUMMARY	CD44 could stabilize ERBB2 and co-activate p-p38 expression.
Literature	GENERIF_SUMMARY	These results indicate that higher membrane JAGGED1 expression may be used to either predict response to anti-HER2 therapy or for detection of NOTCH-sensitive CSCs posttherapy. Sequential blockade of HER2 followed by JAGGED1 or NOTCH could be more effective than simultaneous blockade to prevent drug resistance and tumor progression.
Literature	GENERIF_SUMMARY	Prior to therapy, across all three arms, N9831 patients had similar mean anti-HER2 IgG levels. Following treatment, the mean levels of antibodies increased in the trastuzumab arms but not the chemotherapy-only arm.
Literature	GENERIF_SUMMARY	HER2 overexpression is associated with Gastric Cancer.

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Literature	GENERIF_SUMMARY	Data suggest that erb-b2 receptor tyrosine kinase 2 (ERBB2) and erb-b2 receptor tyrosine kinase 3 (ERBB3) might be promising targets for treatment of SLC3A2-NRG1-positive tumors.
Literature	GENERIF_SUMMARY	HER2 over-expression is induced by p53 mutants through HER2 transcriptional activation.
Literature	GENERIF_SUMMARY	High HER2 expression is associated with invasion and lymph node metastasis in gastric cancer.
Literature	GENERIF_SUMMARY	Basal HER2 phenotype showed poor DFS, but equivalent pCR rate after concurrent neo-adjuvant chemotherapy with trastuzumab. A different treatment approach to basal-HER2 type is needed even for cases that achieved adequate clinical response after neo-adjuvant chemotherapy.
Literature	GENERIF_SUMMARY	Results showed that HER2 and FGFR2 are regulated by DDX6 at the post-transcriptional step in gastric cancer.
Literature	GENERIF_SUMMARY	HER2 amplification status in mucinous adenocarcinoma, gastric type (GAS) of the uterine cervix was studied.
Literature	GENERIF_SUMMARY	These findings suggest that breast cancers classified as HER2(+) only by the alternative probe method are biologically distinct from those classified as HER2(+) by traditional criteria, and raises questions as to whether or not they derive the same benefit from HER2-targeted therapies
Literature	GENERIF_SUMMARY	In the current settings, HER2/neu is not found to be a prognostic marker in head-and-neck cancers.
Literature	GENERIF_SUMMARY	These results may be commented as HE4 expression rises in patients with HER2/neu amplification.
Literature	GENERIF_SUMMARY	The contribution of chromosome aneuploidy to evolutionary variation of cHER2 phenotypes on CTCs may provide additional perspectives with respect to understanding the mechanisms of developing therapeutic resistance in malignant carcinomas.
Literature	GENERIF_SUMMARY	HER2 over-expression was associated with a better disease-free survival in breast cancer patients at risk for recurrence
Literature	GENERIF_SUMMARY	It has been demonstrated that the heterogeneity of HER2 expression accelerated the development of metastases which caused the poor survival of mice with heterogeneous HER2 expression (HER2-60).
Literature	GENERIF_SUMMARY	HER2 positivity was found in a minority of rectal cancer patients and was not significantly associated with clinicopathologic and molecular characteristics.
Literature	GENERIF_SUMMARY	Her-2/neu amplification increases with increasing grade of breast cancer. A high proportion of Her-2/neu gene amplified cases indicates aggressive disease in that area and need for FISH testing on large scale, which is the gold standard for equivocal cases on

Main Category	Category	Terms
		immunohistochemistry.
Literature	GENERIF_SUMMARY	Study demonstrates that HER2 mRNAs post-transcriptionally up-regulate HER3 via the sequestration of miR-125a/b, contributing to enhanced breast cancer growth and acquired anti-HER2 resistance.
Literature	GENERIF_SUMMARY	This study confirmed that biosimilar trastuzumab improves the overall response rate when combined with chemotherapy for HER2+ breast cancer
Literature	GENERIF_SUMMARY	Activating HER2 mutation is present in about 3% of bone metastases from breast cancers, with significantly higher rates in the pleomorphic subtype of lobular cancer.
Literature	GENERIF_SUMMARY	Protein kinase C mediated internalization of ErbB2 is independent of clathrin, ubiquitination and Hsp90 dissociation.
Literature	GENERIF_SUMMARY	Study of breast cancer specimens from women affected with neurofibromatosis type 1 demonstrates that both germline and somatic NF1 mutations associate with an increase in HER2 expression, resulting from either gene amplification or transcriptional upregulation. This finding suggests a synergistic interplay between these two genes, potentially driving the development of breast cancer.
Literature	GENERIF_SUMMARY	estimates for gene-expression (GE) were added for ER (gene ESR1), for PGR (gene PGR) and for HER2 (gene ERBB2). Combined results were obtained in each patient via a scoring system based on all three receptors
Literature	GENERIF_SUMMARY	HER2 gene amplification occurred during the early stages of gastric cancer and showed heterogeneity in several cases. HER2 gene amplification may be involved in tumor progression in early gastric cancer.
Literature	GENERIF_SUMMARY	The Stop & Go study aimed to determine if an intermittent treatment regimen with paclitaxel plus bevacizumab was not inferior in terms of PFS to a continuous regimen, both in combination with bevacizumab maintenance treatment, in first-line treatment of patients with HER2-negative, advanced breast cancer.
Literature	GENERIF_SUMMARY	Clinical confirmation of our machine learned Bayesian networks will have significant impact on our understanding of the role of NRF1 as a valuable biomarker for breast cancer diagnosis and prognosis as well as provide strong rationale for future studies to develop NRF1 signaling-based therapeutics to target HER2+ breast cancer.
Literature	GENERIF_SUMMARY	NAC significantly prevented A. salmonea-induced HER-2/neu depletion and PI3K/AKT inactivation, indicating that A. salmonea-triggered apoptosis is mediated by ROS-inhibited HER-2/neu signaling cascades. To our knowledge, this is the first report describing the anticancer activity of this potentially

Main Category	Category	Terms
		beneficial mushroom against human ovarian carcinom
Literature	GENERIF_SUMMARY	The survival rates in this study are equal to the documented global rates; nodal disease burden emerged as the most important prognostic factor. In addition, in EBCs, a lack of hormone receptor expression and in LABC, Her2neu overexpression appear to worsen the outcome.
Literature	GENERIF_SUMMARY	The Authors reveal a gender difference in the prognostic value of concomitant AIB1 and HER2 copy number gain (CNG) in glioma patients which were barely noticed before. These observations indicated that genetic alterations synergistic with essential respects of sex determination influence glioma biology and patients outcomes.
Literature	GENERIF_SUMMARY	Overexpression of HER2 was observed in 48% (n = 12) of gallbladder carcinoma adenocarcinomas, 58% (n = 7) of high-grade dysplasia, 47% (n = 8) of low-grade dysplasia, and 74% (n = 25) of intestinal metaplasia. Ki-67 index increases in a non-linear fashion as the precursor lesions progress toward malignancy. These markers might be used as a prognostic biomarker in early neoplastic transformation of gallbladder epithelium.
Literature	GENERIF_SUMMARY	TGFbeta signaling can shift from inhibiting to promoting breast cancer development via HER2/EGFR AKT-mediated phosphorylation of Smad3 at S208, enhancing its nuclear accumulation and upregulation of EMT-related genes
Literature	GENERIF_SUMMARY	Study supports the involvement of EGFR, HER2 and HER3 in BCC aggressiveness of and in tumor differentiating towards different histological subtypes.
Literature	GENERIF_SUMMARY	Breast cancer patients with the HER2 germline mutation A270S had a worse survival, especially in HER2-negative patients.
Literature	GENERIF_SUMMARY	finding of 5 mutations in RAS-expanded genes allowed us to understand the resistance to cetuximab in 33% of KRAS WT exon 2 nonresponder patients. HER2 does not seem to be a potential biomarker for cetuximab-targeted therapy.
Literature	GENERIF_SUMMARY	Data showed a high rate of discordance in matched pairs of primary tumors and metastases, suggesting that the accurate evaluation of proto-oncogene protein HER-2 (HER2) status is essential before any therapeutic decision.
Literature	GENERIF_SUMMARY	T-DM1 showed a signal of activity in patients with HER2-overexpressing (IHC 3+) advanced non-small cell lung cancer
Literature	GENERIF_SUMMARY	We demonstrated that multimodality imaging is a reliable approach for selecting potential antibodies and found that 61 IgG manifested significant tumor accumulation with elevated internalization rate thus could be a suitable candidate for further development

Main Category	Category	Terms
		of new HER2-targeted therapies.
Literature	GENERIF_SUMMARY	Although ST6GalI overexpression increased HER2 sialylation, corresponding to decreased HER2 phosphorylation, high alpha2,6sialylation enhanced Akt and ERK phosphorylation levels compared to those in the vector cell line; ST6GalI knockdown had the opposite effects. Collectively, these results implicated a functional role of ST6GalI in promoting tumor cell progression and trastuzumab resistance.
Literature	GENERIF_SUMMARY	Results show mutation in ERBB2-exon17 was associated with worse survival results in patients with pancreatic neoplasm. [review]
Literature	GENERIF_SUMMARY	Expression levels of ERBB2 and CD34 antigen and estrogen receptor were up-regulated in breast cancer patients.
Literature	GENERIF_SUMMARY	Absence of HER2 Expression of Circulating Tumor Cells is associated with Non-Metastatic Esophageal Cancer.
Literature	GENERIF_SUMMARY	High HER2/EIF2C1 ratio is associated with breast cancer.
Literature	GENERIF_SUMMARY	Our findings indicate an overlap of T790M, HER2 amplification, and HER2 mutation, which is rarely reported. Moreover, HER2 mutation was identified during the development of resistance, suggesting that HER2 mutation may cause resistance to first-generation EGFR-TKIs.
Literature	GENERIF_SUMMARY	The expression level of EGFR increased along with higher stages and pathologic grades of BTCC, and the obviously increased expression of HER-2 was statistically associated with clinical stages and tumor recurrence. In addition, the expression level of HER-2 increased along with the higher clinical stage of BTCC. EGFR expression and HER-2 levels were positively associated in BTCC samples.
Literature	GENERIF_SUMMARY	Findings show a network of deregulated exosomal miRNAs with specific expression patterns in exosomes of HER2-positive and TNBC patients that are also associated with clinicopathological parameters and pCR within each BC subtype.
Literature	GENERIF_SUMMARY	analysis of the HER2 mutation and amplification co-occurrence in breast cancer, as well as the antitumor activity of the HER tyrosine kinase inhibitor neratinib
Literature	GENERIF_SUMMARY	Study reports novel somatic mutations of ERBB2 in gallbladder cancer, and its therapeutic implication in the presence and absence of KRAS G12 V and G13D mutations and demonstrates that ERBB2 heterodimerizes with EGFR to constitutively activate the ErbB signaling pathway in gallbladder cancer cells.
Literature	GENERIF_SUMMARY	HER2 copy numbers from FISH showed a significant positive correlation with detected densities from qSMLM.

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Literature	GENERIF_SUMMARY	ERBB2 mutations hyperactivate the HER3/PI3K/AKT/mTOR axis, leading to antiestrogen resistance in ER+ breast cancer. Dual blockade of the HER2 and ER pathways is required for the treatment of ER+/HER2 mutant breast cancers.
Literature	GENERIF_SUMMARY	these results pinpoint transaldolase as a novel metabolic enzyme possessing synthetic lethality with HER2 inhibition.
Literature	GENERIF_SUMMARY	Statistical analysis performed in this study did not reveal the significant relationship between HER2 overexpression on the tumor cells and microvessel density in the tumor stroma.
Literature	GENERIF_SUMMARY	HER2 is generally upregulated in human meningiomas, but in an activated state only.
Literature	GENERIF_SUMMARY	This abnormally sensitive electrochemical sensing performance resulting from anionic porphyrin for DNA sequences specific to HER2 gene will offer considerable promise for tumor diagnosis and treatment
Literature	GENERIF_SUMMARY	In this study, we focused on patients with gastric cancer who received preoperative chemotherapy and aimed to examine the changes in HER2 expression status and amplification of EGFR and MET, not only after HER2-targeted therapy, but also after cytotoxic chemotherapy alone.
Literature	GENERIF_SUMMARY	Fluoro-Chromogenic Labelling for Detection of MCM2 to Assess Proliferation Activity in HER2-amplified Breast Carcinomas.
Literature	GENERIF_SUMMARY	silencing Her2 may inhibit the proliferation and angiogenesis of human meningioma cells.
Literature	GENERIF_SUMMARY	Study shows that brain metastases in patients with GI primary malignancies have a relatively high likelihood of being HER2 positive despite HER2 amplification or overexpression being less commonly found in matched tissue from prior sites of disease.
Literature	GENERIF_SUMMARY	Major state population centers showed the highest HER2 testing.
Literature	GENERIF_SUMMARY	HER2-mediated GLI2 stabilization promotes anoikis resistance and metastasis of breast cancer cells.
Literature	GENERIF_SUMMARY	Structural modeling and analysis of ERBB2 showed that transmembrane domain and juxtamembrane domain mutations function by improving the active dimer interface or stabilizing an activating conformation.
Literature	GENERIF_SUMMARY	HER2 plays a prognostic and predictive role in bladder cancer [review]
Literature	GENERIF_SUMMARY	Inhibition of ezrin synergizes with lapatinib in a PKCalpha-dependent fashion to inhibit proliferation and promote apoptosis in HER2-positive breast cancer cells.

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Literature	GENERIF_SUMMARY	Data show that the caveolin-1 (CAV1) protein is involved in epidermal growth factor receptor 2 (HER2) cell membrane dynamics within the context of receptor endocytosis.
Literature	GENERIF_SUMMARY	This novel molecular axis (CUL3/KCTD10/RhoB) positively regulates the activity of Rac1 in HER2-positive breast cancers.
Literature	GENERIF_SUMMARY	Resistance was overcome by combining ER-directed therapy with the irreversible HER2 kinase inhibitor neratinib.
Literature	GENERIF_SUMMARY	the estrogen receptor (ER) and progesterone receptor (PR), as well as human epidermal growth factor receptor 2 (Her2neu) expressions in breast tumor cells, were evaluated.
Literature	GENERIF_SUMMARY	KDM5 inhibition has potential in the treatment of HER2(+) breast cancer.
Literature	GENERIF_SUMMARY	The role of HER2 amplification testing in metastatic colorectal cancer.
Literature	GENERIF_SUMMARY	Study demonstrated that ErbB2-driven Irf6 down-regulation in ErbB2-overproducing breast epithelial cells blocks their anoikis and promotes their anchorage-independent growth.
Literature	GENERIF_SUMMARY	colorectal cancer patients with cetuximab resistance harbored a somatic mutation R784G at the exon 20 region of HER2 coding sequence
Literature	GENERIF_SUMMARY	Metaplastic breast cancer (MpBC) group had significantly more cases with no hormone receptors (ER, PR) and HER2 overexpression/gene amplification compared with the other invasive breast cancer types group (ER-, 69% versus 23%, p
Literature	GENERIF_SUMMARY	Our data also suggests that PID quantitation of HER2 protein may offer an improvement over conventional HER2 testing in the selection of patients who will be the most likely to benefit from HER2-targeted therapy. Further studies with a larger cohort are warranted.
Literature	GENERIF_SUMMARY	When given as first-line treatment for HER2-positive metastatic breast cancer, PF-05280014 plus paclitaxel demonstrated equivalence to trastuzumab-EU plus paclitaxel in terms of objective response rate
Literature	GENERIF_SUMMARY	HER2 expression in breast cancer was positively correlated with TNM stage, tumor size, and lymph node metastasis, and negatively correlated with tissue grade and estrogen receptor (ER)/progesterone receptor (PR) expression (p
Literature	GENERIF_SUMMARY	WBP2 expression correlates with the response of HER2-positive breast cancer to trastuzumab-based neoadjuvant chemotherapy.
Literature	GENERIF_SUMMARY	Pyrotinib showed activity against non-small-cell lung cancer with HER2 exon 20 mutations in both patient-

Main Category	Category	Terms
		derived organoids and a patient-derived xenografts model. In the clinical trial, pyrotinib showed promising efficacy
Literature	GENERIF_SUMMARY	Low HER2 expression is associated with osteosarcoma.
Literature	GENERIF_SUMMARY	EGFR/ERBB2 germline mutations were found to be rare in Chinese lung cancer patients with more diversity other than the previously reported EGFR-T790M, with EGFR-K757R being the most common EGFR germline mutation.
Literature	GENERIF_SUMMARY	We demonstrated a positive prognostic impact of HER2 overexpression in a large cohort of esophageal adenocarcinoma (EAC), contrary to other solid malignancies including gastric cancer and breast cancer, but consistent to the results of a large study on EAC from 2012.
Literature	GENERIF_SUMMARY	Authors used six whole-exome-sequenced primary HGSOC/USC cell-lines and three xenografts overexpressing HER2/neu and harboring mutated or wild-type PIK3CA/PIK3R1 genes to evaluate the role of PI3K-mutations as potential mechanism of resistance to afatinib.
Literature	GENERIF_SUMMARY	an involvement of CD151 in regulation of ErbB2/ErbB3 heterodimerization and its impact on cell response to Herceptin, is reported.
Literature	GENERIF_SUMMARY	Transferred-tissue Microarray for Fluorescence In Situ Hybridization Test for Human Epidermal Growth Factor Receptor 2 in Breast Cancer.
Literature	GENERIF_SUMMARY	TILs are an independent prognostic factor for HER2-positive early breast cancer patients treated with adjuvant chemotherapy and trastuzumab and may refine the ability to identify patients at low risk of relapse eligible for de-escalated adjuvant therapy.
Literature	GENERIF_SUMMARY	Data show that based on microfluidic experimentation and image processing, both epidermal growth factor receptor 2 (HER2) protein overexpression and HER2 gene amplification could be detected at the cellular level.
Literature	GENERIF_SUMMARY	serum HER-2 extracellular domain could be a biomarker to identify the subgroup of poorer outcome among HER-2 overexpression breast cancer patients.
Literature	GENERIF_SUMMARY	The results suggested a tumorigenic mechanism by which E2 promotes tumor cell proliferation via HER2/COX2 signaling. The present study provided evidence for the molecular impact of E2 on breast tumorigenesis, and suggested possible strategies for preventing and treating human breast cancer.
Literature	GENERIF_SUMMARY	The HER2 immunohistochemistry results obtained by the image analysis method correlated well with those obtained by the manual scoring method.

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Literature	GENERIF_SUMMARY	HER2 overexpression in MCF7 cells leads to an increase in HIF-2alpha but not HIF-1alpha expression in normoxia and an increased upregulation of HIF-2alpha in hypoxia. Global gene expression analysis showed that HER2 overexpression in these cells promotes an exaggerated transcriptional response to both short-term and long-term hypoxia, with increased expression of numerous hypoxia response genes.
Literature	GENERIF_SUMMARY	The accompany HER2 blockade has hindered the therapeutic effect.
Literature	GENERIF_SUMMARY	Results find that high HER2 expression is an adverse prognostic factors for tumor recurrence and progression in non-muscle-invasive bladder cancer (NMIBC) and these patients should be monitored thoroughly for disease recurrence and progression.
Literature	GENERIF_SUMMARY	Resistance to trastuzumab has become a limiting factor for therapeutic efficacy of human epidermal growth factor 2 (HER2)-positive breast cancer.
Literature	GENERIF_SUMMARY	HER2 expression pattern is linked with tumor stage and overall survival; the transforming function of HER2 is found more relevant through beta-catenin and SMAD3.
Literature	GENERIF_SUMMARY	Drugs that target HER2/EGFR protein folding, including DDAs and CsA, have the potential to kill cancers.
Literature	GENERIF_SUMMARY	these findings define HER2-CB2R heteromers as new potential targets for antitumor therapies and biomarkers with prognostic value in HER2-positive breast cancer
Literature	GENERIF_SUMMARY	HER2 positivity may be predictive for a high-risk therapeutic resistance in rectal cancers
Literature	GENERIF_SUMMARY	Gene alterations, such as TP53 mutation, PIK3CA mutation, ERBB2 amplification and CCND1 amplification, and MAPK pathway alteration were associated with pCR in our study.
Literature	GENERIF_SUMMARY	A multi-modal regimen that includes an anti-HER2 agent appears to be a promising treatment for patients with HER2-positive extramammary apocrine carcinoma.
Literature	GENERIF_SUMMARY	The interval between anti-HER2 treatment and the determination of HER2 in second histology is strongly associated with HER2 expression.
Literature	GENERIF_SUMMARY	After case-mix adjustment, mean positivity rates were 87.2% for ER (range 80.4-94.3), 71.3% for PR (62.5-77.5%), and 9.9% for HER2 (5.5-12.7%). Overall, 14 (35.9%), 17 (43.6%) and 11 (28.2%) laboratories showed positivity rates outside the 95% confidence interval for ER, PR and HER2, respectively.
Literature	GENERIF_SUMMARY	HER2 and p95HER2 differentially regulate miRNA expression in MCF-7 breast cancer cells and downregulate MYB proteins through miR-221/222 and miR-503.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	serum level not influenced by pre-eclampsia with HIV infection
Literature	GENERIF_SUMMARY	The landscape of d16HER2 splice variant expression across HER2-positive cancers.
Literature	GENERIF_SUMMARY	Exploration of the Structure and Recognition of a G-quadruplex in the her2 Proto-oncogene Promoter and Its Transcriptional Regulation.
Literature	GENERIF_SUMMARY	Results found Chinese triple-negative breast cancer (TNBC) cases with more CDKN2A loss in the luminal androgen receptor subtype.
Literature	GENERIF_SUMMARY	1year of adjuvant trastuzumab treatment conferred substantial survival benefits and should remain as the preferred treatment for early stage HER2-positive breast cancer.
Literature	GENERIF_SUMMARY	ERBB2 might modulate ATG4B for autophagy induction in oxidative stress-stimulated ARPE-19 cells. ERBB2 knockdown also caused an accumulation of nuclear factor erythroid 2-related factor 2 (NRF2) and enhanced its transcriptional activity.
Literature	GENERIF_SUMMARY	Neither PIK3CA mutations nor Fc-gamma-receptor polymorphisms were associated with sTILs. sTILs may have utility as a prognostic biomarker identifying HER2-positive early breast cancer at low recurrence risk.
Literature	GENERIF_SUMMARY	This study demonstrates these functional fluorescent probes' ability to report metabolic adaptations during primary tumor growth, regression, residual disease, and regrowth in Her2 breast tumors.
Literature	GENERIF_SUMMARY	HER2 Positivity Predicts Unresponsiveness to EGFR-Targeted Treatment in Metastatic Colorectal Cancer.
Literature	GENERIF_SUMMARY	Both erb-b2 receptor tyrosine kinase 2 (ErbB2) and vav guanine nucleotide exchange factor 2 (VAV2)are direct targets of miR-331-3p.
Literature	GENERIF_SUMMARY	Determining the HER2 status of lymph node metastases in breast cancer seems to be rational, but assessing a limited number of metastatic nodes may be sufficient
Literature	GENERIF_SUMMARY	These findings support a role for serum HER2 extracellular domain as a surrogate marker of tissue HER2 status in primary breast cancer.
Literature	GENERIF_SUMMARY	These findings identify a requirement for IL-6 and the IL-6R complex to allow NRG-1 mediated HER2 activation, and a HER2 driven IL-6 production feedback loop.
Literature	GENERIF_SUMMARY	Promoter H3K4me3 was examined for two candidate genes, CCND1 and ERBB2, with chromatin-immunoprecipitation coupled with real-time PCR (ChIP-PCR). These methods and MTS assay were used to verify a relationship between H3K4me3 levels and CCND1 and ERBB2, and to investigate cell viability in

Main Category	Category	Terms
		response to chemotherapeutic drugs in primary cultured pediatric EPN cells
Literature	GENERIF_SUMMARY	miR-770-5p regulates cancer cell proliferation by targeting several genes involved in the HER2 signaling pathway.
Literature	GENERIF_SUMMARY	A biosensor capable of identifying low quantities of breast cancer cells by electrical impedance spectroscopy.
Literature	GENERIF_SUMMARY	miR193a3p was downregulated in HER2 positive breast cancer cells and clinical tissues. Methylationmediated silencing led to decreased expression of miR193a3p in HER2 positive breast cancer. Overexpression of miR193a3p could inhibit proliferation, migration and invasion of breast cancer cells.
Literature	GENERIF_SUMMARY	High HER2 expression is associated with Gastric and Gastroesophageal Junction Adenocarcinoma
Literature	GENERIF_SUMMARY	Oncogenic ERBB2 aberrations and KRAS mutations cooperate to promote pancreatic ductal adenocarcinoma progression.
Literature	GENERIF_SUMMARY	immunohistochemistry is a good screening tool for negative and positive results. Any patient targeted for trastuzumab therapy should undergo confirmation of HER2/neu equivocal status by FISH analysis
Literature	GENERIF_SUMMARY	HER2 upregulated ATF4 expression to promote cell migration in MCF7 cell line.
Literature	GENERIF_SUMMARY	Patients with an HER2-aberrant tumor lung cancer occurred more frequently in females and had a significantly worse prognosis. Among the 724 epidermal growth factor receptor wild-type tumors, 21 (2.9%) were HER2-mutant tumors, including A775-G776insYVMA (n = 15).
Literature	GENERIF_SUMMARY	For CRC tumors that express HER2 protein in the absence of HER2 amplification.
Literature	GENERIF_SUMMARY	Treatment with lapatinib reduced BeWo cell proliferation by inducing apoptosis. Moreover, AREG treatment stimulated BeWo cell proliferation by activating ERK1/2 and PI3K/AKT signaling pathways, which was blocked by lapatinib. Targeting EGFR/HER2 might be a useful therapeutic strategy for human choriocarcinoma
Literature	GENERIF_SUMMARY	Since the initial development of HER2 biomarker testing guidelines, significant spatial intratumoral heterogeneity of HER2 overexpression has been recognized as a major characteristic of this disease. [review]
Literature	GENERIF_SUMMARY	The applicability of combined HER2 and HDAC1 targeting in breast cancer therapy.
Literature	GENERIF_SUMMARY	We provide evidence of a PYK2-driven pro-invasive potential of metformin in pure HER2 cancer therapy and propose that metformin-based therapy should consider

Main Category	Category	Terms
		the molecular heterogeneity of breast cancer to prevent complications associated with cancer chemoresistance, invasion and recurrence in treated patients.
Literature	GENERIF_SUMMARY	Mechanistically, GPAA1 enhanced the levels of metastasis-associated GPI-anchored proteins to increase tumour metastasis and intensified lipid raft formation, which consequently promoted the interaction between EGFR and ERBB2 as well as downstream pro-proliferative signalling.
Literature	GENERIF_SUMMARY	The consistent modulation of these molecules and their possible involvement in the HER2-axis makes them promising new targets to overcome HER2-activation
Literature	GENERIF_SUMMARY	HER2-L755S confers hyperactivation of MAPK and PI3K/AKT/mTOR pathways.
Literature	GENERIF_SUMMARY	Data found that ErbB2 proteins are contained in invasive bladder cancer (BC)-derived exosomes in a CRK-dependent manner and transferred to the distant metastasis of BC.
Literature	GENERIF_SUMMARY	our study demonstrated that HER2 protein expression level (IHC 3+) had significant impact on pCR, along with grade 3 histology, hormone receptor-negative status, and use of trastuzumab and pertuzumab.
Literature	GENERIF_SUMMARY	Against human epidermal growth factor receptor 2 (HER2)-amplified gastric cancer cells.
Literature	GENERIF_SUMMARY	Studied expression levels of erb-b2 receptor tyrosine kinase 2 (Her2) in gastric cancer.
Literature	GENERIF_SUMMARY	esults in light of previous studies relating RTK activity to cholesterol enrichment and/or depletion, and propose a novel model to explain our data that includes the recognition and binding of cholesterol by the Neu transmembrane domain through a putative cholesterol-recognition/interaction amino acid consensus sequence.
Literature	GENERIF_SUMMARY	Frequent amplifications of ERBB2 is associated with primary invasive lobular breast carcinoma.
Literature	GENERIF_SUMMARY	Amplification of HER-2/neu gene, along with high expression of c-Myc, HER-2/neu and cyclin E proteins, are associated with such indices of tumor progression as a low differentiation grade and deep myometrial invasion, suggesting the potential possibility of including these markers in the panel for determining the molecular endometrial carcinomas subtype associated with an aggressive course of the disease.
Literature	GENERIF_SUMMARY	Ezh2 overexpression and activity are pivotal in ErbB2-mediated mammary tumourigenesis.
Literature	GENERIF_SUMMARY	High HER2 expression is associated with gastric adenocarcinoma.
Literature	GENERIF_SUMMARY	Studied association of prognosis and survival in gastric

Main Category	Category	Terms
		cancer patients with regards to phosphatase and tensin homolog (PTEN), epidermal growth factor receptor (EGFR), erb-b2 receptor tyrosine kinase 2 (HER-2), and insulin like growth factor 1 receptor (IGF-1R) expression.
Literature	GENERIF_SUMMARY	Upregulation of HER2 in tubular epithelial cell drives fibroblast activation and renal fibrosis.
Literature	GENERIF_SUMMARY	targeting of the HER2-AKT1 cascade augments damage-induced cellular senescence and apoptosis, and enhances STING-mediated antiviral and antitumour immunity
Literature	GENERIF_SUMMARY	High HER2 expression is associated with Gastric Cancer.
Literature	GENERIF_SUMMARY	The present study indicates that HER2-positive BCCs can exhibit VM in an angiogenic microenvironment after eventually acquiring trastuzumab resistance. The clinical finding supports this in vitro observation. Thus, targeting VM might provide a therapeutic benefit to patients with HER2-positive breast cancer.
Literature	GENERIF_SUMMARY	our analysis of public data revealed that Gpn3 overexpression was associated with a significant decrease in overall survival in patients with estrogen receptor-positive and Human epidermal growth factor receptor 2 (HER2+) breast cancer, supporting our proposal that targeting Gpn3 could potentially benefit patients with breast cancer.
Literature	GENERIF_SUMMARY	The results suggest that high VMP1 expression is a potential marker of poor prognosis in HER2 positive BC. Further studies are needed to elucidate how VMP1 could affect pathways supportive of tumorigenesis.
Literature	GENERIF_SUMMARY	Dual HER2 and CDK7 inhibition induced tumor regression in two HER2iR BC xenograft models in vivo. Our data support the utilization of CDK7 inhibition as an additional therapeutic avenue that blocks the activation of genes engaged by multiple HER2iR kinases.
Literature	GENERIF_SUMMARY	Several mechanisms of endocrine resistance in ER+/ HER2- advanced breast cancers, through ER-alpha itself, receptor tyrosine signaling, or cell cycle pathway, have been identified to be pivotal in endocrine therapy. [review]
Literature	GENERIF_SUMMARY	Mathematical modeling of drug-induced receptor internalization in the HER2-positive SKBR3 breast cancer cell-line.
Literature	GENERIF_SUMMARY	Our data shows that HER-2 characterization on CTCs should be focused on HER-2high expressing CTCs in patient samples containing >/=5 CTCs.
Literature	GENERIF_SUMMARY	HER2-Specific Reduction-Sensitive Immunopolymersomes with High Loading of Epirubicin for Targeted Treatment of Ovarian Tumor.
Literature	GENERIF_SUMMARY	role for HER2 in regulating gastric cancer stem cell

Main Category	Category	Terms
		activity, with Wnt/beta-catenin signaling being mediated via a HER2-dependent pathway.
Literature	GENERIF_SUMMARY	The expressions of ER, PR, HER-2 and Ki-67 in metastatic lesions are associated with the prognosis of breast cancer patients.Their expression discordances between primary and metastatic lesions can guide the treatment and evaluate the risks of recurrence and prognosis.
Literature	GENERIF_SUMMARY	ErbB1 and ErbB2 overexpression in patients with sinonasal inverted papilloma and inverted papilloma with squamous cell carcinoma in China.
Literature	GENERIF_SUMMARY	The role of HER2 targeted therapies in sHER-positive/ tHER2-negative cases upon disease progression are warranted.
Literature	GENERIF_SUMMARY	A substantial discordance in ER, PR, and HER2 status were observed between primary breast cancer tissues and recurrent lesions. Rebiopsy could bring new therapeutic opportunities in the management of patients with recurrent breast cancer.
Literature	GENERIF_SUMMARY	The metastatic patterns in breast cancer strongly correlate with various breast cancer subtypes, mainly designated by ER, PR, and HER2.
Literature	GENERIF_SUMMARY	The level of HER-2/neu mRNA expression can be used as a marker for classified them and their response to different treatment.
Literature	GENERIF_SUMMARY	There was no significant difference in the expression of AR between Her-2 positive and negative cases.
Literature	GENERIF_SUMMARY	High p16 protein expression was associated with ER-positive, PR-negative, and HER2-negative tumors.
Literature	GENERIF_SUMMARY	neurotensin receptor 1 regulates the transactivation of the EGFR and HER2 in a reactive oxygen species-dependent manner
Literature	GENERIF_SUMMARY	A TFAP2C Gene Signature Is Predictive of Outcome in HER2-Positive Breast Cancer.
Literature	GENERIF_SUMMARY	It is recommended to monitor positivity rate of ER, PR, Ki67 and Her2 yearly or periodically to comply with quality assurance requirements.
Literature	GENERIF_SUMMARY	Durable Responses to Afatinib as First-line Therapy for HER2-mutated Metastatic Non-small-cell Lung Cancer.
Literature	GENERIF_SUMMARY	HER-2 positivity in patients with breast carcinoma, namely, overexpression and/or amplification of the HER-2 receptor, is associated with more aggressive tumor phenotypes, malignant transformation of cells, poor clinical outcomes and increased resistance to endocrine therapy.
Literature	GENERIF_SUMMARY	HER2 protein is overexpressed in the gastric adenocarcinomas.

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Literature	GENERIF_SUMMARY	Our findings demonstrate that circERBB2 serves as an important regulator of cancer cell proliferation and shows the potential to be a new therapeutic target of GBC.
Literature	GENERIF_SUMMARY	The potential roles of several oncogenic pathways (phosphatase and tensin homolog (PTEN) and phosphatidylinositol 3-kinase (PI3K)) in relation to HER2 status.
Literature	GENERIF_SUMMARY	The MS-lincRNA landscape reveals a novel lincRNA BCLIN25 that contributes to tumorigenesis by upregulating ERBB2 expression via epigenetic modification and RNA-RNA interactions in breast cancer.
Literature	GENERIF_SUMMARY	Study on HER2 expression shows a low frequency of HER2 gene amplification in all subtypes of cholangiocarcinogenesis (CCA) demonstrating that in a subset of patients HER2 gene amplification already occurs in high-grade biliary intraepithelial neoplasia.
Literature	GENERIF_SUMMARY	Dissecting the molecular recognition of dual lapatinib derivatives for EGFR/HER2.
Literature	GENERIF_SUMMARY	Nectin-4 cis-interacts with ErbB2 and its trastuzumab-resistant splice variants, enhancing their activation and DNA synthesis.
Literature	GENERIF_SUMMARY	Bispecific antibodies enhance tumor-infiltrating T cell cytotoxicity against autologous HER-2-expressing high-grade ovarian tumors.
Literature	GENERIF_SUMMARY	The results reveal a low prevalence of HER2 positivity and highlight HER2 gene amplification as an early, potentially driving event in gallbladder carcinogenesis.
Literature	GENERIF_SUMMARY	YES1 plays an important role in the emergence of resistance to HER2-targeted drugs.
Literature	GENERIF_SUMMARY	We show that regular measurement of sHER2, CA15.3, and CEA levels is useful for predicting the therapeutic response and for monitoring HER2-targeted therapy in patients with HER2-positive metastatic breast cancer.
Literature	GENERIF_SUMMARY	Contribution of synergism between PHF8 and HER2 signalling to breast cancer development and drug resistance.
Literature	GENERIF_SUMMARY	Increased assessment of HER2 in metastatic gastroesophageal cancer patients
Literature	GENERIF_SUMMARY	Genomic profiling of pleomorphic and florid lobular carcinoma in situ reveals highly recurrent ERBB2 and ERRB3 alterations.
Literature	GENERIF_SUMMARY	AR pathway activity correlates with AR expression in a HER2-dependent manner and serves as a better prognostic factor in breast cancer.
Literature	GENERIF_SUMMARY	HER-2 overexpression is a negative predictive factor for recurrence in patients with non-muscle-invasive bladder

Main Category	Category	Terms
		cancer on intravesical therapy.
Literature	GENERIF_SUMMARY	An insertion mutation of ERBB2 enhances breast cancer cell growth and confers resistance to lapatinib through AKT signaling pathway.
Literature	GENERIF_SUMMARY	HER2 Status in Molecular Subtypes of Urothelial Carcinoma of the Renal Pelvis and Ureter.
Literature	GENERIF_SUMMARY	Urothelial cancer harbours EGFR and HER2 amplifications and exon 20 insertions.
Literature	GENERIF_SUMMARY	Structure-Activity Relationship of HER2 Receptor Targeting Peptide and Its Derivatives in Targeted Tumor Therapy.
Literature	GENERIF_SUMMARY	Amplification of the HER2 gene was found in 15% of patients with early recurrence. None of the patients without relapse had HER2-amplified tumors. Post-relapse survival in patients with HER2 gene amplification and/or TP53 mutation in primary tumors was shorter than that in patients without these mutations, especially among postmenopausal women.
Literature	GENERIF_SUMMARY	Targetable ERBB2 mutations identified in neurofibroma/schwannoma hybrid nerve sheath tumors.
Literature	GENERIF_SUMMARY	Dual-Receptor-Targeted (DRT) Radiation Nanomedicine Labeled with (177)Lu Is More Potent for Killing Human Breast Cancer Cells That Coexpress HER2 and EGFR Than Single-Receptor-Targeted (SRT) Radiation Nanomedicines.
Literature	GENERIF_SUMMARY	Expression of Female Sex Hormone Receptors, Connective Tissue Growth Factor and HER2 in Gallbladder Cancer.
Literature	GENERIF_SUMMARY	Association of Germline Genetic Testing Results With Locoregional and Systemic Therapy in Patients With Breast Cancer.
Literature	GENERIF_SUMMARY	A Small Number of HER2 Redirected CAR T Cells Significantly Improves Immune Response of Adoptively Transferred Mouse Lymphocytes against Human Breast Cancer Xenografts.
Literature	GENERIF_SUMMARY	Somatic mutations and copy number variations in breast cancers with heterogeneous HER2 amplification.
Literature	GENERIF_SUMMARY	Association between serum HER2/ErbB2 levels and coronary artery disease
Literature	GENERIF_SUMMARY	HER2-positivity in mucinous cancer might be related to accompanying infiltrating ductal carcinoma histology.
Literature	GENERIF_SUMMARY	Clinicopathological value of ErbB2 gene and protein expression in osteochondroma.
Literature	GENERIF_SUMMARY	ECM1 secreted by HER2-overexpressing breast cancer cells promotes formation of a vascular niche accelerating cancer cell migration and invasion.

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Literature	GENERIF_SUMMARY	Long-term cardiac outcomes of patients with HER2-positive breast cancer treated in the adjuvant lapatinib and/or trastuzumab Treatment Optimization Trial.
Literature	GENERIF_SUMMARY	Cost-effectiveness of a Dual (Immunohistochemistry and Fluorescence In Situ Hybridization) HER2/neu Testing Strategy on Invasive Breast Cancers.
Literature	GENERIF_SUMMARY	HER2 expression and relevant clinicopathological features in esophageal squamous cell carcinoma in a Chinese population.
Literature	GENERIF_SUMMARY	HER2 as a potential biomarker of lymph node metastasis in undifferentiated early gastric cancer.
Literature	GENERIF_SUMMARY	Evaluation of HER2 expression in urothelial carcinoma cells as a biomarker for circulating tumor cells.
Literature	GENERIF_SUMMARY	FDA Approval Summary
Literature	GENERIF_SUMMARY	HER2 as a Predictive Biomarker and Treatment Target in Colorectal Cancer.
Literature	GENERIF_SUMMARY	FOXO3A plays an important role in mediating the effects of trastuzumab, and combination chemotherapy may be a promising treatment for patients with HER2-overexpressing esophageal squamous cell carcinoma .
Literature	GENERIF_SUMMARY	Clinical significance of sHER2ECD and calpain10 expression in tumor tissues of patients with breast cancer.
Literature	GENERIF_SUMMARY	THE STUDY OF THE EXPRESSION OF CDH1, KI67, P53 AND HER2 IN DIFFUSE GASTRIC CARCINOMA.
Literature	GENERIF_SUMMARY	Human Epidermal Growth Factor Receptor 2-Positive Breast Cancer Is Associated with Indigenous American Ancestry in Latin American Women.
Literature	GENERIF_SUMMARY	Biologicals to direct nanotherapeutics towards HER2-positive breast cancers.
Literature	GENERIF_SUMMARY	Incidence and Survival by Human Epidermal Growth Factor Receptor 2 Status in Young Women With Stage I-III Breast Cancer
Literature	GENERIF_SUMMARY	Monitoring the Crosstalk Between the Estrogen Receptor and Human Epidermal Growth Factor Receptor 2 with PET.
Literature	GENERIF_SUMMARY	Patient-reported outcomes from KATHERINE
Literature	GENERIF_SUMMARY	HER-2/neu over expression in gall bladder adenocarcinoma
Literature	GENERIF_SUMMARY	Concurrent ERBB2 missense mutations D769Y and D742 N Are Novel Acquired Mechanism of Gefitinib Resistance but Responds to Gefitinib plus Pyrotinib.
Literature	GENERIF_SUMMARY	The deubiquitylase USP2 maintains ErbB2 abundance via counteracting endocytic degradation and represents a therapeutic target in ErbB2-positive breast cancer.

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Literature	GENERIF_SUMMARY	Correlation between HER2 Expression and Clinicopathological Features of Breast Cancer
Literature	GENERIF_SUMMARY	HER2 intratumoral heterogeneity is independently associated with distal metastasis and overall survival in HER2-positive breast carcinomas.
Literature	GENERIF_SUMMARY	HER2 Overexpression as a Poor Prognostic Determinant in Resected Biliary Tract Cancer.
Literature	GENERIF_SUMMARY	ERBB2/HER2 mutations are transforming and therapeutically targetable in leukemia.
Literature	GENERIF_SUMMARY	PD-L1 and HER2 Expression in Gastroesophageal Cancer
Literature	GENERIF_SUMMARY	Association between HER2 and IL-6 genes polymorphisms and clinicopathological characteristics of breast cancer
Literature	GENERIF_SUMMARY	H2Mab-19 Anti-Human Epidermal Growth Factor Receptor 2 Monoclonal Antibody Therapy Exerts Antitumor Activity in Pancreatic Cancer Xenograft Models.
Literature	GENERIF_SUMMARY	a relatively high rate of HER2 amplifications in gallbladder adenocarcinomas
Literature	GENERIF_SUMMARY	Expression of HER-2/neu in Oral Squamous Cell Carcinoma.
Literature	GENERIF_SUMMARY	Clinical and molecular characteristics of Chinese non-small cell lung cancer patients with ERBB2 transmembrane domain mutations.
Literature	GENERIF_SUMMARY	ERBB2 gene expression silencing involved in ovarian cancer cell migration and invasion through mediating MAPK1/MAPK3 signaling pathway.
Literature	GENERIF_SUMMARY	This meta-analysis indicated that c-erbB-2 overexpression was significantly correlated with the poor prognosis in oral cancer patients. In addition, c-erbB-2 overexpression was associated with occurrence risk, gender, lymph node metastasis, and differentiation grade.
Literature	GENERIF_SUMMARY	Anti-correlation of HER2 and focal adhesion complexes in the plasma membrane.
Literature	GENERIF_SUMMARY	Non-amplification genetic alterations of HER2 gene in non-small cell lung carcinoma.
Literature	GENERIF_SUMMARY	Immunotargeting of the xCT Cystine/Glutamate Antiporter Potentiates the Efficacy of HER2-Targeted Immunotherapies in Breast Cancer.
Literature	GENERIF_SUMMARY	Transcriptional activation of cyclin D1 via HER2/HER3 contributes to EGFR-TKI resistance in lung cancer.
Literature	GENERIF_SUMMARY	A rational mouse model to detect on-target, off-tumor CAR T cell toxicity.

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Literature	GENERIF_SUMMARY	Predictive Value of Molecular Subtypes in Premenopausal Women with Hormone Receptor-positive Early Breast Cancer
Literature	GENERIF_SUMMARY	Nuclear ErbB2 expression in hepatocytes in liver disease.
Literature	GENERIF_SUMMARY	The Expression of p53, CD44, Ki-67, and HER-2/neu Markers in Gastric Cancer and Its Association with Histopathological Indicators
Literature	GENERIF_SUMMARY	Genomic Alterations of NTRK, POLE, ERBB2, and Microsatellite Instability Status in Chinese Patients with Colorectal Cancer.
Literature	GENERIF_SUMMARY	Genetic mutation profile of Chinese HER2-positive breast cancers and genetic predictors of responses to Neoadjuvant anti-HER2 therapy.
Literature	GENERIF_SUMMARY	Differential prognostic value of positive HER2 status determined by immunohistochemistry or fluorescence in situ hybridization in breast cancer.
Literature	GENERIF_SUMMARY	Enhancement of the antitumor effect of HER2-directed CAR-T cells through blocking epithelial-mesenchymal transition in tumor cells.
Literature	GENERIF_SUMMARY	HER2 Testing in Endometrial Serous Carcinoma
Literature	GENERIF_SUMMARY	A novel tumor suppressor ZBTB1 regulates tamoxifen resistance and aerobic glycolysis through suppressing HER2 expression in breast cancer.
Literature	GENERIF_SUMMARY	Human epidermal growth factor 2 (HER2) in early stage uterine serous carcinoma
Literature	GENERIF_SUMMARY	An ErbB2 splice variant lacking exon 16 drives lung carcinoma.
Literature	GENERIF_SUMMARY	WNT5A augments cell invasiveness by inducing CXCL8 in HER2-positive breast cancer cells.
Literature	GENERIF_SUMMARY	Oncogenic mutations within the beta3-alphaC loop of EGFR/ERBB2/BRAF/MAP2K1 predict response to therapies.
Literature	GENERIF_SUMMARY	""""HER2 immunohistochemistry inter-observer reproducibility in 205 cases of invasive breast carcinoma additionally tested by ISH"""" Answer to the statistical issue to avoid misinterpretation."
Literature	GENERIF_SUMMARY	Prognostic and predictive value of the combination of TOP2A and HER2 in node-negative tumors 2 cm or smaller (T1N0) breast cancer.
Literature	GENERIF_SUMMARY	Targetable ERBB2 mutation status is an independent marker of adverse prognosis in estrogen receptor positive, ERBB2 non-amplified primary lobular breast carcinoma
Literature	GENERIF_SUMMARY	Clinicopathologic features of breast cancer reclassified as HER2-amplified by fluorescence in situ hybridization

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		with alternative chromosome 17 probes.
Literature	GENERIF_SUMMARY	Correlation between numerical and categorical immunohistochemical score of Ki-67 and HER2 with clinicopathological parameters of breast cancer.
Literature	GENERIF_SUMMARY	HER2 genomic heterogeneity is a frequent event in gastroesophageal adenocarcinoma and correlates with tumor morphology.
Literature	GENERIF_SUMMARY	PIK3CA Mutation in the ShortHER Randomized Adjuvant Trial for Patients with Early HER2(+) Breast Cancer
Literature	GENERIF_SUMMARY	Canonical ErbB-2 isoform and ErbB-2 variant c located in the nucleus drive triple negative breast cancer growth.
Literature	GENERIF_SUMMARY	Her2 promotes early dissemination of breast cancer by suppressing the p38-MK2-Hsp27 pathway that is targetable by Wip1 inhibition.
Literature	GENERIF_SUMMARY	Differential Expression of HER2 and SKP2 in Benign and Malignant Colorectal Lesions.
Literature	GENERIF_SUMMARY	Arsenic-induced HER2 promotes proliferation, migration and angiogenesis of bladder epithelial cells via activation of multiple signaling pathways in vitro and in vivo.
Literature	GENERIF_SUMMARY	Details of human epidermal growth factor receptor 2 status in 454 cases of biliary tract cancer.
Literature	GENERIF_SUMMARY	Systemic therapy for metastatic HER2-positive breast cancer.
Literature	GENERIF_SUMMARY	Circular HER2 RNA positive triple negative breast cancer is sensitive to Pertuzumab.
Literature	GENERIF_SUMMARY	Risk-based Approaches for Optimizing Treatment in HER2-Positive Early Stage Breast Cancer.
Literature	GENERIF_SUMMARY	Dual blockade of CD47 and HER2 eliminates radioresistant breast cancer cells.
Literature	GENERIF_SUMMARY	Specific targeting of HER2-positive human breast carcinoma SK-BR-3 cells by amygdaline-ZHER2 affibody conjugate.
Literature	GENERIF_SUMMARY	Palbociclib and Trastuzumab in HER2-Positive Advanced Breast Cancer
Literature	GENERIF_SUMMARY	Phosphorylation of STAT3 and ERBB2 mediates hypoxiainduced VEGF release in ARPE19 cells.
Literature	GENERIF_SUMMARY	miR2045p promotes apoptosis and inhibits migration of gastric cancer cells by targeting HER2.
Literature	GENERIF_SUMMARY	Her2 promotes early dissemination of breast cancer by suppressing the p38 pathway through Skp2-mediated proteasomal degradation of Tpl2.

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Literature	GENERIF_SUMMARY	Outcomes of Advanced Gastroesophageal Cancer Patients with Equivocal HER2 Expression with or without ERBB2 Gene Amplification.
Literature	GENERIF_SUMMARY	Case report
Literature	GENERIF_SUMMARY	Intratumoral and intertumoral heterogeneity of HER2 immunohistochemical expression in gastric cancer.
Literature	GENERIF_SUMMARY	Tyrosine Kinase Inhibitors in the Combination Therapy of HER2 Positive Breast Cancer.
Literature	GENERIF_SUMMARY	Direct stimulation of ERBB2 highlights a novel cytostatic signaling pathway driven by the receptor Thr(701) phosphorylation.
Literature	GENERIF_SUMMARY	CDK12 and HER2 coamplification in two urothelial carcinomas with rapid and aggressive clinical progression.
Literature	GENERIF_SUMMARY	Allele-Specific Role of ERBB2 in the Oncogenic Function of EGFR L861Q in EGFR-Mutant Lung Cancers.
Literature	GENERIF_SUMMARY	Calcium based siRNA coating
Literature	GENERIF_SUMMARY	TSTA3 facilitates esophageal squamous cell carcinoma progression through regulating fucosylation of LAMP2 and ERBB2.
Literature	GENERIF_SUMMARY	Expression and significance of Her2 and Ki-67 in gastric adenocarcinoma without distant metastasis
Literature	GENERIF_SUMMARY	Detection of ERBB2 amplification in uterine serous carcinoma by next-generation sequencing
Literature	GENERIF_SUMMARY	Fluorescence in situ hybridisation analysis of human epidermal growth factor receptor 2 status for breast cancer cases in Brunei Darussalam.
Literature	GENERIF_SUMMARY	An analysis of HER2 amplification in cervical adenocarcinoma
Literature	GENERIF_SUMMARY	Prevalence of germline BRCA mutations in HER2-negative metastatic breast cancer
Literature	GENERIF_SUMMARY	The Effects of Low HER2/neu Expression on the Clinicopathological Characteristics of Triple-Negative Breast Cancer Patients.
Literature	GENERIF_SUMMARY	Expression of HER2 and BRCA1 Correlates with Prognosis in Patients with Breast Cancer After Radiotherapy
Literature	GENERIF_SUMMARY	Triple-Positive Breast Carcinoma
Literature	GENERIF_SUMMARY	HER2 Mutated and Nonmutated Non-small Cell Lung Carcinomas Can Harbor Heterogeneous HER2 Gene Amplification and HER2 Protein Expression.
Literature	GENERIF_SUMMARY	Addition of HER2 and CD44 to (18)F-FDG PET-based clinico-radiomic models enhances prediction of neoadjuvant chemoradiotherapy response in

Main Category	Category	Terms
		esophageal cancer.
Literature	GENERIF_SUMMARY	A multivariable prognostic score to guide systemic therapy in early-stage HER2-positive breast cancer
Literature	GENERIF_SUMMARY	[Correlation between HER2 expression before neoadjuvant treatment and RCB classification after surgical treatment in breast cancer].
Literature	GENERIF_SUMMARY	A pan-cancer analysis of HER2 index revealed transcriptional pattern for precise selection of HER2-targeted therapy.
Literature	GENERIF_SUMMARY	CDKL2 Is Associated with HER2 Status and Overall Survival in Gastric Cancer
Literature	GENERIF_SUMMARY	ERBB2-CAR-Engineered Cytokine-Induced Killer Cells Exhibit Both CAR-Mediated and Innate Immunity Against High-Risk Rhabdomyosarcoma.
Literature	GENERIF_SUMMARY	The Role of ERBB2/HER2 Tyrosine Kinase Receptor in the Regulation of Cell Death.
Literature	GENERIF_SUMMARY	EZH2-mediated PP2A inactivation confers resistance to HER2-targeted breast cancer therapy.
Literature	GENERIF_SUMMARY	Aberrant ALOX5 Activation Correlates with HER2 Status and Mediates Breast Cancer Biological Activities through Multiple Mechanisms.
Literature	GENERIF_SUMMARY	Relationship between HER2 expression and tumor interstitial angiogenesis in primary gastric cancer and its effect on prognosis.
Literature	GENERIF_SUMMARY	Third-line treatment of HER2-positive advanced breast cancer
Literature	GENERIF_SUMMARY	EGFR Expression in HER2-Driven Breast Cancer Cells.
Literature	GENERIF_SUMMARY	Correlation study between the magnetic resonance imaging features of breast cancer and expression of immune molecular subtypes.
Literature	GENERIF_SUMMARY	SATB-1 and Her2 as predictive molecular and immunohistochemical markers for urothelial cell carcinoma of the bladder.
Literature	GENERIF_SUMMARY	Heritability of Low ER Staining/HER2-Breast Tumors
Literature	GENERIF_SUMMARY	Human Epidermal Growth Factor Receptor-2 gene polymorphism and breast cancer risk in women from the Northeastern region of Brazil.
Literature	GENERIF_SUMMARY	Clinical significance of HER2 status in T1bN0 breast cancer
Literature	GENERIF_SUMMARY	Th1 cytokine interferon gamma improves response in HER2 breast cancer by modulating the ubiquitin proteasomal pathway.
Literature	GENERIF_SUMMARY	ErbB2 Receptor Tyrosine Kinase 2 is negatively regulated by the p53responsive microRNA31845p in

Main Category	Category	Terms
		cervical cancer cells.
Literature	GENERIF_SUMMARY	Functional Common and Rare ERBB2 Germline Variants Cooperate in Familial and Sporadic Cancer Susceptibility.
Literature	GENERIF_SUMMARY	A feed-forward loop between SorLA and HER3 determines heregulin response and neratinib resistance.
Literature	GENERIF_SUMMARY	Histopathological features of HER2 overexpression in uterine carcinosarcoma
Literature	GENERIF_SUMMARY	A study of Human Epidermal Growth factor receptor-2 [HER-2] in Carcinoma Oesophagus [Single Institutional Tertiary cancer centre study from North-East part of India].
Literature	GENERIF_SUMMARY	Role of HER2neu expression in gastric cancer.
Literature	GENERIF_SUMMARY	Correlation of cyclin D1, HER2, and AMACR expressions with histologic grade in bladder urothelial carcinomas.
Literature	GENERIF_SUMMARY	miR-146a-3p suppressed the differentiation of hAMSCs into Schwann cells via inhibiting the expression of ERBB2.
Literature	GENERIF_SUMMARY	Cancer Vaccines, Treatment of the Future
Literature	GENERIF_SUMMARY	Autophagy Blockade Limits HER2+ Breast Cancer Tumorigenesis by Perturbing HER2 Trafficking and Promoting Release Via Small Extracellular Vesicles.
Literature	GENERIF_SUMMARY	Clinical significance of quantitative categorization of HER2 fluorescent in situ hybridization results in invasive breast cancer patients treated with HER2-targeted agents.
Literature	GENERIF_SUMMARY	Investigating mutations at the hotspot position of the ERBB2 and screening for the novel lead compound to treat breast cancer - a computational approach.
Literature	GENERIF_SUMMARY	Dysregulation of the NRG1/ERBB pathway causes a developmental disorder with gastrointestinal dysmotility in humans.
Literature	GENERIF_SUMMARY	Role of Cholesterol in Transmembrane Dimerization of the ErbB2 Growth Factor Receptor.
Literature	GENERIF_SUMMARY	HER2Ile655Val Single Nucleotide Polymorphism Associated with Early-Onset Breast Cancer Susceptibility
Literature	GENERIF_SUMMARY	HER2-intronic miR-4728-5p facilitates HER2 expression and accelerates cell proliferation and migration by targeting EBP1 in breast cancer.
Literature	GENERIF_SUMMARY	Analysis of plasma HER2 copy number in cell-free DNA of breast cancer patients
Literature	GENERIF_SUMMARY	The multigene classifiers 95GC/42GC/155GC for precision medicine in ER-positive HER2-negative early

Main Category	Category	Terms
		breast cancer.
Literature	GENERIF_SUMMARY	NFIB promotes the progression of gastric cancer by upregulating circMAP7D1 to stabilize HER2 mRNA.
Literature	GENERIF_SUMMARY	Long non-coding RNA DANCR promoted non-small cell lung cancer cells metastasis via modulating of miR-1225-3p/ErbB2 signal.
Literature	GENERIF_SUMMARY	Targeting transcription of MCL-1 sensitizes HER2-amplified breast cancers to HER2 inhibitors.
Literature	GENERIF_SUMMARY	Clinic-Pathological Characteristics and Prognostic Value of PD-L1 and HER2 in Gastric Cancer.
Literature	GENERIF_SUMMARY	Clinical and molecular immune characterization of ERBB2 in glioma.
Literature	GENERIF_SUMMARY	Clinical significance of circulating tumor cells in the response to trastuzumab for HER2-negative metastatic gastric cancer.
Literature	GENERIF_SUMMARY	Knockdown of AKT3 Activates HER2 and DDR Kinases in Bone-Seeking Breast Cancer Cells, Promotes Metastasis In Vivo and Attenuates the TGFbeta/CTGF Axis.
Literature	GENERIF_SUMMARY	Cell specific tumor suppressor effect of Hsa-miR-1226-3p through downregulation of HER2, PIK3R2, and AKT1 genes.
Literature	GENERIF_SUMMARY	A Novel Anti-HER2 Bispecific Antibody With Potent Tumor Inhibitory Effects In Vitro and In Vivo.
Literature	GENERIF_SUMMARY	Role of WT1, B-cell lymphoma 2, Ki-67 (Mib1), and Her2/Neu as diagnostic and prognostic immunomarkers in ovarian serous and endometroid carcinoma.
Literature	GENERIF_SUMMARY	HER-2 Amplification in Uterine Serous Carcinoma and Serous Endometrial Intraepithelial Carcinoma.
Literature	GENERIF_SUMMARY	Structural and dynamic characterization of the C-terminal tail of ErbB2
Literature	GENERIF_SUMMARY	Triple-negative breast lobular carcinoma
Literature	GENERIF_SUMMARY	A single-cell atlas of the healthy breast tissues reveals clinically relevant clusters of breast epithelial cells.
Literature	GENERIF_SUMMARY	Impact of HER2 assessment by CISH in urothelial carcinoma
Literature	GENERIF_SUMMARY	Nectin-4 and p95-ErbB2 cooperatively regulate Hippo signaling-dependent SOX2 gene expression, enhancing anchorage-independent T47D cell proliferation.
Literature	GENERIF_SUMMARY	Correlation between SNPs of PIK3CA, ERBB2 3'UTR, and their interactions with environmental factors and the risk of epithelial ovarian cancer.
Literature	GENERIF_SUMMARY	Dual-Energy CT-Based Nomogram for Decoding HER2

Main Category	Category	Terms
		Status in Patients With Gastric Cancer.
Literature	GENERIF_SUMMARY	Nuclear ErbB2 represses DEPTOR transcription to inhibit autophagy in breast cancer cells.
Literature	GENERIF_SUMMARY	Homogeneous tumor targeting with a single dose of HER2-targeted albumin-binding domain-fused nanobody-drug conjugates results in long-lasting tumor remission in mice.
Literature	GENERIF_SUMMARY	HER2 Splice Site Mutation c.1899-1G>A as the Potential Acquired Resistance to Trastuzumab in a Patient with HER2-Positive Gastric Adenocarcinoma.
Literature	GENERIF_SUMMARY	The role of HER2 and HER3 in HER2-amplified cancers beyond breast cancers.
Literature	GENERIF_SUMMARY	Is there a correlation between HER2 gene amplification level and response to neoadjuvant treatment with trastuzumab and chemotherapy in HER2-positive breast cancer?
Literature	GENERIF_SUMMARY	ST6Gal1 targets the ectodomain of ErbB2 in a site-specific manner and regulates gastric cancer cell sensitivity to trastuzumab.
Literature	GENERIF_SUMMARY	Loss of DLX3 tumor suppressive function promotes progression of SCC through EGFR-ERBB2 pathway.
Literature	GENERIF_SUMMARY	Clinical evaluation of BCL-2/XL levels pre- and post-HER2-targeted therapy.
Literature	GENERIF_SUMMARY	Clinical Outcomes Among Major Breast Cancer Subtypes After Neoadjuvant Chemotherapy
Literature	GENERIF_SUMMARY	Trastuzumab deruxtecan (DS-8201) in patients with HER2-expressing metastatic colorectal cancer (DESTINY-CRC01)
Literature	GENERIF_SUMMARY	Dimeric Her2-specific affibody mediated cisplatin-loaded nanoparticles for tumor enhanced chemo-radiotherapy.
Literature	GENERIF_SUMMARY	Hyperactivation of HER2-SHCBP1-PLK1 axis promotes tumor cell mitosis and impairs trastuzumab sensitivity to gastric cancer.
Literature	GENERIF_SUMMARY	A study of HER2 expression in endometrial carcinoma
Literature	GENERIF_SUMMARY	Cell lineage tracing links ERalpha loss in Erbb2-positive breast cancers to the arising of a highly aggressive breast cancer subtype.
Literature	GENERIF_SUMMARY	USP22 promotes HER2-driven mammary carcinoma aggressiveness by suppressing the unfolded protein response.
Literature	GENERIF_SUMMARY	High MMP-11 expression associated with low CD8+ T cells decreases the survival rate in patients with breast cancer.
Literature	GENERIF_SUMMARY	Urokinase type plasminogen activator receptor (uPAR)

Main Category	Category	Terms
		and human epidermal growth factor receptor 2 (HER2) expression in metastasis of breast cancer.
Literature	GENERIF_SUMMARY	A novel role of ADGRF1 (GPR110) in promoting cellular quiescence and chemoresistance in human epidermal growth factor receptor 2-positive breast cancer.
Literature	GENERIF_SUMMARY	Management of Early-Stage Human Epidermal Growth Factor Receptor 2-Positive Breast Cancer.
Literature	GENERIF_SUMMARY	HER2-PI9 and HER2-I12
Literature	GENERIF_SUMMARY	Induction of EnR stress by Melatonin enhances the cytotoxic effect of Lapatinib in HER2-positive breast cancer.
Literature	GENERIF_SUMMARY	Interaction between HLA-G and NK cell receptor KIR2DL4 orchestrates HER2-positive breast cancer resistance to trastuzumab.
Literature	GENERIF_SUMMARY	Co-occurring gain-of-function mutations in HER2 and HER3 modulate HER2/HER3 activation, oncogenesis, and HER2 inhibitor sensitivity.
Literature	GENERIF_SUMMARY	Clinicopathological response to neoadjuvant therapies and pathological complete response as a biomarker of survival in human epidermal growth factor receptor-2 enriched breast cancer - A retrospective cohort study.
Literature	GENERIF_SUMMARY	Assessment of 25-Year Survival of Women With Estrogen Receptor-Positive/ERBB2-Negative Breast Cancer Treated With and Without Tamoxifen Therapy
Literature	GENERIF_SUMMARY	The prognostic relevance of HER2-positivity gain in metastatic breast cancer in the ChangeHER trial.
Literature	GENERIF_SUMMARY	Locoregional infusion of HER2-specific CAR T cells in children and young adults with recurrent or refractory CNS tumors
Literature	GENERIF_SUMMARY	Clinicopathological Features, Tumor Mutational Burden, and Tumour-Infiltrating Lymphocyte Interplay in ERBB2-Mutated Breast Cancer
Literature	GENERIF_SUMMARY	Topomer-CoMFA proposed as a tool to construct dual EGFR/HER-2 models.
Literature	GENERIF_SUMMARY	Efficacy and safety of trastuzumab emtansine in older patients with HER2-positive advanced breast cancer
Literature	GENERIF_SUMMARY	Circular RNA circ-ERBB2 Elevates the Warburg Effect and Facilitates Triple-Negative Breast Cancer Growth by the MicroRNA 136-5p/Pyruvate Dehydrogenase Kinase 4 Axis.
Literature	GENERIF_SUMMARY	Human epidermal growth factor receptor 2 overexpression in gastric and gastroesophageal junction adenocarcinoma in patients seen at the University Teaching Hospital, Lusaka, Zambia.
Literature	GENERIF_SUMMARY	Comparison of the multigene panel test and OncoScan for the determination of HER2 amplification in breast

Main Category	Category	Terms
		cancer.
Literature	GENERIF_SUMMARY	HER2-Positive Breast Cancer
Literature	GENERIF_SUMMARY	Dendritic Cells Pulsed with Cytokine-Adjuvanted Tumor Membrane Vesicles Inhibit Tumor Growth in HER2-Positive and Triple Negative Breast Cancer Models.
Literature	GENERIF_SUMMARY	The immune microenvironment in EGFR- and ERBB2-mutated lung adenocarcinoma.
Literature	GENERIF_SUMMARY	Novel Nitrogen-Based Chalcone Analogs Provoke Substantial Apoptosis in HER2-Positive Human Breast Cancer Cells via JNK and ERK1/ERK2 Signaling Pathways.
Literature	GENERIF_SUMMARY	Human epidermal growth factor receptor 2 is frequently over-expressed in gastric carcinoma in north Indian patients.
Literature	GENERIF_SUMMARY	Cortactin and HER2 as potential markers for dural-targeted therapy in advanced gastric cancer.
Literature	GENERIF_SUMMARY	Breast Cancer, HER2 Mutations, and Overcoming Drug Resistance.
Literature	GENERIF_SUMMARY	14-3-3zeta promoted invasion and lymph node metastasis of breast invasive ductal carcinoma with HER2 overexpression.
Literature	GENERIF_SUMMARY	Clinical Relevance of Tissue and Serum Human Epidermal Growth Factor Receptor 2 Expression in Patients With Esophageal Squamous Cell Carcinoma.
Literature	GENERIF_SUMMARY	CMBs carrying PTX and CRISPR/Cas9 targeting CerbB2 plasmids interfere with endometrial cancer cells.
Literature	GENERIF_SUMMARY	The Effects of HER2 on CDK4/6 Activity in Breast Cancer.
Literature	GENERIF_SUMMARY	HER2 promotes epithelial-mesenchymal transition through regulating osteopontin in gastric cancer.
Literature	GENERIF_SUMMARY	Comparison of hormonal receptor expression and HER2 status between circulating tumor cells and breast cancer metastases.
Literature	GENERIF_SUMMARY	Spatial deconvolution of HER2-positive breast cancer delineates tumor-associated cell type interactions.
Literature	GENERIF_SUMMARY	An Elaborate New Linker System Significantly Enhances the Efficacy of an HER2-Antibody-Drug Conjugate against Refractory HER2-Positive Cancers.
Literature	GENERIF_SUMMARY	Sertoli Cells Improve Myogenic Differentiation, Reduce Fibrogenic Markers, and Induce Utrophin Expression in Human DMD Myoblasts.
Literature	GENERIF_SUMMARY	IKZF3 deficiency potentiates chimeric antigen receptor T cells targeting solid tumors.
Literature	GENERIF_SUMMARY	Clinicopathological features and BRCA1 and BRCA2

Main Category	Category	Terms
		mutation status in a prospective cohort of young women with breast cancer.
Literature	GENERIF_SUMMARY	Opposing transcriptional programs of KLF5 and AR emerge during therapy for advanced prostate cancer.
Literature	GENERIF_SUMMARY	Cdc25A inhibits autophagy-mediated ferroptosis by upregulating ErbB2 through PKM2 dephosphorylation in cervical cancer cells.
Literature	GENERIF_SUMMARY	Structures of the HER2-HER3-NRG1beta complex reveal a dynamic dimer interface.
Literature	GENERIF_SUMMARY	Quantification of EGFR-HER2 Heterodimers in HER2-Overexpressing Breast Cancer Cells Using Liquid-Phase Electron Microscopy.
Literature	GENERIF_SUMMARY	Human epidermal growth factor receptor 2 inhibits activating transcription factor 7 to promote breast cancer cell migration by activating histone lysine demethylase 1.
Literature	GENERIF_SUMMARY	LASP-1 interacts with ErbB2 in ovarian cancer cells.
Literature	GENERIF_SUMMARY	Moonlight human ribosomal protein L13a downregulation is associated with p53 and HER2/neu expression in breast cancer.
Literature	GENERIF_SUMMARY	High Frequency of Juxtamembrane Domain ERBB2 Mutation in Gastric Cancer.
Literature	GENERIF_SUMMARY	Histopathologic features and molecular genetic landscape of HER2-amplified endometrial carcinomas.
Literature	GENERIF_SUMMARY	Characteristics of HER2-negative breast cancers with FISH-equivocal status according to 2018 ASCO/CAP guideline.
Literature	GENERIF_SUMMARY	Integrated approach to functional analysis of an ERBB2 variant of unknown significance detected by a cancer gene panel test.
Literature	GENERIF_SUMMARY	The initial hormone receptor/HER2 subtype is the main determinator of subtype discordance in advanced breast cancer
Literature	GENERIF_SUMMARY	Evolution of low HER2 expression between early and advanced-stage breast cancer.
Literature	GENERIF_SUMMARY	ERBB2 Copy Number as a Quantitative Biomarker for Real-World Outcomes to Anti-Human Epidermal Growth Factor Receptor 2 Therapy in Advanced Gastroesophageal Adenocarcinoma.
Literature	GENERIF_SUMMARY	Immunoexpression of PTEN, HER2/neu, and Ki-67 in endoscopic gastric carcinoma biopsies; their correlation with histological subtypes and one-year survival.
Literature	GENERIF_SUMMARY	Expression of HER2/neu in gastric adenocarcinoma and its correlation with serum HER2/neu level and E-cadherin expression.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	ErbB2 promotes breast cancer metastatic potential via HSF1/LDHA axis-mediated glycolysis.
Literature	GENERIF_SUMMARY	KIR-HLA Functional Repertoire Influences Trastuzumab Efficiency in Patients With HER2-Positive Breast Cancer.
Literature	GENERIF_SUMMARY	Targetable HER3 functions driving tumorigenic signaling in HER2-amplified cancers.
Literature	GENERIF_SUMMARY	PD-L1 and HER2 expression in gastric adenocarcinoma and their prognostic significance.
Literature	GENERIF_SUMMARY	HER2 Overexpression and Amplification in Uterine Carcinosarcomas With Serous Morphology.
Literature	GENERIF_SUMMARY	Positive predictive value of ERBB2 copy number gain by tissue or circulating tumor DNA next-generation sequencing across advanced cancers.
Literature	GENERIF_SUMMARY	HER2 copy number quantification in primary tumor and cell-free DNA provides additional prognostic information in HER2 positive early breast cancer.
Literature	GENERIF_SUMMARY	HNMT Upregulation Induces Cancer Stem Cell Formation and Confers Protection against Oxidative Stress through Interaction with HER2 in Non-Small-Cell Lung Cancer.
Literature	GENERIF_SUMMARY	Identification of Recurrent Insertions and Deletions in Exon 18 and 19 of Human Epidermal Growth Factor Receptor 2 as Potential Drivers in Non-Small-Cell Lung Cancer and Other Cancer Types.
Literature	GENERIF_SUMMARY	The significance of ErbB2/3 in the conversion of induced pluripotent stem cells into cancer stem cells.
Literature	GENERIF_SUMMARY	Natural History and Characteristics of ERBB2-mutated Hormone Receptor-positive Metastatic Breast Cancer
Literature	GENERIF_SUMMARY	Evaluation of an (131)I-labeled HER2-specific single domain antibody fragment for the radiopharmaceutical therapy of HER2-expressing cancers.
Literature	GENERIF_SUMMARY	A Transcriptional Link between HER2, JAM-A and FOXA1 in Breast Cancer.
Literature	GENERIF_SUMMARY	Significance of HER2 and VEGFR2 in Early-stage Endometrial Cancer.
Literature	GENERIF_SUMMARY	Associations of HER2 Mutation With Immune-Related Features and Immunotherapy Outcomes in Solid Tumors.
Literature	GENERIF_SUMMARY	Natural History of Human Epidermal Growth Factor Receptor 2-Amplified and Human Epidermal Growth Factor Receptor 2 Wild-Type Refractory Metastatic Colorectal Cancer in US Clinical Practice.
Literature	GENERIF_SUMMARY	Serum ErbB2 concentration positively correlated to the glycemic variations in newly diagnosed Type 2 diabetic patients.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	The NF-kappaB/miR-488/ERBB2 axis modulates pancreatic cancer cell malignancy and tumor growth through cell cycle signaling.
Literature	GENERIF_SUMMARY	The status of HER2 in colorectal carcinoma and the relation of HER2 with prognostic parameters and MSI.
Literature	GENERIF_SUMMARY	Changing Trends in Estrogen Receptors/Progesterone Receptors/Human Epidermal Growth Factor Receptor 2 Prevalence Rates Among Jordanian Patients With Breast Cancer Over the Years.
Literature	GENERIF_SUMMARY	ERBB2 S310F mutation independently activates PI3K/AKT and MAPK pathways through homodimers to contribute gallbladder carcinoma growth.
Literature	GENERIF_SUMMARY	Unusual Patterns of HER2 Expression in Breast Cancer
Literature	GENERIF_SUMMARY	FTO mediated ERBB2 demethylation promotes tumor progression in esophageal squamous cell carcinoma cells.
Literature	GENERIF_SUMMARY	Relationship between HER2 overexpression and long-term outcomes of early gastric cancer
Literature	GENERIF_SUMMARY	Clinical significance of ERBB2 S310F mutation in extramammary Paget's disease.
Literature	GENERIF_SUMMARY	The Prognostic Effect of HER2 Gene Amplification in High-Grade Endometrial Carcinomas and its Correlation with Protein Overexpression.
Literature	GENERIF_SUMMARY	Critical roles for EGFR and EGFR-HER2 clusters in EGF binding of SW620 human carcinoma cells.
Literature	GENERIF_SUMMARY	HER2 G776S mutation promotes oncogenic potential in colorectal cancer cells when accompanied by loss of APC function.
Literature	GENERIF_SUMMARY	ERBB2 mutation landscape in non-small cell lung cancer patients in Northeast China.
Literature	GENERIF_SUMMARY	Prognostic and Biologic Significance of ERBB2-Low Expression in Early-Stage Breast Cancer.
Literature	GENERIF_SUMMARY	A Novel miRNA Located in the HER2 Gene Shows an Inhibitory Effect on Wnt Signaling and Cell Cycle Progression.
Literature	GENERIF_SUMMARY	Clinicopathologic significance of Her-2 and P[53] expressions in gastric cancer.
Literature	GENERIF_SUMMARY	Analysis of polymorphisms in EGF, EGFR and HER2 genes in pancreatic neuroendocrine tumors (PNETs).
Literature	GENERIF_SUMMARY	Clinical significance of B7-H3 and HER2 co-expression and therapeutic value of combination treatment in gastric cancer.
Literature	GENERIF_SUMMARY	Umbilicaria esculenta Extract Exhibits Antiwrinkle Activity by Suppressing ErbB2 Phosphorylation.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Both Kras gene mutation and Her2 gene mutation tend to exist in an inverse manner but not mutually exclusive in mucinous ovarian carcinoma
Literature	GENERIF_SUMMARY	HER2-low breast cancer shows a lower immune response compared to HER2-negative cases.
Literature	GENERIF_SUMMARY	EGFR, HER2, and HER3 protein expression in paired primary tumor and lymph node metastasis of colorectal cancer.
Literature	GENERIF_SUMMARY	Molecular Landscape of ERBB2 Alterations in 14,956 Solid Tumors.
Literature	GENERIF_SUMMARY	Urinary HER2, TWEAK and VCAM-1 levels are associated with new-onset proteinuria in paediatric lupus nephritis.
Literature	GENERIF_SUMMARY	The Effect of HER2 Status on Metaplastic Breast Cancer A Propensity Score-Matched Analysis.
Literature	GENERIF_SUMMARY	ErbB2/Her2-dependent downregulation of a cell death-promoting protein BLNK in breast cancer cells is required for 3D breast tumor growth.
Literature	GENERIF_SUMMARY	Reply to Zhu et al.
Literature	GENERIF_SUMMARY	Expression of HER2/NEU in newly diagnosed metastatic carcinoma prostate on TRUS biopsy specimen and its clinical correlation and progression.
Literature	GENERIF_SUMMARY	Interaction between HER2 and ATM predicts poor survival in bladder cancer patients.
Literature	GENERIF_SUMMARY	HER2 c-Terminal Fragments Are Expressed via Internal Translation of the HER2 mRNA.
Literature	GENERIF_SUMMARY	Comparison of Management and Outcomes in ERBB2-Low vs ERBB2-Zero Metastatic Breast Cancer in France.
Literature	GENERIF_SUMMARY	HER2 in Uterine Serous Carcinoma
Literature	GENERIF_SUMMARY	HER2-low inflammatory breast cancer
Literature	GENERIF_SUMMARY	HER2 protein expression correlates with Lauren classification and P53 in gastric cancer patients.
Literature	GENERIF_SUMMARY	Clinical and Genomic Features of HER2 Exon 20 Insertion Mutations and Characterization of HER2 Expression by Immunohistochemistry in East Asian Non-Small-Cell Lung Cancer.
Literature	GENERIF_SUMMARY	ERBB2 exon 20 insertions are rare in Brazilian non-small cell lung cancer.
Literature	GENERIF_SUMMARY	Prognostic value of human epidermal growth factor receptor 2 status and systemic therapy in breast cancer with brain metastases treated with radiotherapy.
Literature	GENERIF_SUMMARY	c-MET positivity and its relationship with histopathological findings in gastric carcinomas exhibiting HER2 gene expression.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Erythroblastic oncogene B-2 status and intracranial metastatic disease in patients with gastrointestinal cancer
Literature	GENERIF_SUMMARY	Real World Characteristics and Clinical Outcomes of HER2-Mutant Non-Small Cell Lung Cancer Patients Detected by Next-Generation Sequencing.
Literature	GENERIF_SUMMARY	The predictive role of ERBB2 point mutations in metastatic colorectal cancer
Literature	GENERIF_SUMMARY	The relevance of EGFR, ErbB receptors and neuregulins in human adipocytes and adipose tissue in obesity.
Literature	GENERIF_SUMMARY	COMMD3-Mediated Endosomal Trafficking of HER2 Inhibits the Progression of Ovarian Carcinoma.
Literature	GENERIF_SUMMARY	Immunohistochemical Study and Clinicopathologic Correlation of Cox-2 and Her-2 Expression in Colorectal Carcinoma
Literature	GENERIF_SUMMARY	Upregulation of SCD1 by ErbB2 via LDHA promotes breast cancer cell migration and invasion.
Literature	GENERIF_SUMMARY	Clinicopathologic significance of human epidermal growth factor receptor 2 expression and gene amplification in gastric carcinoma.
Literature	GENERIF_SUMMARY	The statuses of HER2 expression and mismatch repair in endometrial clear cell carcinoma.
Literature	GENERIF_SUMMARY	Influence of HER2 expression on prognosis in metastatic triple-negative breast cancer-results from an international, multicenter analysis coordinated by the AGMT Study Group.
Literature	GENERIF_SUMMARY	Clinical Utility of Genomic Recurrence Risk Stratification in Early, Hormone-Receptor-Positive, Human Epidermal Growth Factor Receptor 2-Negative Breast Cancer
Literature	GENERIF_SUMMARY	Immunohistochemical Analysis of HER2, EGFR, and Nectin-4 Expression in Upper Urinary Tract Urothelial Carcinoma.
Literature	GENERIF_SUMMARY	Characteristics of the immunogenicity and tumor immune microenvironment in HER2-amplified lung adenocarcinoma.
Literature	GENERIF_SUMMARY	HRCT1, negatively regulated by miR-124-3p, promotes tumor metastasis and the growth of gastric cancer by activating the ERBB2-MAPK pathway.
Literature	GENERIF_SUMMARY	Expression and correlation of PD-L1 and HER2 in oesophageal squamous cell carcinoma.
Literature	GENERIF_SUMMARY	Radiogenomic analysis of prediction HER2 status in breast cancer by linking ultrasound radiomic feature module with biological functions.
Literature	GENERIF_SUMMARY	Association between the Arg72Pro Genotypes in the TP53 Gene and Ile655Val in the HER2 Gene and the Risk of Developing Breast Cancer in the Population of

Main Category	Category	Terms
		Amapa, Northern Brazil.
Literature	GENERIF_SUMMARY	Immunohistochemical based molecular subtypes of muscle-invasive bladder cancer
Literature	GENERIF_SUMMARY	CD98 regulates the phosphorylation of HER2 and a bispecific anti-HER2/CD98 antibody inhibits the growth signal of human breast cancer cells.
Literature	GENERIF_SUMMARY	The adaptor protein VEPH1 interacts with the kinase domain of ERBB2 and impacts EGF signaling in ovarian cancer cells.
Literature	GENERIF_SUMMARY	HER2 Expression in Bladder Cancer
Literature	GENERIF_SUMMARY	Assessment of HER2 in Gastric-Type Endocervical Adenocarcinoma and its Prognostic Significance.
Literature	GENERIF_SUMMARY	HER2 IHC Expression and Gene Amplification in p53-aberrant High-grade Endometrial Endometrioid Carcinoma Suggests That This Population May Benefit From HER2 Testing and Targeted Therapy.
Literature	GENERIF_SUMMARY	HER2+ endometrioid endometrial cancer possesses distinct molecular and immunologic features associated with a more active immune microenvironment and worse prognosis.
Literature	GENERIF_SUMMARY	Loss of the receptors ER, PR and HER2 promotes USP15-dependent stabilization of PARP1 in triple-negative breast cancer.
Literature	GENERIF_SUMMARY	HER2-low in gastro-oesophageal adenocarcinoma
Literature	GENERIF_SUMMARY	Frequency of HER2 protein overexpression and HER2 gene amplification in endometrial clear cell carcinoma.
Literature	GENERIF_SUMMARY	TP53 somatic mutations in Asian breast cancer are associated with subtype-specific effects.
Literature	GENERIF_SUMMARY	Discordances in expression of human epidermal growth factor receptor 2 between primary and metastatic uterine carcinosarcoma
Literature	GENERIF_SUMMARY	Somatic mutation profiling, tumor-infiltrating leukocytes, tertiary lymphoid structures and PD-L1 protein expression in HER2-amplified colorectal cancer.
Literature	GENERIF_SUMMARY	Interferon-gamma is quintessential for NOS2 and COX2 expression in ER[-] breast tumors that lead to poor outcome.
Literature	GENERIF_SUMMARY	HER2 Low Breast Cancer
Literature	GENERIF_SUMMARY	Clinical significance of microinvasive breast cancer across the different subtypes and human epidermal growth factor receptor 2 expression levels.
Literature	GENERIF_SUMMARY	Correlation of PKM2 Expression With HER2/neu and Additional Breast Cancer Biomarkers and its Prognostic Significance.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Correlation between androgen receptor expression and pathological response rate in pre-operative HER2-positive breast cancer patients.
Literature	GENERIF_SUMMARY	Genetic Alterations Detected by Circulating Tumor DNA in HER2-Low Metastatic Breast Cancer.
Literature	GENERIF_SUMMARY	HER2 in uterine serous carcinoma
Literature	GENERIF_SUMMARY	Breast Cancer Mutations HER2V777L and PIK3CAH1047R Activate the p21-CDK4/6-Cyclin D1 Axis to Drive Tumorigenesis and Drug Resistance.
Literature	GENERIF_SUMMARY	Low co-expression of PD-L1 and oncogenic receptor tyrosine kinases HER2 and cMET in urothelial carcinoma is associated with discordant expression between primary and metastatic sites.
Literature	GENERIF_SUMMARY	The ERBB2 Exonic Variant Pro1170Ala Modulates Mitogen-Activated Protein Kinase Signaling Cascades and Associates with Allergic Asthma.
Literature	GENERIF_SUMMARY	ERBB2-PTGS2 axis promotes intervertebral disc degeneration by regulating senescence of nucleus pulposus cells.
Literature	GENERIF_SUMMARY	Upregulated Expression of ERBB2/HER2 in Multiple Myeloma as a Predictor of Poor Survival Outcomes.
Literature	GENERIF_SUMMARY	HER2/ ERBB2 Immunohistochemical Expression and Copy Number Status in Ovarian Mucinous Tumors.
Literature	GENERIF_SUMMARY	Clinical Outcome and Morphology-Based Analysis of p53 Aberrant and Mismatch Repair Protein-Deficient Ovarian Clear Cell Carcinoma and Their Association With p16, HER2, and PD-L1 Expression.
Literature	GENERIF_SUMMARY	Immunohistochemical Expression Of Human Epidermal Growth Factor Receptor-2 (Her-2) In Common Salivary Gland Carcinomas.
Literature	GENERIF_SUMMARY	The dynamics of HER2-low expression during breast cancer progression.
Literature	GENERIF_SUMMARY	Frequent upregulation of HER2 protein in hormone-receptor-positive HER2-negative breast cancer after short-term neoadjuvant endocrine therapy.
Literature	GENERIF_SUMMARY	Analysis of prognosis in different subtypes of invasive lobular carcinoma using the Japanese National Cancer Database-Breast Cancer Registry.
Literature	GENERIF_SUMMARY	Clinical significance of CD166 and HER-2 in different types of gastric cancer.
Literature	GENERIF_SUMMARY	HER-2 overexpression in female genital tract clear cell carcinomas
Literature	GENERIF_SUMMARY	A systematic review and meta-analysis for human epidermal growth factor receptor 2 on upper tract urothelial carcinoma patients.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	HER2 Genetic Intratumor Heterogeneity Is Associated With Resistance to Trastuzumab and Trastuzumab Emtansine Therapy in Recurrent High-Grade Endometrial Cancer.
Literature	GENERIF_SUMMARY	Is HER2 ultra-low breast cancer different from HER2 null or HER2 low breast cancer? A study of 1363 patients.
Literature	GENERIF_SUMMARY	Rare HER2 L796P missense mutation promotes the growth and oncogenic signaling in breast cancer cells.
Literature	GENERIF_SUMMARY	Exosomal Linc00969 induces trastuzumab resistance in breast cancer by increasing HER-2 protein expression and mRNA stability by binding to HUR.
Literature	GENERIF_SUMMARY	The role of Her-2 in penile squamous cell carcinoma progression and cisplatin chemoresistance and potential for antibody-drug conjugate-based therapy.
Literature	GENERIF_SUMMARY	Characterisation of luminal and triple-negative breast cancer with HER2 Low protein expression.
Literature	GENERIF_SUMMARY	Mesothelin expression in gynecologic carcinosarcoma
Literature	GENERIF_SUMMARY	Molecular landscape of ERBB2/HER2 gene amplification among patients with gynecologic malignancies; clinical implications and future directions.
Literature	GENERIF_SUMMARY	Assessment of HER1 (rs11543848) and HER2 (rs1136201) polymorphism and their association with colorectal cancer susceptibility in Khyber Pakhtunkhwa, Pakistan.
Literature	GENERIF_SUMMARY	The genomic and immune landscapes of gastric cancer and their correlations with HER2 amplification and PD-L1 expression.
Literature	GENERIF_SUMMARY	Site-specific glycosylation analysis of epidermal growth factor receptor 2 (ErbB2)
Literature	GENERIF_SUMMARY	Characterizing the Genomic Landscape of the Micropapillary Subtype of Urothelial Carcinoma of the Bladder Harboring Activating Extracellular Mutations of ERBB2.
Literature	GENERIF_SUMMARY	Mitochondrial HER2 stimulates respiration and promotes tumorigenicity.
Literature	GENERIF_SUMMARY	HER2-positive is an independent indicator for predicting pathological complete response to neoadjuvant therapy and Ki67-changed after neoadjuvant chemotherapy predicts favorable prognosis in Chinese women with locally advanced breast cancer.
Literature	GENERIF_SUMMARY	ERBB2-amplified lobular breast carcinoma exhibits concomitant CDK12 co-amplification associated with poor prognostic features.
Literature	GENERIF_SUMMARY	miRNA-449c-5p regulates the JAK-STAT pathway in inhibiting cell proliferation and invasion in human breast

Main Category	Category	Terms
		cancer cells by targeting ERBB2.
Literature	GENERIF_SUMMARY	Potential Role of Nrf2, HER2, and ALDH in Cancer Stem Cells
Literature	GENERIF_SUMMARY	HER2 Affects the Biological Behaviours of Bladder Cancer Cells and is Closely Associated with the Progression and Prognosis of Bladder Cancer.
Literature	GENERIF_SUMMARY	EGFR and ERBB2 Exon 20 Insertion Mutations in Chinese Non-small Cell Lung Cancer Patients
Literature	GENERIF_SUMMARY	Targeting HER2/HER3 co-mutations in metastatic breast cancer
Literature	GENERIF_SUMMARY	Frequency and Nature of Genomic Alterations in ERBB2-Altered Urothelial Bladder Cancer.
Literature	GENERIF_SUMMARY	Human epidermal growth factor receptor-2 expression and subsequent dynamic changes in patients with ovarian cancer.
Literature	GENERIF_SUMMARY	Exercise-induced beta2-adrenergic Receptor Activation Enhances the Antileukemic Activity of Expanded gammadelta T-Cells via DNAM-1 Upregulation and PVR/Nectin-2 Recognition.
Literature	GENERIF_SUMMARY	The prevalence and clinical significance of HER2 expression in prostate adenocarcinoma.
Literature	GENERIF_SUMMARY	Human Epidermal Growth Factor Receptor 2 Overexpression/Amplification in Primary Ovarian Endometrioid Carcinoma.
Literature	GENERIF_SUMMARY	Immunohistochemical Human Epidermal Growth Factor Receptor 2 (HER2) Expression Pattern in Gastric Adenocarcinomas in a Nigerian Tertiary Hospital.
Literature	GENERIF_SUMMARY	Her2 amplification, Rel-A, and Bach1 can influence APOBEC3A expression in breast cancer cells.
Literature	GENERIF_SUMMARY	Mechanism of ERBB2 gene overexpression by the formation of super-enhancer with genomic structural abnormalities in lung adenocarcinoma without clinically actionable genetic alterations.
Literature	GENERIF_SUMMARY	HER2 overexpression in urothelial carcinoma with GATA3 and PPARG copy number gains.
Literature	GENERIF_SUMMARY	Promotion of stem cell-like phenotype of lung adenocarcinoma by FAM83A via stabilization of ErbB2.
Literature	GENERIF_SUMMARY	[Clinicopathological features of breast cancer with HER2 low expression
Literature	GENERIF_SUMMARY	A susceptibility gene signature for ERBB2-driven mammary tumour development and metastasis in collaborative cross mice.
Literature	GENERIF_SUMMARY	A Comparison of Immunohistochemical Expression of Epidermal Growth Factor Receptor and Human Epidermal Growth Factor Receptor 2 in Dental Follicles

Main Category	Category	Terms
		with Different Radiographic Sizes.
Literature	GENERIF_SUMMARY	Targeting HER2 in solid tumors
Literature	GENERIF_SUMMARY	SON is an essential RNA splicing factor promoting ErbB2 and ErbB3 expression in breast cancer.
Pathways	KEGG_PATHWAY	EGFR tyrosine kinase inhibitor resistance
Pathways	KEGG_PATHWAY	Endocrine resistance
Pathways	KEGG_PATHWAY	Platinum drug resistance
Pathways	KEGG_PATHWAY	MAPK signaling pathway
Pathways	KEGG_PATHWAY	ErbB signaling pathway
Pathways	KEGG_PATHWAY	Calcium signaling pathway
Pathways	KEGG_PATHWAY	HIF-1 signaling pathway
Pathways	KEGG_PATHWAY	PI3K-Akt signaling pathway
Pathways	KEGG_PATHWAY	Focal adhesion
Pathways	KEGG_PATHWAY	Adherens junction
Pathways	KEGG_PATHWAY	Tight junction
Pathways	KEGG_PATHWAY	Pathways in cancer
Pathways	KEGG_PATHWAY	Proteoglycans in cancer
Pathways	KEGG_PATHWAY	MicroRNAs in cancer
Pathways	KEGG_PATHWAY	Pancreatic cancer
Pathways	KEGG_PATHWAY	Endometrial cancer
Pathways	KEGG_PATHWAY	Prostate cancer
Pathways	KEGG_PATHWAY	Bladder cancer
Pathways	KEGG_PATHWAY	Non-small cell lung cancer
Pathways	KEGG_PATHWAY	Breast cancer
Pathways	KEGG_PATHWAY	Gastric cancer
Pathways	KEGG_PATHWAY	Central carbon metabolism in cancer
Pathways	REACTOME_PATHWAY	Signaling by ERBB2
Pathways	REACTOME_PATHWAY	Signaling by ERBB2 in Cancer
Pathways	REACTOME_PATHWAY	SHC1 events in ERBB2 signaling
Pathways	REACTOME_PATHWAY	PLCG1 events in ERBB2 signaling
Pathways	REACTOME_PATHWAY	PIP3 activates AKT signaling
Pathways	REACTOME_PATHWAY	Developmental Biology
Pathways	REACTOME_PATHWAY	GRB7 events in ERBB2 signaling

Main Category	Category	Terms
Pathways	REACTOME_PATHWAY	Downregulation of ERBB2:ERBB3 signaling
Pathways	REACTOME_PATHWAY	Signal Transduction
Pathways	REACTOME_PATHWAY	Disease
Pathways	REACTOME_PATHWAY	GRB2 events in ERBB2 signaling
Pathways	REACTOME_PATHWAY	PI3K events in ERBB2 signaling
Pathways	REACTOME_PATHWAY	Negative regulation of the PI3K/AKT network
Pathways	REACTOME_PATHWAY	Generic Transcription Pathway
Pathways	REACTOME_PATHWAY	PI3K/AKT Signaling in Cancer
Pathways	REACTOME_PATHWAY	Constitutive Signaling by Aberrant PI3K in Cancer
Pathways	REACTOME_PATHWAY	Semaphorin interactions
Pathways	REACTOME_PATHWAY	Sema4D in semaphorin signaling
Pathways	REACTOME_PATHWAY	Sema4D induced cell migration and growth-cone collapse
Pathways	REACTOME_PATHWAY	Axon guidance
Pathways	REACTOME_PATHWAY	Diseases of signal transduction by growth factor receptors and second messengers
Pathways	REACTOME_PATHWAY	RAF/MAP kinase cascade
Pathways	REACTOME_PATHWAY	MAPK family signaling cascades
Pathways	REACTOME_PATHWAY	MAPK1/MAPK3 signaling
Pathways	REACTOME_PATHWAY	ERBB2 Regulates Cell Motility
Pathways	REACTOME_PATHWAY	PI5P, PP2A and IER3 Regulate PI3K/AKT Signaling
Pathways	REACTOME_PATHWAY	RNA Polymerase II Transcription
Pathways	REACTOME_PATHWAY	Gene expression (Transcription)
Pathways	REACTOME_PATHWAY	ERBB2 Activates PTK6 Signaling
Pathways	REACTOME_PATHWAY	Signaling by PTK6
Pathways	REACTOME_PATHWAY	Downregulation of ERBB2 signaling
Pathways	REACTOME_PATHWAY	Transcriptional regulation by the AP-2 (TFAP2) family of transcription factors
Pathways	REACTOME_PATHWAY	TFAP2 (AP-2) family regulates transcription of growth factors and their receptors
Pathways	REACTOME_PATHWAY	Intracellular signaling by second messengers
Pathways	REACTOME_PATHWAY	Signaling by Non-Receptor Tyrosine Kinases
Pathways	REACTOME_PATHWAY	Signaling by Receptor Tyrosine Kinases
Pathways	REACTOME_PATHWAY	Constitutive Signaling by Overexpressed ERBB2

Main Category	Category	Terms
Pathways	REACTOME_PATHWAY	Drug-mediated inhibition of ERBB2 signaling
Pathways	REACTOME_PATHWAY	Signaling by ERBB2 KD Mutants
Pathways	REACTOME_PATHWAY	Drug resistance in ERBB2 KD mutants
Pathways	REACTOME_PATHWAY	Resistance of ERBB2 KD mutants to trastuzumab
Pathways	REACTOME_PATHWAY	Resistance of ERBB2 KD mutants to sapitinib
Pathways	REACTOME_PATHWAY	Resistance of ERBB2 KD mutants to tesevatinib
Pathways	REACTOME_PATHWAY	Resistance of ERBB2 KD mutants to neratinib
Pathways	REACTOME_PATHWAY	Resistance of ERBB2 KD mutants to osimertinib
Pathways	REACTOME_PATHWAY	Resistance of ERBB2 KD mutants to afatinib
Pathways	REACTOME_PATHWAY	Resistance of ERBB2 KD mutants to AEE788
Pathways	REACTOME_PATHWAY	Resistance of ERBB2 KD mutants to lapatinib
Pathways	REACTOME_PATHWAY	Signaling by ERBB2 ECD mutants
Pathways	REACTOME_PATHWAY	Signaling by ERBB2 TMD/JMD mutants
Pathways	REACTOME_PATHWAY	Drug resistance in ERBB2 TMD/JMD mutants
Pathways	REACTOME_PATHWAY	Nervous system development
Pathways	REACTOME_PATHWAY	Developmental Cell Lineages
Pathways	REACTOME_PATHWAY	Developmental Cell Lineages of the Integumentary System
Pathways	REACTOME_PATHWAY	Developmental Lineages of the Mammary Gland
Pathways	REACTOME_PATHWAY	Developmental Lineage of Mammary Gland Luminal Epithelial Cells
Pathways	REACTOME_PATHWAY	Developmental Lineage of Mammary Gland Myoepithelial Cells
Pathways	REACTOME_PATHWAY	Developmental Lineage of Mammary Stem Cells
Pathways	WIKIPATHWAYS	miR targeted genes in muscle cell
Pathways	WIKIPATHWAYS	Leptin signaling
Pathways	WIKIPATHWAYS	Prolactin signaling
Pathways	WIKIPATHWAYS	Glioblastoma signaling
Pathways	WIKIPATHWAYS	Polycystic kidney disease pathway
Pathways	WIKIPATHWAYS	Mammary gland development puberty stage 2 of 4
Pathways	WIKIPATHWAYS	Mammary gland development pregnancy and lactation stage 3 of 4
Pathways	WIKIPATHWAYS	Bladder cancer
Pathways	WIKIPATHWAYS	Extracellular vesicle mediated signaling in recipient cells

Main Category	Category	Terms
Pathways	WIKIPATHWAYS	Focal adhesion
Pathways	WIKIPATHWAYS	RAC1 PAK1 p38 MMP2 pathway
Pathways	WIKIPATHWAYS	Pathways affected in adenoid cystic carcinoma
Pathways	WIKIPATHWAYS	Physico chemical features and toxicity associated pathways
Pathways	WIKIPATHWAYS	Endometrial cancer
Pathways	WIKIPATHWAYS	Non small cell lung cancer
Pathways	WIKIPATHWAYS	Breast cancer pathway
Pathways	WIKIPATHWAYS	Pancreatic adenocarcinoma pathway
Pathways	WIKIPATHWAYS	EGF EGFR signaling
Pathways	WIKIPATHWAYS	Head and neck squamous cell carcinoma
Pathways	WIKIPATHWAYS	EGFR tyrosine kinase inhibitor resistance
Pathways	WIKIPATHWAYS	2q11 2 copy number variation syndrome
Pathways	WIKIPATHWAYS	17q12 copy number variation syndrome
Pathways	WIKIPATHWAYS	Cancer pathways
Pathways	WIKIPATHWAYS	Targeted therapy in breast cancer
Pathways	WIKIPATHWAYS	Cyclin dependent kinase 4 6 inhibitors in breast cancer
Pathways	WIKIPATHWAYS	ErbB signaling
Pathways	WIKIPATHWAYS	DNA damage response only ATM dependent
Protein Domains	INTERPRO	Rcpt_L-dom
Protein Domains	INTERPRO	Prot_kinase_dom
Protein Domains	INTERPRO	Ser-Thr/Tyr_kinase_cat_dom
Protein Domains	INTERPRO	Furin-like_Cys-rich_dom
Protein Domains	INTERPRO	Furin_repeat
Protein Domains	INTERPRO	Tyr_kinase_AS
Protein Domains	INTERPRO	Growth_fac_rcpt_cys_sf
Protein Domains	INTERPRO	Kinase-like_dom_sf
Protein Domains	INTERPRO	Tyr_kinase_EGF/ERB/XmrK_rcpt

Main Category	Category	Terms
Protein Domains	INTERPRO	Protein_kinase_ATP_BS
Protein Domains	INTERPRO	Tyr_kinase_cat_dom
Protein Domains	INTERPRO	GF_recep_IV
Protein Domains	INTERPRO	Rcpt_L-dom_sf
Protein Domains	INTERPRO	TM_ErbB1
Protein Domains	INTERPRO	RTK
Protein Domains	INTERPRO	CWI-Assembly_Regulator
Protein Domains	PIR_SUPERFAMILY	TyrPK_EGF-R
Protein Domains	SMART	TyrKc
Protein Domains	SMART	FU
Protein Domains	UP_KW_DOMAIN	Signal
Protein Domains	UP_KW_DOMAIN	Transmembrane
Protein Domains	UP_KW_DOMAIN	Transmembrane helix